India Energy Stack (IES) — Technical Documentation

India Energy Stack

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Getting Started

The **India Energy Stack (IES)** is an initiative of the **Ministry of Power**, Government of India, to build unified digital rails for the power sector — a common set of open specifications that lets any two systems in the sector share verified energy data without bespoke, pair-by-pair integration. **REC** is the nodal agency and **FSR Global** is the knowledge partner.

IES works the way UPI works for payments: it holds no data of its own. Data stays in the systems that already hold it — DISCOM software, metering platforms, vendor databases — and IES specifies how those systems identify each other, find each other, and exchange data in a common, verifiable form.

How IES works — Register, Discover, Exchange

Every IES interaction follows the same three steps:

- **Register** — every participant gets a verifiable digital identity (a W3C DID) and a listing in a shared directory (DeDi). Done once.
- **Discover** — before an exchange, the two systems look each other up, verify each other, and agree terms over the Beckn protocol. No bilateral arrangement needed.
- **Exchange** — data moves over the same signed Beckn channel that Discover established, shaped by published IES schemas built on open standards (e.g. DLMS/COSEM, IEEE 2030.5, OpenADR etc). Where a durable record is needed, the exchange produces a W3C Verifiable Credential the holder keeps — in DigiLocker, for consumers. The IES Exchange flows also support bulk data transfer using both paginated inline transfer or via supported channels like signed urls, SFTP, kafka streams etc.

The specifications cover five building blocks: **Identifiers**, **Registries**, **Exchange**, **Verifiable Credentials**, and the **Security**, **Consent**, **Machine Readable Rules** posture that runs through all of them. IES selects the right open standard for each and publishes a specification that builds on it — IES does not write new standards, and it is not a platform, a database, or a product.

What this Documentation is

This documentation is the **technical reference** for building on IES: the schema definitions with their canonical URLs, per-use-case implementation guides with step-by-step checklists, and role-based adoption pathways. If you are here to build something, go straight to:

- **Schemas Overview** — what each schema is for, in plain language
- **Schemas** — the developer catalog: field references, canonical URIs, versions
- **Use Case Implementation Guides** — end-to-end build instructions with checklists: Consumer Energy Passport, Consumer Meter Digest, Smart Meter Data Exchange, DER Visibility, P2P Energy Transaction
- **Pathways** — where to start, by the kind of organisation you are

Before you build

Whatever you are building, every participant completes the same three prerequisites first: **register your organisation in the DeDi directory, create your `did:web` identity, and stand up your Beckn ONIX adapter**.

Each implementation guide starts its checklist from these steps, and the **Concepts** section walks through each one in detail.

The full story

This reference deliberately stays lean. For a detailed understanding of IES — the concepts, the in-depth architecture, the acceleration and adoption strategy, governance, and the pilot record — read the **IES report** at **REPORT URL TO BE UPDATED**.

Where things stand: the specifications are published and versioned in this repository. Four pilot DISCOMs — PVVNL (Uttar Pradesh), APEPDCL (Andhra Pradesh), DGVCL (Gujarat) and Tata Power (Maharashtra) — completed a 30-day DISCOM Challenge, each building an IES adapter and demonstrating the first four use cases (DER Visibility, Consumer Energy Passport, Consumer Meter Digest, Smart Meter Data Exchange) against the specifications and the pilot sandbox.

Printable version: the deliverable subset of this reference is published as a single PDF — **ies-report.pdf**.

Get in touch

- **IES Secretariat** — `ies.secretariat@fsrglobal.org`
- **REC (Nodal Agency)** — `ies@recindia.com`
- **Issues, discussions, contributions** — github.com/India-Energy-Stack/ies-accelerator

For more information and Updates on IES

- **IES Homepage** — `ies.smartgridobservatory.org`

Schemas Overview

Before the field-by-field technical reference, each IES schema gets one plain-language page here — what it is, who issues it, what standards it follows, and where it still has open questions. This is the same **IES Documentation Template** already used for every use-case guide (sections 1–7 in plain language, a condensed Schedule I/II, Annexures A–C), applied to the schema itself rather than to a use case built on top of it.

Read one of these before the field reference in Schemas if you want the *why* before the *what*.

Schema	What it is	Status
ElectricityCredential v1.2	W3C Verifiable Credential — a consumer connection's static facts and the energy resources behind the meter	Stable — In Pilot
MeterData v0.6	Compact telemetry payload — nine record shapes (eight telemetry profiles plus a shared descriptor object) for smart-meter readings, events and alarms	Stable — In Pilot
MeterDataCredential v0.6	W3C Verifiable Credential wrapping a MeterData payload to attest its provenance	Stable — In Pilot
MeterDataRequest v0.6	Query and authorisation payload for asking a provider for telemetry	Stable — In Pilot
MeterDataRequestCredential v0.1	W3C Verifiable Credential wrapping a MeterDataRequest to prove the requester's authorisation	Stable — In Pilot
ArrFiling v0.5	Structured Aggregate Revenue Requirement filing — DISCOM to SERC	**Work in progress**
OutageNotification v0.1	Planned/unplanned outage record — pull feed + CAP-aligned push	**Work in progress**

The full field-level reference for each of these families lives in the **Schemas** section (directly below this one in the navigation), which also carries the canonical URLs of the machine-readable files (`schema.json`, `context.jsonld`, examples).

Where the P2P and flexibility schemas live (external)

Some IES use cases build on schemas that IES does **not** publish itself — they are maintained upstream in the **Digital Energy Grid (DEG)** project, canonical at schema.nfh.global. If you are looking for the schemas behind **P2P Energy Transaction** (`P2PTrade`, `DEGContract`, `EnergyTradeOffer`, …) or demand-side flexibility (`DemandFlexNeed`, `DemandFlexBuyOffer`, …), you will not find them among the families above — they are mirrored, with a field reference,

in **External Schemas** under the Schemas section. The pages above cover only the schema families IES itself stewards.

How each page is organised

Every page follows the same eleven numbered sections as a use-case guide, minus the use-case-specific extras:

1. **Scope and Purpose** — the problem, in plain words
2. **What It Records / Covers** — what is captured
3. **How Each Item is Identified** — DIDs, identifier patterns
4. **Definitions** — terms and acronyms
5. **Basis of Standards** — the standards this schema follows (BIS -> CEA -> IEC -> IEEE precedence)
6. **Where Indian Standards Do Not Yet Exist** — gaps and the international standards used
7. **The Record(s)** — what the schema actually produces
8. **Schedule I — Field Reference (summary)** — block names, linking to the full auto-generated table
9. **Schedule II** — whether this schema wraps or depends on another
10. **How It Fits Together** — how this schema relates to others and to use cases
11. **Points for Confirmation** — genuinely open questions

- **Annexure A — Standards Referenced**, **Annexure B — Example Payloads**, **Annexure C — JSON Schema** (all linking back to the schema's own files rather than duplicating them)

Where this fits

This section sits directly above the **Schemas** developer catalog in the navigation — the field references and canonical file URLs these pages introduce. The Exchange building block these schemas travel over is covered in **Exchange in depth** under Concepts.

ElectricityCredential

A W3C Verifiable Credential, issued per meter or connection by a distribution licensee, that carries the static facts of a consumer's connection and the energy resources behind the meter.

Field	Value
Applicability	Issued by distribution licensees (DISCOMs); consumed by consumers (holder-bound variant), grid operators and aggregators (bearer variant), banks, subsidy portals and DER marketplaces (as verifiers)
This version	v1.2 replaces every power/energy/voltage scalar field with a `QuantitativeValue {value, unit}` pair and introduces the composable `energyResources[]` hierarchy of seven discriminated kinds, documented in schema.json

---

1. Scope and Purpose

The stakeholder is the distribution licensee (DISCOM) and whoever needs a trustworthy, checkable record of one electricity connection and the assets behind it. Without this schema, a licensee has no common, machine-checkable way to state what is connected at a premises, at what capacity, in what condition — and a consumer, grid operator or aggregator has no way to verify those facts except by trusting an unstructured letter or a manual site visit.

This document explains the ElectricityCredential v1.2 schema. It does not define a new schema — it explains the existing one, which is authoritative and IES-hosted (a verbatim mirror of the upstream `beckn/DEG` specification).

The schema itself is deliberately thin at the top: `attributes.yaml` defines only the credential envelope and the `CustomerProfile` / `CustomerDetails` container shapes directly. Everything else — the `EnergyResource` discriminated union, the `ConsumptionProfile` tariff block, the `IdRef` identity reference, and the base `EnergyCredential` — is pulled in by reference from five sibling schemas (`EnergyCredential/v2.0`, `EnergyResource/v2.1`, `MeterServiceProfile/v1.1`, `CustomerDetails/v1.0`, `IdRef/v1.0`). ElectricityCredential v1.2 is best understood as the assembly point where those building blocks are composed for the electricity distribution use case, not as a monolith that defines every field itself.

2. What It Records / Covers

For one connection, the credential records:

Block	What it records	Source
`customerProfile` (required, non-PII)	A utility customer account (CA) number (`customerNumber`), an optional external identity reference (`idRef`), the `energyResources[]` array (required, minimum one entry), and an optional `consumptionProfiles[]` array	ElectricityCredential v1.2 (`CustomerProfile`)
`energyResources[]`	One or more resources behind the meter — the meter itself, generators (solar, wind, hydro, biogas, CHP, fuel cell), storage (battery/BESS), EV chargers (including vehicle-to-grid), inverters, controllable loads (smart HVAC, water heaters, generic controllable loads), and network equipment (distribution transformers, busbars, feeders, microgrids) — each with make, model, rated and maximum import/export power, commissioning date, location, serial number, an inspection/safety record, and, where relevant, a third-party aggregator enrolment	`EnergyResource/v2.1` (`beckn/DEG`)
`consumptionProfiles[]` (one entry per meter, linked by `meterId`)	Tariff category code, sanctioned load and sanctioned export load, contract maximum demand, billing cycle day, premises type, connection type (single-/three-phase), payment mode (postpaid/prepaid) and service status (active/suspended/closed)	ElectricityCredential v1.2 (`ConsumptionProfile`)
`customerDetails` (optional, PII)	The personally identifiable facts — full name, an optional care-of reference for disambiguating people who share a name in a locality, installation address, service connection date — present only when the credential is issued to the consumer	`CustomerDetails/v1.0` (`beckn/DEG`)

The attribute bag shared by every energy-resource kind (`EnergyResourceCommonAttributes`) explicitly distinguishes

`ratedPower` (manufacturer nameplate value, kept only for backward compatibility) from `maxExport` (maximum power injected to the grid — for a BESS or V2G charger, its maximum discharge rate) and `maxImport` (maximum power drawn from the grid — for a BESS or V2G charger, its maximum charge rate). Both `maxExport` and `maxImport` are defined as always non-negative, so direction is carried by which field is populated, not by a signed value. A real meter entry, for example, carries no power rating at all — only capability and protocol fields — because a meter measures rather than converts power:

```
{
  "id": "did:web:ies.discom.example:assets:meter:MET-UNIT-101",
  "type": "METER",
  "attributes": {"meterCapability": "AMR", "energyDirection": "Forward", ...},
  "parentResources": ["did:web:ies.discom.example:assets:meter:MET-BLDG-001"]
}
```

It records identity, capacity, ratings, status and installation facts. It does not record live meter readings — those belong to a separate telemetry schema.

3. How Each Item is Identified

The credential itself carries an `id` that is a `urn:uuid`. The issuer is identified by `issuer.id`, a `did:web` anchored to the DISCOM's own domain — the examples use both a generic form (`did:web:example-utility.com`) and DISCOM-style domains (`did:web:ies.discom.example`), each paired in `issuer.idRef` with a reference to the state regulator that licensed it (for example `{"issuedBy": "did:web:serc.example", "subjectId": "serc.example:AABPC12345"}` for the DISCOM under the SERC).

Two issuance variants use `credentialSubject.id` differently:

- in the **bearer** variant, `credentialSubject.id` is absent — whoever holds the signed JSON is treated as the subject, so no consumer identifier appears at all;
- in the **holder-bound** variant, `credentialSubject.id` is the consumer's own DID (the shipped examples use a `did:example:` placeholder for the wallet-generated identifier), and `customerProfile.idRef` carries a reference to a separate identity authority (one example references `did:web:ssa.gov` as the issuing authority for a government ID, illustrating the pattern rather than prescribing a specific Indian identity source).

Each entry in `energyResources[]` carries its own `id` — either a `did:web` path under the issuer's domain (for example `did:web:ies.discom.example:assets:meter:MET-UNIT-101`, or `did:web:example-utility.com:assets:solar-plant:DER-SOLAR-001` for a generator) or, where that is more conventional, the meter's own serial number. The schema documents this explicitly as convention rather than a hard rule: "for METER resources the meter serial number is conventional," leaving DISCOMs latitude in how they mint identifiers. Either way, the DISCOM's existing numbering — CIS/CA account numbers, meter serial numbers — is always preserved as an alias inside or alongside the identifier; it is never replaced. `serialNumber` (the manufacturer nameplate value) and `id` (the network-issued identifier) are kept as two distinct fields precisely so one substitution does not erase the other.

Topology between resources is carried by two link arrays rather than by nesting: `parentResources[]` (ids of resources upstream, e.g. the meter or feeder a DER sits behind) and `subResources[]` (child resources, which may be given either as bare id strings or as fully inline `EnergyResource` objects). The shipped submetering example shows this in practice: a building's main meter (`MET-BLDG-001`) is the `parentResources` target for two tenant sub-meters (`MET-UNIT-101`, `MET-UNIT-102`), and a rooftop solar DER in turn names one tenant's sub-meter as its own parent — a three-level chain expressed entirely through parent references, with no nested object structure required.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered.
- **did:web** — a DID method anchored to a domain the participant controls; the DID document (`did.json`) is fetched over HTTPS from that domain.
- **Verifiable Credential (VC)** — a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer.

- **DeDi** — the decentralised registry infrastructure (dedi.global) referenced by the shipped examples' `credentialStatus` block as a revocation registry (`type: dediregistry`), queried by the credential's own UUID.
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers; the example payloads name illustrative DISCOMs rather than real ones.
- **QuantitativeValue** — a `{value, unit}` pair used for every power, energy, apparent-power, reactive-power and voltage field in this schema (`QVPower`, `QVEnergy`, `QVApparentPower`, `QVReactivePower`, `QVVoltage` in the field reference), each mapped to QUDT IRIs via the JSON-LD context (e.g. `kW` -> `qudt:KiloW`, `kWh` -> `qudt:KiloW-HR`).
- **CIM (Common Information Model)** — the IEC data model (IEC 61968 for distribution/metering, IEC 61970 for transmission/generation/DER) that every `energyResources[]` kind is explicitly aligned to at the class level, down to individual attributes (for example `GeneratingUnit.maxOperatingP`, `BatteryUnit.ratedE`, `PowerElectronicsConnection.maxQ`).
- **Bearer credential** — the issuance variant with no `credentialSubject.id`; whoever holds the signed document is treated as its subject.
- **Holder-bound credential** — the issuance variant where `credentialSubject.id` is set to a specific holder's wallet DID.
- **SunSpec DER Model** — a numbered family of standard Modbus/point-map definitions for distributed energy resources (SunSpec Alliance); this schema cites specific model numbers (702, 703, 705, 711) as the source for individual inverter fields.
- **ESPI** — the Energy Services Provider Interface (NAESB REQ.21), a US utility-data standard cited here for the shape of two fields (`energyDirection`, `paymentMode`) where no equivalent Indian standard is named.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. This schema follows, directly and field by field:

Standard	What it governs here
IS 16444	The meter application protocol — the field reference states `DLMS_COSEM` (IEC 62056) is "mandatory for India AMI per BIS IS 16444"
CEA Connectivity Regulations, 2013 (amended 2018) + IEEE 1547-2018 Clause 11	The `inspection` object on every energy resource (asset commissioning and safety inspection at energisation and re-certification)
IEC 61968-9 (CIM, metering)	`EnergyResourceMeter` and common device attributes — `EndDeviceInfo.ratedPower`/`.serialNumber`, `AmiBillingReadyKind` (`meterCapability`), `EndDeviceFunction` (`functions[]`), `FlowDirectionKind` (`energyDirection`, paired with **ESPI NAESB REQ.21**)
IEC 61970-301/302 (CIM, transmission & DER)	Generators, storage, EV chargers, inverters, controllable loads, network equipment — e.g. `GeneratingUnit.maxOperatingP`/`.nominalP`, `BatteryUnit.ratedE`, `PowerElectronicsConnection.maxP/minQ/maxQ/ratedS/inverterMode`, `EnergyConsumer/ConformLoad`, `BaseVoltage.nominalVoltage`
IS 16221 + IEC 61727	`dcArrayCapacity` — DC-side nameplate capacity of a solar array at Standard Test Conditions (industry "kWp"), typically larger than AC-side `maxExport` due to inverter clipping
IEC 61968-1 + GeoJSON RFC 7946 + schema.org PostalAddress	The PII block in `CustomerDetails` — `Customer.name` (`fullName`), `ServiceLocation` (`installationAddress`, `serviceConnectionDate`); the `geo` coordinate pair; optional `address` fields
IEC 62196-2 (+ `CCS1`/`CCS2`, `CHAdeMO`, **GB/T 20234**, **SAE J3400**/NACS)	EV connector types `Type1` (J1772) and `Type2` (Mennekes), alongside the non-IEC industry connectors — a genuinely mixed international field

Standard	What it governs here
ISO 15118-20 + OCPP 2.1 BPT	The `EV_V2G` kind's vehicle-to-grid control protocol

6. Where Indian Standards Do Not Yet Exist

- **Asset commissioning and safety inspection across all DER kinds** — no single Indian standard covers this end-to-end; the schema falls back to IEEE 1547-2018 Clause 11 alongside the CEA Connectivity Regulations 2013 (amended 2018).
- **Third-party aggregator enrolment for flexibility/demand-response** — no Indian standard yet defines this; the `aggregator` object on every resource kind is instead based on IEEE 2030.5 and IEC 61850-7-420 (DER control roles). The schema documents that "controllability" is asset-level and independent of observability — an asset can be observable by an aggregator without being controllable by it.
- **Battery round-trip efficiency measurement** — `roundTripEfficiencyPct` uses IEC 62933-2-1 as its performance test method, and the field reference is explicit that this is distinct both from `stateOfHealthPct` (a cumulative life indicator) and from inverter conversion efficiency.
- **Inverter ride-through categorisation and advanced grid-support functions** — `rideThroughCategory` uses the IEEE 1547-2018 abnormal-operating-performance categories (Category I basic, II enhanced for distribution-connected DER, III advanced for large/transmission-connected DER); `voltVarEnabled` and `freqDroopEnabled` fall back to IEEE 2030.5 and SunSpec DER Models 705 and 711 respectively, with no Indian equivalent named.
- **Demand-response control protocols for smart loads** — `controlProtocol` on `EnergyResourceLoad` draws on OpenADR 2.0b, OCPP 2.0.1, SunSpec Modbus and EEBus — an international, vendor-driven set, not an Indian standard.
- **Prepaid/postpaid billing modality and service-connection lifecycle status** — `paymentMode` and `serviceStatus` are based on CIM's `AmiBillingReadyKind`/ESPI and `UsagePoint.status` respectively; no Indian metering-billing standard is cited as an alternative.

7. The Record(s)

ElectricityCredential v1.2 is a single top-level record: a W3C Verifiable Credential. It is not wrapped by any other credential. It is issued by a DISCOM per meter or connection, and it is comparatively static — it changes when an asset is commissioned, a rating changes, or the connection's tariff terms change, not on every reading.

Within the one credential, `energyResources[]` is a composable hierarchy: every entry is discriminated by its `type` field into one of seven kinds — `EnergyResourceMeter` (`METER`), `EnergyResourceGenerator` (`SOLAR_PV`, `WIND`, `HYDRO`, `BIOGAS`, `CHP`, `FUEL_CELL`, plus the deprecated `SOLAR` alias), `EnergyResourceStorage` (`BESS`, plus the deprecated `BATTERY` alias), `EnergyResourceEVCharger` (`EV_CHARGER`, `EV_V2G`), `EnergyResourceInverter` (`INVERTER`), `EnergyResourceLoad` (`SMART_HVAC`, `SMART_WATER_HEATER`, `CONTROLLABLE_LOAD`), and `EnergyResourceNetwork` (`DT`, `BUS`, `FEEDER`, `MICROGRID`) — each carrying kind-specific attributes on top of the shared `EnergyResourceCommonAttributes` set (make, model, rated/import/export power, telemetry provider, commissioning date, location, serial number, inspection record, aggregator enrolment).

The shipped `example.json` illustrates why the hierarchy needs to be a discriminated union rather than one flat shape: it carries one meter, a `SOLAR_PV` generator (with `dcArrayCapacity` distinct from `maxExport`), a `WIND` generator, and two separate `BESS` entries side by side — one fully specified with `storageType`, `stateOfHealthPct`, `roundTripEfficiencyPct` and an `aggregator` enrolment, the other with only the minimum fields populated, showing that most kind-specific attributes are optional rather than mandated even within the same kind.

Two issuance variants of this one record exist, distinguished only by whether `credentialSubject.id` is set and by what is populated in `customerDetails`:

- the **bearer** variant, used for grid-side, non-PII issuance (the DER Visibility use case);
- the **holder-bound** variant, issued to a consumer's own wallet, documented end-to-end in its own use-case guide, Consumer Energy Passport.

This page describes the schema and its structure; readers who need the holder-bound issuance walkthrough — identity proofing, wallet delivery, selective disclosure — should go to that use-case guide rather than this one.

8. Schedule I — Field Reference (summary)

The schema is built from these logical blocks:

Block	What it contains
Envelope	`id`, `type`, `issuer`, `validFrom`, `validUntil`, `credentialStatus`, `credentialSubject`, `proof` — the standard W3C VC wrapper fields.
CustomerProfile	the required, non-PII block: `customerNumber`, optional `idRef`, the `energyResources[]` array (minimum one entry), and optional `consumptionProfiles[]`.
CustomerDetails	the optional, PII block: `fullName`, optional `careOf`, `installationAddress` (GeoJSON `geo` plus optional `address`), `serviceConnectionDate`.
EnergyResource kinds	seven typed entries (`EnergyResourceMeter`, `EnergyResourceGenerator`, `EnergyResourceStorage`, `EnergyResourceEVCharger`, `EnergyResourceInverter`, `EnergyResourceLoad`, `EnergyResourceNetwork`), each inheriting `EnergyResourceCommonAttributes` and adding its own kind-specific fields, plus the `subResources[]` / `parentResources[]` topology links common to every kind.
ConsumptionProfile	`meterId`, `sanctionedLoad`, optional `sanctionedExportLoad`, `billingCycleDay`, `contractMaxDemand`, `tariffCategoryCode`, `premisesType`, `connectionType`, `paymentMode`, `serviceStatus` — one entry per meter.
QuantitativeValue types	`QVPower` (`W`/`kW`/`MW`), `QVEnergy` (`kWh`/`MWh`), `QVApparentPower` (`kVA`/`MVA`), `QVReactivePower` (`kVAR`/`MVAR`), `QVVoltage` (`V`/`kV`) — the `{value, unit}` pairs used throughout the other blocks.

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json, with each standards-derived field's description prefixed "Based on") is at ElectricityCredential v1.2 — Field reference.

9. Schedule II

Aspect	Detail
Stand-alone?	Not applicable as a wrapping relationship in the usual sense — ElectricityCredential v1.2 is explicitly a **composition**, not a self-contained monolith.
Defines directly	Its own `attributes.yaml` defines only the envelope and the `CustomerProfile`/`CustomerDetails` shapes.
Pulls in by `$ref`	`EnergyCredential/v2.0` (as its base, via `allOf`), `EnergyResource/v2.1` (for the seven-kind discriminated union), `MeterServiceProfile/v1.1` (aliased as `ConsumptionProfile`), `CustomerDetails/v1.0`, and `IdRef/v1.0`.
Wrapped by another credential?	No — it does not wrap, and is not wrapped by, any other top-level *credential*; it remains the one Verifiable Credential issued and verified as a single signed document.

10. How It Fits Together

ElectricityCredential v1.2 is the one schema behind two different use cases, distinguished by issuance variant and audience. As the **Consumer Energy Passport**, it is issued holder-bound to a consumer's wallet, carrying

customerDetails and a consumer-scoped credentialSubject.id, and gives the consumer a portable, verifiable proof of their connection and assets. As **DER Visibility**, it is issued bearer, scoped to a feeder, substation or the licensee as a whole, carrying only energyResources[] with no consumer PII, and gives grid operators and aggregators a checkable view of what is connected on the network.

The shipped examples show a third pattern in between: **parallel metering**, where one premises has two sibling meters under a single customer profile — an import meter (energyDirection: Forward) measuring grid consumption and a separate export meter (energyDirection: Reverse) measuring gross solar feed-in under its own tariff code (FIT-SOLAR-01), with the solar DER's parentResources pointing at the export meter rather than the import meter. This is the same credential and the same schema as the single-meter case; the topology links (parentResources, meterId) are what let one credential represent arrangements from a single domestic meter up to a multi-meter building with tenant sub-metering.

In all cases, the credential carries only static, structural facts; live interval readings are handled by a separate telemetry schema referenced by the same meter and asset identifiers, not by this credential.

11. Points for Confirmation

1. The choice between a did:web path identifier and a bare meter serial number for energyResources[].id is documented in the schema's own field reference as convention ("for METER resources the meter serial number is conventional") rather than a fixed rule — implementers should confirm which their DISCOM will use consistently before onboarding assets. In practice, all three shipped examples use the did:web path form exclusively (e.g. did:web:ies.discom.example:assets:meter:MET-UNIT-101), with the bare serial number appearing only as the trailing path segment (as in did:web:example-utility.com:assets:meter:MET2025789456123) or duplicated in the separate serialNumber field — none of the examples uses an unqualified serial number as the id on its own, so the bare-serial-as-id option remains a documented allowance rather than a demonstrated pattern.
2. The deprecated type aliases SOLAR (-> SOLAR_PV) and BATTERY (-> BESS) remain valid for backward compatibility; consumers of this schema should confirm their own tooling accepts both the deprecated and current values during the transition period.
3. ratedPower is kept on every resource kind for backward compatibility, with maxExport (and, for load-drawing resources, maxImport) preferred; implementers should confirm which field their downstream systems read until ratedPower is fully retired.
4. dcArrayCapacity on solar generators and maxExport are both expressed in the same QVPower unit family (kW/MW), but carry different physical meanings (DC nameplate at STC versus AC-side grid injection limit) — the schema notes this distinction only in prose, not in the unit string itself; implementers should confirm downstream systems do not conflate the two when only one is populated.
5. Several fields central to distributed-energy-resource management — aggregator enrolment, ride-through category, volt-VAR and frequency-droop flags, demand-response control protocol — have no Indian standard behind them at all today (Section 6); as India's own DER/DR regulatory framework matures, these fields may need a follow-up revision once a CEA regulation or BIS standard exists to reference in their place.
6. This schema is a composition over five separately versioned sibling schemas (EnergyCredential/v2.0, EnergyResource/v2.1, MeterServiceProfile/v1.1, CustomerDetails/v1.0, IdRef/v1.0); a version bump in any of those sibling schemas could change ElectricityCredential's effective shape without a corresponding version bump here — implementers should confirm which exact sibling versions are pinned before treating "v1.2" as a complete specification of the wire format.

Annexure A — Standards Referenced

Standard	Scope
IS 16444	DLMS/COSEM (IEC 62056) meter application protocol, mandatory for India AMI
CEA Connectivity Regulations, 2013 (amended 2018)	Asset commissioning and safety inspection
IEEE 1547-2018	DER interconnection: commissioning (Cl. 11), abnormal-operation ride-through categories
IEC 61968-9	CIM — metering (EndDeviceInfo, AmiBillingReadyKind, EndDeviceFunction, FlowDirectionKind, UsagePoint)

Standard	Scope
IEC 61968-1	CIM — customer and service location (Customer.name, ServiceLocation)
IEC 61970-301	CIM — loads and network equipment (EnergyConsumer/ConformLoad, PowerTransformer, BusbarSection, Feeder, BaseVoltage)
IEC 61970-302	CIM — generation, storage, EV chargers, inverters (GeneratingUnit, BatteryUnit, PowerElectronicsConnection)
IS 16221	PV module qualification (DC array capacity, "kWp")
IEC 61727	PV grid interface
ESPI (NAESB REQ.21)	Energy flow direction and billing modality field shapes
SunSpec DER Models 702, 703, 705, 711	Inverter apparent/reactive power, ramp time, volt-VAR, frequency-droop
IEC 62933-2-1	Battery round-trip efficiency test method
IEC 61850-7-420	DER control roles (aggregator enrolment)
IEEE 2030.5	DER aggregator/flexibility control roles; volt-VAR/ frequency-droop fallback
IEC 62196-2	EV connector types (Type 1/J1772, Type 2/Mennekes)
GB/T 20234; SAE J3400 (NACS)	Non-IEC EV connector standards in common use
ISO 15118-20; OCPP 2.1 BPT	Vehicle-to-grid (V2G) bidirectional charging control
GeoJSON RFC 7946; schema.org PostalAddress	Installation address geometry and postal fields

Annexure B — Example Payloads

Example payloads for this schema are at schemas/ElectricityCredential/v1.2/examples/ — a single-meter household with mixed generation and storage (`example.json`), a multi-tenant building with main-meter/sub-meter topology and tenant-level rooftop solar (`example-submetering.json`), and a parallel import/export metering arrangement for a solar feed-in tariff (`example-parallel-metering.json`).

Annexure C — JSON Schema

The machine-readable schema definitions are at schema.json and attributes.yaml.

MeterData

A lightweight, high-throughput telemetry payload schema for exchanging smart-meter readings and events over the IES Data Exchange (Beckn) wire, without cryptographic wrapping.

Field	Value
Applicability	Issued by AMISPs, HES/MDM systems and DISCOMs; consumed by DISCOMs, regulators (SERC/CERC), TSPs and consented third parties over IES Data Exchange
This version	Covers the nine compact profile shapes and the shared Data Descriptor Engine, defined in schema.json

1. Scope and Purpose

The stakeholders are the AMISP or HES/MDM system that holds smart-meter data, and the DISCOM, regulator, TSP or consented third party that needs it. Without a common shape for telemetry, each of these parties would exchange meter readings, billing resets, events and alarms in a bespoke format, and every new integration would require a fresh parser on both sides.

MeterData v0.6 is designed around a specific architectural fact: Head-End Systems handle millions of meters, so the wire format has to be cheap to parse and cheap to store, while billing handoffs need clarity and an auditable proof trail. The schema's own documentation (`UserGuide.md`) frames this as two different "forms" of the same nine record shapes — a compact matrix form for HES-to-MDM bulk telemetry, and an elaborated, self-describing form for MDM-to-billing handoffs where auditability matters more than bytes.

This document explains the **MeterData v0.6** schema — it does not define a new schema; it walks through the schema that already exists at `schemas/MeterData/v0.6/schema.json`.

2. What It Records / Covers

MeterData v0.6 captures, across nine record shapes:

Record shape	What it records	Source
Identity (`BaseProfile`)	Which meter(s), customer(s) and service delivery point(s) a payload is about, via the shared identity fields (`meterRefs`, `customerRefs`, `serviceDeliveryPointRefs`) that every telemetry profile inherits	MeterData v0.6
`CustomerProfile`	Slow-changing customer and service-point metadata — the customer, tariff category, sanctioned load, payment mode, billing cycle day, contract maximum demand, sanctioned export load, and the meters, service delivery points and `Association` records (feeder/DT topology, telemetry provider, commissioning date, DER capacity) on their connection. A `CustomerDetails` sub-object carries the name and is marked a **PII sub-object to be omitted entirely for anonymised or pseudonymous exchange**	MeterData v0.6
`IntervalProfile`	High-resolution interval readings — block load survey data at intervals such as 15 or 30 minutes (`PT15M`/ `PT30M`)	IS 15959 / DLMS-COSEM (OBIS)
`DailyProfile`	Daily accumulated readings — a daily load survey summary (`P1D`), structurally identical to `IntervalProfile`	IS 15959 / DLMS-COSEM
`MonthlyProfile`	Monthly billing resets — cumulative registers, time-of-use buckets and maximum demand at each billing cycle; *structurally forbidden* from the compact matrix form (requires `readings` and `touBuckets` directly) because a monthly snapshot is sparse and highly variable	IS 15959 / DLMS-COSEM

Record shape	What it records	Source
`BillDetails`	Computed billing details — bill amount, bill number, due dates, currency, prepaid balance and payment status, plus a breakdown into `energyCharges`, `fixedCharges` and `otherCharges`	MeterData v0.6 (ISO 4217)
`InstantaneousProfile`	Instantaneous electrical snapshots — voltages, currents, and active/reactive power at one moment in time	IS 15959
`EventProfile`	Event history — diagnostic and tamper events matching IS 15959 codes, each with an integer `eventId`, phase, and duration	IS 15959
`AlarmProfile`	Active alarms — real-time alerts (tamper, low prepayment credit, voltage sag, overload), each with a `status` (`ACTIVE`/`CLEARED`) and `severity` (`CRITICAL`/`WARNING`/`INFO`)	IS 15959

A ninth, non-telemetry shape — `PayloadDescriptorProfile` — carries no readings itself; it is the shared dictionary described in Section 7.

It does not carry a cryptographic signature or proof of provenance — that is deliberately left out of this schema and handled separately (see Section 9). A concrete illustration of how compact the wire format gets: a `DailyProfile` interval row can be as small as

```
{ "id": 0, "payloads": [24512.4, 25134.8] }
```

with the two numbers' meaning (which OBIS code, what unit, import or export) declared once in a shared `PayloadDescriptorProfile` rather than repeated per row.

3. How Each Item is Identified

MeterData uses a generic `Identifier{scheme, value, namespace}` object rather than a single fixed identifier type. The `scheme` field names how the value should be read, and is one of: `METER_SERIAL`, `METER_BADGE`, `MRID`, `OBIS`, `SHORT_CODE`, `CONSUMER_NUMBER`, `SERVICE_DELIVERY_POINT`, `DID`, `ORG`, `OTHER`. Fields that reference an entity by ID (`meterRefs`, `customerRefs`, `serviceDeliveryPointRefs`, `parentResources`, `Customer.id`, `Meter.id`, `ServiceDeliveryPoint.id`) use `IdentifierList` — a non-empty array whose first element is the primary identifier, with any additional elements as alternates under different schemes.

Most schemes carry existing identifiers verbatim — a meter serial number, a consumer number, a service-delivery-point code — reused as-is rather than replaced. Two schemes come from named external standards: `MRID` follows the Common Information Model (IEC 61968/61970), and `DID` follows W3C DID Core where a verifiable digital identifier is used instead of a local account number. In practice, the example payloads consistently use `DID` as the working identifier scheme in the shape `did:web:<discom-domain>:<entity-class>:<local-id>` — for instance `did:web:discom.example:consumers:CONSUMER-ACME-8888`, `did:web:discom.example:assets:meter:METER-GENUS-01`, and `did:web:discom.example:assets:feeder:FDR-11KV-BLR-04` for grid-topology parents — showing the `did:web` naming convention used consistently across consumer, meter, service-delivery-point, feeder and transformer references, not just meters.

Individual readings are named either by their OBIS code (for example `1.0.1.8.0.255`) or by a short human-readable name (for example `kWh imp`); the crosswalk between the two forms lives in a canonical registry file, `IES codes.json`, shipped alongside the schema. That registry currently lists 75 distinct register codes plus a 25-entry IS 15959 event taxonomy (IDs 1–305+, each with a phase and a kind — `occurrence`, `restoration`, `configurationChange` or `control`) and an 8-entry alarm register (e.g. Voltage Sag, Voltage Swell, Over Current, Magnetic Tamper). Each code entry in the registry carries its own `source` citation down to the table or clause — for example the meter serial number

register (`0.0.96.1.0.255`) is sourced to "**IS 15959 Part 1 Table 30; Part 2 Table A12; Part 3 Table 12**", and the manufacturer name register to "**IS 15959 Part 2 Table A12**" — so the crosswalk is not just OBIS-to-label but OBIS-to-clause.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered. MeterData's examples consistently render these under the `did:web` method.
- **DeDi** — the decentralised registry infrastructure (dedi.global) IES uses for namespaces, network membership, public keys and revocation; not an identifier method itself (identifiers stay `did:web`). It is where a `MeterDataCredential`'s `credentialStatus` (revocation) is checked.
- **Verifiable Credential (VC)** — a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer.
- **Beckn Protocol** — the open interaction protocol IES uses for discovery, negotiation and confirmation between parties (search / select / init / confirm / status and their `on_` callbacks); MeterData travels as a payload on this wire during Data Exchange.
- **AMISP** — an Advanced Metering Infrastructure Service Provider, the metering agency that operates smart-meter data collection on a DISCOM's behalf, and a typical issuer of MeterData payloads.
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers, and a typical consumer of MeterData payloads.
- **SERC/CERC** — State/Central Electricity Regulatory Commission, the tariff and licensing regulator for the power sector, and a downstream consumer of MeterData for regulatory reporting.
- **OBIS code** — the object identification system, standardised in India via IS 15959, used to name a specific register or reading, such as `1.0.1.8.0.255`.
- **Data Descriptor Engine** — the mechanism (the `PayloadDescriptorProfile` object, its `payloadDescriptors` and `compactSequences`, described in Section 7) that lets bulk numeric telemetry avoid repeating metadata inline.
- **READING vs. USAGE** — two modes a telemetry value can carry: `READING` is the physical register value at a point in time (strictly increasing for cumulative registers); `USAGE` is the delta or consumed amount over a period. The schema enforces mode-dependent field rules: `openingValue`/`closingValue` on a `Reading` **must only be provided when the associated payload descriptor specifies `reportedMode=USAGE`** — they are structurally excluded otherwise.
- **AccumulationBehaviour** — a register-level classification (`CUMULATIVE`, `DELTA`, `INSTANTANEOUS`, `SUMMATION`, `INDICATING`) distinct from but related to `TelemetryMode`; for example cumulative energy registers use `CUMULATIVE` behaviour and support both `READING` and `USAGE` modes, while Maximum Demand registers use `SUMMATION`/`INDICATING` behaviour and are strictly `USAGE`.
- **Common Information Model (CIM)** — the IEC 61968/61970 data model that defines `MRID`, one of the identifier schemes MeterData reuses.
- **MeterDataCapabilities / MeterDataRequest / MeterDataAuthorisation** — the companion `MeterDataRequest v0.6` schema family (not this schema) through which a Data Provider advertises what it can share, a Data Consumer scopes a query, and a utility issues a signed authorisation grant — the discovery-and-consent machinery that sits in front of a MeterData exchange.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. MeterData ties individual fields to specific clauses rather than citing the standard only at schema level:

Standard	What it governs here
IS 15959	The OBIS-code and event-ID conventions for Indian AMI. `IES codes.json` cites the exact Part and table per register — e.g. meter serial number / device ID -> **Part 1 Table 30 / Part 2 Table A12 / Part 3 Table 12**; other registers cite **Part 2 Clauses 13/14, Tables A1/A4/A8** — tracking which meter category (`A`–`D4`) each applies to

Standard	What it governs here
IS 16444	The overarching AC smart meter specification the profile shapes carry data for
IS 15959 event/diagnostic tables	The `EventProfile`/`MeterEvent` shape and the 25-entry event taxonomy in `IES codes.json` (IDs 1–305+, phase-qualified; occurrence/restoration/ configurationChange/control kinds)

6. Where Indian Standards Do Not Yet Exist

The one gap documented for this schema is the identifier scheme: the `MRID` identifier scheme is sourced from the IEC Common Information Model (IEC 61968/61970) because no Indian standard defines an equivalent network-asset identifier scheme. MeterData reuses `MRID` as one value of its generic `IdentifierScheme` enum rather than inventing a parallel Indian scheme.

More broadly, MeterData's `Meter.serviceKind` field already anticipates gas, water and heat meters (`ELECTRICITY`/`GAS`/ `WATER`/`HEAT`) alongside electricity, but IS 15959 and IS 16444 are electricity-specific — there is no equivalent Indian AMI standard cited in this schema's registry for the other utility kinds, so `serviceKind` values beyond `ELECTRICITY` currently travel without a documented Indian standards basis in this schema.

7. The Record(s)

MeterData v0.6 defines nine record shapes, discriminated by a `profileType` field:

- **CustomerProfile** (`profileType: CUSTOMER`) — slow-changing. Groups a `Customer` object (id, name, `consumerCategory`, `sanctionedLoadKw`, `billingCycleDay`, `paymentMode`, `connectionType`, `contractMaxDemandKw`, `tariffCategoryCode`, `sanctionedExportLoadKw`) with the customer's `ServiceDeliveryPoint`, `Meter` and `Association` records. `customer`, `serviceDeliveryPoints`, `meters` and `associations` are all required. Issued/updated by the DISCOM or AMISP whenever customer or connection details change. The schema's own guide notes this profile can also be issued in a **PII-redacted form** — customer name and account PII omitted, consumers referenced only by public DID — for anonymised grid-topology lookups by TSPs (see `Anonymised_Topology_Example.json`).
- **IntervalProfile** (`INTERVAL`) — high-frequency block load survey data at fixed intervals (e.g. `PT15M`/`PT30M`), required field `intervalPeriod`. Issued by the AMISP or HES on each read cycle, typically HES/MDM-originated and deliberately **omitting `customerRefs`** to decouple technical metering data from the commercial billing domain.
- **DailyProfile** (`DAILY`) — daily accumulated load survey (`P1D` cadence), same shape as `IntervalProfile`. Issued once per day.
- **MonthlyProfile** (`MONTHLY`) — cumulative registers, time-of-use buckets (`touBuckets`) and maximum demand at each billing reset; required fields `timePeriod` and `readings`. Structurally cannot carry `intervals` or `compactSequenceRef` (see Section 2). Issued once per billing cycle — or more than once, if a mid-cycle meter swap or an ad-hoc demand reset produces multiple `MonthlyProfile` records chained under the same consumer account (see `Billing_MeterChange.json` and `MonthlyProfile_MultipleResets.json` in the examples directory).
- **BillDetails** (`BILL_DETAILS`) — computed billing outcome for a cycle (`amountDue` and `timePeriod` required; `billNumber`, `billDate`, `dueDate`, `currency`, `energyCharges`, `fixedCharges`, `otherCharges`, `prepaidBalance`, `paymentStatus` optional). Issued by the billing/CIS system, and may carry `source: CIS_COMPUTED` on its readings to show they were derived rather than metered directly.
- **InstantaneousProfile** (`INSTANTANEOUS`) — a real-time snapshot of electrical quantities at one moment; required fields `timestamp` and `readings`. Issued on demand or on a short polling cycle.
- **EventProfile** (`EVENT`) — a log of `MeterEvent` entries (diagnostic and tamper events, matching IS 15959 codes); required fields `timePeriod` and `events`. Each `MeterEvent` has a required `timestamp` and integer `eventId`, plus optional `eventName`, `phase`, `sequence`, `magnitude` and `duration`. Issued as events occur — the guide is explicit that empty telemetry blocks should not be transmitted; the `events` array should only contain diagnosed instances.
- **AlarmProfile** (`ALARM`) — a log of `MeterAlarm` entries (immediate-state conditions: tamper, low prepayment credit, voltage sag, overload); required fields `timestamp` and `alarms`. Each `MeterAlarm` requires `timestamp`, `alarmId` and `status` (`ACTIVE`/`CLEARED`), with optional `alarmName` and `severity` (`CRITICAL`/`WARNING`/`INFO`). Issued as alarms activate or clear.

- **PayloadDescriptorProfile** (`DESCRIPTOR`) — not a telemetry profile itself; a shared dictionary object carried alongside the other profiles in a payload array. Required fields are `id` and `payloadDescriptorSets`, where each `PayloadDescriptorSet` bundles `payloadDescriptors` (register metadata: `readingType`, optional canonical `obis` code, `unit`, `flowDirection`, `category`, `reportedMode`, `touZone`, `multiplier`, `accuracy`) and `compactSequences` (named column layouts of dense numeric matrices, each `SequenceItem` mapping a `readingType` to one of five `attribute` kinds: `value`, `occurredAt`, `openingValue`, `closingValue`, `validationStatus`). This is the Data Descriptor Engine referred to elsewhere in this page.

Seven of the nine `profileType` records inherit from a common `BaseProfile` — `INTERVAL`, `DAILY`, `MONTHLY`, `BILL_DETAILS`, `INSTANTANEOUS`, `EVENT` and `ALARM`, the metered profiles. `CUSTOMER` and `DESCRIPTOR` do **not**: they declare their own required fields and carry no `meterRefs`, because neither addresses a metering point. `BaseProfile` — as of v0.6 — carries *only* identity and routing fields (`meterRefs` required; `customerRefs`, `serviceDeliveryPointRefs`, `payloadDescriptorSetRef` optional). This is a deliberate architectural tightening: the schema's changelog describes `BaseProfile` as having been "purged of all telemetry-specific properties (`readings`, `intervals`, `timestamp`, `timePeriod`)" so that each derived profile can strictly declare only the fields relevant to its own cadence via JSON Schema `allOf` composition — an `IntervalProfile`, for instance, can never carry a bare `timestamp` or `touBuckets` field, because those belong to other profiles.

Within these profiles, individual readings distinguish `READING` (the physical register value) from `USAGE` (the delta over a period), and each `Reading` or `Override` carries a `validationStatus` (`VALID`/`ESTIMATED`/`MANUAL`/`SUSPECT`/`REJECTED`) and a `source` (`METER`/`HES`/`ESTIMATED`/`MANUAL`/`IMPORT`/`MDM_COMPUTED`/`CIS_COMPUTED`). In the compact matrix form, an `Interval.payloads` array holds positionally-aligned `number`/`string`/`boolean` values against the active `compactSequence`, and a separate sparse `overrides` array — keyed by `descriptorIndex` — is reserved specifically for quality deviance (a `changeMethod` or `failCode` on one bad interval), not for routine per-row metadata like timestamps, which now travel as dense columns instead.

8. Schedule I — Field Reference (summary)

MeterData v0.6 is built from these logical blocks:

Block	What it contains
BaseProfile	shared identity/routing fields (`meterRefs` required; `customerRefs`, `serviceDeliveryPointRefs`, `payloadDescriptorSetRef` optional), inherited by the seven metered profiles only — not by `CustomerProfile` or `PayloadDescriptorProfile`.
CustomerProfile / Customer / ServiceDeliveryPoint / Meter / Association / CustomerDetails	customer, connection and meter-installation metadata, including feeder/DT topology (`parentResources`, replacing older `feederId`/`dtId` fields), lightweight DER capacity fields (`generationCapacityKw`, `storageCapacityKw`) on `Association`, a multi-utility `serviceKind` on `Meter`, and a PII-isolated `CustomerDetails` sub-object.
IntervalProfile, DailyProfile, MonthlyProfile	the three load-survey cadences, plus `TouBucket` for time-of-use segmentation.
BillDetailsProfile (`BillDetails` in the schema)	computed billing outcome with a charges breakdown.
InstantaneousProfile	real-time electrical snapshot.
EventProfile / MeterEvent	diagnostic and tamper event log.
AlarmProfile / MeterAlarm	active alert state.
PayloadDescriptorProfile / PayloadDescriptorSet / PayloadDescriptor / CompactSequence / SequenceItem	the shared descriptor dictionary that powers the Data Descriptor Engine.
Reading / Override / TimePeriod / IntervalPeriod / Interval	the shared reading and time-window primitives used across profiles.
Identifier / IdentifierList	the generic `{scheme, value, namespace}` object and its non-empty-array wrapper, used throughout (Section 3).

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) is at MeterData v0.6 — Field reference.

9. Schedule II

Aspect	Detail
Stand-alone?	Yes — MeterData v0.6 is a stand-alone payload schema; it does not itself wrap or depend on another schema. It carries no wrapper, no `issuer`, and no `proof` block.
Wrapped when?	When a use case needs cryptographic proof of provenance (a signed, durable record rather than a raw telemetry exchange), a MeterData payload is instead wrapped *inside* the separate **MeterDataCredential v0.6** schema.
Wrapper's base	`MeterDataCredential` is itself a subclass of **`EnergyCredential v2.0`** (from the Beckn DEG specification) rather than a bespoke envelope.
Wrapper inherits	`issuer`, `validFrom`/`validUntil`, `credentialStatus` (checked against a DeDi registry for revocation) and `proof` (an Ed25519Signature2020 signature) from `EnergyCredential`.
Wrapper defines	only `credentialSubject.meterData` itself.

10. How It Fits Together

MeterData is the payload that travels on the IES Data Exchange (Beckn) wire whenever raw smart-meter telemetry needs to move between an AMISP, a DISCOM, a regulator or a consented third party — this is its role in the **Smart Meter Data Exchange** use case, where it is the primary telemetry payload. When a consumer instead wants a durable, verifiable record of their own readings — for example to share with a bank or another verifier — the same MeterData payload shape is wrapped inside a **MeterDataCredential**, which is the pattern used in the **Consumer Meter Digest** use case. In both cases MeterData defines what the data looks like; it is the surrounding exchange (Beckn) or wrapper (MeterDataCredential) that determines how it is discovered, requested and, where needed, signed.

The companion **MeterDataRequest v0.6** family (`MeterDataCapabilities`, `MeterDataRequest`, `MeterDataAuthorisation`) sits in front of the exchange: a provider advertises what profile types, OBIS registers, multipliers and modes it supports via a `MeterDataCapabilities` document; a consumer scopes a query against that capability set; and a utility issues a signed `MeterDataAuthorisation` naming the grantee, grantor, validity window, purpose and the granted capability subset — before any `MeterData` payload is returned. The schema's own `UserGuide.md` documents a five-step TSP access flow (Discovery -> Authorisation -> User Consent -> Data Request -> Data Retrieval) that ends with the utility returning a cryptographically signed `MeterData` payload.

A documented "anonymised directory lookup" pattern also uses `CustomerProfile` itself as a PII-redacted topology index: a utility can return a `CustomerProfile` with customer name omitted and consumers referenced only by DID, so a TSP can resolve which meters sit under a given feeder or transformer (via `Association.parentResources`) purely for grid-planning purposes, without ever seeing billing PII.

MeterData also has a documented, not-yet-reconciled overlap with the **ElectricityCredential** schema (see Section 11), since both schemas separately describe some overlapping consumer and DER-capacity concepts for different purposes.

11. Points for Confirmation

1. **Internal version label lags the directory name.** `attributes.yaml`'s `info.version` still reads `0.5.0` and its `info.description` still describes "six compact profile shapes (CUSTOMER, INTERVAL, DAILY, BILLING, INSTANTANEOUS, EVENT)" even though the directory is `v0.6` and the schema now defines nine shapes (adding `MONTHLY`, `ALARM` and `DESCRIPTOR`, and renaming `BILLING` to `BILL_DETAILS`). Reviewers should confirm whether this is a stale label needing a fix, or an intentional versioning convention explained elsewhere.

2. **meterType vs. meterCategory.** MeterData keeps both fields on `Meter`: `meterType` (the meter's communication/technology capability, e.g. AMR/AMI/Prepaid/NetMeter/Bidirectional) and `meterCategory` (the IS 15959 regulatory category, A/B/C/D1–D4). Both are retained deliberately; no consolidation is planned, but reviewers should confirm this is still the intended design.
3. **premisesType vs. consumerCategory.** ElectricityCredential v1.1's `attributes.yaml` defines `premisesType` (Residential/Commercial/Industrial/Agricultural) directly as a source-local field, whereas `v1.0` and `v1.2` reach the same field only indirectly — their `attributes.yaml` source instead references `consumptionProfiles[]` items from an external schema, with `premisesType` appearing only once that reference is bundled/inlined into the compiled schema artifact — while `MeterData` separately defines `consumerCategory` (the utility tariff category, e.g. LT2A/NDS/HT/HT-1).
4. **DER capacity modelling divergence.** `ElectricityCredential v1.2` models every physical DER asset richly via a typed, discriminated `energyResources[]` array (seven kinds: `EnergyResourceMeter`, `EnergyResourceGenerator`, `EnergyResourceStorage`, `EnergyResourceEVCharger`, `EnergyResourceInverter`, `EnergyResourceLoad`, `EnergyResourceNetwork`, each inheriting a common attribute bag), while `MeterData` only carries three lightweight capacity fields — `sanctionedExportLoadKw` (on `Customer`), and `generationCapacityKw` / `storageCapacityKw` (on `Association`). This structural divergence is explicitly documented as pending alignment in a future major version.
5. **serviceKind beyond electricity has no cited Indian standard in this schema.** `Meter.serviceKind` already includes `GAS`, `WATER` and `HEAT` alongside `ELECTRICITY`, but every standard cited in this schema (IS 15959, IS 16444) is electricity-specific. It is unclear whether multi-utility metering under `MeterData` is an intentional forward-looking extension point or an artifact of reusing a shared enum, and whether a separate standards basis is expected before non-electricity `serviceKind` values are used in production.
6. **Address representation in `ServiceDeliveryPoint` diverges between schema and examples.** `attributes.yaml` defines `ServiceDeliveryPoint.address` as a reference to the external Beckn `Address/v2.0` schema and `.geo` as a `GeoJSONGeometry`, but the shipped example payloads (e.g. `CustomerProfile.json`, `Billing_MeterChange.json`) instead place flat `addressLine`, `city`, `postalCode`, `latitude` and `longitude` fields directly on the service delivery point. Since the schema is permissive (extra properties are not rejected), both forms currently validate, but reviewers should confirm which representation is the intended canonical one going forward.

Annexure A — Standards Referenced

Standard	Scope
IS 15959 (Parts 1–3)	OBIS-code and event-ID conventions for Indian AMI; the schema's `IES codes.json` registry cites specific Part/Table/Clause references per register (e.g. Part 1 Table 30 / Part 2 Table A12 / Part 3 Table 12 for the meter serial number register; Part 2 Clause 13 and Clause 14 for others)
IS 16444	AC smart meter specification
IEC 61968 / IEC 61970	Common Information Model (CIM) — source of the `MRID` identifier scheme
W3C DID Core	Basis of the `DID` identifier scheme, used throughout examples under the `did:web` method
W3C VC Data Model 2.0 / Beckn EnergyCredential v2.0	Not used by MeterData itself, but the basis of the companion `MeterDataCredential` wrapper (Section 9)

Annexure B — Example Payloads

Example payloads for each profile type, including edge cases (mid-cycle meter swaps, multiple monthly resets, anonymised topology lookups, OBIS vs. short-code representations, and multi-meter bulk datasets), are available in `schemas/MeterData/v0.6/examples/`.

Annexure C — JSON Schema

The authoritative machine-readable definitions are `schemas/MeterData/v0.6/schema.json` and `schemas/MeterData/v0.6/attributes.yaml`.

MeterDataCredential

A signed, tamper-evident wrapper that attests who delivered a set of smart-meter readings, and that the readings have not been altered since.

Field	Value
Applicability	Issued by data providers (AMISPs, MDM systems, or a DISCOM acting in that role); consumed by DISCOMs, consumers' wallets, and any downstream verifier of delivered telemetry
This version	Wraps a `MeterData v0.6` payload (single profile or array of profiles) inside a W3C Verifiable Credential envelope, defined in `schema.json`

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1. Scope and Purpose

The stakeholders are the parties on either side of a smart-meter data exchange: a provider — an AMISP or MDM system (or a DISCOM performing that role) — that has telemetry to deliver, and a receiver — a DISCOM, a consumer's wallet, or any other downstream system — that needs to trust it. Once meter readings leave the metering system and travel over a network as a message payload, the receiver has no inherent way to know who actually produced the data, whether it arrived unaltered, or whether the provider has since withdrawn it (for example after a meter fault or a privacy request). Without a mechanism for this, every data exchange has to fall back on trusting the channel itself, which does not scale across many providers and receivers.

This document explains the **MeterDataCredential v0.6** schema — it does not define a new schema; it describes the existing one, which wraps a MeterData v0.6 payload in a verifiable envelope so its authenticity and provenance can be checked independently of the channel it travelled over. The schema's own description (in `attributes.yaml`) is explicit that it is used specifically in Beckn **`on_status`** flows: the provider attaches the credential in `dataPayload`, "cryptographically binding the delivered meter data to the provider's identity and the originating `MeterDataRequest`."

2. What It Records / Covers

MeterDataCredential does not itself record meter readings. Structurally it is thin by design — the entire schema-specific surface is one object, `MeterDataCredentialSubject`, with exactly two properties (`id` and `meterData`); everything else is inherited from EnergyCredential v2.0. What it records:

Records	Detail	Source
Issuing provider identity	The `issuer` block (id, name, `licenseNumber`, e.g. `SERC-AMISP-2025-007`), inherited wholesale from EnergyCredential	`EnergyCredential/v2.0`
Subject identity	`credentialSubject.id`, a DID naming the consumer or asset entity the data is about (e.g. `did:web:discom.example:consumers:RR-1234`)	`EnergyCredential/v2.0`
Validity window	`validFrom` / `validUntil`, inherited from EnergyCredential (worked examples use a one-year window)	`EnergyCredential/v2.0`
Revocation status	`credentialStatus`, a pointer to a registry where the provider can mark the credential invalid if the data is later recalled	`EnergyCredential/v2.0`

Records	Detail	Source
Attested payload	`credentialSubject.meterData` — the one field the subject object `requires` — carrying the real MeterData v0.6 profile or set of profiles	MeterData v0.6
Cryptographic proof	`proof`, binding all of the above together so any change to the payload invalidates the signature	W3C VC Data Model

A short illustrative snippet from the customer-profile example makes the wrapping concrete — this is the whole schema-specific contribution, everything above and below it is inherited envelope:

```
"credentialSubject": {
  "id": "did:web:discom.example:consumers:RR-1234",
  "meterData": {
    "@type": "CustomerProfile",
    "profileType": "CUSTOMER",
    "customer": { "name": "Acme Industries Pvt Ltd", "consumerCategory": "NDS", "sanctionedLoadKw": 15.0 },
    "meters": [ { "meterType": "AMI", "meterCategory": "B", "serviceKind": "ELECTRICITY" } ]
  }
}
```

`meterData` accepts either a single profile object or — as in the interval-profile example — an array that opens with a `PayloadDescriptorProfile` (the dictionary defining `readingType`, `unit`, `flowDirection`, `obis` code and column layout for the dense numeric payload that follows) and then one or more data profiles that reference it via `payloadDescriptorSetRef`/`compactSequenceRef`. MeterData v0.6 itself defines **nine** discriminated profile shapes reachable this way (`profileType` is the discriminator): `PayloadDescriptorProfile`, `CustomerProfile`, `IntervalProfile`, `DailyProfile`, `MonthlyProfile`, `BillDetails`, `InstantaneousProfile`, `EventProfile`, and `AlarmProfile` — one more than the eight the MeterData v0.6 per-version README's prose enumerates, because the descriptor profile itself is also a valid array member, not just supporting metadata.

It covers identity, provenance, validity and tamper-evidence of delivered telemetry. It does not cover the meaning of the readings themselves — that is the concern of the MeterData schema it wraps.

3. How Each Item is Identified

The credential's `issuer` is identified by a DID (Decentralised Identifier) — the provider's verifiable digital identifier, resolvable to confirm the issuer and detect tampering. In every worked example this is a `did:web` identifier (`did:web:amisp.discom.example`), with the actual signing key referenced from `proof.verificationMethod` as a DID URL fragment (`did:web:amisp.discom.example#key-1`).

The `credentialSubject.id` is a DID naming the consumer or asset entity the delivered data is about — in the examples, a `did:web` identifier scoped under the issuing DISCOM's own domain (`did:web:discom.example:consumers:RR-1234`), using the same path-based convention as the issuer's own `did:web`.

Revocation status is tracked through a DeDi-hosted registry referenced in `credentialStatus`, rather than through any bespoke IES revocation mechanism. Concretely, the examples show `credentialStatus.type` as `"dediregistry"`, with `id` and `statusListCredential` both pointing at `https://dedi.global/dedi/query|lookup/<issuer-did>/vc-revocation-registry` and `statusPurpose` set to `"revocation"` — i.e. the registry entry is namespaced under the *issuer's* own DID, so each provider effectively runs its own revocation list.

The credential does not mint new identifiers for meters, service points, or readings — those are carried inside the wrapped MeterData payload using its own `Identifier` scheme (`{scheme, value[, namespace]}`, where `scheme` is one of `METER_SERIAL`, `METER_BADGE`, `MRID`, `OBIS`, `SHORT_CODE`, `CONSUMER_NUMBER`, `SERVICE_DELIVERY_POINT`, `DID`, `ORG`, or `OTHER`). In the worked examples, meters, service delivery points and customers are all identified via `DID`-scheme references (e.g. `did:web:discom.example:assets:meter:DISCOM-SM-2025-654321`), keeping the identification model consistent end-to-end even though the credential wrapper itself only ever mints DIDs for issuer and subject.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered.
- **Verifiable Credential (VC)** — a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer.
- **DeDi** — the decentralised registry infrastructure (dedi.global) used for Beckn subscriber registries and credential revocation registries; not an identifier method itself. The schema's revocation registry is queried and looked up at `https://dedi.global/dedi/{query|lookup}/<issuer-did>/vc-revocation-registry`.
- **Beckn Protocol** — the open interaction protocol IES uses for discovery, negotiation and confirmation between parties (search / select / init / confirm / status and their `on_` callbacks). MeterDataCredential specifically rides in the `on_status` callback.
- **AMISP** — an Advanced Metering Infrastructure Service Provider, the metering agency that operates smart-meter data collection on a DISCOM's behalf. In the worked examples the issuer is "DISCOM AMISP – Smart Metering Data Platform" with regulatory licence `SERC-AMISP-2025-007`.
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers.
- **Data Descriptor Engine** — the MeterData v0.6 mechanism (described in that schema's README) that separates a `PayloadDescriptorProfile` dictionary (defining `readingType`, `unit`, `flowDirection`, `obis` code and compact-sequence column layout) from the dense numeric `payloads` arrays that follow it, avoiding repeating metadata on every interval reading.
- **IES Cell** — the governance body being constituted under the Central Electricity Authority (CEA) with representation from across the sector; owns the schema specifications and their versioning.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. MeterDataCredential inherits its envelope rather than redefining it, so the credential-level standard is the W3C VC framework; any Indian- or IEC-standard references for the metering data itself belong to the wrapped MeterData v0.6 payload.

Standard	Role here
W3C Verifiable Credential Data Model 2.0	The credential envelope — issuer, validity period, credential status, proof
W3C DID Core	Identifying issuer and subject
DEG EnergyCredential v2.0	The superclass this schema subclasses (`allOf: [$ref: …/EnergyCredential/v2.0]`), itself a subclass of `beckn: Credential`
Ed25519Signature2020	`proof.type` in the worked examples
IS 15959 / IEC 62056 (OBIS)	*Transitively*, via the inline-wrapped MeterData payload — e.g. `EventProfile` is an "IS 15959 event log"; `meterCategory` (`A`, `B`, `C`, `D1`–`D4`) is the IS 15959 classification; OBIS codes identify physical quantities in `payloadDescriptors`, resolved against `IES codes.json`

The `@context` chain a valid instance must declare is three deep: `https://www.w3.org/ns/credentials/v2` -> `https://schema.nfh.global/EnergyCredential/v2.0/context.jsonld` -> this schema's own `context.jsonld`, each layer adding only what it defines.

6. Where Indian Standards Do Not Yet Exist

No Indian standard governs the verifiable-credential envelope itself, so the W3C VC Data Model 2.0 and W3C DID Core are used directly for issuer identity, signing, validity and revocation. This is the only gap documented for the credential wrapper; the standards applicable to the metering content itself are addressed in the MeterData v0.6 schema, not here (MeterData already anchors event logging and meter categorisation to IS 15959, and physical-quantity coding to OBIS/IEC 62056, so that layer is not a standards gap in the same sense).

The MeterDataCredential family's top-level `README.md` (the version index one directory up from `v0.6/`) already agrees with the v0.6 files on this point: it states DeDi-registry revocation (`credentialStatus.type: dediregistry`), consistent with the actual `attributes.yaml`, compiled `schema.json`, and all three example payloads, which likewise implement `credentialStatus` as a bespoke `"dediregistry"` type pointing at a DeDi lookup URL.

7. The Record(s)

MeterDataCredential defines one record: a Verifiable Credential instance that is issued each time a provider delivers a batch of meter data. It is not a static, long-lived record like a connection or asset credential — a new instance is issued for each delivery, with a validity window scoped to that delivery (one year in the worked examples), and it is superseded rather than amended when fresh data is needed.

The schema's own `required` list at the subject level is deliberately minimal: only `meterData` is required on `MeterDataCredentialSubject` (`id` — the subject DID — is optional at the schema level, though every worked example populates it). This means the schema permits, in principle, a credential attesting a meterData payload without naming a specific subject DID — useful for aggregate or feeder-level data that is not about one consumer, though none of the three example payloads exercise that case; all three show a named consumer subject.

It is issued by the provider — an AMISP, an MDM system, or a DISCOM performing that role — as the response to a confirmed request. Its counterpart on the request side is **MeterDataRequestCredential**, a separate schema (currently at v0.1) that is issued by the requester (typically the DISCOM) to prove authorisation to ask for the data, and is used at `confirm` time — specifically attached in `commitmentAttributes.ies:meterDataRequest` — in the Beckn exchange. MeterDataCredential is issued by the provider to attest what was actually delivered, and is used at `on_status` time, attached in `dataPayload`. Downstream, MeterDataCredential can be re-issued holder-bound to a consumer's own wallet, so the consumer can hold and re-present their verified meter history without going back to the DISCOM each time.

8. Schedule I — Field Reference (summary)

MeterDataCredential is built from two logical blocks:

Block	What it contains
credential envelope	inherited from EnergyCredential v2.0: `issuer` (id, name, licence number), `validFrom`/`validUntil`, `credentialStatus`, and `proof`; none of these are redefined in this schema's own `attributes.yaml` — they arrive entirely via the `allOf` reference.
`credentialSubject`	defined by this schema: the subject `id` (DID of the consumer or asset entity, optional) and `meterData` (required), the attested MeterData v0.6 payload (a single profile object or an array of profile objects, per the `oneOf` in MeterData's own root schema).

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) is at MeterDataCredential v0.6 — Field reference.

9. Schedule II

Aspect	Detail
Stand-alone?	No — MeterDataCredential is not stand-alone. It wraps a MeterData v0.6 payload inside `credentialSubject.meterData`.

Aspect	Detail
Division of labour	The credential provides the verifiable envelope (identity, validity, proof, revocation), and the MeterData schema provides the substance (customer, interval, daily, monthly, bill-details, instantaneous, event, alarm, or descriptor profiles).
How coupled	The `$ref` in `attributes.yaml` points at MeterData's own `attributes.yaml` component directly (`.../MeterData/v0.6/attributes.yaml#/components/schemas/MeterData`), so the two schemas are compiled together rather than loosely coupled.
Validation	Validating a MeterDataCredential instance requires checking both — the credential envelope and the embedded MeterData payload against its own schema.
Inherits from	One level up, the credential envelope itself is inherited from (wrapped by reference to) EnergyCredential v2.0, so a complete validation chain touches three schema documents: EnergyCredential v2.0, MeterDataCredential v0.6, and MeterData v0.6.

10. How It Fits Together

In a Beckn data-exchange flow, a receiver first sends a confirmed request carrying a **MeterDataRequestCredential** to prove it is authorised to ask for specific meter data — the provider's own catalog entry, at `catalog/publish` time, has already advertised both `ies:dataSchema` (pointing at MeterData v0.6, the schema used for response payloads) and `ies:capabilityProfile` (a `MeterDataRequest` describing the maximum scope the provider can serve), plus an `offerAttributes.policy.requiredCredentials` clause naming `MeterDataRequestCredential` as a precondition. The provider (an AMISP, MDM system, or DISCOM) then responds at `on_status` with the requested telemetry wrapped in a **MeterDataCredential**, binding the data to its own identity, a validity window, and a revocation path through the DeDi registry.

The receiving party checks that the credential type is `MeterDataCredential`, that `validUntil` has not passed, that the credential has not been revoked (per `credentialStatus`), and that the embedded `meterData` profiles fall within the scope originally contracted in the `MeterDataRequest` — concretely, per the sibling MeterDataRequestCredential's own README, this means resolving the issuer DID, retrieving the public key at `verificationMethod`, verifying the `proof` signature over the canonical serialisation, and finally validating `meterData` against the MeterData v0.6 schema itself. A monthly-profile example illustrates the kind of payload a receiver would be checking scope against — readings plus time-of-use buckets for a single billing month:

```
"meterData": {
  "@type": "MonthlyProfile", "profileType": "MONTHLY",
  "timePeriod": { "start": "2026-01-01T00:00:00+05:30", "duration": "P1M" },
  "readings": [ { "readingType": "kWh imp", "value": 487.3, "validationStatus": "VALID", "source": "METER"
  ↪  } ],
  "touBuckets": [ { "zone": 1, "readings": [ { "readingType": "kWh imp", "value": 195.2 } ] } ]
}
```

In the **Consumer Meter Digest** use case, this same schema is re-issued holder-bound to the consumer's own wallet DID, letting the consumer re-present their verified meter history to a third party (a bank, marketplace, or installer) without the DISCOM being asked again. More generally, it is the provenance-attestation layer for any Smart Meter Data Exchange flow where the receiver needs proof of where delivered telemetry came from.

11. Points for Confirmation

1. The schema's changelog carries a single entry — v0.6, dated 2026-06-06, "Initial draft — aligns with MeterData v0.6" — while the family README lists the schema as Stable — In Pilot; the changelog has not been extended past that initial entry to record the later status.

2. **Revocation mechanism**: the MeterDataCredential family's top-level version-index README and the schema's own `attributes.yaml`, compiled `schema.json`, and all three example payloads all agree — `credentialStatus` is a `"dediregistry"` type against a DeDi lookup URL.
3. The internal `title`/`version` metadata inside the *wrapped* MeterData schema's own `attributes.yaml` still reads `version: 0.5.0` even though it lives under the `MeterData/v0.6/` directory and is what MeterDataCredential v0.6 references — worth confirming whether this is a versioning oversight in MeterData's own source file (out of scope for this page to fix, since it lives in the read-only MeterData schema directory, but relevant to anyone tracing MeterDataCredential's dependency chain).
4. The revocation flow in practice — how quickly a provider is expected to revoke a MeterDataCredential after a meter fault or privacy request, and how downstream holders (including consumer wallets in the Consumer Meter Digest use case) are notified — is not detailed in the schema files.
5. Provenance scope-checking (that embedded `meterData` profiles fall within what was originally requested) is described as a receiver-side responsibility in both this schema's and MeterDataRequestCredential's documentation; how strictly this is enforced, and by whom, is not yet specified.
6. `credentialSubject.id` (the subject DID) is optional at the schema level even though every worked example populates it — whether an unscoped-subject credential (e.g. for feeder-level aggregate data with no single consumer) is an intended, exercised case, or simply an artefact of the schema not yet marking it required, is not stated.

Annexure A — Standards Referenced

Standard	Scope
W3C VC Data Model 2.0	Verifiable Credential envelope — issuer, validity, credential status, proof
W3C DID Core	Decentralised identifiers for issuer and credential subject
DEG EnergyCredential v2.0 (beckn.io)	Parent credential type MeterDataCredential subclasses; supplies `issuer`/`licenseNumber`, `validFrom`/`validUntil`, `credentialStatus`, `proof`
IS 15959 (via wrapped MeterData v0.6)	Event-log diagnostic/tamper codes (`EventProfile`) and meter regulatory category (`meterCategory`: A, B, C, D1–D4)
IEC 62056 / OBIS (via wrapped MeterData v0.6)	Object identification system for physical-quantity codes in `payloadDescriptors`

Annexure B — Example Payloads

Example credential instances (a customer-profile attestation, an interval-profile attestation with a `PayloadDescriptorProfile`, and a monthly-profile attestation with ToU buckets — all issued by the same worked-example issuer, "DISCOM AMISP") are in schemas/MeterDataCredential/v0.6/examples.

Annexure C — JSON Schema

The authoritative machine-readable definitions are schema.json and attributes.yaml.

MeterDataRequest

The query, capability and authorisation shapes a party uses to ask for smart-meter telemetry — not the telemetry itself.

Field	Value
Applicability	DISCOMs and AMISPs (as providers/grantors); AMISPs, TSPs and other authorised third parties (as requesters/grantees)

Field	Value
This version	Covers the three composable request-layer models — MeterDataCapabilities, MeterDataAuthorisation, MeterDataRequest — defined in `schema.json`

1. Scope and Purpose

The stakeholders are the party that holds smart-meter data — typically a DISCOM or the AMISP acting on its behalf — and the party that wants a defined slice of it, such as another AMISP, a TSP, or a consented third party. Today, before any data can move, the two sides have to work out three things in an ad hoc way: what the provider is even able to serve, who has actually been allowed to ask for it, and exactly what window and scope a given query covers. Without a shared shape for these three things, each integration re-invents its own capability list, its own authorisation record, and its own query format.

This document explains the **MeterDataRequest v0.6** schema. It does not define a new schema — it walks through the existing one, ahead of the auto-generated field reference in the schema's own README.

Version 0.6 is a structural break from v0.5. The v0.5 schema (still on disk for reference) was a single flat `MeterDataRequest` object: a requester listed `resources[]`, a `scope`, a `from`/`duration` window, and an `includeDetails[]` array of `ProfileRequest` objects (`profileType` + free-text `values[]` + `requestedMode`) — with no way to describe what the provider actually supported, and no separate authorisation record at all. v0.6 splits that single object into three composable models — `MeterDataCapabilities`, `MeterDataAuthorisation`, `MeterDataRequest` — so "what can be served," "who is allowed to ask," and "what is being asked for right now" become distinct, independently reusable records rather than one undifferentiated request blob.

2. What It Records / Covers

MeterDataRequest v0.6 is not a Verifiable Credential. It is a `oneOf` payload envelope — the schema's root type (`MeterDataRequest` in `schema.json`, titled "MeterDataRequest Payload") explicitly states a document conforming to it "can be a MeterDataRequest, a MeterDataAuthorisation, or a MeterDataCapabilities document" — and it covers three distinct things:

Document type	What it records	Required fields
`MeterDataCapabilities` — what a provider can serve	The profile types it supports (`CustomerProfile`, `IntervalProfile`, `DailyProfile`, `MonthlyProfile`, `BillDetails`, `InstantaneousProfile`, `EventProfile`, `AlarmProfile`), optionally down to specific registers or readings (`ProfileCapability.readings[]`), which query scopes it can answer (`supportedScopes[]`), and the longest historical window (`maxHistoryDuration`)	`profiles[]` only (scope and history limit optional)
`MeterDataAuthorisation` — who is authorised, for what, how long	A grant naming the authorising entity (`grantor`), the authorised entity (`grantee`), the `purpose` as free text, a `validFrom`/`validUntil` window, and the specific slice of `capabilities` granted	all six: `grantor`, `grantee`, `purpose`, `validFrom`, `validUntil`, `capabilities`

Document type	What it records	Required fields
`MeterDataRequestObject` — the actual query	Which `consumers[]` or `resources[]` it targets (both optional and independent), whether it covers just that resource or its children (`scope`), the `from` start and `duration`, `consumerConsent[]` references, the `authorisation` backing it, the `capabilitiesRequested` slice, and an optional `maxRecordsShared` cap	`from`, `duration`, `capabilitiesRequested`

The profile-type enumeration is drawn from the DLMS-COSEM metering data model; all three document types belong to the MeterDataRequest v0.6 family.

A short real example makes the capability slice concrete. The shipped `Anonymised_Telemetry_Request.json` asks for two DISCOM meters' interval telemetry without naming any consumer at all:

```
"resources": [
  "did:web:discom.example:meters:DISCOM-SM-2025-654321",
  "did:web:discom.example:meters:DISCOM-SM-2025-999999"
],
"scope": "ResourceOnly",
"capabilitiesRequested": {
  "profiles": [{ "profileType": "IntervalProfile",
    "readings": [{ "value": "kWh imp block", "mode": "USAGE" }, { "value": "kWh exp block", "mode": "USAGE"
    ↪  }] }]
}
```

3. How Each Item is Identified

The schema targets consumers and resources by URI/DID: `consumers[]` is a list of consumer URIs or DIDs, and `resources[]` is a list of resource URIs or DIDs — for example meter DIDs or service-point URIs. The real examples use the `did:web` method throughout — `did:web:ies.discom.example:customer:IN-MH-CUST-80`, `did:web:ies.discom.example:meter:IN-MH-MTR-89721`, `did:web:ies.discom.example:discom:DISCOM`, `did:web:ies.discom.example:tsp:ACME-TSP` — showing a consistent `did:web:<domain>:<role>:<local-id>` shape, though the schema itself only constrains these fields to `format: uri` and does not mandate any particular DID method. `MeterDataAuthorisation.grantor` and `.grantee` are likewise URIs/DIDs identifying the authorising and authorised entities. `consumerConsent[]` is looser still — plain `text` strings rather than `uri`-formatted, and the example payload uses a consent identifier (`did:web:ies.discom.example:consent:CN-89021-SIGNED-BY-USER`) rather than a hash, though the field description only calls for "consent references/receipts."

Individual registers within a capability are identified by `ValueCapability.value`, which is deliberately dual-form: either an OBIS code (e.g. `1.0.1.8.0.255`) or a human-readable short code (e.g. `kWh imp`). All six example payloads shipped with this schema use the short-code form exclusively (`kWh imp`, `kWh imp block`, `kVAh imp block`, `V_R`, `I_R`, `MD kW`, `MD kVA`) — none of them exercise the raw-OBIS form in `value`, even though the field description allows it. The authoritative mapping between these short codes and their underlying OBIS codes, permitted profiles, and supported telemetry modes lives outside this schema, in the sibling `MeterData` family's `IES codes.json` registry (see section 5).

The schema does not mint a new identifier scheme of its own; it consumes whatever DID or URI already identifies the consumer, meter, service point, or organisation elsewhere in IES.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered.
- **Verifiable Credential (VC)** — a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer.

- **Beckn Protocol** — the open interaction protocol IES uses for discovery, negotiation and confirmation between parties (search / select / init / confirm / status and their `on_` callbacks).
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers.
- **AMISP** — an Advanced Metering Infrastructure Service Provider, the metering agency that operates smart-meter data collection on a DISCOM's behalf.
- **SERC/CERC** — State/Central Electricity Regulatory Commission, the tariff and licensing regulator for the power sector.
- **MeterDataCapabilities** — the provider's advertised data-access envelope: which profile types and registers it can serve (`profiles[]`, required), which query scopes it supports (`supportedScopes[]`), and the longest history window (`maxHistoryDuration`).
- **MeterDataAuthorisation** — the grant/token recording who authorised whom (`grantor`/`grantee`), for what `purpose`, for how long (`validFrom`/`validUntil`), over a stated `capabilities` slice. All six fields are required.
- **MeterDataRequest** (`MeterDataRequestObject` in the schema definitions) — the actual query payload: target `consumers[]`/`resources[]`, `scope`, `from`/`duration` time window, `consumerConsent[]` references, the backing `authorisation`, the `capabilitiesRequested` slice, and an optional `maxRecordsShared` cap.
- **ProfileCapability** — one entry in `MeterDataCapabilities.profiles[]`, naming a `profileType` (one of the eight enumerated profile types, e.g. `IntervalProfile`) and optionally the specific `readings[]` (registers) supported under it — if `readings` is omitted, the description states "all readings under this profile are supported."
- **ValueCapability** — one entry in `ProfileCapability.readings[]`: the register's `value` (OBIS or short code), its telemetry `mode`, and two rarely-populated optional fields — `multiplier` (a decimal scaling factor, e.g. `0.001` or `1000`) and `accuracy` (an accuracy class or precision value). Only the shipped `Capabilities_Example.json` and `MDM_Capabilities_Example.json` populate `multiplier`/`accuracy` (both `1.0` / `0.5` for interval-profile registers); every other example omits them.
- **ScopeType** — the three-value enum (`ResourceOnly`, `ResourceAndChildren`, `ChildrenOnly`) shared by `MeterDataRequestObject.scope` and `MeterDataCapabilities.supportedScopes[]`, describing whether a query or capability applies to the named resource alone, the resource and everything under it, or only its children (e.g. a feeder's meters but not the feeder itself).
- **TelemetryMode** — the two-value enum (`READING`, `USAGE`) naming whether a register is being asked for as a raw cumulative reading or as a derived usage/consumption figure over the window.
- **OBIS code** — the register identifier used inside `ValueCapability.value` to name a specific meter quantity (e.g. `1.0.1.8.0.255`), drawn from the same code space as IS 15959 / IEC 62056.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. MeterDataRequest v0.6 is a request/authorisation envelope, not a metering data model, so `attributes.yaml` carries no `x-standard` field annotations of its own; its one point of contact with a standard is indirect — `ValueCapability.value` names a register using an OBIS/short code from the same register space the sibling `MeterData` schema follows.

Standard	Role here
IS 15959 ("Data Exchange for Electricity Meter Reading, Tariff and Load Control")	Anchors the register space. The `MeterData v0.6` family's `IES codes.json` gives 75 register entries, each citing a specific IS 15959 part/table/clause — e.g. `0.0.96.1.0.255` (meter serial) -> Part 1 Table 30 / Part 2 Table A12 / Part 3 Table 12; `0.0.1.0.0.255` (clock) -> Part 2 Table A1; current/voltage/energy registers -> Part 2 Clauses 13/14
IEC 62056 (OBIS / DLMS-COSEM)	Second-order reference behind IS 15959 in the order of preference — the international harmonisation of the OBIS code space, not cited directly by this schema

Each register entry also states its valid `profiles[]` and `supportedModes[]` (`READING`/`USAGE`), which this schema's `ProfileCapability.profileType` and `ValueCapability.mode` enums are checked against (see section 7).

6. Where Indian Standards Do Not Yet Exist

No gap is currently documented for this schema. The request/capability/authorisation shapes (which profiles a provider serves, who is authorised, what window a query covers) are an IES-specific query envelope; they do not correspond to a metering data-model area where an Indian standard would otherwise be expected. The one area with any standards content — the OBIS/short-code register space named in `ValueCapability.value` — is already anchored to specific IS 15959 parts, tables and clauses in the `MeterData` family's code registry, not left undocumented.

7. The Record(s)

MeterDataRequest v0.6 defines three related, composable records, all carried as plain JSON payloads rather than credentials:

- **MeterDataCapabilities** — a relatively static record. It is published by a provider (a DISCOM or AMISP) describing what it can serve, and it changes only when the provider's supported profiles, registers, scopes, or history window change. In practice it is advertised as a Beckn catalog entry, so a requester can discover it before asking for anything. The shipped examples show real variety in how granular a capabilities record can be: `Billing_Capabilities_Example.json` advertises a `CustomerProfile` and a `BillDetails` profile with no `readings` at all (meaning "everything under these profiles") alongside a `MonthlyProfile` with four explicit billing registers (`kWh imp`, `kWh exp`, `MD kW`, `MD kVA`) and a ten-year `maxHistoryDuration: "P10Y"`; `MDM_Capabilities_Example.json` instead advertises five profile types including `AlarmProfile` and `EventProfile` (each with no readings listed), a six-register `InstantaneousProfile` naming individual phase voltages/currents (`V_R`, `V_Y`, `V_B`, `I_R`, `I_Y`, `I_B`), all three `supportedScopes` values, and only a three-year `maxHistoryDuration: "P3Y"`.
- **MeterDataAuthorisation** — a grant record with a defined validity window (`validFrom`/`validUntil`). It is issued by the grantor (e.g. the DISCOM) to a grantee (e.g. a TSP) for a stated purpose, and it names a specific `MeterDataCapabilities` subset the grantee is allowed to draw on. It changes whenever access is newly granted, narrowed, or expires. The shipped `Authorisation_Example.json` is concrete about what a real grant narrows down to: the DISCOM (`did:web:ies.discom.example:discom:DISCOM`) grants ACME-TSP (`did:web:ies.discom.example:tsp:ACME-TSP`) a one-year window (`2026-05-01` to `2027-05-01`) for "ESG Carbon Accounting & Trend Analysis" — restricted to just two registers (`kWh imp block` under `IntervalProfile`, `kWh imp` under `DailyProfile`), a single scope (`ResourceOnly`), and a one-year history cap (`P1Y`) — visibly narrower than the provider's full advertised capability set.
- **MeterDataRequest** — the per-query record, issued by the authorised requester each time it wants telemetry. It carries its own capability slice (`capabilitiesRequested`), references or embeds the backing `MeterDataAuthorisation`, and states the specific window, scope and targets for that one query. It is the most frequently created of the three. Both shipped request examples reference their authorisation **by URI rather than embedding it inline** — `Request_Example.json` and `Anonymised_Telemetry_Request.json` both set `"authorisation"` to a full URL pointing at `Authorisation_Example.json`, even though `authorisation` is defined as a `oneOf` accepting either an inline `MeterDataAuthorisation` object or a `uri` string. Neither shipped example exercises the inline-object form, despite it being schema-valid.

These three records relate to each other by direct composition — a `MeterDataRequest` embeds or references a `MeterDataAuthorisation`, which in turn names a `MeterDataCapabilities` subset — and they relate to the wider IES schema set as the request leg that is answered by a separate MeterData response payload.

8. Schedule I — Field Reference (summary)

MeterDataRequest v0.6 is built from five logical blocks:

Block	What it contains
MeterDataCapabilities	a provider's advertised profiles, supported scopes, and maximum history window. Only `profiles[]` is required.
MeterDataAuthorisation	the grantor, grantee, purpose, validity period, and granted capability slice. All six fields (`grantor`, `grantee`, `purpose`, `validFrom`, `validUntil`, `capabilities`) are required.

Block	What it contains
MeterDataRequest (`MeterDataRequestObject`)	the query itself: targets, scope, time window, consent references, authorisation, requested capability slice, and record cap. Only `from`, `duration` and `capabilitiesRequested` are required.
ProfileCapability	the nested detail inside a capabilities block, naming individual profile types (one of eight enumerated values) and, where needed, the specific registers within them.
ValueCapability	the innermost detail: a register's OBIS/short code, telemetry mode, and optional multiplier/accuracy.

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) is at MeterDataRequest v0.6 — Field reference.

9. Schedule II

Aspect	Detail
Stand-alone?	Not applicable as a stand-alone dependency in this direction — MeterDataRequest v0.6 is a plain payload schema; it does not wrap or depend on another schema. But MeterDataRequest is itself depended on.
Optionally wrapped	A `MeterDataRequest` payload may be carried inside a **MeterDataRequestCredential** (schema family `MeterDataRequestCredential v0.1`), a W3C Verifiable Credential (VC Data Model 2.0) used to prove the requester's authorisation at Beckn `confirm` time.
Wrapper's base	That wrapper schema is itself declared a **subclass of `EnergyCredential v2.0`** (the Beckn DEG specification's common energy-credential envelope, expressed in its `attributes.yaml` as `allOf: - $ref: https://schema.nfh.global/EnergyCredential/v2.0`), inheriting `issuer`, `validFrom`/`validUntil`, `credentialStatus` and `proof` from that parent rather than redefining them.

See the dedicated MeterDataRequestCredential overview for that wrapping schema's own field reference.

10. How It Fits Together

MeterDataRequest v0.6 is the request/query leg of the Smart Meter Data Exchange use case (see use-cases/smart-meter-data-exchange/README.md). A provider (DISCOM or AMISP) advertises a `MeterDataCapabilities` catalog entry over Beckn; a requester obtains a `MeterDataAuthorisation` naming the capability slice it may draw on; it then issues a `MeterDataRequest` naming its target consumers or resources, the window it wants, and the capability slice it is asking for. That request is answered by a separate MeterData response payload carrying the actual readings.

Where the requester needs to prove its authorisation cryptographically at Beckn `confirm` time rather than relying on the plain JSON alone, the `MeterDataRequest` is wrapped in a `MeterDataRequestCredential`. The credential's own README describes the check the provider performs on receipt: verify the credential type is `MeterDataRequestCredential`, that `validUntil` has not passed, that `credentialStatus` (checked against the DeDi registry) is not revoked, and that the embedded `MeterDataRequest.resources` fall within the contracted scope. The same credential also carries, in its `resourceAttributes`, an `ies:capabilityProfile` — a `MeterDataRequest`-shaped object describing the *maximum* scope the provider can serve — so the check on `confirm` is really "is the requester's embedded `MeterDataRequest` a subset of the provider's advertised `capabilityProfile`," tying all three of this schema's models (capabilities, authorisation, request) directly into the credential-verification step.

The schema's own semantic validator (`validation/validator.py`, sharing core logic with `MeterData/v0.6/validation/validator.py`) illustrates where structural JSON Schema validation stops and semantic checking begins: it re-parses every `ValueCapability.value` string against the `MeterData` family's `IES codes.json` register table to confirm the register is real, that it is permitted under the stated `profileType` (the validator's own test fixture `invalid_request_profile_mismatch.json` requests `kWh imp` — a register permitted under `DAILY`, `MONTHLY`, and `INSTANTANEOUS` per the code table, but not `IntervalProfile` — under `IntervalProfile`, which the JSON Schema alone cannot catch), that the requested `mode` is one of the register's `supportedModes` (`invalid_request_mode_unsupported.json` requests `READING` mode for a register whose code-table entry only supports `USAGE`), and that any embedded `MeterDataAuthorisation`'s `validUntil` is strictly after its `validFrom` (`invalid_authorisation_expired.json` sets `validUntil` one hour *before* `validFrom`). None of these four checks are expressible in `schema.json` alone — they depend on the external code registry and on comparing two date-time fields against each other.

11. Points for Confirmation

1. The split into three composable models (MeterDataCapabilities / MeterDataAuthorisation / MeterDataRequest) is a structural break from v0.5's single flat request object; confirm downstream consumers have migrated off the v0.5 shape.
2. The relationship between an inline `MeterDataAuthorisation` object and a URI reference to one (both are valid for the `authorisation` field) needs a documented convention for when each form is expected — notably, both shipped examples (`Request_Example.json`, `Anonymised_Telemetry_Request.json`) use the URI form exclusively, so the inline-object path is currently unexercised by any real example, particularly for audit and revocation checking.
3. `ValueCapability.value` accepts either a raw OBIS code or a human-readable short code, but every shipped example uses only the short-code form — worth confirming whether the OBIS form is expected to appear in production traffic or is effectively vestigial for this schema.
4. The semantic checks that actually give `ValueCapability.value` + `mode` + `profileType` their meaning — resolvability against the `MeterData` family's `IES codes.json`, profile-membership, and mode-support — live entirely in an external validator script and a sibling schema's code table, not in `schema.json` itself; worth confirming whether that dependency should be declared explicitly (e.g. as a schema reference or `x-standard` annotation) rather than left implicit in a validation script.
5. The OBIS/short-code touchpoint in `ValueCapability.value` ties this schema to IS 15959 (via the `MeterData` code table's part/table/clause citations), but that link is carried only through the external code registry, not declared as a formal standard reference on this schema itself — worth confirming whether it should be made explicit.

Annexure A — Standards Referenced

Standard	Scope
IS 15959 (Parts 1–3)	Register/OBIS code space used to name individual registers in `ValueCapability.value`; cited at the individual register level (specific Part/Table/Clause per code) in the related `MeterData` family's `IES codes.json`, referenced here only indirectly
IEC 62056	International harmonisation of the OBIS code space underlying IS 15959; second in the fixed IES order of preference, not cited directly by this schema
W3C Verifiable Credentials Data Model 2.0	Basis of the optional `MeterDataRequestCredential` wrapper (a separate schema family) used to carry a `MeterDataRequest` cryptographically at Beckn `confirm` time

Annexure B — Example Payloads

Example payloads for all three composable models are at schemas/MeterDataRequest/v0.6/examples/, including a general capabilities example, a billing-focused capabilities example, an MDM (meter data management) capabilities example with alarm/event profiles, an authorisation example, a full request example, and an anonymised (no-consumer) telemetry request example.

Annexure C — JSON Schema

The JSON Schema and attribute source are at schemas/MeterDataRequest/v0.6/schema.json and schemas/MeterDataRequest/v0.6/attributes.yaml.

MeterDataRequestCredential

A W3C Verifiable Credential that a data requester attaches to a Beckn request to prove it is authorised to ask for smart-meter telemetry.

Field	Value
Applicability	Issued by a data requester (e.g. a DISCOM, or a third-party aggregator such as a Technical Service Provider) that attaches it in `commitmentAttributes.ies:meterDataRequest` at Beckn `confirm`; consumed by the metering data provider (e.g. an AMISP or MDM system) that receives it
This version	Defines the credential envelope and its one substantive field, `credentialSubject`, whose `meterDataRequest` property wraps a `MeterDataRequest` object from the separate **MeterDataRequest v0.6** schema family, as specified in attributes.yaml

1. Scope and Purpose

The stakeholders are a data requester – typically a DISCOM, though `attributes.yaml` explicitly also names a "third-party aggregator" as a possible requester – and a metering data provider – typically an AMISP or an MDM (Meter Data Management) system. In a Beckn data-exchange flow, the provider cannot safely hand over smart-meter telemetry to a seeker based on an unverified claim of identity and scope: it needs to know, cryptographically and without a call back to anyone, who is asking, whether they are still authorised, and exactly what they are allowed to ask for.

The schema's own description states the purpose precisely: the credential "binds the requester's identity to a scoped, time-bounded query expressed as a MeterDataRequest." It exists specifically to satisfy a provider policy that requires "a credential-backed, verifiable request before fulfilling data delivery" – meaning attaching it is not universal practice but something a specific provider's offer can mandate.

This document explains the **MeterDataRequestCredential v0.1** schema. It does not define a new schema; it explains the existing one.

2. What It Records / Covers

The credential records:

Records	Detail	Source
Requesting-entity identity	A DID at both `issuer.id` and `credentialSubject.id`, plus the issuer's human-readable `name` and a regulatory `licenseNumber` (the worked example uses `SERC-DISCOM-2025-001`, a State Electricity Regulatory Commission licence format)	`EnergyCredential/v2.0`

Records	Detail	Source
Scoped data request	An embedded `MeterDataRequest` object (not redefined inline) carrying which `consumers` and `resources` (meter or feeder DIDs) the request covers, a hierarchical `scope`, the `from`/`duration` window, `consumerConsent` references, an `authorisation` (inline `MeterDataAuthorisation` or a URI), the `capabilitiesRequested` (profile types, registers and telemetry mode), and an optional `maxRecordsShared` cap	MeterDataRequest v0.6
VC envelope	The standard verifiable-credential envelope — `issuer`, `validFrom`/`validUntil`, `credentialStatus`, and `proof` — inherited rather than redefined	`EnergyCredential/v2.0` (W3C VC)

It does not carry the telemetry itself. It only authorises a request for that telemetry.

The worked example in `examples/example.json` grounds this concretely: the DISCOM (`did:web:ies.discom.example`, licence `SERC-DISCOM-2025-001`) issues a credential valid for the single month `2026-04-01` to `2026-04-30`, whose embedded `MeterDataRequest` asks – for feeder `did:web:ies.discom.example:feeder:IN-KA-BLR-ZONE-A` and everything under it (`scope: ResourceAndChildren`) over a three-month window (`from: 2026-01-01`, `duration: P3M`) – for four profile types at once: `CustomerProfile` (no reading restriction, i.e. all registers), `IntervalProfile` restricted to `kWh imp block` in `USAGE` mode, `DailyProfile` restricted to `kWh imp` in `READING` mode, and `MonthlyProfile` restricted to `kWh imp` in `USAGE` mode – capped at `maxRecordsShared: 10000`. This shows the credential's core discipline in miniature: a short, hard-expiring validity window (30 days) authorising a much longer historical query (three months of data), scoped down to named registers and explicit reading modes rather than "give me everything."

3. How Each Item is Identified

The requesting entity is identified by a DID, carried both as the credential's `issuer.id` (inherited from EnergyCredential) and as `credentialSubject.id`; `attributes.yaml` marks `credentialSubject.id` with `format: uri` and an explicit description, "DID of the requesting entity (e.g., the DISCOM)." The issuer also carries a `licenseNumber` (inherited from EnergyCredential) alongside its DID, so a provider can tie the DID back to a recognised, licensed entity – in the example, a SERC DISCOM licence number.

Consumers and meter/feeder resources referenced inside the embedded `MeterDataRequest` are identified as DIDs too: `consumers` and `resources` are both typed `array` of `string, format: uri` in the MeterDataRequest v0.6 `attributes.yaml`, and the example instantiates `resources` with a feeder DID (`did:web:ies.discom.example:feeder:IN-KA-BLR-ZONE-A`) rather than an individual meter DID, relying on `scope: ResourceAndChildren` to sweep in every meter under that feeder. `consumerConsent` is typed as an array of plain strings (not `format: uri`), so it is meant to carry consent references or receipt identifiers rather than necessarily being a DID.

This schema does not introduce its own identifier scheme – it reuses the DID-based identifiers already defined for the entities and resources it references. Revocation status is checked against a DeDi registry referenced in `credentialStatus`; the example's `credentialStatus.type` is `dediregistry` and both `id` and `statusListCredential` point at `https://dedi.global/dedi/query/...` and `https://dedi.global/dedi/lookup/...` endpoints scoped to the issuer's own DID (`did:web:ies.discom.example/vc-revocation-registry`), meaning each issuer maintains its own revocation list rather than a shared central one.

4. Definitions

- **DID (Decentralised Identifier)** – a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that

it has not been altered.

- **Verifiable Credential (VC)** – a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer.
- **DeDi** – the decentralised registry infrastructure (dedi.global) used for Beckn subscriber registries and credential revocation registries; not an identifier method itself. Each issuer's revocation registry is addressed by its own DID (e.g. `.../did:web:ies.discom.example/vc-revocation-registry`).
- **Beckn Protocol** – the open interaction protocol IES uses for discovery, negotiation and confirmation between parties (search / select / init / confirm / status and their `on_` callbacks).
- **DISCOM** – a distribution licensee (electricity distribution company) serving retail consumers.
- **AMISP** – an Advanced Metering Infrastructure Service Provider, the metering agency that operates smart-meter data collection on a DISCOM's behalf.
- **MDM** – a Meter Data Management system; `attributes.yaml` names it alongside AMISP as an example data provider.
- **TSP (Technical Service Provider)** – a third-party aggregator that can also act as a requester under this credential per the schema's own description ("e.g., DISCOM, third-party aggregator"); the sibling MeterDataRequest schema's own worked `Authorisation_Example.json` shows the DISCOM (`did:web:ies.discom.example:discom:DISCOM`) as grantor authorising a TSP (`did:web:ies.discom.example:tsp:ACME-TSP`) as grantee, for the stated purpose "ESG Carbon Accounting & Trend Analysis" – illustrating the kind of downstream delegation this credential's `authorisation` field is built to carry.
- **OBIS code** – an IEC-standard register-naming scheme (see Section 5) used inside the embedded `MeterDataRequest.capabilitiesRequested` to name specific meter registers, e.g. `1.0.1.8.0.255` for cumulative imported active energy; the schema also accepts a "short code" alias such as `kWh imp` for the same register.
- **Scope (`ResourceOnly` / `ResourceAndChildren` / `ChildrenOnly`)** – the three-value enum (`ScopeType` in MeterDataRequest v0.6) controlling whether a request against a hierarchy node (e.g. a feeder or DISCOM) covers only that node, that node and everything beneath it, or only the children and not the node itself.
- **Telemetry mode (`READING` / `USAGE`)** – the two-value enum (`TelemetryMode` in MeterDataRequest v0.6) distinguishing a raw register reading (a cumulative meter dial value) from a derived usage/consumption figure (a difference between two readings over an interval).

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. MeterDataRequestCredential v0.1 is a credential envelope; it defines no new engineering/metering standards. No IS or CEA/IEGC citation appears anywhere in this schema family — the envelope is pure W3C/DEG, and the wrapped request payload's only cited standard is the IEC OBIS/DLMS-COSEM register scheme.

Standard	Role here
W3C Verifiable Credential Data Model 2.0	Followed directly for the credential envelope
DEG EnergyCredential v2.0	Superclass (`allOf: - $ref: …/EnergyCredential/v2.0`) supplying issuer, validity window, revocation status and proof — the same inheritance pattern as sibling `ConsumptionProfileCredential` / `GenerationProfileCredential`
IEC 62056 (OBIS / DLMS-COSEM)	*Via the wrapped MeterDataRequest v0.6* — `ValueCapability.value` is "the OBIS code (e.g. 1.0.1.8.0.255) or short code (e.g. kWh imp) representing the register"

The engineering substance this credential authorises access to — registers, profile types, telemetry modes — lives in the wrapped **MeterDataRequest v0.6** schema, whose validator enforces register/profile/mode rules against `IES codes.json` at runtime (e.g. `kWh imp`, defined `accumulationBehaviour: CUMULATIVE`, may not be requested under `IntervalProfile`, which expects the per-interval `kWh imp block`, defined `DELTA`). None of this validation lives inside the credential itself — it is inherited by reference (see section 7).

6. Where Indian Standards Do Not Yet Exist

No gap is currently documented for the credential envelope itself: it is built on the W3C Verifiable Credential Data Model and the DEG EnergyCredential v2.0 base, neither of which has an Indian-standard counterpart to compare against because they are protocol/format concerns, not engineering ones.

The wrapped **MeterDataRequest v0.6** payload is where a gap is more visible on inspection, even though the schema's own files do not name it as an open item: its `ProfileCapability.profileType` enum (`CustomerProfile`, `IntervalProfile`, `DailyProfile`, `MonthlyProfile`, `BillDetails`, `InstantaneousProfile`, `EventProfile`, `AlarmProfile`) and its register-naming convention (OBIS codes per IEC 62056) are drawn entirely from the international DLMS-COSEM metering data model. No IS standard is cited for these profile classes or register codes in the schema source; Indian metering practice (e.g. CEA's Smart Meter functional/communication requirements) is referenced elsewhere in the IES documentation set for meter-side data, but this credential-and-request pairing does not itself point to an IS equivalent for the specific profile/register taxonomy it uses.

7. The Record(s)

MeterDataRequestCredential defines a single record: a signed, time-bounded Verifiable Credential that changes with every request it authorises – it is not a long-lived static record like a connection credential. It is issued by the data requester (e.g. the DISCOM) at the point it wants to ask a provider (e.g. an AMISP or MDM system) for telemetry, and it is presented, not re-issued, each time that same authorisation is used within its validity window. The worked example shows a deliberately short window – 30 days (`2026-04-01` to `2026-04-30`) – authorising a request against a substantially longer historical span (three months back to `2026-01-01`), which illustrates that the credential's *validity* period and the *data window* it requests are two independent time ranges that must both be checked.

It relates to other IES records in two ways:

1. **It wraps** a `MeterDataRequest` object inside `credentialSubject.meterDataRequest` – the actual scoped, time-bounded ask. That object is itself one of three composable models the MeterDataRequest v0.6 schema defines (its root schema is a `oneOf` over `MeterDataRequestObject`, `MeterDataAuthorisation`, and `MeterDataCapabilities`): the request specifies `consumers/resources/scope/from/duration/consumerConsent/authorisation/capabilitiesRequested/maxRecordsShared`; the `authorisation` field can itself be an inline `MeterDataAuthorisation` object (grantor, grantee, purpose, validity window, granted capabilities) or a URI reference to one, giving MeterDataRequestCredential an optional second, nested layer of delegation evidence beyond its own VC proof.
2. **It is the request-side counterpart to MeterDataCredential.** The MeterDataCredential v0.6 README states this relationship explicitly in its own comparison table: MeterDataRequestCredential is issued by the requester and wraps a `MeterDataRequest`, used at Beckn `confirm` to prove the right to ask, with the requesting entity as its subject; MeterDataCredential is issued by the provider and wraps a `MeterData` payload, used at Beckn `on_status` to attest what was delivered, with the consumer/asset as its subject. The two credentials bracket the same exchange from opposite ends.

8. Schedule I — Field Reference (summary)

The schema is built from these logical blocks:

Block	What it contains
inherited **EnergyCredential v2.0 envelope**	`issuer` (`id`, `name`, `licenseNumber`), `validFrom`/`validUntil`, `credentialStatus` (DeDi registry reference), and `proof` (the worked example uses `Ed25519Signature2020`)
credentialSubject (`MeterDataRequestCredentialSubject`) embedded **MeterDataRequest object** (defined in the separate MeterDataRequest v0.6 family, required fields `from`, `duration`, `capabilitiesRequested`)	the DID of the requesting entity (`id`) and its one required, substantive field, `meterDataRequest` `consumers`, `resources`, `scope`, `from/duration`, `consumerConsent`, `authorisation`, `capabilitiesRequested` (itself built from `MeterDataCapabilities` -> `ProfileCapability` -> `ValueCapability`, down to individual OBIS/short-code registers and telemetry modes), and the optional `maxRecordsShared` cap

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) for this credential is at MeterDataRequestCredential v0.1 – Field reference; the fields of the object it wraps are documented at MeterDataRequest v0.6 – Field reference.

9. Schedule II

MeterDataRequestCredential does not stand alone in either direction.

Aspect	Detail
Wraps	a `MeterDataRequest` payload inside its `credentialSubject`
Wrapped by	(a subclass of) the DEG-published EnergyCredential v2.0, from which it inherits its envelope rather than redefining it
Report template	There is no separate Schedule II report template for this schema

10. How It Fits Together

MeterDataRequestCredential is the optional authorisation attachment on the request leg of the Beckn data-exchange flow described in Smart Meter Data Exchange. Concretely, per the schema's own README:

- **Provider side, `catalog/publish`**: the provider advertises `ies:dataSchema` (a URL to the MeterData schema used for response payloads), `ies:capabilityProfile` (a `MeterDataRequest`-shaped object describing the maximum scope the provider can serve – its own data-access envelope), and `offerAttributes.policy.requiredCredentials` declaring that a valid `MeterDataRequestCredential` must be attached at `confirm`.
- **Seeker side, `confirm`**: the seeker includes the signed credential in `commitmentAttributes.ies:meterDataRequest`. The embedded `MeterDataRequest` must be a subset of the provider's advertised `capabilityProfile` – e.g. a seeker cannot request `EventProfile` registers if the provider's capability profile never listed `EventProfile` among its `profiles`.
- **Provider verification**: the provider checks that the credential's type is `MeterDataRequestCredential`, that `validUntil` has not passed, that its status is not revoked (via the DeDi registry named in `credentialStatus`), and that the embedded `MeterDataRequest.resources` and requested registers fall within the scope and register set the provider has actually agreed to serve.
- **Provider side, `on_status`**: the response is attested separately by **MeterDataCredential**, whose embedded `meterData` payload must cover the profile types and time window originally requested. So the two credentials bracket the same exchange from opposite ends: MeterDataRequestCredential proves the right to ask, MeterDataCredential attests what was delivered.

11. Points for Confirmation

1. The schema's own changelog records v0.1 (2026-05-26) only as "Initial draft", while the family README lists it as Stable — In Pilot; the changelog has not been updated to reflect the later status.
2. This credential is optional in the Smart Meter Data Exchange use case – when a provider's offer policy requires it versus when a request may proceed without it is a per-provider policy decision that this document does not resolve.
3. The `consumerConsent` field is typed as a plain string array (not `format: uri`), unlike `consumers` and `resources` which are DIDs – the schema does not specify whether consent entries must be DIDs, opaque receipt hashes, or some other format, leaving consent-evidence interoperability between DISCOMs and TSPs undefined at the schema level.
4. `authorisation` may be an inline `MeterDataAuthorisation` object or a bare URI reference to one; the schema does not specify how a provider is expected to dereference and independently verify a URI-referenced authorisation (e.g. whether it must itself be signed, or how staleness/revocation of that referenced grant is checked), which matters once a TSP-style delegation chain (DISCOM -> TSP) is involved rather than a direct DISCOM-to-provider request.
5. Register-level validation (OBIS/short-code resolution, profile-type restriction, telemetry-mode compatibility) is enforced entirely by the separate `MeterDataRequest v0.6` validator against an `IES codes.json` lookup that lives outside this credential's own files – so a reader of MeterDataRequestCredential alone cannot see which specific

registers or profile-mode combinations are actually valid without also consulting the MeterDataRequest v0.6 validation tooling.

6. No IS or CEA/IEGC standard is cited anywhere in this schema family for the profile-type taxonomy or register-naming scheme it relies on (both are IEC/DLMS-COSEM-derived); whether an Indian-specific smart-meter data standard should eventually take precedence per the IES order of preference is not addressed in the source.

Annexure A — Standards Referenced

Standard	Scope
W3C Verifiable Credential Data Model 2.0	The credential format MeterDataRequestCredential is expressed in
DEG EnergyCredential v2.0	The base schema MeterDataRequestCredential subclasses for its envelope (issuer, validity, status, proof)
IEC 62056 (DLMS-COSEM object identification / OBIS codes)	The register-naming scheme used by the wrapped MeterDataRequest's `ValueCapability.value` field (e.g. `1.0.1.8.0.255`)

Annexure B — Example Payloads

A complete example instance is at schemas/MeterDataRequestCredential/v0.1/examples (the DISCOM requesting Q1 2026 customer, interval, daily and monthly profiles for all meters under feeder `IN-KA-BLR-ZONE-A`). The wrapped `MeterDataRequest` object's own worked examples – covering a plain request, an inline authorisation grant, and provider capability declarations – are at schemas/MeterDataRequest/v0.6/examples.

Annexure C — JSON Schema

The machine-readable definitions are at schema.json and attributes.yaml. The wrapped object's own definitions are at MeterDataRequest v0.6 schema.json and attributes.yaml.

ArrFiling

A structured, machine-readable version of the Aggregate Revenue Requirement (ARR) filing a distribution licensee submits to its state electricity regulator.

Field	Value
Applicability	Issued (filed) by a distribution licensee (DISCOM); consumed by the State Electricity Regulatory Commission (SERC) or Joint Electricity Regulatory Commission that receives the filing.
This version	Covers the ARR filing payload only — filing metadata, per-fiscal-year basis, and line items — defined in ArrFiling v0.5 schema.json. Work in progress: the line-item `category`/`subCategory` enumeration is an IES superset still converging across SERCs and is expected to gain additions in this version.

1. Scope and Purpose

The stakeholders are the DISCOM that prepares and files its revenue requirement, and the SERC that receives, reviews and rules on it. Today, every state uses its own spreadsheet format for this filing, so a SERC that wants to compare cost structures or trends across DISCOMs — or across states — cannot do so without manually re-keying each filing, and a DISCOM operating (or a body analysing DISCOMs) across states cannot reuse a single extract from its finance system.

The schema's own design notes (in `attributes.yaml`) name four concrete format mismatches it was built to unify: multi-year tariff (MYT) wide-format filings (the working example is a DISCOM's filing to its SERC), annual single-year filings that carry historical SERC-approved data going back over a decade (the working example is a filing to a Joint ERC), differing line-item granularity (some DISCOMs file one consolidated O&M line, others split it into Employee Costs / Admin & General / Repair & Maintenance; "Return" appears in different filings as Return on Net Fixed Assets, Return on Equity, or Return on Capital Base depending on which SERC regulation applies), and line items that evolve across years as new cost heads appear and old ones get consolidated. ArrFiling v0.5 is built to hold all four patterns in the same shape rather than forcing every filer into one fixed table.

This document explains the **ArrFiling v0.5** schema: it does not define a new schema, it explains the existing one.

2. What It Records / Covers

For one regulatory filing, ArrFiling captures:

Records	Detail	Source
Filing identity	A filing reference number (`filingId`, e.g. `"SERC/ARR/DISCOM/MYT/2024-29"`), the filing date, and `filingType`: `MYT` (multi-year control-period, mixing actual and proposed years), `ANNUAL` (single year or a run of historical approved data), `TRUE_UP` (reconciliation of actuals against previously approved amounts), or `REVISED` (an amended filing)	ArrFiling v0.5
Who is filing, and to whom	The licensee's full name (`licensee`) and short code (`licenseeCode`), the state (`stateProvince`), and the receiving `regulatoryCommission` (e.g. `"SERC"`)	ArrFiling v0.5
Period covered	`controlPeriodStart` / `controlPeriodEnd`, populated only for MYT filings (e.g. `"FY 2024-25"` to `"FY 2028-29"`)	ArrFiling v0.5
How amounts are expressed	Currency (fixed to `INR`) and a mandatory `unitScale` (`CRORE`, `LAKH`, or `ABSOLUTE`) applied to every amount — set once for the whole document, no per-line override	ArrFiling v0.5
Status in the regulatory process	`DRAFT`, `SUBMITTED`, `UNDER_REVIEW`, `APPROVED`, or `REJECTED`	ArrFiling v0.5
Regulatory form traceability	An optional `formReference` at filing level (e.g. `"Form 1"`) and line-item level (e.g. `"Form 1.1"`), plus a free-text `notes` array for footnotes and regulatory-order references that make a cost line legally precise	ArrFiling v0.5
Per-year detail	For each fiscal year, a `yearType` (`BASE_YEAR`, `CONTROL_PERIOD`, `HISTORICAL`) crossed with an `amountBasis` (`AUDITED`, `APPROVED`, `PROPOSED`, `TRUED_UP`, `NOT_FILED`) — it is the *combination* that carries meaning; `NOT_FILED` marks placeholder years inside a filed control period that have no data yet	ArrFiling v0.5

Records	Detail	Source
Cost, revenue and adjustment lines	Each with a `category` (`VARIABLE`, `FIXED`, `INCOME`, `SUB_TOTAL`, `ARR`, `ADJUSTMENT`), an optional `subCategory` (twelve values, from `POWER_PURCHASE` and `O_AND_M` through `NON_TARIFF_INCOME` and `NET_ARR`), the `head` and `particulars` as on the form, the `amount` (nullable, or negative for a credit/deduction), and roll-up into subtotals via `componentOf` and a human-readable `formula` on subtotal/ARR rows	ArrFiling v0.5

A concrete illustration of the amount and adjustment conventions, drawn from a real MYT filing's proposed year: `power-purchase-cost` (`VARIABLE`, `11673.43`), an `ADJUSTMENT` row `fppca-adjustment` with amount `-1434.67` ("Prior-Year Fuel Cost Adjustment"), and another ADJUSTMENT row `pp-cost-adjustment` at `-60.95` ("Power Purchase Cost Adjustment per Tariff Order for FY 2025-26") — both negative because they reduce the supply-cost subtotal they roll into, exactly as the schema's field description for `amount` specifies ("Negative values represent credits, deductions, or adjustments that reduce ARR").

3. How Each Item is Identified

ArrFiling does not use decentralised identifiers (DIDs) for the objects inside the payload itself. Instead it uses two plain, stable reference schemes grounded directly in regulatory practice:

- The **filing** as a whole carries a `filingId` — the regulatory filing reference number already used by the DISCOM and SERC — plus a `licenseeCode` identifying the DISCOM. Sampled `filingId` values follow a state-specific convention already in use on paper, e.g. `"SERC/ARR/DISCOM/MYT/2024-29"` for a multi-year filing and `"SERC/ARR/DISCOM/HISTORICAL/2012-2024"`-style references for a long-run historical annual filing; the schema does not impose its own ID grammar, it just carries whatever reference the regulator already assigns.
- Each **line item** carries a `lineItemId` — a stable, kebab-case identifier (for example `power-purchase-cost`, `interest-working-cap`) that is reused across fiscal years so the same cost line can be tracked and compared year over year — and a `serialNumber` matching the display order on the regulatory form itself.
- A line item's position in the roll-up hierarchy is carried by `componentOf` (the `lineItemId` of the parent subtotal or ARR row it feeds into) rather than by nesting. Sampled data shows `componentOf` can point either at an intermediate `SUB_TOTAL` (e.g. five network-cost lines all set `componentOf: network-and-sldc-cost`) or, in a simpler historical filing with only one subtotal, directly at the final `ARR` row (`non-tariff-income`'s `componentOf` is `net-arr` with no intervening subtotal). `formula` is only meaningful on `SUB_TOTAL` and `ARR` rows, and even there it is not universally populated: one sampled historical year's `total-revenue-requirement` subtotal carries no `formula` string at all, while the `net-arr` row above it does (`"total-revenue-requirement + non-tariff-income"`) — so `formula` should be treated as an optional, human-readable annotation for cross-checking, not a guaranteed computable expression on every aggregation row.

The payload schema itself defines no DID scheme for these identifiers. However, when an ArrFiling payload is exchanged over Beckn Data Exchange, the envelope carrying it is signed using the DISCOM's `did:web` key, so the filing as a whole is tied to a verifiable issuer identity at the transport level even though the identifiers inside the payload are plain regulatory references.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered.
- **did:web** — a DID method anchored to a domain the participant controls; the DID document (`did.json`) is fetched over HTTPS from that domain. Used here to sign the Beckn envelope carrying an ArrFiling payload.

- **Verifiable Credential (VC)** — a tamper-evident, cryptographically signed digital document (W3C VC Data Model 2.0) that a verifier checks offline using the issuer's published key, with no callback to the issuer. ArrFiling itself is not a VC; only the transport envelope is signed.
- **Beckn Protocol** — the open interaction protocol IES uses for discovery, negotiation and confirmation between parties (search / select / init / confirm / status and their `on_` callbacks). ArrFiling payloads travel over Beckn Data Exchange.
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers; the filer of an ArrFiling.
- **SERC/CERC/JERC** — State/Central Electricity Regulatory Commission, or a Joint Electricity Regulatory Commission serving more than one state/UT, the tariff and licensing regulator for the power sector; the recipient of an ArrFiling. The schema's `regulatoryCommission` field is deliberately free text (not an enum) precisely so it can name any of these interchangeably.
- **MYT (Multi-Year Tariff)** — a filing type (`filingType: MYT`) spanning a control period of several years, mixing actual and proposed data; carries `controlPeriodStart`/`controlPeriodEnd`.
- **True-up** — a filing type (`filingType: TRUE_UP`) that reconciles actuals against amounts previously approved by the SERC; at the fiscal-year level, the corresponding `amountBasis` value is `TRUED_UP`.
- **ARR (Aggregate Revenue Requirement)** — the total revenue a DISCOM is entitled to recover through tariffs; the final net line item in the schema's line-item hierarchy, carrying `category: ARR` and (per the sampled examples) a `subCategory` of `NET_ARR`.
- **FPPCA** — Fuel and Power Purchase Cost Adjustment: a category of `ADJUSTMENT` line item the schema explicitly calls out (alongside true-up corrections and other pass-throughs) as the kind of correction the `ADJUSTMENT` category exists to hold.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. ArrFiling does not follow a metering or asset-modelling standard in this order — it is a regulatory-filing structure, not a technical/electrical schema, and `attributes.yaml` carries no `x-standard` annotations. Instead:

Standard	Role here
SERC tariff regulations (per state)	The filing form, cost categories and timetable (each commission's MYT / Annual Tariff Regulations). ArrFiling's `category`/`subCategory` is a superset across these, absorbing cross-DISCOM variation — e.g. "Return" as Return on Net Fixed Assets, Return on Equity, or Return on Capital Base depending on the SERC (`return-on-nfa` and `return-on-equity` both appear as `lineItemId` values)
Beckn Protocol	Discovery and delivery on the wire
W3C VC Data Model 2.0 / W3C DID Core	Issuer key and signature on the Beckn envelope that carries the payload

6. Where Indian Standards Do Not Yet Exist

The JSON shape that carries an ARR filing is itself an IES specification — no Indian or international standard predates a machine-readable ARR filing format. The gap this schema fills is structural, not electrical: SERCs' cost taxonomies vary state to state, so ArrFiling's `category`/`subCategory` enums are an IES-defined superset, with per-SERC mapping notes tracked in the schema. The two sampled example filings make the scale of this variation concrete rather than abstract: one DISCOM's own historical filings shift cost-head granularity release over release — splitting O&M into `employee-costs` / `admin-general-expenses` / `repair-maintenance` in an early year, then consolidating to a single `o-and-m` line from the next year onward, while later years introduce line items (`incentive-disincentive`, `om-sharing-gain-loss`) that simply did not exist on the form in earlier years. This is the live area of work referenced in the Status row above, rather than a case of an international standard substituting for a missing Indian one.

7. The Record(s)

ArrFiling defines one filing built from three nested, relational record types:

- **ArrFiling (root)** — issued once per regulatory submission by the DISCOM. It changes each time the DISCOM files, resubmits or revises (`filingType` covers MYT, ANNUAL, TRUE_UP and REVISED). It is the parent record: it carries the filing's identity, the parties, the currency/scale, and the overall status as the filing moves through the regulatory process (draft through submitted, under review, approved or rejected). Only `objectType`, `filingId`, `licensee`, `regulatoryCommission`, `currency`, `unitScale`, and `fiscalYears` are required at this level — `filingDate`, `filingType`, `licenseeCode`, `stateProvince`, `controlPeriodStart/End`, `status`, `formReference` and `notes` are all optional, which is what lets a sparse historical filing (no `filingDate`, no control period, no notes) validate just as cleanly as a fully-annotated MYT filing.
- **ArrFiscalYear** — one entry per fiscal year covered by the filing, nested inside ArrFiling (`fiscalYears`, minimum one entry). It changes as a year moves from proposed, to trued-up, to historical across successive filings. Only `fiscalYear`, `amountBasis` and `lineItems` are required; `yearType` itself is optional, so a filing can assert what the numbers *are* (audited, approved, proposed…) without always asserting *where in the tariff cycle* the year sits.
- **ArrLineItem** — one entry per cost, revenue, subtotal or adjustment row, nested inside each fiscal year (`lineItems`, minimum one entry per year). Line items are the most granular and most frequently referenced record: their `lineItemId` is designed to stay stable across years so the same cost line can be compared and tracked over time, and they reference each other (via `componentOf` and `formula`) to express how subtotals and the final ARR are computed. Only `lineItemId`, `category`, `head` and `amount` are required — `subCategory`, `particulars`, `formReference`, `serialNumber`, `componentOf` and `formula` are all optional, which is exactly why the two sampled filings can carry line items of very different richness (one filing's notes tie individual lines to specific corrigenda; the other's historical years carry none of that annotation) while both validate against the same schema.

A worked example of the full roll-up chain, taken from a real MYT filing's proposed year: five `FIXED`/`NETWORK_COST` lines (`transmission-cost`, `sldc-cost`, `net-distribution-cost`, `pgcil-expenses`, `uldc-charges`) each set `componentOf: network-and-sldc-cost`; the `network-and-sldc-cost` `SUB_TOTAL` row then carries `formula: "transmission-cost + sldc-cost + net-distribution-cost + pgcil-expenses + uldc-charges"`. A second group of eleven lines (power purchase, distribution cost attributable to supply, interest lines, provisions, and true-up/FPPCA adjustments) all set `componentOf: supply-cost`, and `supply-cost` carries the corresponding eleven-term formula. Finally the `aggregate-arr` row (`category: ARR`) sets `formula: "network-and-sldc-cost + supply-cost"` — a two-level roll-up expressed entirely through plain-text `lineItemId` references rather than nested objects.

ArrFiling relates to other IES records at the transport level: it is carried as the payload inside a Beckn Data Exchange envelope, alongside signed contract and receipt artefacts that together form the non-repudiable audit trail of the submission — but those envelope-level artefacts are not part of the ArrFiling schema itself.

8. Schedule I — Field Reference (summary)

ArrFiling is built from three logical blocks:

Block	What it contains
ArrFiling	global filing metadata: filing identity, licensee and regulator, control period, currency/scale, status, and the list of fiscal years
ArrFiscalYear	one fiscal year's classification (base year, control-period year, or historical) and amount basis (audited, approved, proposed, trued-up, or not filed), plus its list of line items
ArrLineItem	one cost, revenue, subtotal or adjustment row: its category and sub-category, its form heading and description, its amount, and how it rolls up into other line items

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) is at ArrFiling v0.5 — Field reference.

9. Schedule II

Wrapping / dependency	Detail
Not applicable	ArrFiling is stand-alone. It is a payload schema in its own right and does not wrap or depend on another IES schema.
Not a Verifiable Credential	It defines no VC wrapper; when it travels over Beckn Data Exchange, the signature is applied to the transport envelope, not to a credential structure defined by ArrFiling itself.

10. How It Fits Together

ArrFiling is the payload at the centre of the DISCOM Regulatory Filing use case: a DISCOM prepares an ArrFiling v0.5 object from its finance-system extract, and delivers it as a signed, schema-validated payload over Beckn Data Exchange to the SERC, which ingests it directly rather than receiving a PDF or spreadsheet. In that exchange, the DISCOM acts as the provider (BPP) and the SERC as the consumer (BAP) — the reverse of the roles in the Smart Meter Data Exchange use case, though the same identity, signing and registry infrastructure is reused on both sides. ArrFiling itself carries only the filing content; the surrounding Beckn envelope (discovery, contract, signed receipt) is what makes the submission a non-repudiable regulatory record.

Both sampled examples publish their JSON-LD wiring the same way — an `@context` pointing at the canonical `context.jsonld` URL and an `@type` of `ArrFiling` alongside the payload's own `objectType: ARR_FILING` — so a filing can be resolved as linked data by generic tooling while still validating as a plain JSON document against `schema.json`. Supporting workbooks, where a filing needs them, ride as separate signed datasets in the same exchange, linked back to the filing by `filingId`, rather than being embedded inside ArrFiling.

11. Points for Confirmation

1. **Cost-category convergence.** The `category`/`subCategory` enumeration on `ArrLineItem` is an IES superset across SERC tariff regulations, still converging; further additions are expected as more states are mapped, rather than a settled final list. The two sampled filings alone already use different `lineItemId` naming for functionally similar rows across years from the same DISCOM (`return-on-nfa` vs `return-on-equity`; `employee-costs`/`admin-general-expenses`/`repair-maintenance` vs a single `o-and-m`), which is a live illustration of the convergence problem rather than a settled mapping.
2. **Per-SERC category mapping.** Each DISCOM's mapping from its own finance-system categories to the schema's `category`/`subCategory` enums needs to be agreed and signed off (regulatory affairs and finance) before the schema can be relied on for cross-DISCOM comparison.
3. **`formula` is optional and inconsistently populated even within one filing.** One sampled historical filing has a `SUB_TOTAL` row (`total-revenue-requirement`) with no `formula` string at all, while the `ARR` row above it does carry one. Any tooling built to recompute or audit roll-ups from `formula` needs to handle the case where a subtotal's formula is simply absent, rather than assuming every `SUB_TOTAL`/`ARR` row is computably self-describing.
4. **Large workbook attachments.** The schema and its examples cover the structured filing itself; how supporting workbooks are packaged and referenced alongside an ArrFiling payload in the Beckn envelope is a delivery-layer detail outside this schema's scope, not fully specified here.

Annexure A — Standards Referenced

Standard	Scope
SERC tariff regulations (state-specific MYT / Annual Tariff Regulations)	Define the filing form, cost categories and filing timetable that ArrFiling's enumerations are built to cover as a superset
Beckn Protocol	Discovery, negotiation and delivery of the ArrFiling payload over IES Data Exchange
W3C VC Data Model 2.0; W3C DID Core	Issuer key (`did:web`) and signature applied to the Beckn envelope carrying the payload

Annexure B — Example Payloads

Example ArrFiling payloads are at schemas/ArrFiling/v0.5/examples/. The bundled file contains two contrasting filings: an MYT control-period filing (six fiscal years spanning audited, approved, proposed and not-yet-filed years, with a rich multi-level subtotal roll-up and adjustment lines) and a long-run historical annual filing (thirteen consecutive fiscal years from the same DISCOM, showing how line-item granularity and naming shift release over release even without any change to the schema itself).

Annexure C — JSON Schema

The authoritative schema definitions are schemas/ArrFiling/v0.5/schema.json and schemas/ArrFiling/v0.5/attributes.yaml.

OutageNotification (WIP)

A machine-readable payload a distribution licensee publishes to describe one planned or unplanned electricity outage, usable both as a public outage-map feed and as a push alert to affected consumers.

Field	Value
Applicability	Issued by a distribution licensee (DISCOM), or a system acting on its behalf (an OMS, MDMS, SCADA, or a real-time data-acquisition system such as the DISCOM's RTDAS); consumed by outage-map/website feeds, subscribed consumers, mass-alerting networks, and downstream reliability reporting
This version	v0.1 covers the outage notice itself — class, cause, affected assets and area, timing, impact, response, public-facing text, and provenance back to the detecting smart-meter signal — defined in the companion JSON Schema file `schema.json`. It is grounded directly in a real DISCOM system of record (an Outage Management System, `discom.example` in the examples) and a published public artifact (a DISCOM's "Detail of Planned Shutdown" PDF), not designed in the abstract.

1. Scope and Purpose

The stakeholders are the distribution licensee that must tell the public and its own field teams what is out, where, and for how long, and the consumers, subscribers and outage-map viewers who currently cannot get that information in a consistent, machine-readable form. A licensee's outage-management system (OMS) knows the moment a feeder trips or a shutdown is scheduled, but without a shared schema that knowledge stays locked inside the OMS — it cannot be republished automatically as a map feed, pushed as a subscriber alert, or linked back to the smart-meter reading that first detected it.

The design note behind this schema is explicit that no single dominant "outage publication" schema exists internationally — mature practice *composes* several standards, each contributing one layer (data model, publication pattern, push envelope, integration flow, reliability reporting). OutageNotification v0.1 is the result of that composition exercise, decoded from three authoritative inputs: the live OMS screens of a DISCOM (Breakdown-Shutdown entry, Main Dashboard, Workforce Supply-Complaints dashboard, Breakdown Details, Lineman-Allocation), the DISCOM's published planned-shutdown PDF (confirmed to be a flattened public projection of the same OMS data — nothing in the PDF is not already in the OMS), and the global standards survey below.

This document explains one schema, **OutageNotification v0.1**. It does not define a new schema — it is a plain-language walkthrough of the schema as published, ahead of the auto-generated field reference.

2. What It Records / Covers

For a single outage event, the schema captures:

Records	Detail	Standard / source
Identity	A stable id reused across updates so a later message can refer back to the original. In the source data this is a Breakdown/Shutdown ID (e.g. `DISCOM-BD-6357257`); one ID commonly spans *multiple feeders*, which is why `affectedAssets` is an array	DISCOM OMS
Classification	`outageClass` (PLANNED, BREAKDOWN, SCHEDULED_ROSTERING, EMERGENCY_ROSTERING — from the OMS "Down Info" value set) and a separate lifecycle `status` (SCHEDULED, ACTIVE, PARTIALLY_RESTORED, RESTORED, CANCELLED); restoration is a *status transition*, not a class	DISCOM OMS
Message semantics (push delivery)	`msgType` (ALERT/UPDATE/ CANCEL), `references` (prior notice ids updated or cancelled), and `severity` (EXTREME/SEVERE/ MODERATE/MINOR/ UNKNOWN), plus a coarse local `category` kept separate from the analytic `cause.category`	OASIS CAP v1.2 (`alert/msgType`, `info/severity`)
Cause	Three layers: a standardized `category` + `subcategory`; the vendor's own fault-reason `code` carried verbatim; an asset/voltage `faultType` (e.g. `11KV`); and free text (`text`) that may be localized	IEEE 1782-2022 (§4.4/§4.5)
Network context	The DISCOM's organisational hierarchy — **Discom -> Zone -> Circle -> Division -> Subdivision -> Substation -> Feeder/DT** (the `Subdivision` level surfaced only from the live OMS screens)	DISCOM OMS
Affected assets	Which feeders, DTs, substations or line segments are out (`assetLevel`), each with a `voltageLevel` (33kV/ 11kV/LT/DT), an optional `consumerCategory` (from the DISCOM "Feeder Type" field), a `meterRef` to the reporting smart meter, and optional map geometry	OutageNotification v0.1
Affected area	The geography affected, as free text (`areaDesc`), named administrative areas (villages/wards), and geometry	OASIS CAP v1.2; GeoJSON (RFC 7946)

Records	Detail	Standard / source
Impact	How many customers are affected (`customersAffected`), and which connection points are de-energized or confirmed energized, as `UsagePoint` references — reserved for an authenticated consumer tier, not the public feed	IEC 61968-3 (CIM)
Timing	The outage window as one `TimePeriod{start, duration}`, the estimated restoration time, the actual restoration time once known (feeds reliability indices), and an SLA target in minutes (`slaTargetMinutes`)	IEEE 1366
Field response	Whether partial back-feed has been given via an alternate feed, the resource/repair status (PENDING, RESOURCE_ALLOCATED, IN_PROGRESS — after the OMS lineman-allocation states), and a linked complaint count	DISCOM OMS
Public-facing text	A localized headline, description and instruction for the consumer, mirroring CAP's `info` block, for both the outage-map feed and the push alert	OASIS CAP v1.2; BCP-47 (`language`)
Provenance	Where the outage was auto-detected — a link to the originating smart-meter signal (raw digital-input port and vendor status code) and the specific alarm(s) that triggered it	MeterData v0.6
Extensions	A namespaced space for a DISCOM's own additional fields, e.g. `{"discom": {"breakdownId": "6357257", "downType": "FEEDER"}}`, preserving vendor-local concepts without promoting them into the shared schema	OutageNotification v0.1

A real planned-outage example from the schema's own examples directory shows several of these blocks together — one Breakdown ID spanning four rural 11kV feeders under a single substation, with polygon geometry for the whole affected area and a point geometry for one feeder:


```
"id": { "scheme": "OTHER", "value": "DISCOM-BD-6357257", "namespace": "discom.example" },
"outageClass": "PLANNED", "status": "SCHEDULED",
"affectedAssets": [
  { "id": { "scheme": "FEEDER", "value": "barehta" }, "assetLevel": "FEEDER",
    "voltageLevel": "11kV", "consumerCategory": "RURAL",
    "geo": { "type": "Point", "coordinates": [77.45, 29.41] } },
  { "id": { "scheme": "FEEDER", "value": "Gosaiganj" }, "assetLevel": "FEEDER", "voltageLevel": "11kV",
    ↪ "consumerCategory": "RURAL" }
],
"timing": { "period": { "start": "2026-06-22T06:43:00+05:30", "duration": "PT5H" }, "slaTargetMinutes": 240
 ↪ }
```

3. How Each Item is Identified

OutageNotification reuses a single `Identifier{scheme, value, namespace}` object throughout, rather than issuing DIDs for every asset. The notice itself carries a stable `id` of this form, reused across UPDATE and CANCEL messages so a viewer or subscriber can tell that a later message concerns the same outage — in the live examples this is literally the DISCOM's breakdown/shutdown ID carried as `scheme: OTHER, value: "DISCOM-BD-6562139", namespace: "discom.example"`, unchanged from ALERT through the RESTORED UPDATE.

Assets (feeders, substations, distribution transformers), meters, and network-context entries (e.g. `discom`, `substation`) are referenced the same way — as an `Identifier`, ideally a stable GIS feature id or an MRID where one exists, so the record can be joined to a map or to another IES record. The `Identifier.scheme` enum itself is of mixed origin: `MRID` is drawn from CIM (IEC 61968/61970), `DID` from W3C DID Core, and the remaining scheme values (`METER_SERIAL`, `GIS_FEATURE_ID`, `SERVICE_DELIVERY_POINT`, `FEEDER`, `SUBSTATION`, `DT`, `CONSUMER_NUMBER`, `ORG`, `OTHER`) are local vendor or DISCOM master keys carried as-is — the schema's own inline `$comment` on this field says so explicitly: *"MRID borrowed from IEC 61968/61970 (CIM); DID borrowed from W3C DID Core; remaining values local."*

The design note's GIS-readiness principle sharpens why this matters: every asset carrying a stable `Identifier` (ideally `GIS_FEATURE_ID` or `MRID`) lets a consumer that already holds the DISCOM's own GIS layer *join on id and pull geometry from GIS* rather than have the notice restate it — keeping notices small while GIS stays the single authoritative geometry source. Inline `geo` on the asset or affected area is the fallback for a consumer with no GIS of its own.

The schema does not mint DIDs for outages, assets or consumers itself; where a DID is already the identifier of record (for example a DISCOM's own `did:web`), that identifier is expressed through the same `Identifier` object with `scheme = DID`.

4. Definitions

- **DID (Decentralised Identifier)** — a W3C-standard verifiable digital identifier (W3C DID Core) that names one thing (an organisation, a person, an asset) and can be checked electronically to confirm the issuer and that it has not been altered.
- **DISCOM** — a distribution licensee (electricity distribution company) serving retail consumers.
- **AMISP** — an Advanced Metering Infrastructure Service Provider, the metering agency that posts the real-time feeder energized/de-energized signal (derived from a substation meter's digital-input port) into the DISCOM's MDMS/OMS.
- **RTDAS** — Real-Time Data Acquisition System, a DISCOM-specific detection system named directly in the `provenance.source` enum alongside MDMS/OMS/SCADA/AMI/MANUAL.
- **SERC/CERC** — State/Central Electricity Regulatory Commission, the tariff and licensing regulator for the power sector.
- **CIM (Common Information Model)** — the IEC 61968/61970 family's shared data model for electricity network objects. The 2021 edition of IEC 61968-3 split the older, single `OutagesAndFaults` package into five: `PlannedOutages`, `PlannedOutageNotifications`, `UnplannedOutages`, `EquipmentFaults`, and `LineFaults` — this schema's `outageClass` (PLANNED vs BREAKDOWN) mirrors that same planned/unplanned split, and `UnplannedOutages`' lists of energized/de-energized `UsagePoints` are the direct model for this schema's `impact.energized`/`impact.deEnergized`.
- **CAP (Common Alerting Protocol)** — OASIS CAP v1.2 (also referenced as ITU-T X.1303), the global all-hazard public-warning standard, explicitly inclusive of power outages. This schema borrows its `alert -> info -> area` envelope: message type, references between messages, severity, the affected area (with polygon/circle geometry), and headline/instruction text.
- **ODIN** — the Outage Data Initiative Nationwide, a US Department of Energy Office of Electricity / Oak Ridge National Laboratory publication pattern for utility outage data. It contributes the two-tier publishing model this schema follows: a coarse, anonymous public API/feed versus a finer-grained authenticated API for designated stakeholders (mirrored here as the public vs. authenticated tiers for `impact.deEnergized`), a subscribe model for push, and the pragmatic convention of a coded cause *plus* free text because real-world causes exceed any closed enum.
- **MultiSpeak** — a NRECA (US rural electric cooperative) utility-industry interoperability specification. Its outage interfaces — OD (outage detection from AMI), OA (outage analysis/OMS), CB/CH (customer calls), and the `ODEventNotification` device-status message — describe exactly the upstream flow the DISCOM's OMS screens implement (AMI/MDMS detects a feeder outage from meter alarms, the OMS confirms it, a notice is published), and validate this schema's provenance link from alarm to outage.

- **GeoJSON** — RFC 7946, the standard JSON format for geographic geometry (points, lines, polygons) in WGS84/EPSG:4326 coordinates, used here for GIS-ready outage maps and directly emittable as a `FeatureCollection` for map frontends.
- **IEEE 1366** — the IEEE guide defining electric power distribution reliability indices (SAIDI, SAIFI, CAIDI, MAIFI), which this schema's `actualRestoration` and `impact.customersAffected` fields feed as source data.
- **RDSS** — the Revamped Distribution Sector Scheme, India's smart-metering mandate; named in the design note as the concrete policy lever this schema's smart-meter-driven detection pipeline directly supports.
- **IEEE 1782-2022** — the IEEE guide "for Collecting, Categorizing, and Utilizing Information Related to Electric Power Distribution Interruption Events," providing a standardized taxonomy of ten outage cause categories (§4.4) and their subcategories (§4.5), used directly for this schema's `cause.category` and `cause.subcategory`. It is a *guide* (recommended practice), and its text is copyrighted — this schema cites its section numbers rather than reproducing its content.
- **MDMS/OMS/SCADA** — Meter Data Management System, Outage Management System, and Supervisory Control and Data Acquisition — the licensee-side systems that detect, record and manage outages.
- **UsagePoint** — a CIM (IEC 61968-3) concept for a point of connection to the network, used here to record which connections are de-energized or energized.
- **SACHET** — India's NDMA/C-DOT Common Alerting Protocol-based mass-alerting platform (paired with the NDMA Cell Broadcast system); named in the implementation guide as the concrete India-specific channel this schema's CAP-shaped push output is designed to feed for wide-area events.
- **IS 15959 / DLMS-COSEM (IEC 62056)** — the Indian Standard (aligned to IEC 62056) for smart-meter data exchange, cited as the standardized layer that the *vendor's own* feeder-status and alarm codes should ultimately be validated against; `MeterData`'s `AlarmProfile.alarmId` is the field in the companion schema that aligns to it.

5. Basis of Standards

The IES order of preference is fixed: **IS -> CEA Regulations/IEGC -> IEC -> IEEE**. No Indian standard for outage-notification data exists to occupy the first two tiers (the one IS-tier standard the pipeline touches — IS 15959/DLMS-COSEM — governs the *upstream* feeder-status signal, not this schema's own fields; see section 6). The schema composes several standards, each occupying a distinct layer:

Standard	What it governs here
IEC 61968-3 (CIM), 2021	Outage and UsagePoint semantics, and the planned/unplanned split `outageClass` mirrors; `impact.deEnergized/energized` -> CIM `Outage.deEnergizedUsagePoint/energizedUsagePoint`
OASIS CAP v1.2 (ITU-T X.1303)	The push-alert envelope — `msgType`, `references`, `severity`, `publicInfo.*`, `affectedArea.text/geo` map to CAP `alert/info/area` fields
IEEE 1366 (+ RDSS, CEA reliability reporting)	Restoration-time and customer-count fields feeding SAIDI/SAIFI/CAIDI/MAIFI indices
IEEE 1782-2022 §4.4–4.5	The standardized outage cause category (ten categories) and subcategory taxonomy, used with IEEE 1366
ODIN (US DOE/ORNL) *(convention)*	The public/authenticated two-tier API pattern and coded-plus-free-text cause convention
MultiSpeak *(convention)*	The AMI/MDMS-to-OMS detection flow and provenance chain
GeoJSON (RFC 7946)	All GIS geometry, in WGS84

The design note names further de-facto references that shaped the delivery pattern without being cited as data-model standards: the **UK Power Networks "Live Faults"** open dataset (the reference pattern for a public, near-real-time, GIS-ready outage feed), and the **US DOE OE-417/EIA** regime and **ENTSO-E Transparency Platform** (both confirming a regulatory need to publish outages in structured form).

6. Where Indian Standards Do Not Yet Exist

The schema itself documents, field by field, exactly where a published standard applies and where it does not — this is the clearest account of the gap for this domain:

- **Cause classification** — no Indian standard defines a cause taxonomy for outage reporting. The schema uses IEEE 1782-2022 (§4.4 for `cause.category`, §4.5 for `cause.subcategory`), used together with IEEE 1366. Load-shedding/rostering has no standard category at all; scheduled rostering is mapped to the nearest category (PLANNED) and emergency rostering to OTHER, with the real nature of the event carried separately in `outageClass`. The gap is not merely theoretical: the schema's `fault_reason_crosswalk.json` maps all 31 codes of a real DISCOM OMS fault-reason pick-list (`FORM_ID 8312`) onto the IEEE taxonomy, and documents that the source pick-list itself is "operational and India-specific — not a published standard," and contains outright duplicates (code `13 Overload` and code `26 Overloading` map to the same category/subcategory; `20 System improvement work` and `29 System Improvement Work/Deposit Works` likewise; transformer faults are split unevenly across four separate codes, `3`, `4`, `10`, `11`, and `23`). The crosswalk further records non-obvious classification calls the DISCOM had to make, e.g. an 11kV (and LT/ABC) line fault is classified `EQUIPMENT` because it is a distribution feeder *downstream* of the substation, whereas IEEE's `POWER_SUPPLY` category is reserved for the system that *feeds* the substation (so a 33kV sub-transmission fault is `POWER_SUPPLY/SUBTRANSMISSION` while an 11kV feeder fault is `EQUIPMENT/OTHER`); a fire inside a control room or switchyard is `EQUIPMENT` (utility-originated) while an external fire reaching the line is `PUBLIC/FIRE_POLICE`.
- **Push-alert structure** — no Indian standard defines a consumer alert envelope. OASIS CAP v1.2 is used for `msgType`, `references`, `severity`, and the area/geometry block. India does have a national CAP-based channel — SACHET (NDMA/C-DOT) paired with the NDMA Cell Broadcast system — which the implementation guide names as the intended downstream consumer of this schema's CAP-shaped output for wide-area events, but SACHET is an alerting *platform*, not a data-format standard this schema itself conforms to beyond CAP.
- **Identifier scheme for assets** — no Indian standard exists here either; the `Identifier.scheme` enum is of mixed origin (`MRID` from CIM, `DID` from W3C DID Core, the rest local).
- **Reliability reporting maturity** — the design note cites an independent review of Indian DISCOM outage reporting finding that only about 17 of 36 DISCOMs currently publish SAIFI/SAIDI at all, and that most of the rest still publish bare interruption date/time lists with no affected-customer count. A schema that captures `impact.customersAffected` and `actualRestoration` as first-class, structured fields is presented explicitly as a direct upgrade path for India's reliability reporting, and as well-aligned with RDSS's smart-metering direction — but this is a gap the schema is designed to help close, not one it can be said to have already closed.
- **Vendor detection-signal codes** — the raw feeder-status codes coming off a substation meter's digital-input port (e.g. `FEEDER_STATUS=102`) are vendor/platform-specific, not a standard enum; the design note identifies IS 15959 Part 1/2 (DLMS-COSEM, aligned to IEC 62056) — specifically its OBIS codes and standard event/alarm code list — as the standardized layer these vendor codes should ultimately be validated against, the same layer `MeterData`'s `AlarmProfile.alarmId` aligns to. Mapping a specific vendor's raw codes to that standard is left as an open task for each deployment (see section 11).
- **Everything the schema marks as purely local** — `feederStatus`, `outageClass`, `status`, `category` (the display/filtering category, deliberately not CAP's `info/category`), `assetLevel`, `consumerCategory`, and `source` (the detecting system) have no applicable published standard at all today. `outageClass` in particular is taken directly from the DISCOM OMS "Down Info" value set — a local, additive enum, not a published standard. `consumerCategory` is sourced from the DISCOM's own "Feeder Type" field, which in the source PDF also includes values (`TEHSEEL`, `INDEPENDENT`) not yet represented in the schema's enum — the design note flags these as candidates to fold into `OTHER` or add as future additive values.

7. The Record(s)

OutageNotification defines one record type, the outage notice itself. It is not static: one signed record exists per outage, and it is updated in place over the life of that outage rather than replaced by an unrelated new object. A NEW record is issued at detection (or at scheduling, for a planned outage); one or more UPDATE records follow as the restoration estimate or field status changes; and a RESTORED or CANCELLED record closes it out. Each version is independently signed.

The real three-message lifecycle in the schema's own examples makes this concrete: an ALERT is issued the moment RTDAS detects a de-energized 11kV feeder (`status: ACTIVE`, `severity: SEVERE`, `provenance.source: RTDAS`, carrying the raw digital-input signal `feederStatus: DE_ENERGIZED, rawCode: "102"`); roughly three hours later an UPDATE closes the same notice id (`status: RESTORED`, `msgType: UPDATE`, `severity` downgraded to `MINOR`, `timing.`

`actualRestoration` now set, and the signal block updated to `feederStatus: ENERGIZED, rawCode: "101"`) — the same `id` value (`DISCOM-BD-6562139`) ties both messages together, and the elapsed time between `detectedAt` and `actualRestoration` (2 hours 43 minutes here) is exactly the duration IEEE 1366 indices are computed from.

The record is issued by the distribution licensee or a system acting on its behalf (an OMS, MDMS or SCADA system, an RTDAS, or a manual entry — the `provenance.source` enum names all of these explicitly, with MANUAL reserved for outages entered by a human rather than detected automatically). Where the outage was auto-detected, the record's `provenance` block relates it directly to another IES record — the MeterData v0.6 schema's `AlarmProfile` — by carrying references into that record's alarms (`meterRef`, `alarmId`, `timestamp`), so a viewer can trace an outage notice back to the smart-meter signal that triggered it. The provenance block also carries the raw detection signal itself (`OutageSignal`: `eventId`, normalized `feederStatus`, the `diPort` the signal arrived on, and the vendor's `rawCode`) and a `detectionRef` pointing at the real-time detection record (e.g. the DISCOM's `RTDAS_DATA_ID`) — the schema notes that an absent `detectionRef`, or a `"0"` sentinel value, is itself the signal that an outage was entered manually rather than auto-detected.

8. Schedule I — Field Reference (summary)

OutageNotification v0.1 is built from the following logical blocks:

Block	What it contains
OutageNotification	the root object: identity, class, status, message semantics, cause, timing window reference, network context, affected assets and area, impact, response, public-facing text, provenance, and extensions
OutageCause	the standardized cause category and subcategory, vendor fault type and code, and free-text reason
OutageAsset	one affected network asset (feeder, substation, DT, line segment, or service point), its level, and optional map geometry
OutageNetworkContext	the DISCOM's own organisational hierarchy (zone, circle, division, subdivision, substation, district)
OutageAffectedArea	the affected geography as free text, named administrative areas, and GeoJSON geometry
OutageImpact	customers affected and the de-energized/energized connection points
OutageTiming	the outage window, estimated restoration, actual restoration, and SLA target
OutageResponse	back-feed status, field resource status, and linked complaint count
OutagePublicInfo	the localized headline, description and instruction shown to consumers
OutageProvenance	the detecting system, the AMISP code, the raw detection signal (`OutageSignal`), and references into the originating smart-meter signal and MeterData alarms (`OutageAlarmRef`)

The full field-by-field reference (Field / Type / Description, auto-generated from schema.json) is at OutageNotification v0.1 — Field reference.

9. Schedule II

Aspect	Detail
Report template	Not applicable as a populated report template

Aspect	Detail
Wrapping / dependency	OutageNotification is stand-alone: it is not a Verifiable Credential and does not wrap another IES schema as its payload
Cross-reference	It does, however, reference another IES record — its `provenance.alarmRefs` field points into MeterData v0.6's `AlarmProfile` records where an outage was auto-detected — but this is a cross-reference between two independent records, not a wrapping relationship
Optional future wrapping	The design note separately notes an optional, not-yet-adopted possibility: wrapping the notice as a W3C Verifiable Credential (`OutageNotificationCredential`, issued by the DISCOM's DID) for tamper-evidence and cross-network trust, mirroring the existing `MeterDataCredential` pattern — but v0.1 as published is signed only at the Beckn envelope level, and bare JSON is described as sufficient for ordinary web publishing

10. How It Fits Together

OutageNotification is signed at the Beckn envelope level when exchanged, not issued as a W3C Verifiable Credential the way ElectricityCredential is. It sits downstream of a two-layer pipeline that the design note deliberately keeps separate: a thin **detection/ingest** layer and a **publication** layer. In the detection layer, an AMISP-operated substation meter's digital-input port reports a raw energized/de-energized signal (the companion `FeederStatusIngest.openapi.yaml` contract: bearer-token auth, a single `discomCode` tenant, client-supplied `eventId` for idempotency, and batch array input because one substation trip can cascade across many feeders at once); the licensee's MDMS/OMS/SCADA correlates consecutive signals per feeder, enriches them against master data (network hierarchy, GIS geometry, customer counts, the fault-reason crosswalk), and raises an `OutageNotification`. Where that detection path exists, this schema's `provenance` block links the resulting notice back to the specific MeterData `AlarmProfile` record and raw signal that triggered it, giving a reviewer a traceable chain from raw digital-input signal to public notice.

The same notice object serves two audiences without change: published as-is it becomes a pull feed for a website or outage map — the design note specifies a `GET /outages.geojson` mode emitting a GeoJSON `FeatureCollection`, one `Feature` per affected asset or area, with outage attributes in `properties` so it drops directly into a Leaflet/MapLibre/Mapbox map and can be weighted into a heatmap by `impact.customersAffected` — and pushed as-is it becomes a CAP-shaped alert to subscribed or affected consumers, with a companion `GET /outages.cap.xml` mode and an explicit downstream path into India's SACHET/NDMA Cell Broadcast mass-alerting infrastructure for wide-area events. A `GET /outages/lookup?sdp=...` or `?lat=&lon=` mode is also described, letting a single consumer ask "is my connection affected right now?" — the geospatial complement to the map and push views. The public feed and the push channel both draw a line, ODIN-style, between a coarse public tier (no PII) and an authenticated tier that adds the affected `UsagePoint` list.

There is currently no IES use-case guide built on this schema — it exists today as a published schema only, ahead of any use case that would combine it with identity, discovery or credential-issuance steps.

11. Points for Confirmation

1. The schema is explicitly marked Work In Progress by its own README and `attributes.yaml`; field names, enums and interpretation may change before a stable release, so nothing here should be treated as fixed for production integration.
2. The design note itself flags that the DISCOM dashboard shorthand "SD/BD/RS" needs confirmation from the DISCOM that RS specifically means Roster Shutdown (rotational load-shedding) and not another controlled-outage category — the schema's interpretation (restoration modelled as `status=RESTORED`, a transition, not as its own class) is stated as the working assumption, not yet verified with the source utility.
3. Whether the MDMS emits the smart-meter-alarm-to-unplanned-outage linkage fully automatically, or whether the OMS's "Create New Fault" step is a manual operator action, is an open question the design note raises

directly — it determines which fields can be trusted to auto-populate versus which still need human entry.

4. No IES use-case guide yet exists for OutageNotification — how it will be combined with Register/Discover/Exchange steps in a full use case is not yet documented.
5. Several enums are explicitly local with no applicable published standard (`feederStatus`, `outageClass`, `status`, `category`, `assetLevel`, `consumerCategory`, `source`); whether any of these should be proposed for standardization, or left as DISCOM-local values, is open. `consumerCategory` in particular does not yet cover two values (`TEHSEEL`, `INDEPENDENT`) that appear in the source DISCOM PDF.
6. The vendor fault-reason code (`cause.code`) is deliberately left unstandardized and carried verbatim, with a separate crosswalk file mapping it to the IEEE 1782 taxonomy; the design note poses this directly as an open question — adopt the OMS-seeded code list as a canonical IES vocabulary, or keep the field purely free-text for v0.1 — and separately recommends the source DISCOM de-duplicate its own 31-code master list (which currently contains at least three duplicate pairs) before that decision is made.
7. Raw vendor feeder-status and digital-input codes (e.g. `FEEDER_STATUS=102`) are not yet mapped to the IS 15959/DLMS-COSEM (IEC 62056) standard event and OBIS code list that the design note identifies as the correct standardized target — each deployment currently has to obtain its own meter vendor's code legend and build that mapping itself.
8. The granularity of `impact.deEnergized[]` exposed on the public tier versus the authenticated tier is called out as an open privacy question, not yet settled by a published policy.

Annexure A — Standards Referenced

Standard	Scope
IEC 61968-3:2021 (CIM)	Outage/UsagePoint data semantics; the 2021 split into `PlannedOutages/PlannedOutageNotifications/UnplannedOutages/EquipmentFaults/LineFaults` underlies `outageClass`; `Outage.deEnergizedUsagePoint/energizedUsagePoint` underlie `impact`
OASIS CAP v1.2 (ITU-T X.1303)	Push-alert envelope: `msgType`, `references`, `severity`, `area`/geometry, `headline/description/instruction`
ODIN (US DOE/ORNL)	Public/authenticated two-tier publication pattern; subscribe/push model; coded-plus-free-text cause convention
MultiSpeak (NRECA)	AMI/MDMS-to-OMS detection flow (OD/OA/CB/CH interfaces, `ODEventNotification`) and provenance chain
GeoJSON (RFC 7946)	GIS geometry in WGS84/EPSG:4326 for outage maps; direct `FeatureCollection` emission
IEEE 1366 (with RDSS and CEA reliability reporting)	Restoration time and customer-count fields feeding SAIDI/SAIFI/CAIDI/MAIFI reliability indices
IEEE 1782-2022	Outage cause category (§4.4, ten categories) and subcategory (§4.5) taxonomy
W3C DID Core	`Identifier.scheme = DID` values, where a DID is already the identifier of record
IS 15959 / DLMS-COSEM (IEC 62056)	Target standard for vendor feeder-status/meter event codes upstream of this schema (not yet mapped in v0.1)

Annexure B — Example Payloads

Example OutageNotification payloads are available in schemas/OutageNotification/v0.1/examples/ — including a multi-feeder planned-shutdown notice, a three-message unplanned-breakdown lifecycle (ALERT -> RESTORED UPDATE) with full detection provenance, and the DISCOM's published planned-shutdown set transformed into schema form.

Annexure C — JSON Schema

The full machine-readable definitions are at schemas/OutageNotification/v0.1/schema.json and schemas/OutageNotification/v0.1/attributes.yaml.

Use Case Overviews

Consumer Energy Passport

A DISCOM-issued, wallet-held proof of what sits behind a consumer's meter. The Passport is the IES Electricity-Credential v1.2 issued holder-bound to the consumer's wallet as a W3C Verifiable Credential: the connection, its tariff and sanctioned load, and every meter, solar array, battery and EV charger behind it. A bank, subsidy portal or marketplace verifies it offline against the issuer's published key — no DISCOM letter.

Implementation Guide ->

Field	Value
Applicability	All distribution licensees
This version	Consumer Energy Passport *variant* of ElectricityCredential v1.2. Static credential only; live interval data uses MeterData, separately.

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1. Scope and Purpose

The stakeholders are the distribution licensee (DISCOM) and the consumer it serves. As rooftop solar, batteries and EV charging grow behind consumers' meters, the licensee often cannot see what's connected, at what capacity, or where; and the consumer cannot prove their connection and assets to a bank, subsidy portal or marketplace without a manual DISCOM letter.

This document defines the **Consumer Energy Passport** — the holder-bound issuance of the ElectricityCredential v1.2, carrying the static facts of a consumer's connection and the energy resources behind their meter. It does not define a new schema: ElectricityCredential v1.2 is the schema; the Passport profiles how it's shaped, issued and delivered when the consumer is the audience (`credentialSubject.id` = wallet DID; `customerProfile.idRef` = a government-ID reference).

2. What It Records / Covers

Records	Detail	Source
Consumer & issuing licensee	The consumer and the DISCOM that issues the credential	ElectricityCredential v1.2 (`customerProfile`, `issuer`)
Service connection	Tariff category and sanctioned load	ElectricityCredential v1.2 (`consumptionProfiles[]`)
Meters	The net meter, and a generation meter where used	ElectricityCredential v1.2 (`energyResources[]`, type `METER`)
Distribution transformer	Where known	ElectricityCredential v1.2 (`energyResources[]`, network equipment)

Records	Detail	Source
Energy resources behind the meter	Solar, battery, EV charger, inverter, controllable load — with capacity, inspection status and equipment details	ElectricityCredential v1.2 (`energyResources[]`)

It records identity, capacity and status — **not live readings**, which are the MeterData record, linked by asset identifier (see Consumer Meter Digest).

3. How Each Item is Identified

Every item — consumer, connection, each meter, each asset — carries a DID (Decentralized Identifier, W3C DID Core). Existing licensee numbers are reused as aliases inside identifiers.

Subject	Identifier method	Example
DISCOM (issuer)	`did:web` on owned domain	`did:web:ies.discom.example`
Regulator	`did:web` on owned domain	`did:web:ies.serc.example`
Consumer (holder)	`did:key` (wallet) or `tel:` URI	`did:key:z6MkjVQ8r4f3rPuY…`
Meter / inverter / DER / transformer	`did:web` under issuer domain	`did:web:ies.discom.example:assets:meter:NM-44091234`
Existing CIS consumer number	Plain string, kept verbatim	`1102004567` -> `customerProfile.customerNumber`

DeDi (Decentralised Directory) is used only for Beckn subscriber registries and credential revocation — not as an identifier method for consumers, meters or assets.

4. Definitions

- **DER** — any device behind a meter that generates (solar, wind), stores (battery) or controllably consumes (EV charger) electricity.
- **DID** — verifiable digital identifier (W3C DID Core), detailed in §3.
- **Verifiable Credential (VC)** — a tamper-evident, signed document (W3C VC Data Model 2.0) a verifier checks offline against the issuer's published key.
- **Energy Resource** — any physical asset in the credential, one typed entry in `energyResources[]`.
- **QuantitativeValue** — a `{value, unit}` pair for every power/energy/capacity figure, mapped to QUDT.
- **Holder-bound** — issued to a specific holder's wallet, vs. a bearer credential.
- **Net Meter** — the bidirectional billing meter; the source of truth.
- **Aggregator** — a third party enrolling controllable resources for demand response.

5. Basis of Standards

Fixed order of preference: **IS -> CEA Regulations / IEGC -> IEC -> IEEE**, recorded in the field tables below.

Standard	Role here
IS 16444	Smart meter specification
IS 15959	DLMS/COSEM; OBIS codes for metering
CEA (Installation and Operation of Meters) Regulations, 2006	The net meter as the source of truth

6. Where Indian Standards Do Not Yet Exist

- **Grid asset data model** — IEC CIM (IEC 61970-301, IEC 61968-11) underlies the `energyResources[]` kinds.

- **DER electrical attributes** — IEEE 1547-2018 (ride-through, anti-islanding, volt-var, freq-watt); CEA's own limits are retained where CEA sets them (harmonics per IEEE 519-2014; DC injection <=0.5% under CEA Connectivity Regs 2013, am. 2019).
- **PV module rating** — DC array capacity (kWp) follows **IS 14286** (= IEC 61215).

7. The Record

One record: a static Verifiable Credential, changed only on commissioning, capacity change or ownership transfer, and re-issued (with revocation of the old) on material change. It is holder-bound — the consumer holds it in DigiLocker or a DID wallet and discloses fields selectively. Live interval data is a separate MeterData record, linked by identifier.

8. Schedule I — Consumer Energy Passport (Static Record)

Schedule I is a standards-and-status reference table over the real ElectricityCredential v1.2 schema. Each row's **Normative Path** column is an actual JSON path in `schemas/ElectricityCredential/v1.2/schema.json`; **Schema Requires** reflects what the JSON Schema itself enforces. The **Type** column states the CEP profile's expected format for the field (e.g. "did:key", "text", "QuantitativeValue (kW)") — it is a schema-level constraint only where **Schema Requires** says so (e.g. an `enum`, a `required` object shape, a `pattern`); where **Schema Requires** is silent on format, **Type** is CEP profile expectation, not something `validate_schema.py` enforces. The **Standard** and **CEP Profile Guidance** columns are *informative annotations* — they record which standard governs the concept and whether the Consumer Energy Passport profile expects issuers to populate the field. They do not add fields to the schema. Where the earlier draft used a conceptual path with no EC v1.2 equivalent, it is retired to §8.6 as *not represented*, rather than mapped to something that doesn't exist.

8.1 Holder, Issuer and Customer Identity

Normative Path (EC v1.2)	**Type**	**Schema Requires**	**Standard** *(informative)*	**CEP Profile Guidance** *(informative)*
`credentialSubject.id`	did:key (or did:web)	Optional	W3C DID Core	Mandatory — the holder's wallet DID
`credentialSubject.customerProfile`	object	Required	—	Mandatory — the container for customer identity and assets
`credentialSubject.customerProfile.customerNumber`	text (billing number)	Required	CEA Meter Regs; DISCOM billing	Mandatory
`credentialSubject.customerProfile.energyResources`	array, min 1	Required	—	Mandatory
`credentialSubject.customerProfile.idRef.issuedBy` / `.subjectId`	IdRef {did/uri, string}	Object optional; if present, both `issuedBy` and `subjectId` are required	IES IdRef	Optional — government-ID reference (e.g. DigiLocker), see §11.1
`credentialSubject.customerDetails.fullName` / `.installationAddress` / `.serviceConnectionDate`	text / GeoJSON+address / date-time	Optional object; if present, all three required	CIM (IEC 61968-1)	Mandatory — kept out of `customerProfile` since it is PII
`issuer.id`	did:web	Required	IES registry	Mandatory — the DISCOM's operational identifier
`issuer.name`	text	Required	—	Mandatory

Normative Path (EC v1.2)	Type	Schema Requires	Standard *(informative)*	CEP Profile Guidance *(informative)*
`issuer.idRef.issuedBy` / `.subjectId`	IdRef {did:web, string}	Optional	IES registry	Optional — regulator-issued DISCOM registration reference

8.2 Service Connection and Metering

EC v1.2 has no standalone "service connection" object. Tariff and load facts for a connection live in `consumptionProfiles[]`, linked to its net meter by `meterId`; the meter itself is one entry in `energyResources[]`.

Normative Path	Type	Schema Requires	Standard *(informative)*	CEP Profile Guidance *(informative)*
`credentialSubject.customerProfile.consumptionProfiles[].meterId`	text (= a `METER` `energyResources[].id`)	Required, within each entry	CIM UsagePoint	Mandatory
`consumptionProfiles[].tariffCategoryCode`	text	Required, within each entry	SERC tariff order	Mandatory
`consumptionProfiles[].sanctionedLoad`	QuantitativeValue (kW)	Required, within each entry	CEA; SERC supply code	Mandatory
`consumptionProfiles[].sanctionedExportLoad`	QuantitativeValue (kW)	Optional	CEA	Optional
`consumptionProfiles[].contractMaxDemand`	QuantitativeValue (kW)	Optional	CEA	Optional
`consumptionProfiles[].serviceStatus`	enum `active`/ `suspended`/`closed`	Optional	CIM UsagePoint.status	Optional
`consumptionProfiles[].connectionType`	enum `Single-phase`/ `Three-phase`	Optional	CIM UsagePoint.phaseCode	Optional
`consumptionProfiles[].premisesType`	enum Residential/ Commercial/ Industrial/ Agricultural	Optional	—	Optional
`consumptionProfiles[].paymentMode`	enum `POSTPAID`/ `PREPAID`	Optional	CIM; ESPI	Optional
`consumptionProfiles[].billingCycleDay`	integer 1–31	Optional	—	Optional
`energyResources[]` (type `METER`) `.id` — net meter	did:web	Required, per resource	IS 16444; CIM Meter	Mandatory

Normative Path	Type	Schema Requires	Standard *(informative)*	CEP Profile Guidance *(informative)*
`energyResources[]` (type `METER`) `.attributes.serialNumber`	text	Optional	IS 16444	Optional
`energyResources[]` (type `METER`) `.attributes.meterCapability`	enum Electromechanical/ CMRI/AMR/AMI	Optional	CIM AmiBillingReadyKind	Optional
`energyResources[]` (type `METER`) `.attributes.energyDirection`	enum Forward/ Reverse/ Bidirectional/Net	Optional	CIM FlowDirectionKind; ESPI	Optional — `Reverse` distinguishes a generation meter
`energyResources[]` (type `METER`, generation meter) `.parentResources[]`	array of resource ids	Optional	IES topology	Optional — points at the net meter's id
`energyResources[]` (type `INVERTER`) `.id`	did:web	Required, per resource	IEEE 1547; CIM PowerElectronicsConnection	Optional
`energyResources[]` (type `INVERTER`) `.attributes.serialNumber`	text	Optional	equipment nameplate	Optional
`energyResources[]` (type `DT`) `.id` — distribution transformer	did:web	Required, per resource	CIM PowerTransformer	Optional
`energyResources[]` (type `DT`) `.attributes.serialNumber`	text	Optional	equipment nameplate	Optional
`energyResources[]` (type `DT`) `.attributes.nominalVoltage`	QuantitativeValue (kV)	Optional	CIM BaseVoltage	Optional

8.3 Energy Resources — Common Fields (every entry in `energyResources[]`, any type)

Normative Path	Type	Schema Requires	Standard *(informative)*	CEP Profile Guidance *(informative)*
`energyResources[].id`	did:web	Required, per resource	IES; CIM	Mandatory
`energyResources[].type`	enum, per kind — see §8.4	Required, per resource	IEC 61850-7-420; CIM	Mandatory
`energyResources[].parentResources[]`	array of resource id strings only	Optional	IES topology	Optional — carries the DER<->inverter<->meter<->DT topology, see §10

Normative Path	Type	Schema Requires	Standard *(informative)*	CEP Profile Guidance *(informative)*
`energyResources[].subResources[]`	array; each item is *either* a resource id string *or* an inline `EnergyResource` object	Optional	IES topology	Optional — carries the DER<->inverter<->meter<->DT topology, see §10
`energyResources[].attributes.make` / `.model` / `.serialNumber`	text	Optional	equipment nameplate	Optional
`energyResources[].attributes.maxExport` / `.maxImport` / `.ratedPower`	QuantitativeValue (kW)	Optional	IEEE 1547-2018; CEA	Mandatory for generation/storage/EV-charger resources
`energyResources[].attributes.commissioningDate`	date-time	Optional	—	Optional
`energyResources[].attributes.inspection.date` / `.result` / `.inspectorId`	date / enum pass,fail,conditional / text	Optional	IEEE 1547; CEA	Optional
`energyResources[].attributes.aggregator.id` / `.name` / `.controllable` / `.enrolledOn`	did:web / text / boolean / date	Optional	IEC 61850-7-420; IEEE 2030.5	Optional — see §8.5

8.4 Energy-Resource Type Discriminators and Type-Specific Attributes

DER Concept	`energyResources[].type` value(s)	Type-specific attributes	Standard *(informative)*
Solar (PV)	`SOLAR_PV` (`SOLAR` deprecated)	`attributes.dcArrayCapacity` (kW, "kWp" by convention), `.nominalPower` (kW), `.efficiency` (%)	IS 16221; IS 14286; IEC 61727
Battery	`BESS` (`BATTERY` deprecated)	`attributes.storageCapacity` (kWh), `.storageType` (enum), `.stateOfHealthPct`, `.roundTripEfficiencyPct`	IEC 62933; IEC 62619
EV charger	`EV_CHARGER` / `EV_V2G` (bidirectional)	`attributes.connectorType` (enum), `.controlProtocol` (enum), `.v2xProtocol` (schema-permitted on either type; CEP profile guidance restricts population to EV_V2G resources, not schema-enforced)	IS 17017; IEC 62196; OCPP; ISO 15118

DER Concept	energyResources[].type value(s)	Type-specific attributes	Standard *(informative)*
Wind	WIND	attributes.nominalPower / .maxExport (kW)	IEC 61400
Other generation (hydro, biogas, CHP, fuel cell)	HYDRO / BIOGAS / CHP / FUEL_CELL	attributes.nominalPower (kW), .efficiency (% — most relevant for CHP/ FUEL_CELL)	CIM GeneratingUnit (IEC 61970-301)
Inverter	INVERTER (const)	attributes. ratedApparentPower (kVA), .maxReactivePower / . minReactivePower (kVAR), .rideThroughCategory, . operatingMode, . voltVarEnabled, . freqDroopEnabled, . enterServiceRampTimeSec (seconds to ramp 0->rated power after reconnection)	IEEE 1547-2018; SunSpec DER Models
Distribution transformer / bus / feeder / microgrid	DT / BUS / FEEDER / MICROGRID	attributes. nominalVoltage (kV), .zone, .substationId, .feederCode	CIM (IEC 61970-301)
Controllable load	SMART_HVAC / SMART_WATER_ HEATER / CONTROLLABLE_ LOAD	attributes. controlProtocol, . loadCategory	IEC 61850-7-420; OpenADR 2.0b

8.5 Aggregator Enrolment and Controllability (informative)

EC v1.2 has no separate array for "devices an aggregator can control." Any energyResources[] entry — a BESS, EV charger, or controllable load — carries its own attributes.aggregator object (id, name, controllable boolean, enrolledOn). Whether an asset is dispatchable is answered per-resource, not via a separate list, and a device that is both a DER and separately dispatchable (e.g. an aggregator-enrolled BESS) is one energyResources[] entry, not two.

8.6 Concepts Not Represented in ElectricityCredential v1.2

Conceptual field (earlier draft)	Disposition
serviceConnection.did — a DID for the "service connection" itself	Not represented. The connection has no separate identifier; it is addressed via consumptionProfiles[]. meterId pointing at the net meter's energyResources[]. id.
discom.trustDid — a second "trust" DID for the issuer, distinct from its operational DID	Not represented. issuer carries a single id (did:web). A regulator-issued registration reference belongs in issuer.idRef.
der[].profile.inverterDids[] — an explicit list field linking a DER to its inverter(s)	Not represented as a field. Use parentResources[] on the DER entry, pointing at the inverter's id — see the topology diagram in §10.
der[].profile.generationMeterDid	Not represented as a field on the DER. A generation meter is its own energyResources[] entry (type METER), linked into the topology via its own parentResources[].
der[].profile.oemNameplate.maker (as a did:web) / .deviceId (composite model.serial)	Not represented. Nameplate identity is flat text: attributes.make, .model, .serialNumber — there is no manufacturer DID field and no composite device-id field.
controllableAssets[] (separate array) / .linkedDerDid	Not represented as a separate array — see §8.5.

Conceptual field (earlier draft)	Disposition
EV charger `operator`	Not represented — no operator field on `EnergyResourceEVChargerAttributes`.
`consumer.customerReference` — a separate consumer-facing reference distinct from the billing number	Not represented. `customerProfile.customerNumber` (the utility CA number) is the only customer-identity field; there is no second `customerReference` field.
`der[].profile.identity.imei` / `.m2mSimId` — device-level cellular/M2M identity metadata on a DER	Not represented as schema fields on any `EnergyResource` kind. `m2mSimId` is an out-of-band MeterData transport concept, documented in DER Visibility §9.2 — not a Schedule I / EC v1.2 field.
`der[].profile.battery.chargePower` / `.dischargePower` — dedicated battery charge/discharge-power telemetry fields	Not represented as named fields. A BESS resource's charge/discharge limits are carried on the common `attributes.maxImport` (charge) / `.maxExport` (discharge) fields inherited from `EnergyResourceCommonAttributes` — there are no BESS-specific `chargePower`/`dischargePower` fields.

8.7 Example and Validation

Full worked example: `schedule-i-example.json`. It is an illustrative payload fixture — the `proof` block carries a placeholder value and is **not** a cryptographically valid signature.

Validate it structurally against ElectricityCredential v1.2:

```
python -X utf8 -B scripts/validate_schema.py schemas/ElectricityCredential/v1.2/schema.json
 ↪  use-cases/consumer-energy-passport/examples/schedule-i-example.json
```

9. Schedule II

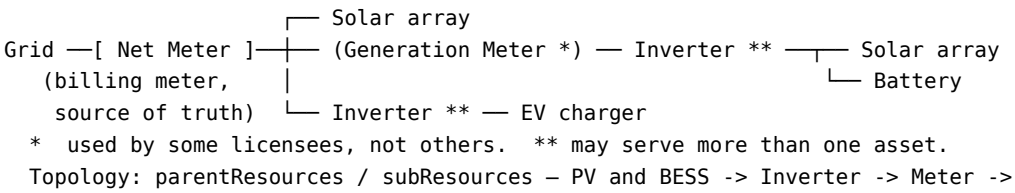
Not applicable. The Consumer Energy Passport carries no time-dependent data. Every field in Schedule I is fixed at issuance and changes only on re-issuance after a material change, so there is no live half to map.

Live data that references this credential is exchanged separately, under its own use case:

Live record	Where it is specified	How it links back
The consumer's own meter readings	Consumer Meter Digest — Schedule II	`meterRefs` carries the net meter's `energyResources[].id` from Schedule I
Telemetry from behind-the-meter DER (inverter, PV, battery)	DER Visibility — Schedule II	`meterRefs` carries the inverter's `energyResources[].id` from Schedule I

Neither is carried inside the Passport, and neither is required to read it.

10. How It Fits Together



Generation is measured at the net meter, a generation meter, or the inverter's own live data (MeterData). All assets live in one `energyResources[]` array; the credential carries identity and ratings, MeterData carries live readings.

11. Points for Confirmation

1. **Holder-binding method.** Confirmed for the DigiLocker Pull URI channel: the `DigiLockerId` from the pull request is carried into `customerProfile.idRef` as the identity binding (see DigiLocker Integration — Identity Binding). Non-DigiLocker methods (offline-KYC XML, in-person) remain open, to be documented for privacy review.
2. **Selective-disclosure profile** with first verifiers (SD-JWT-VC typical).
3. **Re-issuance triggers** on material change, and revocation into the DeDi registry.
4. **Schema-host provenance** — served from `india-energy-stack.github.io/ies-accelerator`; a custom IES/gov domain is a governance decision.

Schemas Used in This Use Case

A single schema — **ElectricityCredential v1.2** (W3C VC). No additional schemas: Schedule I is the whole record and there is no Schedule II (§9).

Live readings that reference this credential use MeterData v0.6 under their own use case — the consumer's own meter via Consumer Meter Digest, behind-the-meter DER via DER Visibility.

Value Unlock

The consumer gets a portable, tamper-evident proof of their connection and assets that banks, subsidy portals and marketplaces verify offline — no DISCOM callback. The DISCOM cuts verification load and gets a clean asset register as a by-product. Regulators get a consistent, auditable asset record across licensees. Selective disclosure lets the consumer reveal only the fields a verifier needs.

Annexure A — Standards Referenced

Standard	Scope
IS 16444 (Parts 1, 2)	AC smart meter — specification
IS 15959 (Parts 1–3)	DLMS/COSEM companion spec; OBIS codes
IS 14286 (= IEC 61215)	PV module design qualification and type approval
IS 16221 (Parts 1, 2) (= IEC 62109-1/-2)	Inverter safety (inverter resource only)
IS 16270	Solar PV batteries
IS 17017 (series) (from IEC 61851 / 62196)	EV conductive charging
CEA (Installation and Operation of Meters) Regs, 2006	Metering legal framework; net meter as source of truth
CEA (Connectivity of DG Resources) Regs, 2013, am. 2019	Connectivity below 33 kV; PCC; DC-injection; harmonics
IEC 61968-1 / -9 / -11	CIM — customer, metering, distribution model
IEC 61970-301 / -302	CIM — network model; DER / power-electronics / battery
IEC 61850-7-420	DER control roles
IEC 61727 / IEC 62116	PV utility interface; anti-islanding
IEC 62933 / 62933-2-1 / IEC 62619	Storage systems; performance test; battery safety
IEC 62196	EV connectors
IEC 62056	DLMS/COSEM; OBIS
IEEE 1547-2018	DER interconnection and interoperability
IEEE 519-2014	Harmonic control (CEA-referenced)
IEEE 2030.5	DER communications / aggregator roles
W3C VC Data Model 2.0; W3C DID Core	Credential envelope; identifiers
GeoJSON (RFC 7946); schema.org PostalAddress	Location and postal address
OCPP; ISO 15118	EV charger control / V2G

Annexure B — Example Payload

A single-phase residential connection (5 kW sanctioned load) with a 3.6 kWp PV array, a small wind turbine and two batteries — one a 10 kWh aggregator-enrolled BESS — all parented to the net meter, which hangs off its feeder. Holder-bound; identifiers illustrative.

- **`examples/example.json`**
- **`example-submetering.json`** — building main meter + tenant sub-meters + rooftop solar
- **`example-parallel-metering.json`** — import + export meter (solar FIT)

Annexure C — JSON Schema

Canonical: `https://india-energy-stack.github.io/ies-accelerator/schemas/ElectricityCredential/v1.2/` — `schema.json` (Draft 2020-12), `context.jsonld`, `vocab.jsonld`. ElectricityCredential v1.2 is one of two schema-version directories (of eleven across IES) that annotates individual fields with their governing standard, via the singular `x-standard` property; the §5 order of preference (`IS -> CEA Regulations / IEGC -> IEC -> IEEE`) applies regardless of whether a given schema carries that annotation.

Consumer Meter Digest

A consumer's own meter readings, DISCOM-signed and packaged for sharing. The Digest is a MeterDataCredential v0.6 issued on consumer demand, holder-bound to their wallet (W3C Verifiable Credential), carrying readings for a specified period. A bank, installer or energy app verifies it offline instead of trusting an emailed PDF bill.

Implementation Guide ->

Field	Value
Applicability	All distribution licensees
This version	Consumer Meter Digest *variant* of MeterDataCredential v0.6, holder-bound. Wraps MeterData v0.6 profiles.

---

1. Scope and Purpose

The stakeholder is the consumer sharing verified consumption history with a bank, marketplace, energy app, housing society or installer, and the DISCOM that issues the digest. Today the consumer prints and emails PDF bills; verifiers can't confirm authenticity, so most call the DISCOM anyway.

This document defines the **Consumer Meter Digest** — a verifier-friendly, DISCOM-signed bundle of the consumer's telemetry. Not a new credential type: MeterDataCredential v0.6 is the schema, profiled for a consumer audience.

2. What It Records / Covers

Records	Detail	Source
Meter & service-delivery point	`meterRefs` and the service-delivery-point reference	MeterDataCredential v0.6 wrapping MeterData v0.6
Period	The window the readings cover	MeterData v0.6
Readings	Profile-typed data (`intervals` for `INTERVAL`/`DAILY` cadence, `readings` for other profile types), discriminated by `profileType`	MeterData v0.6 (IS 15959 / DLMS-COSEM)
Data quality	Estimation flags, missing intervals	MeterData v0.6
Issuer & proof	Issuing DISCOM (`did:web`) and cryptographic proof	MeterDataCredential v0.6 (W3C VC)

Granularity: `DAILY`, `MONTHLY`, or `INTERVAL` (15-minute interval data expressed as `profileType: INTERVAL` with `intervalPeriod.duration: PT15M`). Typical max period: 24 months for `MONTHLY`, 90 days for 15-minute `INTERVAL` data.

3. How Each Item is Identified

Reuses the Consumer Energy Passport scheme:

Subject	Identifier method	Example
DISCOM (issuer)	`did:web` on owned domain	`did:web:ies.discom.example`
Consumer (holder)	`did:key` (wallet)	`did:key:z6MkjVQ8r4f3rPuY…`
Meter	`did:web` under issuer domain	`did:web:ies.discom.example:assets:meter:NM-44091234`

The meter identifier matches the one in the consumer's Consumer Energy Passport — a verifier sees the two as a coherent pair.

4. Definitions

- **Holder-bound** — `credentialSubject.id` set to the holder's wallet DID.
- **READING** — register value at a point in time; cumulative series must be non-decreasing.
- **USAGE** — delta / consumed amount over a period.
- **OBIS** — Object Identification System code (IEC 62056 / IS 15959) for a meter register.
- **Summary** — a derived aggregate computed from raw readings by downstream analytics; not itself a field defined by the MeterData v0.6 schema.

See **IES Meter Data Model** for the underlying meter-data terminology.

5. Basis of Standards

Fixed IES order of preference: **IS -> CEA Regulations / IEGC -> IEC -> IEEE**.

Standard	Role here
IS 15959	DLMS/COSEM; OBIS codes — metering reads
IS 16444	Smart meter specification
W3C VC Data Model 2.0	The credential envelope

6. Where Indian Standards Do Not Yet Exist

The credential envelope (W3C VC) and the underlying compact-profile data model have no Indian equivalent; the MeterData v0.6 shapes are an IES specification on top of IEC 62056 / IS 15959.

7. The Record

One record: a Verifiable Credential wrapping a MeterData profile. Unlike the Passport, it's a **point-in-time snapshot** with a short `validUntil` (hours to days) — the consumer re-requests when fresh data is needed. Holder-bound; presentable to multiple verifiers within its validity window. Revocation rarely matters in practice, but the same DeDi-hash flow is available.

8. Schedule I — Static Fields of the Credential

Schedule I is the Consumer Meter Digest profile view over the real MeterDataCredential v0.6 wrapper and its embedded MeterData v0.6 payload. **Normative Path** names an actual path in those schemas. **Schema Requires** records schema-requiredness; **CMD Guidance** is informative profile guidance and does not add constraints to either schema. The complete field references remain canonical for fields not listed here.

8.1 Credential Envelope and Holder Binding

Normative Path	Type	Schema Requires	Standard *(informative)*	CMD Guidance *(informative)*
`@context`	array of context URIs	Required by the externally referenced W3C Credential branch	W3C VC Data Model 2.0	Include the W3C, EnergyCredential and MeterDataCredential contexts
`id`	credential URI / URN	Optional; permitted by the open credential envelope	W3C VC Data Model 2.0	Unique per issued digest
`type`	array including `MeterDataCredential`	Required by the externally referenced W3C Credential branch	W3C VC Data Model 2.0	Identify this credential family explicitly
`issuer`	issuer object	Optional at the EnergyCredential root; if present, `id`, `name` and `licenseNumber` are required	W3C VC; DID Core	Mandatory for this profile; use the DISCOM or authorised provider's registered `did:web`
`validFrom` / `validUntil`	date-time	Not required by the generated local wrapper schema	W3C VC Data Model 2.0	Mandatory profile convention; use a short validity window appropriate to a point-in-time digest
`credentialStatus`	status object	Optional at the EnergyCredential root; if present, `id` and `type` are required	DeDi registry	Optional operational revocation/ suspension reference
`proof`	proof object	Optional at the EnergyCredential root; required members apply if present	W3C VC Data Model 2.0	Mandatory for an issued digest; sign the complete credential
`credentialSubject.id`	URI / DID	Optional when `credentialSubject` is present	DID Core	Mandatory for this holder-bound profile; use the consumer's verified holder identifier
`credentialSubject.meterData`	one MeterData profile or non-empty array of profiles	Required inside `credentialSubject`; the wrapper does not currently require `credentialSubject` itself	MeterData v0.6	Mandatory; carry the descriptor plus the requested digest profiles when compact payloads are used

Repository-local structural validation uses a permissive stub for the external EnergyCredential reference. Implementations must therefore enforce the envelope and CMD profile requirements above in addition to validating the embedded MeterData payload.

8.2 Digest Profiles and Time Windows

Normative Path	Type / Allowed Value	Schema Requires	Standard *(informative)*	CMD Guidance *(informative)*
`credentialSubject.meterData[].profileType`	`DESCRIPTOR`	Required on a descriptor profile	MeterData v0.6	Include when another profile uses `payloadDescriptorSetRef` or compact `payloads[]`
`credentialSubject.meterData[].profileType`	`INTERVAL`	Required on an interval profile	IS 15959 / DLMS-COSEM	Use for 15- or 30-minute blocks; 15-minute cadence is `intervalPeriod.duration: PT15M`
`credentialSubject.meterData[].intervalPeriod.start` / `.duration`	date-time / ISO 8601 duration	Required on `INTERVAL`	ISO 8601	Defines the first interval and cadence
`credentialSubject.meterData[].intervals[]`	interval objects	Optional in the schema	MeterData v0.6	Populate for compact interval delivery; each `intervals[].id` is required and strictly increases within the profile
`credentialSubject.meterData[].profileType`	`DAILY`	Required on a daily profile	IS 15959 / DLMS-COSEM	Use for daily digest records
`credentialSubject.meterData[].intervalPeriod.start` / `.duration`	date-time / ISO 8601 duration	Required on `DAILY`	ISO 8601	Use the requested daily window and cadence
`credentialSubject.meterData[].profileType`	`MONTHLY`	Required on a monthly profile	IS 15959 / DLMS-COSEM	Use for billing-cycle or multi-month history
`credentialSubject.meterData[].timePeriod.start` / `.duration`	date-time / ISO 8601 duration	Required on `MONTHLY`	ISO 8601	Defines the month or billing period represented
`credentialSubject.meterData[].readings[]`	array of `Reading`	Required on `MONTHLY`; optional on `INTERVAL` / `DAILY`	MeterData v0.6	Use explicit readings where the compact descriptor/sequence form is not used
`credentialSubject.meterData[].touBuckets[]`	time-of-use buckets	Optional on `MONTHLY`	SERC tariff order; MeterData v0.6	Populate only when the requested digest includes ToU segmentation

8.3 Meter and Descriptor Fields

The identity of the metering point and the dictionary that gives the readings meaning. Both are fixed for the digest; the readings themselves are in Schedule II.

Normative Path	Type / Allowed Value	Schema Requires	Standard *(informative)*	CMD Guidance *(informative)*
`meterRefs[]`	`Identifier {scheme, value, namespace?}`	Required on every telemetry profile	IS 16444; IES identifiers	Point to the same meter used by the Consumer Energy Passport
`serviceDelivery PointRefs[]`	list of `Identifier`	Optional	CIM (IEC 61968-9)	Include when the verifier must bind the reading to a service point
`customerRefs[]`	list of `Identifier`	Optional	CIM (IEC 61968-9)	Minimise disclosure; the holder binding may be sufficient
`payloadDescriptor SetRef` / `compactSequenceRef`	text references	Optional	MeterData v0.6	References must resolve to the accompanying `DESCRIPTOR` profile when compact payloads are used
`payloadDescriptor Sets[].payload Descriptors[]. readingType`	text / governed short code	Required per descriptor	IS 15959 / OBIS	Declare every reading type carried by the compact sequence
`payload Descriptors[].obis` / `.unit` / `. reportedMode`	OBIS text / unit enum / `READING` or `USAGE`	Optional	IEC 62056; IS 15959	Use canonical OBIS and mode metadata where available

9. Schedule II — Meter Readings (Live Record)

Schedule II is the **live half** of the digest — the measured values themselves. They are the reason the credential exists; everything in Schedule I describes which meter they came from, over what window, and under whose binding.

A digest is a *snapshot*: once issued, the readings inside one credential no longer change. The fields are still live-class — a new reading arrives every block at the meter — and a later digest carries later values.

9.1 Reading and Quality Fields

Normative Path	Type / Allowed Value	Schema Requires	Standard *(informative)*	CMD Guidance *(informative)*
`readings[]. readingType` / `.value`	text / number	Both required per reading	IS 15959 / OBIS	Use the descriptor set's canonical reading type
`readings[]. openingValue` / `. closingValue`	number	Optional; semantic rules apply to `USAGE`	MeterData v0.6	Include together when the digest needs auditable block-delta arithmetic
`readings[]. occurredAt` / `.timePeriod`	date-time / period object	Optional	ISO 8601	Use the field appropriate to point-in-time or period data
`readings[]. validationStatus`	`VALID`, `ESTIMATED`, `MANUAL`, `SUSPECT`, `REJECTED`	Optional	MeterData v0.6	Populate so the verifier can distinguish measured and estimated values

Normative Path	Type / Allowed Value	Schema Requires	Standard *(informative)*	CMD Guidance *(informative)*
`readings[].source`	`METER`, `HES`, `ESTIMATED`, `MANUAL`, `IMPORT`, `MDM_COMPUTED`, `CIS_COMPUTED`	Optional	MeterData v0.6	Populate when provenance affects verifier decisions

10. How It Fits Together

```
Consumer (wallet)                    DISCOM
  |── request via DigiLocker /  ──────>| verify entitlement (Passport)
  |   consented app                    | pull readings from MDM
  |                                    | build MeterData/v0.6 payload
  |                                    | wrap: credentialSubject.id = wallet DID
  |<── signed Digest ──────────────────|  validUntil = 24h–7d
  |
  |── present Digest ──> Verifier (offline verify)
```

11. Points for Confirmation

1. **Maximum-range policy** by granularity — to be tightened per use case.
2. **Summary-derivation formula** for any downstream analytics built on these readings, especially ToD bucket boundaries (may vary by SERC) — out of scope for the credential schema itself.
3. **Latency budget** — MDM read-path performance is the binding constraint.

Schemas Used in This Use Case

Holder-bound issuance of **MeterDataCredential v0.6**, wrapping a **MeterData v0.6** payload. Entitlement is typically proven by a Consumer Energy Passport.

Value Unlock

The consumer proves actual consumption history, DISCOM-signed, verifiable without a callback. The DISCOM cuts bill-verification calls and gains a clean, consented data-sharing surface — a higher-trust input than emailed PDF bills.

Annexure A — Standards Referenced

Standard	Scope
IS 15959 (Parts 1–3)	DLMS/COSEM companion spec; OBIS codes
IS 16444 (Parts 1, 2)	AC smart meter — specification
IEC 62056	DLMS/COSEM; OBIS
IEC 61968-9	CIM — meter reading and control
W3C VC Data Model 2.0; W3C DID Core	Credential envelope; identifiers
RFC 3339 / ISO 8601	Date-time format for `intervalPeriod` / `timePeriod`

Annexure B — Example Payload

- **`example-customer-profile.json`**
- **`example-interval-profile.json`**
- **`example-monthly-profile.json`**

Annexure C — JSON Schema

Canonical: `https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataCredential/v0.6/` — `schema.json`, `context.jsonld`, `vocab.jsonld`. The payload schema: MeterData v0.6.

Annexure D — Derived Views

Computed downstream by the recipient from the Schedules above. None is a field of MeterDataCredential v0.6, and none is exchanged.

Derived View	Source	Schema Status	Treatment
Period total	`readings[]` or compact `intervals[].payloads[]` for the requested window	Not a MeterDataCredential / MeterData v0.6 field	Compute downstream and label the method and source period
Peak demand	Demand readings plus `occurredAt` / interval position	Not a schema field	Compute downstream; retain the source reading and timestamp for audit
Time-of-use breakdown	`touBuckets[]` or interval readings joined to tariff periods	`touBuckets[]` is native for `MONTHLY`; a rendered summary is derived	Keep native buckets in the signed payload; render labels and totals downstream
Missing-interval count	Expected cadence versus received interval IDs	Not a schema field	Compute downstream; do not insert a synthetic `dataQuality` object into the credential
Verifier-facing PDF / dashboard	Any of the above	Presentation only	May accompany the credential, but the signed JSON remains the authoritative record

Smart Meter Data Exchange

A standard, audit-trailed way to move smart-meter telemetry between an AMISP, a DISCOM, a State regulator, and consented third parties. Discovery, contracting and payload delivery all run over Beckn — the rail behind the IES Discover and Exchange steps; the payload on the wire is MeterData v0.6. Integrate once, and the same request/response pattern works against every adopting DISCOM.

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Field	Value
Applicability	All AMISPs, DISCOMs and SERCs
This version	Built on MeterData v0.6, MeterDataRequest v0.6 and (optional) MeterDataRequestCredential v0.1 over Beckn.

> For consumer-pull of *their own* meter data, see **Consumer Meter Digest**. This is the bulk, scheduled, machine-to-machine flow.

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1. Scope and Purpose

The stakeholders are the DISCOM (data fiduciary), the AMISP (data processor) and any authorised third party. Today every DISCOM-AMISP pair builds a bespoke integration — a one-time dump, then incremental files over FTP, or a project-specific API — negotiated and built from scratch each time.

This document defines **Smart Meter Data Exchange** — a one-to-many, standard data shape (MeterData v0.6) carried over a one-to-many discovery, contracting and exchange layer (Beckn). A TSP integrates once and the same pattern works across DISCOMs. IES standardises **the interface where parties exchange data** — it does not change how any party stores or moves data internally.

2. What It Records / Covers

Four things, and only these four:

Records	Detail	Source
The contract	How a request is discovered, agreed and carried	Beckn protocol
Consent & scope	How consent, scope and duration are recorded	MeterDataRequest v0.6 / MeterDataRequestCredential v0.1
The data shape	The MeterData wire format	MeterData v0.6 (DLMS-COSEM / IS 15959)
Receipts & audit	Proof of what was exchanged	Beckn protocol; W3C VC

The meter keeps speaking DLMS-COSEM / IS 15959; the head-end, MDM and AMISP's systems are unchanged.

Eight MeterData v0.6 compact profiles cover every cadence:

Profile	Carries
`CUSTOMER`	Slow-changing customer / service-point / meter installation metadata
`INTERVAL`	Block load survey at 15- or 30-min resolution
`DAILY`	Daily accumulated load survey
`MONTHLY`	Monthly billing resets (cumulative kWh, ToU, MD)
`BILL_DETAILS`	Utility-side billing computed details
`INSTANTANEOUS`	Real-time snapshot of V/A/P/Q
`EVENT`	IS 15959 diagnostic / tamper events
`ALARM`	Real-time active alerts

3. How Each Item is Identified

Subject	Identifier method	Example
DISCOM (data fiduciary)	`did:web` on owned domain	`did:web:ies.discom.example`
AMISP / TSP (data processor)	`did:web` on owned domain	`did:web:np.example.com`
Meter / DT / Feeder	`did:web` under DISCOM domain	`did:web:ies.discom.example:assets:meter:NM-44091234`

No new IDs to allocate. Existing meter SLNOs, DT codes and feeder codes stay exactly as they are; the `did:web` wrapper reuses them. Adoption is *nice to have* for a first deployment — bare IDs work in payloads initially.

4. Definitions

- **DLMS-COSEM** — the meter<->head-end wire protocol (IS 15959 in India).
- **OBIS** — Object Identification System code identifying a meter register.
- **HES** — Head-End System, the AMI layer talking to meters.
- **MDM/MDMS** — Meter Data Management System, the DISCOM's system of record.
- **AMISP** — the entity deploying and operating smart meters and the HES for a DISCOM.
- **READING vs USAGE** — the physical register value vs. the delta/consumed amount over a period.

5. Basis of Standards

Fixed IES order of preference: **IS -> CEA Regulations / IEGC -> IEC -> IEEE**.

Standard	Role here
IS 16444	Smart meter specification (followed directly)
IS 15959 / IEC 62056 (DLMS-COSEM)	Meter readings (followed directly)
MeterData v0.6	IES specification standardising the JSON shape that carries those readings
IEC 61968 / -100	HES<->MDMS interoperability (CEA/RDSS guidance, alongside MultiSpeak)
CIM (IEC 61968-9)	Master data

6. Where Indian Standards Do Not Yet Exist

The compact-profile **JSON shape** is an IES specification — no predating Indian or international standard. Beckn (discovery, contracting and payload delivery) is also an IES choice. Event codes follow **IS 15959** allocations directly.

7. The Record

No Verifiable Credential by default. Each exchange produces: a **signed Beckn contract** (discovery, scope, parties, time-bound authorisation), a **signed MeterData v0.6 payload** (inline or signed-URL), and a **signed receipt**. Together: a verifiable audit trail for DPDP accountability and dispute resolution. For a durable, holder-bound record, wrap in **MeterDataCredential v0.6** — see Consumer Meter Digest.

8. Schedule I — Static Fields of the Data Exchange

Schedule I tabulates the static contracts that frame this exchange: MeterDataRequest v0.6 selects scope and capabilities, and the optional MeterDataRequestCredential v0.1 makes a request portable and verifiable. The MeterData v0.6 response itself is the live half — see Schedule II. **Schema Requires** is normative for the named schema; **SMDX Guidance** is informative deployment guidance.

8.1 MeterDataRequest v0.6 — Query and Scope

Normative Path	**Type / Allowed Value**	**Schema Requires**	**Standard** *(informative)*	**SMDX Guidance** *(informative)*
`consumers[]`	array of URI / DID	Optional	DID Core; DPDP consent scope	Use only for consumer-scoped requests
`resources[]`	array of resource URI / DID	Optional	IES identifiers	Use for meters, service points, feeders or other registered resources
`scope`	`ResourceOnly`, `ResourceAndChildren`, `ChildrenOnly`	Optional	MeterDataRequest v0.6	State the hierarchy rule explicitly for feeder/DT requests
`from`	date-time	Required	ISO 8601	Start of the requested data window
`duration`	duration	Required	ISO 8601	Requested window length; the provider still enforces its advertised maximum

Normative Path	Type / Allowed Value	Schema Requires	Standard *(informative)*	SMDX Guidance *(informative)*
`consumerConsent[]`	array of consent references	Optional	DPDP accountability	Required by policy when the request exposes consumer-linked data
`authorisation`	inline `MeterData Authorisation` or URI	Optional	MeterDataRequest v0.6	Bind the requester, purpose, validity and allowed capabilities
`capabilities Requested`	`MeterData Capabilities` object	Required	MeterDataRequest v0.6	Request only profiles/registers the provider advertises
`maxRecordsShared`	integer >= 1	Optional	MeterDataRequest v0.6	Use as a batch/page cap, not as a substitute for the time window

8.2 Capabilities and Authorisation

Normative Path	Type / Allowed Value	Schema Requires	SMDX Guidance *(informative)*
`capabilitiesRequested. profiles[]`	array of `ProfileCapability`	Required	Enumerate each requested telemetry profile
`profiles[].profileType`	`CustomerProfile`, `IntervalProfile`, `DailyProfile`, `MonthlyProfile`, `BillDetails`, `InstantaneousProfile`, `EventProfile`, `AlarmProfile`	Required per entry	`DESCRIPTOR` is not requestable; it accompanies the response when compact data needs a dictionary
`profiles[].readings[]`	array of `ValueCapability`	Optional	Omit to request all supported registers in that profile; otherwise list only required registers
`readings[].value`	OBIS code or governed short code	Required per value capability	Prefer the canonical code published by the provider
`readings[].mode`	`READING` or `USAGE`	Optional	Match the physical meaning of the register and response descriptor
`readings[].multiplier` / `.accuracy`	number / number	Optional	Request only when scaling or accuracy is material
`capabilitiesRequested. supportedScopes[]`	array of scope enums	Optional	Relevant in a provider capability advertisement; the request must stay within it
`capabilitiesRequested. maxHistoryDuration`	ISO 8601 duration	Optional	Provider-advertised history ceiling
`authorisation.grantor` / `.grantee` / `.purpose`	URI / URI / text	All required in an inline authorisation	Identify who granted access, who receives it, and why
`authorisation.validFrom` / `.validUntil`	date-time / date-time	Both required	Reject expired or not-yet-valid grants

Normative Path	Type / Allowed Value	Schema Requires	SMDX Guidance *(informative)*
`authorisation.capabilities`	`MeterDataCapabilities`	Required	Must cover the requested profile, scope and registers

8.3 Optional MeterDataRequestCredential v0.1

Normative Path	Type	Schema Requires	SMDX Guidance *(informative)*
`@context` / `type`	W3C credential context / type array	Required by the externally referenced W3C Credential branch	Include the W3C, EnergyCredential and MeterDataRequestCredential contexts and the concrete credential type
`id`	credential URI / URN	Optional; permitted by the open credential envelope	Assign a unique request-credential identifier
`issuer`	EnergyCredential issuer object	Optional at the EnergyCredential root; if present, `id`, `name` and `licenseNumber` are required	Mandatory for a portable authorisation; identify the issuing requester/ authority
`validFrom` / `validUntil` / `credentialStatus` / `proof`	credential envelope fields	Not required at the wrapper root; nested status/proof requirements apply if those objects are present	Apply the validity, revocation and signature rules required by the exchange policy
`credentialSubject.id`	requester URI / DID	Optional when `credentialSubject` is present	Identify the authorised requesting entity
`credentialSubject.meterDataRequest`	MeterDataRequest v0.6 payload	Required inside `credentialSubject`; the wrapper does not currently require `credentialSubject` itself	Carry the exact scoped, time-bounded request being authorised

As with MeterDataCredential, repository-local structural validation stubs the external EnergyCredential reference. A provider must enforce the complete credential envelope and exchange-policy requirements separately.

For Indian-terminology mapping, see **IES Meter Data Model**.

9. Schedule II — Meter Readings (Live Record)

Schedule II is the **live half** of this exchange — the fields that keep arriving after the request is agreed, as against the static request, authorisation and credential contracts of Schedule I. Every field below is carried in the MeterData v0.6 response.

Scope is set in Schedule I and enforced here: a response carries only the profiles and period the authorisation in §8.2 permits.

9.1 MeterData v0.6 — Response Profiles

MeterData has **nine total record shapes**: one shared `DESCRIPTOR` dictionary plus the eight requestable telemetry profiles below.

profileType	Schema-required Payload Fields	Purpose in this Use Case	Standards / Basis *(informative)*
DESCRIPTOR	id, payloadDescriptorSets[]	Defines reading types, units, modes and compact sequences used by response profiles; not directly requestable	MeterData v0.6; IS 15959 / OBIS
CUSTOMER	customer, serviceDeliveryPoints[], meters[], associations[]	Slow-changing customer, connection, meter and topology metadata	CIM (IEC 61968-9)
INTERVAL	meterRefs[], intervalPeriod; data in intervals[] or readings[]	15- or 30-minute load survey	IS 15959 / DLMS-COSEM
DAILY	meterRefs[], intervalPeriod; data in intervals[] or readings[]	Daily accumulated survey	IS 15959 / DLMS-COSEM
MONTHLY	meterRefs[], timePeriod, readings[]; optional touBuckets[]	Billing-reset and ToU history	IS 15959; SERC tariff periods
BILL_DETAILS	meterRefs[], timePeriod, amountDue; optional readings, charges and payment fields	Utility-computed billing outcome	Utility billing / CIS
INSTANTANEOUS	meterRefs[], timestamp, readings[]	Real-time V/A/P/Q and related snapshot values	IS 15959 / DLMS-COSEM
EVENT	meterRefs[], timePeriod, events[]	Diagnostic and tamper events	IS 15959 event allocation
ALARM	meterRefs[], timestamp, alarms[]	Active or cleared alert state	MeterData v0.6

9.2 Shared Response Fields and Compact Data

Normative Path	Type / Allowed Value	Schema Requires	SMDX Guidance *(informative)*
meterRefs[]	Identifier {scheme, value, namespace?}	Required on all seven non-customer telemetry profiles	Bind every series to its source meter(s)
customerRefs[] / serviceDeliveryPointRefs[]	arrays of Identifier	Optional	Include only at the authorised disclosure level
payloadDescriptorSetRef / compactSequenceRef	text references	Optional	Resolve against the accompanying DESCRIPTOR profile
payloadDescriptorSets[].payloadDescriptors[]	descriptor objects	Required inside each descriptor set	Declare readingType; add OBIS, unit, flow direction and reportedMode where known
intervals[].id	integer	Required per interval	Strictly increasing within a profile
intervals[].payloads[]	compact primitive array	Optional	Arity and order must match the referenced compact sequence
intervals[].readings[] / profile readings[]	array of Reading	Optional except where the profile requires it	Each reading requires readingType and numeric value

Normative Path	Type / Allowed Value	Schema Requires	SMDX Guidance *(informative)*
`readings[].validationStatus` / `.source`	governed enums	Optional	Preserve estimation, rejection and source provenance from HES/MDM
`events[].timestamp` / `.eventId`	date-time / integer	Both required per event	Use the IS 15959 event allocation where applicable
`alarms[].timestamp` / `.alarmId` / `.status`	date-time / integer / `ACTIVE` or `CLEARED`	All required per alarm	Keep alarm lifecycle state explicit

10. How It Fits Together

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Today: DISCOM —bespoke— AMISP-1 / AMISP-2 / analytics / third party  (n × m integrations)
With IES: DISCOM ── one MeterData v0.6 over Beckn ──► AMISP-1 / AMISP-2 / analytics / third party  (n + m)
```

Roles don't change; the record makes them explicit. The DISCOM remains the data fiduciary (authorises a TSP once, scoped, time-bound); the AMISP remains the processor (IES doesn't alter ownership); the TSP integrates once (same discovery/contract/shape across every adopting DISCOM). Every exchange leaves a signed receipt for DPDP accountability.

11. Points for Confirmation

1. **Chunking convention** for bulk historical pulls — being agreed across deployments.
2. **Quality flags** — the v0.6 `validationStatus` enum is stable; the per-cadence recommended profile is being tightened.
3. **Aggregator-controlled DERs** — the consent/control flow for *acting* on (not just reading) a DER is specified separately.

Schemas Used in This Use Case

Schema	Role
MeterData v0.6	The payload — eight compact profiles
MeterDataRequest v0.6	The query / capabilities shape
MeterDataRequestCredential v0.1 *(optional)*	Seeker authorisation VC

Value Unlock

DISCOMs/AMISPs — one interface replaces bespoke pair-by-pair integration. **Regulators/analytics** — one consistent format with a verifiable audit trail, so analysis is comparable across DISCOMs. **Consumers** — consented third parties can read meter data on standard rails, the basis for services like green loans, solar quotes and demand-shift offers.

Annexure A — Standards Referenced

Standard	Scope
IS 16444 (Parts 1, 2)	AC smart meter — specification
IS 15959 (Parts 1–3)	DLMS/COSEM companion spec; OBIS codes; event codes
IEC 62056	DLMS/COSEM

Standard	Scope
IEC 61968-9	CIM — meter reading and control
IEC 61968-100	Web service implementation profile (CEA AMI guidance)
CEA (Installation and Operation of Meters) Regs, 2006	Metering legal framework
RDSS	Policy context for current AMI deployment
DPDP Act 2023	Consent and accountability framework
W3C VC Data Model 2.0; W3C DID Core	(Optional — MeterDataRequestCredential)

Annexure B — Example Payloads

25 example payloads covering every profile shape and derived reports at **`schemas/MeterData/v0.6/examples/`** — highlights: `CustomerProfile.json`, `IntervalProfile.json`, `DailyProfile.json` / `MonthlyProfile.json`, `MultiMeterBulkDataset*.json`, `EventProfile.json` / `AlarmProfile.json`, `AggregatedFeeder.json`.

Annexure C — JSON Schema

Canonical: `https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/` — `schema.json`, `context.jsonld`, `vocab.jsonld`, `IES codes.json` (OBIS registry).

Annexure D — Derived Views

Computed downstream by the recipient. None is an exchanged record or an additional IES schema.

Derived View	**Inputs**	**Schema Status**	**Treatment**
Feeder-aggregated profile	Meter/SDP topology plus interval, daily or monthly profiles	Derived; no separate schema	Aggregate only within the authorised scope and retain source profile references
Anonymised response	Any requested profile after identifier/PII transformation	Derived; examples only	Record the anonymisation method and never imply that a transformed identifier is the original meter ID
Billing summary	`MONTHLY` and/or `BILL_DETAILS` profiles	Native source profiles; rendered summary is derived	Preserve the signed/raw response as the audit source
Data-quality dashboard	`validationStatus`, `source`, missing interval IDs and cadence	Derived; no `dataQuality` summary object in MeterData v0.6	Compute rates and exceptions downstream
Exchange receipt	Beckn contract, signed response and acknowledgement	Protocol evidence, not a MeterData report	Retain with the request/response identifiers for DPDP and dispute audit

Example payloads for these response patterns are under `MeterData/v0.6/examples/`.

DER Visibility

*A DISCOM's view of every distributed energy resource (DER) behind its meters, in two halves: what is connected (Schedule I, ElectricityCredential v1.2) and what it is doing now (Schedule II, MeterData v0.6). Both halves are executable today at the per-consumer / per-DER level; the grid-side per-locus **aggregate** of Schedule I remains an illustrative future profile — see §1 and §8.2.*

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Field	Value
Applicability	All distribution licensees
This version	Executable today: the per-consumer ElectricityCredential v1.2 (Energy Passport). Illustrative, future: a grid-side, PII-free per-locus profile conceptually reusing `energyResources[]` + `consumptionProfiles[]` (the building blocks of ElectricityCredential v1.2), with the network locus — not a consumer — as subject; see §1.

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1. Scope and Purpose

The stakeholders are the DISCOM (issuer), its grid operator, and any aggregator enrolling controllable resources. As rooftop solar, batteries and EV charging spread, the licensee often can't answer: what's connected on feeder F-02, at what capacity, and is it controllable?

This document defines **DER Visibility** — a DISCOM's view of the distributed energy resources connected behind its meters, for grid operators and aggregators to ingest directly. It carries **no consumer names, addresses or contact details** in any tier.

Two delivery tiers. Not all of this use case carries the same privacy weight, and the distinction is load-bearing:

Tier	What it carries	Who may read it
Open	Counts and capacity totals per feeder or substation — no resource identifiers, no meter identifiers	Anyone
Authenticated	The per-resource records of Schedule I and the per-DER telemetry of Schedule II, keyed by resource and meter identifiers	Grid operators and enrolled aggregators, under a stated lawful basis

Identifier-keyed records are **pseudonymous, not anonymous**: a meter or resource identifier resolves back to a single consumer through the licensee's own systems. They are therefore personal data, and they sit in the authenticated tier. Only the open tier is publishable without access control. This tiering is the *proposed* answer to the privacy review still open at §11.3 — it is not yet a settled position.

What is executable today. The only currently executable ElectricityCredential v1.2 path is the **per-consumer credential** — the Consumer Energy Passport — whose `credentialSubject.customerProfile.customerNumber` is a required field. It carries the same `EnergyResource` and `ConsumptionProfile` building blocks referenced below, issued to one consumer at a time. See the validated Schedule I example. Today, a grid operator or aggregator wanting per-connection detail would consume that credential (with consumer consent), not a feeder-level aggregate.

A note on the aggregate (illustrative, future). ElectricityCredential is issued per consumer connection — its subject is one `customerProfile` with one customer number — so it **cannot combine multiple consumers' credentials** into a feeder- or substation-level record; each consumer keeps their own Energy Passport. A PII-free, per-locus view is described below as an **illustrative future profile**: an array of `EnergyResource` entries (with their topology links) plus matching `ConsumptionProfile` entries (sanctioned load, export limits, keyed by meter) for the locus, subject to the feeder / substation / licensee DID. It may conceptually reuse the `EnergyResource` / `ConsumptionProfile` structures ElectricityCredential composes, but it **does not validate as EC v1.2** (which requires a single `customerProfile` / `customerNumber`), is **not a signed EC v1.2 credential**, and has **no canonical executable example** today. It remains pending a separately governed schema / credential-subject contract — see §11.4.

2. What It Records / Covers

Illustrative — describes the future per-locus aggregate profile (§1); not a validated ElectricityCredential v1.2 payload.

Records	Detail	Source
Issuing licensee & locus	The DISCOM and the network locus the record covers	ElectricityCredential v1.2 (`issuer`)
Distribution transformers	Those in scope, where known	ElectricityCredential v1.2 (`energyResources[]`, network equipment)
Energy resources behind those DTs	Solar, battery, EV charger, inverter, controllable load — with capacity, inspection status, equipment	ElectricityCredential v1.2 (`energyResources[]`)
Topology	Parent/sub-resource chain (PV and BESS -> Inverter -> Meter -> DT)	ElectricityCredential v1.2 (`parentResources[]` / `subResources[]`)
Aggregator binding	Optional third-party aggregator enrolment	ElectricityCredential v1.2

Consumer identity is omitted, and consumers are never merged. One consumer's credential is never combined with another's: the record carries only `energyResources[]` and `consumptionProfiles[]` entries for the locus — no `customerDetails`, no customer numbers. Each consumer's own credential exists separately as the Energy Passport, held by the consumer.

That removes *direct* identifiers, not *all* linkage. `consumptionProfiles[].meterId` and each `energyResources[].id` remain resolvable to one consumer inside the licensee's systems, which is why these records sit in the authenticated tier (§1) rather than the open one.

3. How Each Item is Identified

Identical to Consumer Energy Passport §3 for the executable per-consumer path. *Illustrative, future (§1):* in the illustrative per-locus aggregate, the `credentialSubject.id` would differ by scope:

Scope	Example
Per feeder	`did:web:ies.discom.example:assets:feeder:F02`
Per substation	`did:web:ies.discom.example:assets:substation:SS-11KV-12`
Licensee-wide	`did:web:ies.discom.example`

4. Definitions

See Consumer Energy Passport §4 for shared terms (DER, DID, VC, EnergyResource, QuantitativeValue, Net Meter, Aggregator, DISCOM).

5. Basis of Standards

Identical to Consumer Energy Passport §5: IS -> CEA -> IEC -> IEEE; metering follows IS 16444 / IS 15959 directly; the net meter is the source of truth under CEA (Installation and Operation of Meters) Regulations, 2006.

6. Where Indian Standards Do Not Yet Exist

Identical to Consumer Energy Passport §6 — IEC CIM for the asset model; IEEE 1547 for DER attributes (CEA's own limits retained where set); IS 14286 (= IEC 61215) for PV module rating.

7. The Record

This use case produces **two records, not one** — the split the Schedules follow:

Record	Schedule	Nature	Status
The DER asset register — what is connected, where, at what capacity	Schedule I	Static; changes only on commissioning, capacity change or transfer	Executable today per consumer (§8.1); the PII-free per-locus aggregate is illustrative (§8.2)
DER telemetry — what those resources are doing now	Schedule II	Live; arrives continuously	Executable today, per DER

Illustrative (§1). In the future per-locus profile, each locus would be one signed Verifiable Credential per refresh cycle. Unlike the consumer-held Passport, it would be **published rather than held** — grid operators and aggregators would ingest it from the DISCOM's BPP catalogue. Open publication applies only to the open tier defined in §1; any locus record carrying resource or meter identifiers is served through the authenticated tier instead. Re-issuance would be regular (weekly for growth areas, monthly otherwise) or on material change; revocation would use the same DeDi flow as the Passport.

8. Schedule I — Static Fields of the Credential

Schedule I separates the executable per-consumer contract from the illustrative future aggregate. Rows in §8.1 are real ElectricityCredential v1.2 paths. Rows in §8.2 are **conceptual mappings, not normative JSON paths**: no current schema permits a PII-free multi-consumer locus subject, and no canonical aggregate example exists.

8.1 Executable Today — Per-consumer ElectricityCredential v1.2

Record Surface	Normative EC v1.2 Path	Schema Requires	DER Visibility Treatment
Issuer	`issuer.id` / `issuer.name`	Both required	The DISCOM is the credential issuer
Consumer subject	`credentialSubject.customerProfile`	Required	One credential covers one consumer connection only
Customer number	`credentialSubject.customerProfile.customerNumber`	Required	Prevents this schema from representing a PII-free multi-consumer aggregate
Energy resources	`credentialSubject.customerProfile.energyResources[]`	Required, minimum one	Carries meters, DER, inverters and network resources; see Passport §8.3–8.4
Topology	`energyResources[].parentResources[]` / `.subResources[]`	Optional	Links DER -> inverter -> meter -> DT/feeder where known
Capacity and controls	`energyResources[].attributes`	Optional	Carries rated/import/export limits, inspection, location and aggregator enrolment
Connection/load context	`credentialSubject.customerProfile.consumptionProfiles[]`	Optional array; required fields apply per entry	Links tariff/load/export facts to the relevant meter
Consumer PII	`credentialSubject.customerDetails`	Optional	Do not disclose to a grid operator unless separately authorised

Record Surface	Normative EC v1.2 Path	Schema Requires	DER Visibility Treatment
Executable example	`schedule-i-example.json`	Validated fixture	Use for the current per-connection ingestion path

8.2 Illustrative Future — PII-free Per-locus Aggregate

The following table is a design inventory for a separately governed subject contract. It deliberately does not claim validation against ElectricityCredential v1.2.

Conceptual Field *(not a current schema path)*	Expected Shape	Reused Building Block / Basis	Status and Guidance
Locus subject	feeder, substation or licensee DID/URI	`credentialSubject.id`; IES identifier patterns	Required by the future profile; the locus replaces the consumer as subject
Issuing licensee	issuer DID/URI and name	EC v1.2 `issuer`	Required; issuer remains the DISCOM
`energyResources[]`	array of typed resources	EC v1.2 `EnergyResource` union	Required; include only resources inside the stated locus
`energyResources[].id` / `.type`	stable resource identifier / governed discriminator	EC v1.2 common resource fields	Required per resource
`energyResources[].parentResources[]` / `.subResources[]`	arrays of resource identifiers or permitted inline children	EC v1.2 topology	Use to express DER -> inverter -> meter -> DT/feeder relationships
`energyResources[].attributes.ratedPower` / `.maxExport` / `.maxImport`	unit-bearing quantities	EC v1.2 common attributes	Populate where capacity is known and material to operations
`energyResources[].attributes.inspection`	date, result and inspector reference	EC v1.2 inspection object	Optional commissioning/ status evidence
`energyResources[].attributes.aggregator`	id, name, controllable flag, enrolment date	EC v1.2 aggregator object	Optional; records enrolment without duplicating the resource
`consumptionProfiles[]`	per-meter load/tariff context	EC v1.2 `ConsumptionProfile`	Include only where sanctioned-load/ export-limit context is needed
`consumptionProfiles[].meterId`	resource identifier	EC v1.2 meter link	Required per included consumption profile
`consumptionProfiles[].sanctionedLoad` / `.sanctionedExportLoad`	unit-bearing quantities	EC v1.2 load fields	Operational context, not measured telemetry
Consumer identity	no `customerNumber`; no `customerDetails`	Privacy boundary in §1–2	Must be absent from the future aggregate
Refresh/version metadata	separately governed	No current EC v1.2 aggregate field set	Open design item; must be specified before an executable fixture is published

9. Schedule II — DER Telemetry (Live Record)

Schedule II is the **live half** of DER Visibility — the time-dependent fields exchanged for this use case, as against the static asset facts in Schedule I. Where Schedule I records what is connected and how big it is, Schedule II records what it is doing now.

It is a hybrid mapping over MeterData v0.6: a native electrical subset that validates directly against the schema and its semantic validator, plus explicitly informative extensions for concepts MeterData v0.6 does not carry natively, plus out-of-band transport and security requirements. **The complete Schedule II record does not validate natively as MeterData v0.6** — only the canonical subset in §9.1 does (see §9.4).

Two things distinguish Schedule II from the rest of this use case:

- **It is executable today, per DER.** Unlike the illustrative per-locus aggregate in §8.2, the mapping below validates against a shipped schema with a canonical fixture (§9.4).
- **It arrives on a different rail.** DER telemetry travels inverter -> MQTT -> the national MNRE M2M platform (§9.3) — not over Beckn, and not through the Smart Meter Data Exchange path that carries net-meter readings. See §10.

Privacy. Per-inverter telemetry is traceable to one consumer's premises. It is *pseudonymous, not anonymous*, and belongs behind the authenticated tier described in §1 — it is never part of an openly published aggregate. See §11.3.

9.1 Native MeterData v0.6 Mapping — Electrical Subset

MeterData v0.6 profiles are per-meter. `meterRefs` (an `Identifier` with `scheme: "DID"`) references the metering point — here the inverter's `energyResources[].id` from Schedule I. Semantic validation (`validator.py`) applies different rules to `reportedMode: READING` descriptors (cumulative, per-timestamp values — no `openingValue`/`closingValue`, and cumulative values must not decrease across a series) versus `reportedMode: USAGE` descriptors (block-incremental values, where `closingValue - openingValue` must match the reported `value`). The canonical fixture keeps these in separate profile objects accordingly: **INSTANTANEOUS** for READING-mode snapshots, **INTERVAL** for USAGE-mode block energy.

DER Concept	Native `readingType`	OBIS	Unit	`reportedMode` / profile	Standard *(informative)*
Voltage, R/Y/B phase	`V_R` / `V_Y` / `V_B`	1.0.32.7.0.255 / 1.0.52.7.0.255 / 1.0.72.7.0.255	V	READING — INSTANTANEOUS	IEC 61724-1; IEC 61850
Current, R/Y/B phase	`I_R` / `I_Y` / `I_B`	1.0.31.7.0.255 / 1.0.51.7.0.255 / 1.0.71.7.0.255	A	READING — INSTANTANEOUS	IEC 61850
Power factor (3-phase)	`PF_3P`	1.0.13.7.0.255	ratio (no unit code)	READING — INSTANTANEOUS	IEC 61850
Frequency	`Freq`	1.0.14.7.0.255	Hz	READING — INSTANTANEOUS	IEEE 1547; IEC 61850
Active power import	`P_Import`	1.0.1.7.0.255	kW	READING — INSTANTANEOUS	IEC 61724-1; IEC 61850
Reactive power (lag)	`Q_Lag`	1.0.3.7.0.255	kvar	READING — INSTANTANEOUS	IEEE 1547; IEC 61850
Apparent power	`S_Total`	1.0.9.7.0.255	kVA	READING — INSTANTANEOUS	IEC 61850
Active energy import — cumulative	`kWh imp`	1.0.1.8.0.255	kWh	READING — INSTANTANEOUS	IEC 61724-1; OBIS (IS 15959)
Active energy export — cumulative *(accepted in lieu of a generation meter)*	`kWh exp`	1.0.2.8.0.255	kWh	READING — INSTANTANEOUS	IEC 61724-1; OBIS (IS 15959)

DER Concept	Native `readingType`	OBIS	Unit	`reportedMode` / profile	Standard *(informative)*
Active energy import — block incremental	`kWh imp block`	1.0.1.29.0.255	kWh	USAGE — INTERVAL	IEC 61724-1; OBIS
Active energy export — block incremental	`kWh exp block`	1.0.2.29.0.255	kWh	USAGE — INTERVAL	IEC 61724-1; OBIS

9.2 Explicitly Informative Extensions (not native MeterData v0.6 fields)

Concept	Why it is informative, not native	Suggested treatment
Harmonic distortion *(at 11 kV and above)*	`%` is not in MeterData v0.6's `UnitOfMeasure` enum	Report out-of-band, or as an unvalidated custom descriptor with no `unit` set; standard remains IEEE 519-2014 (CEA-referenced)
DC voltage / current / power	MeterData v0.6 has no AC/DC discriminator field; `PayloadDescriptor.category` is a free string, not a governed enum	If reported, use a custom `category` (e.g. `"dc"`) on a V/A/kW-unit descriptor — this is a convention, not a defined native concept
Inverter state / fault code	Status/alarm codes are not a `Reading.value` (number). Binary fault conditions partially overlap the native `AlarmProfile`/`MeterAlarm` shape (`status`: ACTIVE/CLEARED, `severity`), but a general inverter operating-state code has no native field	Use `AlarmProfile` for binary fault conditions; treat richer state codes as an extension
Irradiance (plane of array), module/ambient temperature	`W/m²` and `°C` are not in the `UnitOfMeasure` enum	Report out-of-band or as an unvalidated custom descriptor
Battery state of charge / state of health	`%` is not in the `UnitOfMeasure` enum	Report out-of-band
EV session ID, connector status	String values, not `Reading.value` (number). Charging power and energy delivered *are* representable as native kW/kWh readings	Session/connector metadata stays out-of-band or in Schedule I's DER/aggregator attributes
`m2mSimId`, `tlsCertFingerprint` (device security certificate)	No device-credential fields exist in MeterData v0.6	Out-of-band (see §9.3)

9.3 Transport and Security (out-of-band — MNRE M2M framework)

Requirement	Standard
Transport: MQTT to the national MNRE M2M platform	MNRE M2M
Device identity: `m2mSimId`	MNRE M2M
Transport security: TLS, with device certificate fingerprint	MNRE (TLS; IEC 62443 guidance)
Data residency: held in India	MNRE M2M

These are transport/security requirements on the channel carrying MeterData v0.6 payloads, not fields inside the payload itself.

9.4 Example and Validation

Executable canonical-subset fixture: `schedule-ii-example.json` — one `DESCRIPTOR` profile plus one `INSTANTANEOUS` and one `INTERVAL` profile, covering the §9.1 electrical subset only.

Validate structurally against MeterData v0.6:

```
python -X utf8 -B scripts/validate_schema.py schemas/MeterData/v0.6/schema.json
 ↪  use-cases/der-visibility/examples/schedule-ii-example.json
```

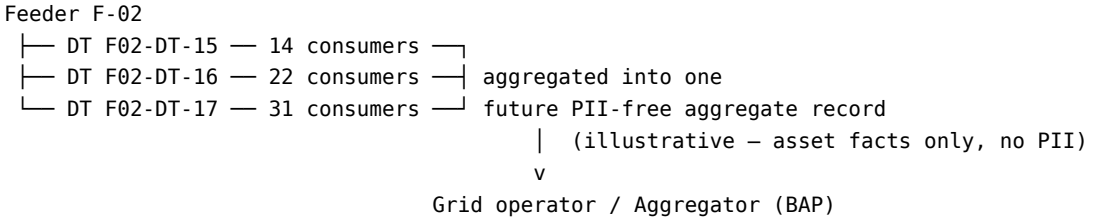
Validate semantics:

```
python -X utf8 -B schemas/MeterData/v0.6/validation/validator.py
 ↪  use-cases/der-visibility/examples/schedule-ii-example.json
```

This fixture exercises OBIS-to-readingType resolution against the canonical descriptor set, mode/profile matching (each `readingType`'s `reportedMode` and permitted profile shape), allowed-attribute type checks on each interval payload, interval-id monotonicity (strictly increasing `id` within a profile), and compact-sequence arity (payload count per interval matching the referenced sequence's item count). It does **not** exercise cumulative non-decrease (its `INTERVAL` profile carries USAGE-mode block-incremental values, not READING-mode cumulative ones) or opening/closing math (no reading in this fixture carries `openingValue`/`closingValue`).

10. How It Fits Together

Illustrative / non-normative — describes the future aggregate profile (§1); the boxes below are not a signed EC v1.2 credential.



Each consumer's own per-consumer ElectricityCredential v1.2 (the Energy Passport) is built from the same source-of-truth (CIS / DERMS / inspection register) that would back this future aggregate — the two stay conceptually in sync because they'd read the same data and reuse the same building blocks.

10.1 Where each Schedule's data comes from

DER Visibility is the one IES use case fed by **more than one exchange**. The two Schedules do not share a rail, a counterparty or a security model:

Schedule	Data	Source	Transport
I — static	Asset register: resources, topology, capacity, inspection	DISCOM's own CIS / DERMS / inspection register	Internal; surfaced per consumer as an EC v1.2 credential (§8.1)
I — static	Aggregator enrolment and controllability	The aggregator	Per-resource `attributes.aggregator`, written into the register
II — live	DER telemetry (inverter, PV, BESS)	The inverter or its gateway	MQTT -> national MNRE M2M platform (§9.3)
II — live	Net-meter readings, where used in place of inverter data	AMISP / MDM	Beckn, via Smart Meter Data Exchange

DER Visibility is therefore not a single-schema use case. ElectricityCredential v1.2 carries the static register; the live half arrives over two other channels entirely. The resource and meter identifiers established in Schedule I are what join them.

11. Points for Confirmation

1. **Refresh cadence per locus** — to be tuned per pilot, once the aggregate profile is formalised.
2. **Aggregator binding** — the exact `telemetryProvider` field and the proof an aggregator presents to claim a resource.
3. **Privacy review** — confirmation that the two-tier split proposed in §1 meets DPDP grid-side disclosure norms. Both `consumptionProfiles[].meterId` (Schedule I) and per-inverter telemetry (Schedule II) are keyed to identifiers that resolve to one consumer — pseudonymous rather than anonymous — so both sit in the authenticated tier, and only aggregate counts and capacity totals are proposed for open publication. **This is a proposal, not a settled position**, and it is the open item on which Schedule II's disclosure model depends.
4. **Aggregate record shape** — ElectricityCredential requires a single `customerProfile` with one customer number, so it cannot represent a PII-free, multi-consumer aggregate. That aggregate needs its own credential-subject shape, formalised upstream through separate governance; until then it remains an illustrative future profile with no canonical executable example.

Schemas Used in This Use Case

Two schemas, one per Schedule:

Schedule I (static) — ElectricityCredential v1.2. Executable today, issued per consumer as the Energy Passport — see the validated Schedule I example.

Schedule II (live) — MeterData v0.6. Executable today, per DER — see the validated Schedule II example. The canonical electrical subset (§9.1) validates natively; the extensions in §9.2 and the transport requirements in §9.3 do not and are marked informative.

Illustrative, future: a per-locus aggregate of the Schedule I register that would conceptually reuse the `EnergyResource` and `ConsumptionProfile` structures ElectricityCredential composes. It is not itself an EC v1.2 payload and has no schema of its own yet; formalising one is tracked in §11.4.

Value Unlock

Illustrative, future (§1) — describes the value case for the aggregate profile once it exists; the per-consumer Energy Passport is the only path executable today.

Grid operator — first-class feeder-level visibility for forecasting, planning, dispatch and outage analysis. **Aggregators** — a signed discovery surface for controllable resources; enrolment becomes mechanical. **DISCOM** — the same data backing every consumer Passport, republished once, with names and addresses never disclosed at either tier. **Regulators** — a consistent, auditable DER register across licensees.

The live half (Schedule II) is what makes forecasting and dispatch real rather than notional: the static register says a 5 kW array exists, and only telemetry says what it is generating now.

Annexure A — Standards Referenced

Identical to Consumer Energy Passport — Annexure A.

Annexure B — Example Payload

Schedule I (static): the validated Consumer Energy Passport Schedule I example, for the executable per-consumer path.

Schedule II (live): the validated `schedule-ii-example.json` — one `DESCRIPTOR` profile plus one `INSTANTANEOUS` and one `INTERVAL` profile, covering the §9.1 electrical subset. Validation commands are in §9.4.

No canonical example exists for the illustrative per-locus aggregate (§1, §11.4) — it remains a conceptual future profile pending the separately governed credential-subject contract.

Annexure C — JSON Schema

Schedule I: the EC v1.2 `schema.json`, `context.jsonld` and `vocab.jsonld` referenced in Consumer Energy Passport — Annexure C apply to the per-consumer Energy Passport credential — see §1.

Schedule II: MeterData v0.6 — `schema.json`, with the semantic validator at `schemas/MeterData/v0.6/validation/validator.py`.

The illustrative future per-locus aggregate has no schema, context or vocab of its own — it is not an EC v1.2 payload and there is nothing to validate it against yet; formalising a schema is tracked in §11.4.

Annexure D — Derived Views

Computed downstream from the Schedules. None is a schema, a populated template, or an exchanged record — they are what a consumer of this use case builds after ingestion.

Operational View	Inputs	Schema Status	Treatment
Connected DER inventory	Schedule I `energyResources[]` grouped by type and locus	Derived	Count and capacity totals are computed from the source records; this is the open tier of §1
DER growth over time	successive inventory snapshots	Derived	Compare versioned snapshots; do not present one credential as a time series
Feeder/DT loading study	Schedule I topology and sanctioned load/export limits, joined to Schedule II telemetry	Derived	Static facts do not substitute for measured load profiles
Controllability register	per-resource aggregator and controllable attributes	Derived	Preserve the underlying resource identifier and enrolment evidence
Exception list	missing topology, inspection or capacity fields	Derived	Report absence as an evidence gap, not as zero capacity
Future signed aggregate	conceptual §8.2 record	**Not currently executable**	Requires an approved schema/credential-subject contract and canonical fixture before publication

P2P Energy Transaction

*Two prosumers — consumers who can both inject and consume energy — on different DISCOMs execute a direct, signed energy trade over the same Beckn wire that carries IES dataset exchanges; the payload is a contract and its fulfilment, not a dataset. Each DISCOM is represented in the protocol by a regulated **Ledger Provider**, and settlement is computed by signed Rego policy, with no central exchange.*

Implementation Guide ->

Applicability	Trading platforms, regulated Ledger Providers, DISCOMs
This version	Built on the DEG `P2PTrade` / `DEGContract` / `BecknTimeSeries` family (canonical at schema.nfh.global) over Beckn, with signed Rego policies governing the network and contract rules. Mirrored in External Schemas — Energy Trading.

Initial topology. The architecture supports one Ledger Provider per DISCOM, but the starting configuration connects **all trading platforms to a single Ledger Provider** at `ies-p2p-energy-ledger.beckn.io` — the two LPs in the diagrams below collapse into one (the intra-DISCOM topology; same protocol, fewer hops). The network namespaces are `indiaenergystack.in/test-ies-p2p-trading-network` (test) and `indiaenergystack.in/ies-p2p-trading-network` (production).

---

1. Scope and Purpose

The **stakeholders** are two prosumers (buyer and seller) on potentially different DISCOMs, their respective trading platforms (TPs), and the regulated Ledger Provider (LP) contracted by each DISCOM. Today, peer-to-peer energy trade requires bespoke bilateral integrations, ad-hoc settlement spreadsheets, and a central exchange-style intermediary — none of which exist in the form Indian DISCOMs need.

This document defines **P2P Energy Transaction** — a one-to-many discovery, contracting and settlement pattern carried over the same Beckn wire that carries dataset exchanges. The contract is a `DEGContract` with a `P2PTrade` body. Allocation and reconciliation flow as `BecknTimeSeries` inside the same envelope. Network rules are enforced by a **signed Rego network policy** in the adapter, and settlement terms by the seller-DISCOM's **contract policy** — a signed Rego bundle published on DeDi. Any participant evaluates locally; no central exchange.

A trading platform integrates once. The same pattern works inter-DISCOM (two LPs, one peer leg) and intra-DISCOM (one LP, the two LPs collapse).

2. What It Records / Covers

For one peer-to-peer trade the records carry:

Records	Detail	Source
The contract	Agreed quantity, price per kWh, delivery window, the four roles (buyer / seller / buyer's DISCOM / seller's DISCOM), and the `policyUrl` for the Rego bundle in force	`DEGContract` / `P2PTrade` (DEG)
The offer	The seller's price, available quantity, validity window, source-type constraint (e.g. no GRID-sourced energy)	`EnergyTradeOffer` (DEG)
Per-interval time series	`PRICE_PER_KWH`, `AVAILABLE_QTY`, `REQUESTED_QTY`, `BUYER_DISCOM_ALLOC`, `SELLER_DISCOM_ALLOC`, `FINAL_ALLOC`	`BecknTimeSeries` (DEG)
LP<->DISCOM binding	Each LP's `utilityId` and ledger endpoint	`DiscomLedgerProvider` (DEG)
Meter-data sub-transactions	Per-DISCOM actual injected / consumed quantities per interval, supplied during reconciliation by the DISCOM to its contracted LP as input to allocation — rides inside the same `message.contract` envelope as `BecknTimeSeries`, **not** a separate `MeterData` exchange	`BecknTimeSeries` (DEG)
Revenue flows	Computed from the final allocation by the seller-DISCOM's **contract policy** Rego, recorded inside the contract (wire key `revenueFlows`, type `RevenueFlow`; injected by the `contractpolicyenforcer` ONIX step)	`RevenueFlow` (DEG)

Customer PII and raw meter data stay with the customer's own DISCOM and TP. Both LPs record the confirmed contract — including the agreed price — and the cascaded allocation and settled-quantity updates.

3. How Each Item is Identified

Participants are identified by their plain **network subscriber IDs** — the `participantId` under which they are registered in the network registry (DeDi). No `did:web` (or any other DID scheme) is required for participants; the subscriber ID is a hostname-style string and is what appears in `context.bapId` / `context.bppId` and in the contract's `roles[]`.

Subject	Identifier method	Example
Trading platform (BAP / BPP)	Network subscriber ID	`buyerapp.example.com`
Ledger Provider (LP)	Network subscriber ID	`seller-discom-ledger.example.com`
DISCOM	Network subscriber ID; bound to its LP by `utilityId` in the `DiscomLedgerProvider` block	`buyerdiscom.example.com` (`utilityId: TEST_DISCOM_BUYER`)
Buyer / Seller (prosumer)	Represented by their TP — the contract's `roles[buyer/seller]` carry the TP's subscriber ID; the prosumer is pinned by their meter reference	`roles[buyer].participantId = buyerapp.example.com`
Meter / DT / feeder referenced in the trade	Existing utility asset ID wrapped as `did:web:<discom-id>:meters:<meter-number>` (per Register — Identifier patterns)	`did:web:buyerdiscom.example.com:meters:NM-44091234`
Network policy	Rego file loaded by the adapter (`opapolicychecker` step) from `specification/policies/`	`p2p-trading-ies-wave2-networkpolicy.rego`
Contract policy	DeDi-published Rego record URL (`contractAttributes.policy.url`)	`https://api.dedi.global/dedi/lookup/indiaenergystack.in/ies-policies/ies-p2p-network-settlement-rego-policy-v1`

No new identifier scheme. The four-actor topology reuses the same subscriber-registry machinery as every other IES use case; public keys resolve from the same DeDi record the subscriber ID names.

4. Definitions

- **Prosumer** — a consumer who can both inject (sell) and consume (buy) energy.
- **TP** (Trading Platform) — the BAP or BPP that represents a prosumer on the network and runs the matching engine.
- **LP** (Ledger Provider) — a regulated service that holds each DISCOM's slice of the trade record. Each DISCOM contracts exactly one LP; two DISCOMs may share an LP.
- **Inter-DISCOM** — buyer and seller served by different DISCOMs; two LPs involved.
- **Intra-DISCOM** — buyer and seller served by the same DISCOM; the two LPs collapse into one.
- **Discovery service** — the network service that answers `discover` queries against catalogs that provider nodes have listed via `publish-catalog`.
- **`BecknTimeSeries`** — the per-interval payload carrier; declares `payloadDescriptors` (each column's `payloadType` and `insertedBy`) and per-interval `payloads[]`.
- **Cascade** — the auto-routing by which contract and settled-quantity messages reach both TPs and both LPs in the right order; implemented by the `degledgerrecorder` ONIX plugin (see Auto-routing of contracts and allocations in the Implementation Guide).
- **Policy-as-code** — the network and contract rules as signed Rego policies, evaluated locally with OPA.

5. Basis of Standards

IES order of preference: **IS -> CEA -> IEC -> IEEE**. Indian standards do not yet exist for peer-to-peer energy trade as a protocol. The IES choices:

Standard	Role here
Beckn Protocol v2	The discovery / contracting / status lifecycle (`discover` -> `select` -> `init` -> `confirm` -> `status`); providers list offers to the Catalog service via `publish-catalog`, consumers query the Discovery service with `discover`
DEG schema family	`P2PTrade`, `EnergyTradeOffer`, `EnergyTradeDelivery`, `DEGContract`, `DiscomLedgerProvider`, `BecknTimeSeries` — canonical at schema.nfh.global
OPA / Rego	The policy bundle format (standardised by CNCF)
W3C VC Data Model 2.0 / W3C DID Core	Issuer key, signing
JSON-LD 1.1	Wire format and semantic resolution

Meter data referenced by the trade conforms to **IS 16444** and **IS 15959** — the same standards as the Smart Meter Data Exchange.

6. Where Indian Standards Do Not Yet Exist

The whole protocol — the four-actor topology, the `BecknTimeSeries` payload vocabulary for trade negotiation, the cascaded routing, the policy-as-code framework — is an IES choice with no Indian standard predating it. The CERC Innovation Sandbox order (2023) is the regulatory umbrella; CEA / CERC standards specific to peer-to-peer trade are expected and will inform future versions.

7. The Records

The P2P Energy Transaction flow produces three distinct kinds of signed artefact per trade:

1. The **contract** — `DEGContract` carrying a `P2PTrade` body. Recorded by both TPs and both LPs at confirm time (each LP receives it as a blocking `on_confirm` forward from its TP).
2. The **per-interval allocation series** — `BecknTimeSeries` carrying buyer-DISCOM allocation, seller-DISCOM allocation, final allocation. Recorded by both LPs and both TPs as Phase 5 cascades.
3. The **revenue-flow record** — computed from the final allocation by the seller-DISCOM's contract policy via the `contractpolicyenforcer` ONIX step; signed by the policy author and stored in `message.contract.consideration[id=auto-settlement-flows].considerationAttributes` (wire key `revenueFlows`, type `RevenueFlow`).

Together they form a **complete, attributable audit trail** of the trade — from offer to settlement — with no central exchange.

If the use case needs a holder-bound credential — e.g. a prosumer carrying a credit-worthiness attestation in a wallet — that uses the Consumer Energy Passport or Consumer Meter Digest separately.

8. Schedule I — Static Fields of the Exchange

Schedule I is a use-case profile table over DEG's externally governed schema family, canonical at schema.nfh.global and mirrored in External Schemas — Energy Trading. Because these fields are not maintained in this repository, the schema-qualified names below identify the upstream contract rather than local JSON-Schema paths. **Upstream Requires** follows the mirrored v2.0 field tables; **P2P Guidance** is informative use-case guidance.

8.1 Contract Roles and Policy

Upstream Field	Type / Allowed Value	Upstream Requires	P2P Guidance *(informative)*
`P2PTrade`	EnergyContract subclass	Inherits its fields; defines no additional fields in the mirrored table	Contract body identifying the P2P trade profile
`DEGContract.roles[]`	array of role objects	Required	Carry the participant-binding roles used by the trade
`DEGContract.roles[].role`	text	Required per role	Network policy expects buyer, seller, buyer-DISCOM and seller-DISCOM roles
`DEGContract.roles[]. participantId`	text or `null`	Key required per role; value may be null before binding	Bind to the registered subscriber/compound participant ID at select/init
`DEGContract.policy`	policy object	Required	Points to the signed Rego contract policy in force
`DEGContract.policy.url` / `.queryPath`	URI / text	Both required	Resolve only from the configured trusted DeDi policy prefixes
`DEGContract. revenueFlows[]`	array of signed role/value/ currency rows	Optional legacy shape	Wave-2 settlement uses the `RevenueFlow` consideration shape in §8.4 instead
inherited `Contract. participants[]`	Beckn/DEG participant objects	Defined by the parent Contract schema	Carry participant attributes such as meter and utility identity; consult the canonical parent schema

8.2 Offer, Customer and Order Item

Upstream Field	Type	Upstream Requires	P2P Guidance *(informative)*
`EnergyTradeOffer. validityWindow`	time period	Optional	Expire the offer before its earliest delivery interval
`EnergyTradeOffer. contractAttributes`	JSON-LD object	Optional	At catalog publication, declare known roles and the contract-policy terms
`EnergyTradeOffer. contractAttributes.@type`	text	Required when the object is used	Normally `DEGContract`
`EnergyTradeOffer. commitmentAttributes`	`BecknTimeSeries` JSON-LD object	Optional	Carries seller price/ available-quantity columns and their descriptor contract
`EnergyTradeOffer. commitmentAttributes.@ type`	constant `TimeSeries`	Required when the object is used	Use the BecknTimeSeries v1.0 context
`EnergyCustomer.meterId`	text (`der://meter/{id}` in the upstream description)	Required	Pins the represented buyer or seller to a meter
`EnergyCustomer. sanctionedLoad`	number (kW)	Optional	Use only where network/contract policy evaluates the approved load

Upstream Field	Type	Upstream Requires	P2P Guidance *(informative)*
`EnergyCustomer.utilityCustomerId` / `.utilityId`	text / text	Optional	Utility account and service-territory binding; minimise disclosure outside regulated participants
`EnergyCustomer.platformUrl`	base URI	Optional	Base address for cascade sub-transactions
`EnergyOrderItem.providerAttributes`	object	Required	Carries the `EnergyCustomer` identity for the order item
`EnergyOrderItem.fulfillmentAttributes`	object	Optional	Populate only in status/update responses for delivery tracking

9. Schedule II — Trade Actuals and Settlement (Live Record)

Schedule II is the **live half** of a P2P trade — the per-interval values that keep arriving once the contract in Schedule I is agreed. Schedule I fixes who is trading, under which policy, on what terms; Schedule II carries what was actually offered, allocated, delivered and settled, slot by slot.

The wire shape is `BecknTimeSeries`, not MeterData v0.6 — see §9.3.

9.1 Per-interval Negotiation and Allocation (`BecknTimeSeries`)

Upstream Field / Payload Type	Type	Upstream Requires	Writer / Role in the Trade *(informative)*
`intervalPeriod`	ISO 8601 start + duration	Required	Defines the delivery slots
`payloadDescriptors[]`	event/report payload descriptors	Required	Declares every column's type, unit/currency and responsible writer
`intervals[]`	interval rows	Required	Carries the typed values for each slot
`resourceName` / `clientName`	text / text	Optional	Identifies the source resource and reporting participant
`PRICE_PER_KWH`	event payload, INR	Profile-required at offer/confirm	Seller platform
`AVAILABLE_QTY`	event payload, kWh	Profile-required in the published offer	Seller platform
`REQUESTED_QTY`	event payload, kWh	Profile-required after buyer selection	Buyer platform
`BUYER_DISCOM_ALLOC` / `BUYER_DISCOM_STATUS`	report payloads	Required by the reconciliation profile when buyer allocation is reported	Buyer DISCOM
`SELLER_DISCOM_ALLOC` / `SELLER_DISCOM_STATUS`	report payloads	Required by the reconciliation profile when seller allocation is reported	Seller DISCOM
`FINAL_ALLOC`	report payload, kWh	Required by the settled profile	Seller DISCOM; network policy enforces `FINAL_ALLOC <= min(buyer allocation, seller allocation)`

Every `payloadType` used in an interval must be declared by the profile's descriptor contract. The full schema uses OpenADR's open-string payload convention; the governed P2P names above are profile constraints enforced by the network/contract policies, not a closed enum in base `BecknTimeSeries`.

9.2 Settlement Revenue Flow

Upstream Field	Type	Upstream Requires	P2P Guidance *(informative)*
`consideration[].consideration Attributes.@type`	constant `RevenueFlow` in this profile	Required by the P2P settlement profile	Store policy output in the `auto-settlement-flows` consideration entry
`RevenueFlow.revenueFlows[]`	array of role/value/currency objects	Required	Contract-policy output must balance to zero across rows
`revenueFlows[].role`	text	Required	Names the buyer/seller platform or DISCOM role
`revenueFlows[].value`	signed number	Required	Positive receives; negative pays
`revenueFlows[].currency`	ISO 4217 code	Required	Use `INR` for this profile
`revenueFlows[].description`	text	Optional	Explain energy, wheeling, platform or shortfall components

9.3 Resource and Meter-actual Binding

Record Surface	Shape	Contract Status	P2P Treatment
Offered energy resource	`EnergyResource` discriminated union	Upstream shared schema	Minimal P2P use is stable `id` + `type`; add attributes only when policy needs them
Trade-side meter identity	`EnergyCustomer.meterId`	Required upstream field	Identifies the prosumer endpoint used by the trade
Daily injected/consumed actuals	`BecknTimeSeries` report payload types	Governed by the P2P reconciliation profile	Supplied by each DISCOM to its LP inside the contract-status flow
MeterData v0.6	Separate IES telemetry schema	**Not embedded as a MeterData profile in this trade contract**	A DISCOM may derive actuals from its MeterData system, but the trade-side wire shape remains BecknTimeSeries
Ledger-provider binding	`DiscomLedgerProvider` upstream schema	Defined only at schema.nfh.global	Associates a regulated ledger endpoint with its DISCOM/utility ID

10. How It Fits Together

Everything buyer-side mirrors everything seller-side: a buyer prosumer and their trading platform (BAP) on one side, a seller prosumer and their trading platform (BPP) on the other, the two platforms meeting over the `select / init / confirm / status` leg. Each DISCOM is represented by one regulated Ledger Provider, and each LP pulls metered actuals daily from its own DISCOM as the input to allocation. The two LPs — one per DISCOM — never speak to each other directly; the two TPs are the only liaison between them. No central exchange. Discovery goes through the network's Discovery service: the SellerTP lists offers via `publish-catalog`, the BuyerTP queries with `discover`.

The block-topology diagram and the full six-phase sequence diagram (with the automated ONIX legs shaded) live in the **Implementation Guide -> What each actor does, per phase** — placed there so a developer has the topology, the wire sequence and the per-actor build steps in one place.

The six phases

This is the inter-DISCOM flow. Intra-DISCOM (buyer and seller behind the same DISCOM) collapses Phase 2's optional limit check and Phase 5 into a single ledger. The wire sequence is drawn — with the automated (ONIX) legs shaded — in the Implementation Guide; in words:

1. **Discovery** — SellerTP lists an `EnergyTradeOffer` catalog (`publish-catalog`); BuyerTP queries the Discovery service with `discover` and a JSONPath intent filter.
2. **Select and init** — quantity and price refined; optional LP headroom pre-check.
3. **Confirm** — the buyer's `confirm` is answered by the seller's `on_confirm`; as that `on_confirm` travels, each TP's `degledgerrecorder` forwards a rewritten copy to its own LP and **blocks until the LP ACKs** — seller-side before the `on_confirm` leaves for the buyer, buyer-side before the buyer app receives it. The LPs do not emit an `on_confirm` of their own; their sync ACK is the record receipt.
4. **Delivery** — seller injects, buyer consumes.
5. **Allocation and reconciliation** — **daily**, each LP asks its own DISCOM (a `status` request on the LP's `/bap/caller`) for metered actuals covering the meters and intervals that still carry an **unallocated trade**; the DISCOM answers with an `on_status` bearing those actuals, and the LP allocates its side. Either TP may then **poll** for the peer's latest allocation with a `status` request — which the peer TP's adapter auto-cascades to its own ledger before that ledger answers with a fresh `on_status`. Every allocation and settled-quantity `on_status` then cascades **symmetrically** through the two TPs to the opposite LP (an automatic forward at one TP, an automatic record at the other): buyer-side updates run `BuyerLP -> BuyerTP -> SellerTP -> SellerLP`, seller-side updates run the mirror chain, so all four parties converge on `FINAL_ALLOC`. The `contractpolicyenforcer` step computes the revenue flows as the settled `on_status` passes the seller TP.
6. **Billing and settlement** — buyer pays seller (off-ledger via TPs); DISCOM monthly bills are adjusted accordingly.

The allocation logic on each LP can be as simple as **pro-rata across the customer's trades in the delivery window**. The same Phase-5 message flow supports multiple rounds — a provisional allocation, a final allocation after meter-data finalisation, a deviation true-up — by repeating the `/status` round-trip with a fresh `BecknTimeSeries` payload. Iteration is a payload concern, not a protocol concern.

The cascade rules, the four ledger interfaces, and the payload snapshots that make this flow concrete are in the **Implementation Guide**.

11. Points for Confirmation

1. **Wheeling / penalty tariff values** — the rates in force are whatever the seller-DISCOM's published contract policy defines; each DISCOM sets production values per its tariff order by publishing its own policy version.
2. **`contractpolicyenforcer` ONIX step** — ships in the wave-2 seller-TP config (computes revenue flows on `on_status` from the DeDi-resolved contract policy); being aligned across LP implementations.
3. **TEST -> PROD `utilityId` allow-list** — the network bundle's production rules check approved IDs only; the production allow-list is governance-pending.
4. **CERC sandbox graduation** — production-grade network policy bundle awaits CERC sign-off post-sandbox.
5. **Intra-DISCOM topology** — collapsing the two LPs into one is supported and lighter; the configuration convention is in the wave-2 devkit, being formalised.
6. **Daily meter-data pull cadence** — the LP->DISCOM actuals pull is modelled as a daily job over meters and intervals with unallocated trades; the exact cadence and the retry/true-up window are per-DISCOM and being formalised.

Schemas Used in This Use Case

Schema	Role
P2PTrade	The contract `@type` — agreed quantity, price, delivery window, the four roles, the policy URL
EnergyTradeOffer	The seller's offer block
EnergyTradeDelivery	The performance block populated during reconciliation
DEGContract	The envelope — roles, the rego policy URL, computed revenue flows
DiscomLedgerProvider	The LP<->DISCOM binding (`utilityId`, `ledgerUrl`)
BecknTimeSeries	Per-interval payload carrier — declares `payloadDescriptors` and `payloads[]`
ElectricityCredential v1.2 *(optional)*	Seller's attestation of meter / sanctioned-load / DER details backing the offer

A consolidated field reference is in **External Schemas — Energy Trading**.

Value Unlock

For prosumers — a direct, signed channel for selling distributed energy, with no central exchange.

For trading platforms — one integration; the same protocol intra- and inter-DISCOM; allocation and settlement computed by signed Rego, not custom code.

For DISCOMs — wheeling charges and deviation penalties are the output of a signed function over a signed contract, not a bilateral spreadsheet. Visibility into peer-to-peer trade is a by-product, not a separate reporting effort.

For regulators — the network rules are the policy bundle itself. A policy change is a new signed bundle on DeDi, picked up by every participant on the next contract.

Annexure A — Standards Referenced

Standard	Scope
CERC Innovation Sandbox Order, 2023	Regulatory umbrella for peer-to-peer trade pilots
Beckn Protocol v2	Discovery, contracting, payload delivery, status, signed audit
DEG `P2PTrade` / `DEGContract` / `BecknTimeSeries` family	The payload schema family on the wire
OPA / Rego (CNCF)	Policy-as-code format for network and settlement bundles
IS 16444 (Parts 1, 2)	AC smart meter — specification (for trade-side meter quantities)
IS 15959 (Parts 1–3)	DLMS/COSEM data-exchange companion specification; OBIS codes
W3C VC Data Model 2.0; W3C DID Core	Issuer keys; signing
JSON-LD 1.1	Wire format and semantic resolution

Annexure B — Example Payloads

The wave-2 devkit ships example payloads for every phase, per role:

-> **`devkits/p2p-trading-ies-wave2/uc1/`**

Annexure C — JSON Schema

Canonical references at **schema.nfh.global**:

- **P2PTrade/v2.0**
- **DEGContract/v2.0**
- **EnergyTradeOffer/v2.0**
- **EnergyTradeDelivery/v2.0**
- **DiscomLedgerProvider/v2.0**
- **BecknTimeSeries/v1.0**

A consolidated field reference for the trade schemas (except `DiscomLedgerProvider` and `EnergyTradeDelivery`, defined only at schema.nfh.global) is in **External Schemas — Energy Trading**.

Annexure D — Derived Views

Computed downstream from the Schedules above. None is exchanged, and none is a separate populated IES schema.

Derived View	Sources	Schema Status	Treatment
Per-DISCOM monthly bill adjustment	settled `FINAL_ALLOC`, revenue-flow rows and the applicable policy	Derived	Exclude traded volume/include charges according to the signed policy and local billing rules
Trading-platform book of trades	confirmed contracts plus allocation/status history	Derived	Preserve contract and interval identifiers so every row traces to signed evidence
Ledger reconciliation statement	both LP copies, DISCOM allocation series and final allocation	Derived	Differences are exceptions; do not overwrite either signed source
Prosumer statement	price, requested/final quantity and applicable charges	Presentation only	May be rendered for the consumer; the signed contract and revenue flow remain authoritative
Network performance report	trade counts, delivery ratios and policy failures	Derived aggregate	Must not expose raw customer identifiers or meter data
Holder-bound credit/energy credential	Consumer Energy Passport or Consumer Meter Digest	Separate schema/use case	Do not repurpose the P2P contract as a wallet credential

Use Case Implementation Guides

Consumer Energy Passport

In a hurry? Jump to the Checklist. For the standards basis and full field schedule, see the **Overview**.

The Consumer Energy Passport is an ElectricityCredential v1.2 issued holder-bound to a consumer's wallet. It is not a new credential type — it is *how* the existing v1.2 credential is shaped, issued, and delivered when the consumer is the audience.

A DISCOM signs it. The consumer carries it in DigiLocker or a DID wallet. Banks, marketplaces, regulators, subsidy portals, and consented apps verify it offline using the DISCOM's published `did.json`.

Why this use case exists

Consumers carry paper attestations of their connection details — sanctioned load, tariff category, asset list — for KYC at banks, rooftop-solar marketplaces, EV-charger installers, housing societies, and subsidy portals. Every recipient calls or emails the DISCOM to confirm. The Passport replaces that loop with a credential the verifier checks offline, without contacting you.

How it differs from a bearer ElectricityCredential

Same schema, same issuance pipeline. Two issuance-time differences:

Concern	Bearer ElectricityCredential v1.2	Consumer Energy Passport
`credentialSubject.id`	absent — bearer; whoever holds the JSON is treated as the subject	**set** to the consumer's wallet DID (`did:key` or `did:jwk`)
`customerProfile.idRef`	optional	**populated** with a verifiable government-ID reference (e.g. masked Aadhaar, DigiLocker pull receipt)

Everything else (`customerProfile`, `customerDetails`, `energyResources[]`, `consumptionProfiles[]`, `issuer`, `proof`, revocation) is identical to any other ElectricityCredential v1.2. The `type` array stays `["VerifiableCredential", "ElectricityCredential"]` — no new VC type is introduced.

Actors and flow

Role	Who	What they do
Issuer	DISCOM	Signs the Passport once the consumer has been identity-proofed
Holder	Consumer	Stores the Passport in DigiLocker / a DID wallet; chooses when and to whom to present

Role	Who	What they do
Verifier	Bank, marketplace, regulator, subsidy portal, housing society, EV-charger installer	Reads the Passport from the wallet; verifies the issuer's signature, revocation status, and the holder-binding proof

A typical issuance happens once at customer onboarding (and then is re-issued on material change — meter swap, sanctioned-load change, DER commissioning).

Building blocks

Block	Used for
Identifiers and Addressing	DISCOM's `did:web`; consumer's wallet `did:key`; asset / meter / connection DIDs that appear inside the credential
Energy Credentials	The single home for signing, verifying, and revoking — including the Consumer Energy Passport variant under "Credential variants"
Holder binding	Wallet-DID binding pattern; presentation-time challenge / VP proof
DigiLocker delivery	Delivers the credential into the consumer's DigiLocker for easy access and sharing; for many use cases DigiLocker's Aadhaar pull also acts as the identity-binding step

Setup: Register -> Discover -> Exchange

1. **Register** — set up the DISCOM `did:web` and run OpenCred -> Setup Register; Energy Credentials — Set up OpenCred.
2. Decide your identity-proofing method (DigiLocker pull, offline-KYC XML, in-person KYC, record-match) and document it for privacy review — see Identifiers — Identity-proofing at issuance.
3. **Exchange** — issue the Passport via Energy Credentials — Issue your first credential, setting `credentialSubject.id` to the wallet DID and populating `customerProfile.idRef` with the government-ID reference.
4. Deliver into the consumer's DigiLocker or wallet -> DigiLocker delivery.
5. Wire revocation into the same flow you use for any v1.2 credential.

DigiLocker integration (DocType `NYCER`)

For Indian consumers, DigiLocker is the delivery channel of choice — the credential lands in a wallet they already use, ready to view and share. The Passport travels as DocType **`NYCER`** through the DigiLocker **Pull URI** mechanism: the consumer searches for the credential, DigiLocker calls your endpoint, your endpoint calls OpenCred to issue the signed v1.2 credential, and DigiLocker stores it. The structures below are the current NYCER shapes; the full delivery flow (API Setu registration, HMAC verification, CIS lookup, the OpenCred issue and package calls, error handling) is in DigiLocker delivery.

Endpoint

```
POST https://{your-domain}/digilocker/pulluri
Content-Type: application/xml
x-digilocker-hmac: <Base64(HMAC-SHA256(api_key, request_body))>
```

Verify the HMAC before any processing; reject with HTTP 401 if invalid. Register the endpoint on API Setu with DocType `NYCER`, `UDF1 Label = Consumer Number`, `UDF2 Label = Registered Mobile Number`.

Inbound request — `PullURIRequest` (NYCER)

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIRequest ver="3.0"
                ts="2026-04-01T10:30:00+05:30"
                txn="TXN-20260401-001"
                orgId="discom">
  <DocDetails>
    <DocType>NYCER</DocType>
    <DigiLockerId>7a3f9b2c-1e4d-4f8a-b6c2-0d5e8f1a2b3c</DigiLockerId>
    <UDF1>DISCOM-2025-001234567</UDF1>   <!-- consumer number -->
    <UDF2>9876543210</UDF2>              <!-- registered mobile number -->
    <FullName>Priya Sharma</FullName>    <!-- optional, from DigiLocker profile -->
  </DocDetails>
</PullURIRequest>
```

Echo `ts` and `txn` in the response. `UDF1` is the primary CIS lookup key; `UDF2` is secondary verification; DigiLocker validates `FullName` against your response's `IssuedTo`.

Identity binding — the DigiLocker ID

NeGD requires the `DigiLockerId` from the request to be carried inside the issued credential's identity binding. Populate `customerProfile.idRef`:

```
holder_idref = {
    "issuedBy":  "did:web:digilocker.gov.in",
    "subjectId": f"digilocker.gov.in:{digilocker_id}",   # the <DigiLockerId> from the request
}
```

This is exactly the identity-proofing step from the setup list above — for DigiLocker delivery, the Aadhaar-backed pull *is* the proofing.

Outbound response — `PullURIResponse`

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIResponse ver="3.0" ts="{ts}" txn="{txn}" orgId="discom">
  <ResponseStatus Status="1" ts="{ts}" txn="{txn}"/>
  <DocDetails>
    <DocType>NYCER</DocType>
    <URI>{uri}</URI>
    <IssuedTo>{issued_to}</IssuedTo>
    <ValidFrom>{valid_from}</ValidFrom>
    <ValidTo>{valid_to}</ValidTo>
    <DocContent format="pdf">{pdf_b64}</DocContent>
    <VcContent format="json-ld">{vc_b64}</VcContent>
    <VcType>application/ld+json</VcType>
  </DocDetails>
</PullURIResponse>
```

Field	Description
`URI`	`{issuer_id}-NYCER-{consumerNo}` — e.g. `in.gov.discom-NYCER-100000012345`. The trailing doc-id segment is required (NeGD rejects a URI ending at the DocType), and the org segment must equal your registered API Setu `issuer_id`. Keyed on consumer number alone, so a refresh overwrites the last — correct for a slow-changing connection credential.

Field	Description
`DocContent`	Base64-encoded PDF — rendered with a **separate** `POST /v1/credentials/package` call (`formats: ["pdf"]`); an inline `packageFormats` field on the issue body is silently ignored
`VcContent`	Base64-encoded W3C VC JSON-LD — the signed ElectricityCredential v1.2 as issued, proof intact
`VcType`	`application/ld+json` — required by NeGD immediately after `VcContent`; a response without it is rejected

For a full, validated v1.2 payload (the `credentialSubject` that goes inside `VcContent`), see the Schedule I example. For errors, return `Status="0"` with a message in `ResponseStatus` — see DigiLocker delivery — error format.

Selective disclosure

The Passport is a regular VC, so any selective-disclosure profile your wallet supports (SD-JWT-VC is the typical choice) works. The DISCOM is not in the disclosure loop — the wallet chooses which fields to present to each verifier.

Checklist

Step 0 — Prerequisites

- [ ] Register your organisation in the DeDi directory and create your `did:web` identity — see Setting up Register
- [ ] Stand up your Beckn ONIX adapter and connect it to the network — see Setting up Discover & Exchange
- [ ] Pass the basic conformance check — see Conformance Checklist

Step 1 — base issuance in place.

- [ ] Complete the Issuing Credentials setup checklist

Step 2 — identity proofing.

- [ ] Method chosen (DigiLocker pull / AADHAAR_OFFLINE_KYC / in-person / DISCOM record match)
- [ ] Privacy review completed; procedure tested with one consumer

Step 3 — wallet delivery.

- [ ] DigiLocker Pull URI registered for DocType `NYCER` -> DigiLocker integration above and DigiLocker delivery
- [ ] Direct DID-push tested for one non-DigiLocker wallet (where relevant)
- [ ] A freshly issued Passport reaches a wallet end-to-end

Step 4 — issuance shape.

- [ ] `credentialSubject.id` = wallet DID; `customerProfile.idRef` = verified government-ID *reference* (not the raw number)
- [ ] `validUntil` set to a sensible horizon (re-issued on material change)
- [ ] Schema validation passes against ElectricityCredential v1.2

Step 5 — verifier interop.

- [ ] Selective-disclosure profile agreed with the first verifiers (SD-JWT-VC typical)
- [ ] Verification rehearsed with one verifier (bank / marketplace / subsidy portal)
- [ ] Revocation tested — revoking invalidates all presentations within minutes

Team. [ ] Customer-ops SPOC (identity proofing) · [ ] IT SPOC (wallet delivery) · [ ] Governance / Compliance SPOC (privacy review)

References

- ElectricityCredential v1.2 schema — the schema this use case rides on
- Overview — Consumer Energy Passport — standards basis, definitions, full field schedule
- Energy Credentials — Credential variants — where the Passport variant is documented

- Identifiers — Holder binding patterns
- DigiLocker delivery

Consumer Meter Digest

In a hurry? Jump to the Checklist. For the standards basis and full field schedule, see the **Overview**.

The Consumer Meter Digest is a MeterDataCredential v0.6 issued, on consumer demand, holder-bound to the consumer's wallet, carrying their own meter readings for a specified period. It is not a new credential type — it is *how* an existing MeterDataCredential is configured when a consumer (rather than another DISCOM or AMISP) is the audience.

The Digest gives a consumer a portable, signed, verifier-friendly bundle of their telemetry — for a loan application, a rooftop-solar quote, a tariff comparison, a marketplace listing, a housing-society compliance check — without those parties having to phone the DISCOM.

Why this use case exists

Consumers regularly need to share their actual electricity-consumption pattern. Today they print PDFs of monthly bills and email scans. Verifiers have no way to confirm the bill is real and unaltered, so most of them call the DISCOM anyway. The Digest replaces that loop with a credential the verifier can check offline against the DISCOM's published key.

How it differs from a B2B MeterDataCredential

Same schema, same issuance pipeline. Three issuance-time differences:

Concern	B2B MeterDataCredential v0.6	Consumer Meter Digest
Triggered by	Beckn `confirm` from a DISCOM consumer-pull	Consumer request (often via DigiLocker or a consented wallet app)
`credentialSubject.id`	absent — bearer; the receiving DISCOM is the implicit subject	**set** to the consumer's wallet DID
`validUntil`	days to weeks	**hours to days** — Digests are point-in-time snapshots

The schema body — `profileType`, `meterRefs`, `intervalPeriod` / `timePeriod`, `intervals` / `readings` (raw `INTERVAL`/`DAILY`/`MONTHLY` profiles), `validationStatus`, issuer, proof — is identical to any other MeterDataCredential v0.6. Derived summary aggregates are downstream analytics, not schema fields. The `type` array stays `[ "VerifiableCredential", "MeterDataCredential"]` — no new VC type is introduced.

Actors and flow

Role	Who	What they do
Holder	Consumer	Initiates the request from their wallet / DigiLocker; stores the returned Digest; presents it to verifiers of their choosing
Issuer	DISCOM	Pulls the relevant readings from its MDM, signs the Digest, delivers it back into the consumer's wallet

Role	Who	What they do
Verifier	Bank, marketplace, energy app, housing society, EV-charger installer	Reads the Digest; resolves the DISCOM's `did:web` and (if cited) the regulator's licensing pointer to validate; reads the `intervals/readings` (any summary is downstream analytics, not part of the credential)

Granularity options today: `DAILY`, `MONTHLY`, or `INTERVAL` (15-minute data, expressed as `profileType: INTERVAL` with `intervalPeriod.duration: PT15M`). Derived summary aggregates, where offered, are downstream analytics rather than schema fields. Maximum period typically 24 months for `MONTHLY`, 90 days for 15-minute `INTERVAL` data.

Building blocks

Block	Used for
Identifiers and Addressing	DISCOM's `did:web`; consumer's wallet `did:key`; meter and connection DIDs that anchor the Digest
Energy Credentials	Issuance, signing, verification, revocation — including the Consumer Meter Digest variant under "Credential variants"
Data Exchange	If the consumer-pull endpoint is fronted by your BPP over Beckn, the BAP/BPP machinery is the same as Smart Meter Data Exchange — only the trigger (consumer, not DISCOM) and the `validUntil` differ
DigiLocker delivery	Delivers the Digest into the consumer's DigiLocker for easy access and sharing

Setup: Register -> Discover -> Exchange

1. **Register.** The consumer must hold a credential proving the right to request data for this meter — typically a Consumer Energy Passport (holder-bound ElectricityCredential v1.2) or a minimal customer credential.
2. **Discover.** Catalogue a "consumer-pull" endpoint on your BPP that accepts a request bearing that credential and returns a MeterDataCredential v0.6 -> Setup Exchange.
3. **Exchange.** Issue the Digest via Energy Credentials — Issue your first credential, setting `credentialSubject.id` to the consumer's wallet DID, `schemaId` to `ies/meter-data-credential/v0.6`, and `validUntil` to a short window matching the use case (24h for loan portals, up to 7d for non-time-sensitive flows).
4. Deliver to the consumer's wallet — DigiLocker delivery, or directly to a known DID inbox if the wallet exposes one.
5. Revocation rarely matters in practice (Digests typically expire faster than they would need revocation), but the same DeDi-hash revocation flow is available if you need it.

DigiLocker integration (DocType `MPLTR`)

For Indian consumers, DigiLocker is the delivery channel of choice — the Digest lands in a wallet they already use, ready to view and share. NeGD has allotted **`MPLTR`** (Meter Document) as the DocType for the Consumer Meter Digest; it registers as an additional record on the **same Pull URI endpoint** a DISCOM already runs for the Consumer Energy Passport (`NYCER`) — one handler, branched by `DocType`. The structures below are the current MPLTR shapes; the full delivery flow (API Setu registration, HMAC verification, the OpenCred issue and package calls, the open asks to DigiLocker) is in DigiLocker delivery.

Endpoint

```
POST https://{your-domain}/digilocker/pulluri
Content-Type: application/xml
```

```
x-digilocker-hmac: <Base64(HMAC-SHA256(api_key, request_body))>
```

Verify the HMAC before any processing; reject with HTTP 401 if invalid. Register the DocType on API Setu with `UDF1 Label = Consumer Number`, `UDF2 Label = Statement Period` (e.g. `2026-05` or `last-12-months`), `UDF3 Label = Registered Mobile Number`.

Inbound request — `PullURIRequest` (MPLTR)

The period travels as a UDF, so the consumer can pull a chosen window:

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIRequest ver="3.0"
                ts="2026-05-15T10:00:00+05:30"
                txn="TXN-20260515-042"
                orgId="discom">
  <DocDetails>
    <DocType>MPLTR</DocType>
    <DigiLockerId>7a3f9b2c-1e4d-4f8a-b6c2-0d5e8f1a2b3c</DigiLockerId>
    <UDF1>DISCOM-2025-001234567</UDF1>   <!-- consumer number -->
    <UDF2>last-12-months</UDF2>          <!-- statement period: YYYY-MM, a range, or a keyword -->
    <UDF3>9876543210</UDF3>             <!-- registered mobile number -->
    <FullName>Priya Sharma</FullName>
  </DocDetails>
</PullURIRequest>
```

`UDF2` accepts `YYYY-MM`, a range, or a keyword like `last-12-months`; a recurring pull with the current month fetches the latest statement. Validate the period and bound it to the scope the originating request authorised.

Outbound response — `PullURIResponse`

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIResponse ver="3.0" ts="{ts}" txn="{txn}" orgId="discom">
  <ResponseStatus Status="1" ts="{ts}" txn="{txn}"/>
  <DocDetails>
    <DocType>MPLTR</DocType>
    <URI>{uri}</URI>
    <IssuedTo>{issued_to}</IssuedTo>
    <ValidFrom>{valid_from}</ValidFrom>
    <ValidTo>{valid_to}</ValidTo>
    <DocContent format="pdf">{pdf_b64}</DocContent>
    <VcContent format="json-ld">{vc_b64}</VcContent>
    <VcType>application/ld+json</VcType>
  </DocDetails>
</PullURIResponse>
```

`DocContent` is the base64 rendered statement PDF (produced with a separate `POST /v1/credentials/package` call, `formats: ["pdf"]`); `VcContent` is the base64 signed `MeterDataCredential` JSON-LD, proof intact; `VcType` must be `application/ld+json`, immediately after `VcContent` — NeGD rejects a response without it.

The document key — period in the URI

This is the one structural difference from the Passport. `NYCER` is keyed on the consumer number alone, so a refresh overwrites the last — correct for a connection credential. The Digest is a statement series: put the **period into the URI** so every statement coexists as a distinct document:

```python
# NYCER — overwrite on refresh (correct for a connection credential)
uri = f"in.gov.discom-NYCER-{consumer.consumer_number}"
```

```
# MPLTR — period in the key, so every statement coexists
uri = f"in.gov.discom-MPLTR-{consumer.consumer_number}-{period}"   # period e.g. "2026-05" or
 ↪  "2025-05_2026-04"
```

URI uniqueness is managed issuer-side; treating each period-keyed URI as a distinct, independently listable/deletable document was agreed in principle with DigiLocker on the 12 June 2026 call, with written confirmation still owed — see DigiLocker delivery — the ask to DigiLocker.

The `VcContent` payload — example twelve-month statement

The JSON-LD that travels in `VcContent` for a twelve-month `MONTHLY`-profile Digest, following the canonical MeterDataCredential v0.6 monthly example — one `MonthlyProfile` object per month, carried as an array in `meterData` (trimmed proof; first month shown in full):

```
{
  "@context": [
    "https://www.w3.org/ns/credentials/v2",
    "https://schema.nfh.global/EnergyCredential/v2.0/context.jsonld",
    "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataCredential/v0.6/context.jsonld"
  ],
  "id": "urn:uuid:9b3c1f0a-2d4e-4ba8-9f1c-1e7a90c8a5d2",
  "type": ["VerifiableCredential", "MeterDataCredential"],
  "issuer": {
    "id": "did:web:ies.discom.example",
    "name": "Example State Distribution Company Limited",
    "licenseNumber": "SERC-DISCOM-2025-007"
  },
  "validFrom": "2026-05-15T10:00:00+05:30",
  "validUntil": "2026-05-22T10:00:00+05:30",
  "credentialStatus": {
    "id": "https://dedi.global/dedi/query/did:web:ies.discom.example/vc-revocation-registry",
    "type": "dediregistry",
    "statusPurpose": "revocation",
    "statusListCredential":
     ↪  "https://dedi.global/dedi/lookup/did:web:ies.discom.example/vc-revocation-registry"
  },
  "credentialSubject": {
    "id": "did:key:z6MkjVQ8r4f3rPuY7CG2D6Lf8WJxJBs5sjkR8d3v2Bv4nP4Z",
    "meterData": [
      {
        "@context":
         ↪  "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/context.jsonld",
        "@type": "MonthlyProfile",
        "profileType": "MONTHLY",
        "meterRefs": [
          { "scheme": "DID", "value": "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }
        ],
        "serviceDeliveryPointRefs": [
          { "scheme": "DID", "value": "did:web:ies.discom.example:connections:SDP-A-12345" }
        ],
        "timePeriod": { "start": "2025-05-01T00:00:00+05:30", "duration": "P1M" },
        "readings": [
          { "readingType": "kWh imp", "value": 391.0, "validationStatus": "VALID", "source": "METER" }
        ]
      },
```

```
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-06-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 412.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-07-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 478.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-08-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 463.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-09-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 421.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-10-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 384.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-11-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 352.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-12-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 339.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-01-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 367.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-02-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 358.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-03-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 405.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-04-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 437.0 }] }
    ]
  },
  "proof": { "type": "Ed25519Signature2020", "proofValue": "z5wQ…" }
}
```


For raw analytics data, the same wrapper carries an `INTERVAL` profile (`PT15M` blocks, ~96 readings a day), typically preceded by a `DESCRIPTOR` profile referenced via `payloadDescriptorSetRef` — `meterData` becomes an array of profiles. Both shapes, and the full `handle_mpltr` issuance code, are in DigiLocker delivery — issuing the Digest.

Two open schema/rendering notes, tracked in DigiLocker delivery: `MeterDataCredential` v0.6 has no `idRef` field on `credentialSubject` yet, so NeGD's DigiLocker-ID identity binding can be rendered only into `DocContent`, not carried as a structured VC field; and configurable rendering of the statement card remains an open ask to DigiLocker.

Checklist

Step 0 — Prerequisites

- [ ] Register your organisation in the DeDi directory and create your `did:web` identity — see Setting up Register
- [ ] Stand up your Beckn ONIX adapter and connect it to the network — see Setting up Discover & Exchange
- [ ] Pass the basic conformance check — see Conformance Checklist

Step 1 — base issuance in place.

- [ ] Complete the Issuing Credentials setup checklist

Step 2 — MDM read path.

- [ ] Read access tested for the granularities you'll support (`DAILY`, `MONTHLY`, `INTERVAL` at `intervalPeriod.duration: PT15M`, and any downstream summary analytics)
- [ ] Max-range policy decided (e.g. 24 months for `MONTHLY`, 90 days for 15-minute `INTERVAL` data)
- [ ] Latency budget understood — consumer flows need a Digest in seconds

Step 3 — consumer-pull endpoint.

- [ ] Beckn `DatasetItem` for the pull endpoint published on your BPP (`accessMethod: INLINE`)
- [ ] `offerAttributes.policy.requiredCredentials` restricts callers to a valid Consumer Energy Passport (or minimal customer credential)
- [ ] Supported granularities and maximum request range match Step 2's policy

Step 4 — issuance shape.

- [ ] `credentialSubject.id` = wallet DID; `schemaId` = `ies/meter-data-credential/v0.6`
- [ ] `validUntil` short — 24h for loan portals, up to 7d for less time-sensitive flows
- [ ] Schema validation passes against MeterDataCredential v0.6 — this is repository-local structural validation; see Conformance Checklist — What this checklist proves for what it does and doesn't establish

Step 5 — wallet delivery.

- [ ] DigiLocker pull tested end-to-end for DocType `MPLTR`, period-keyed URI verified -> DigiLocker integration above and DigiLocker delivery
- [ ] One direct DID-push path tested for non-DigiLocker wallets
- [ ] Consumer sees the Digest within the Step 2 latency budget

Step 6 — verifier interop.

- [ ] Verification rehearsed with one verifier (bank / marketplace / housing society / EV installer)
- [ ] If you emit derived summary analytics alongside the Digest, share their schema + description with verifiers up front — these are downstream outputs, not part of the MeterDataCredential/v0.6 schema
- [ ] Revocation flow tested (even though Digests usually expire before needing it)

Team. [ ] IT SPOC (MDM read path + BPP catalogue) · [ ] Customer-ops SPOC (wallet support) · [ ] Governance / Compliance SPOC (data-disclosure policy)

References

- MeterDataCredential v0.6 schema — the schema this use case rides on
- Overview — Consumer Meter Digest — standards basis, definitions, full field schedule
- Energy Credentials — Credential variants — where the Digest variant is documented
- Smart Meter Data Exchange use case — the B2B sibling of this consumer flow
- DigiLocker delivery

Smart Meter Data Exchange

A standard, audit-trailed way to exchange smart-meter telemetry between an AMISP, a DISCOM, a state regulator, and consented third parties — over IES Data Exchange, carrying the MeterData payload (currently `v0.6`).

You replace bespoke FTP drops, proprietary MDMS APIs, and CSV email attachments with one signed, schema-validated flow. The same surface area works AMISP->DISCOM, DISCOM->SERC, and DISCOM->consented-third-party.

> For consumer-pull of *their own* meter data (one-meter, on-demand, wallet-shareable), see Consumer Meter Digest. This guide is the bulk, scheduled, machine-to-machine flow. For the standards basis and full field schedule, see the **Overview**.

Building blocks used

This use case is a thin composition over four existing building blocks. The setup steps below map one-to-one to them — read the building-block pages for the detail; come back here for what changes for the meter-data case.

What you need	Building block	What it gives you
A name and a signing key your counterparties can verify	Identifiers & Addressing, Registries	A DeDi namespace under your control and your current public key published into it. A formal `did:web` document is **not** required for this use case; the namespace + key are enough for Beckn signing and verification.
A trust boundary — "who can transact on this network"	Registries — IES networks today	Your subscriber record referenced in the IES network registry
The transport that carries discovery, contract, audit, and the payload	Data Exchange	The Beckn lifecycle + ONIX adapter; the same `accessMethod` covers inline payloads and signed-URL handoff
(Optional) Cryptographic proof that the requester is authorised	MeterDataRequestCredential	A signed credential the seeker attaches at `confirm`; provider verifies offline

---

The dataset — `MeterData/v0.6`

The payload is the MeterData compact-profile schema (currently `v0.6`). v0.6 standardises eight profile shapes covering every smart-meter cadence:

Profile	Carries
`CUSTOMER`	Slow-changing customer / service-point / meter installation metadata
`INTERVAL`	Block load survey at 15- or 30-minute resolution
`DAILY`	Daily accumulated load survey
`MONTHLY`	Monthly billing resets (cumulative kWh, ToU buckets, MD)
`BILL_DETAILS`	Utility-side billing computed details
`INSTANTANEOUS`	Real-time snapshot of V / A / P / Q
`EVENT`	IS 15959 diagnostic / tamper events
`ALARM`	Real-time active alerts

For the Indian-terminology mapping (OBIS codes, IS 15959 event IDs, CIM master-data fields) see the companion page IES Meter Data Model.

Optional consent — `MeterDataRequestCredential`

When a third party (a consented app, a researcher, sometimes a regulator) is the BAP, the provider's offer policy can require an authorisation credential at `confirm` time. IES uses `MeterDataRequestCredential` — a W3C VC 2.0 that wraps a `MeterDataRequest` and identifies the requester, scopes the request (which meters, which time window, which profile), and is signed by the authoriser (typically the DISCOM, sometimes the consumer themselves via OpenCred).

```
Seeker (BAP)                                    Provider (BPP)
e.g. consented app, SERC                        e.g. AMISP, DISCOM

  |— confirm  —— { MeterDataRequest } ——
  |               + MeterDataRequestCredential —>|
  |                                               |
  |                                               ├— verify VC: type [x]
  |                                               |            validUntil [x]
  |                                               |            DeDi status [x]
  |                                               |            scope ⊆ contract
  |<— on_confirm / on_status – MeterData/v0.6 ———|
```

The provider checks credential type, expiry, DeDi revocation status, and that the embedded request scope is within the contracted scope — no callback to the issuer.

Setup: Register -> Discover -> Exchange

The order below takes you from zero to a working AMISP -> DISCOM flow on the local devkit, then upgrades to the real IES network.

1. Decide scope

Meter population, geography, granularity (15-min / daily / monthly), counterparties (AMISP->DISCOM only, DISCOM->SERC, DISCOM->third-party). Phase 1 should be one cadence and one counterparty.

2. Register — get a network identity

Both BAP and BPP need a DeDi subscriber record with a published signing key. If you have one already (any other Beckn flow on this network), reuse it. If not, follow **Setup Register**.

You do **not** need to rename anything in your CIS, MDMS, or asset registers. Your existing IDs are reused as the tail of the DID.

3. Discover — stand up the Data Exchange adapters

Both sides run an ONIX adapter -> **Setup Exchange**. The fastest start is the devkit sandbox:

```
git clone https://github.com/beckn/DEG.git
cd DEG/devkits/data-exchange/install
docker compose up -d
```

This brings up `sandbox-bap`, `sandbox-bpp`, and `beckn-router` pre-wired with placeholder identities.

4. Exchange — publish your dataset catalogue (BPP)

Publish a Beckn catalogue entry for the `MeterData/v0.6` dataset you offer — `programID`, geographic scope, refresh cadence, `accessMethod` (`INLINE` for <=MB-scale chunks; `SIGNED_URL` for daily/monthly bulk), and any required credentials (point at `MeterDataRequestCredential` if you require one). Pre-agreed bilateral subscriptions can skip `discover` and go straight to `confirm`.

5. Exercise the flow

The minimal lifecycle is `confirm` -> `on_confirm` -> (`status` -> `on_status` if delivery is asynchronous):

Step	Sender	What happens
`confirm`	BAP	Requests data for a meter list + date range. Optionally attaches a `MeterDataRequestCredential`.
`on_confirm`	BPP	Acknowledges; declares whether delivery is inline or async.
`on_status`	BPP	Delivers `MeterData/v0.6` inline or returns a signed-URL pointer.

The payload sits inside `message.contract.commitments[].resources[].resourceAttributes` qualified with the DDM `DatasetItem/v1.1` context — see Data Exchange — `DatasetItem` and `accessMethod`.

6. Connect your real metering system

Replace the sandbox response with a live connection to your HES / MDMS. Verify the Indian-OBIS -> `MeterData/v0.6` mapping for every reading you ship — IES Meter Data Model is the reference.

7. (Optional, at the end) Adopt the `did:web` convention for meters and assets

There are no new IDs to allocate. Your existing meter SLNOs, DT codes, feeder codes, and substation codes are the IDs of record and stay exactly as they are. The IES convention is a `did:web` wrapper format that reuses those existing IDs:

```
did:web:<discom-domain>:meters:<existing-meter-serial>
did:web:<discom-domain>:dts:<existing-DT-code>
did:web:<discom-domain>:feeders:<existing-feeder-code>
did:web:<discom-domain>:substations:<existing-substation-code>
```

`<discom-domain>` is your own domain — the same one your `did.json` and DeDi namespace hang off (see Setup Register §1.1). The DID method is always `did:web`, so these resolve under your domain exactly like your organisation DID; DeDi acts as the discovery and verification layer for your namespace — see Identifiers — Appendix C. This step is *nice to have*, not a blocker for first deployment — initial flows can use bare IDs inside `MeterData` payloads and adopt the `did:web` form incrementally.

Checklist for your meter-data rollout

Step 0 — Prerequisites

- [ ] Register your organisation in the DeDi directory and create your `did:web` identity — see Setting up Register
- [ ] Stand up your Beckn ONIX adapter and connect it to the network — see Setting up Discover & Exchange
- [ ] Pass the basic conformance check — see Conformance Checklist

For the full network onboarding detail (sandbox->test->prod cut-over, key management), see **Concepts — Before you build**.

The items below are **meter-data-specific** additions on top of the prerequisites:

- [ ] Meter population, geography, granularity, and counterparty for Phase 1 agreed
- [ ] Source system mapped (HES / MDMS endpoint, owner team, SLA)
- [ ] Indian-OBIS -> `MeterData/v0.6` mapping verified for every reading and event you ship (see IES Meter Data Model)
- [ ] Data-quality flags (missing / estimated) handled per the v0.6 `validationStatus` enum
- [ ] Catalogue entry published (`programID`, geographic scope, refresh cadence, `accessMethod`, credential requirements)
- [ ] (If third-party access in scope) `MeterDataRequestCredential` verification wired into the BPP — type, validity, DeDi status, scope contract
- [ ] (Optional) `did:web` convention adopted for meters / DTs / feeders — wrapping (not replacing) existing internal numbers, so VCs can be issued about them

- [ ] IT/data SPOC nominated; Authorised Signatory nominated

Open items

- **Chunking convention** for bulk historical pulls (daily / monthly / quarterly) is being agreed across deployments.
- **Quality flags** — the v0.6 `validationStatus` enum is stable; the recommended profile of allowed values per cadence is being tightened.

References

- MeterData — schema and examples (current: `v0.6`)
- MeterDataRequest — request payload schema (current: `v0.6`)
- MeterDataRequestCredential — authorisation VC (current: `v0.1`)
- IES Meter Data Model — OBIS / IS 15959 / CIM -> MeterData v0.6 mapping
- Overview — Smart Meter Data Exchange — standards basis, definitions, full field schedule
- Data Exchange chapter
- Registries and Directories
- Identifiers and Addressing

IES Meter Data Model

The reference for how Indian smart-meter terminology — OBIS codes, IS 15959 profile buffers, IS 15959 event IDs, CIM master data — maps onto the `MeterData` schema (this page is pinned to the current `v0.6` shape) carried by IES Data Exchange.

If you are implementing Smart Meter Data Exchange, this is the page you keep open while wiring your HES / MDMS to v0.6 payloads.

> `MeterData/v0.6` is the current canonical schema. Earlier drafts of IES used the working name `IES_Report` — treat those references as superseded by `MeterData/v0.6`.

---

1. The shape of `MeterData/v0.6` in one glance

A `MeterData/v0.6` payload is a **JSON array** with two kinds of object:

Object	Role
`PayloadDescriptorProfile`	Dictionary — declares `payloadDescriptors[]` (one per quantity: OBIS, readingType, unit, flowDirection, reportedMode) and `compactSequences[]` (column layout for the dense `payloads` matrix).
One of the profile objects below	Carries actual readings or events, referencing a descriptor set by name.

The eight profile shapes:

Profile	Indian-side equivalent	When it ships
`CUSTOMER`	CIS / Asset Master / GIS master data	On enrolment, on change
`INTERVAL`	IS 15959 Load Profile (`1.0.99.1.0.255`)	15 / 30-min block load survey

Profile	Indian-side equivalent	When it ships
DAILY	IS 15959 Daily Profile (1.0.94.91.0.255)	Once / day
MONTHLY	IS 15959 Billing Profile (0.0.98.1.0.255)	Billing reset (typically month-end)
BILL_DETAILS	Utility-side billing computed details	On bill finalisation
INSTANTANEOUS	OBIS 1.0.x.7.0.255 real-time registers	On poll / push
EVENT	IS 15959 Event Log Profiles (0.0.99.98.x.255)	On event + scheduled pull
ALARM	Real-time alarms (tamper / overload / low credit)	On occurrence

OBIS codes are first-class in v0.6 — they sit inside `payloadDescriptors[].obis` as a string verbatim from the meter. The schema does not translate them, so the mapping below is mostly about *which profile a given OBIS belongs in* and *which `readingType` / `unit` / `reportedMode` to advertise*.

The canonical, machine-readable OBIS catalogue ships as `IES codes.json` (GitHub Pages, served raw) under `schemas/MeterData/v0.6/` — each entry pins `obis`, `name`, `category`, `profiles[]`, `supportedModes[]`, `meterCategories[]`, and the IS 15959 source. **That file is the source of truth.** The tables on this page are the human-readable summary; if a mapping looks wrong, the JSON wins. Browse the source on GitHub.

---

2. IS 15959 profile buffer -> `MeterData/v0.6` profile

The Indian Load Profile / Daily Profile / Billing Profile buffers map one-to-one to v0.6 profile shapes. The `intervalPeriod.duration` distinguishes cadence.

IS 15959 profile (OBIS)	Typical period	`MeterData/v0.6` profile	`intervalPeriod.duration`
Load Profile (1.0.99.1.0.255)	15 / 30 min	INTERVAL	PT15M / PT30M
Daily Profile (1.0.94.91.0.255)	24 h	DAILY	PT24H
Billing Profile (0.0.98.1.0.255)	Monthly	MONTHLY	P1M
Voltage / Current event buffers (0.0.99.98.0.255 / .1.255)	Event-driven	EVENT	`timePeriod` window
Power-failure / Tamper / Control buffers (0.0.99.98.2.255 / .4.255 / .5.255)	Event-driven	EVENT (POWER_FAIL etc.) / ALARM (tamper / control)	`timePeriod` window
Real-time registers (1.0.x.7.0.255)	Polled / pushed	INSTANTANEOUS	Point-in-time

---

3. OBIS register -> `payloadDescriptors[]`

For each profile, the OBIS register is carried in `payloadDescriptors[].obis` and qualified with `readingType`, `unit`, `flowDirection`, and `reportedMode` (`USAGE` / `READING` — the two values of the v0.6 `TelemetryMode` enum; demand registers are `USAGE`).

3.1 Energy registers — Load / Daily / Monthly

Quantity	OBIS (block incremental)	OBIS (cumulative)	Unit	flowDirection	reportedMode	Goes in profile
Active energy import	1.0.1.29.0.255	1.0.1.8.0.255	kWh	IMPORT	USAGE	INTERVAL / DAILY / MONTHLY
Active energy export	1.0.2.29.0.255	1.0.2.8.0.255	kWh	EXPORT	USAGE	INTERVAL / DAILY / MONTHLY
Apparent energy import	1.0.9.29.0.255	1.0.9.8.0.255	kVAh	IMPORT	USAGE	INTERVAL / DAILY / MONTHLY
Reactive energy (lag)	1.0.5.29.0.255	1.0.5.8.0.255	kvarh	IMPORT	USAGE	INTERVAL / MONTHLY
Reactive energy (lead)	1.0.8.29.0.255	1.0.8.8.0.255	kvarh	EXPORT	USAGE	INTERVAL / MONTHLY
ToU import (Zones 1..8)	—	1.0.1.8.<z>.255	kWh	IMPORT	USAGE	MONTHLY
Maximum demand (kW)	—	1.0.1.6.0.255	kW	IMPORT	USAGE	MONTHLY
Maximum demand (kVA)	—	1.0.9.6.0.255	kVA	IMPORT	USAGE	MONTHLY

3.2 Instantaneous registers

Quantity	OBIS	Unit	reportedMode	Profile
Voltage (phase-neutral)	1.0.12.7.0.255	V	READING	INSTANTANEOUS
Phase current	1.0.11.7.0.255	A	READING	INSTANTANEOUS
Neutral current	1.0.91.7.0.255	A	READING	INSTANTANEOUS
Frequency	1.0.14.7.0.255	Hz	READING	INSTANTANEOUS
Signed power factor	1.0.13.7.0.255	—	READING	INSTANTANEOUS
Active power (import)	1.0.1.7.0.255	kW	READING	INSTANTANEOUS
Apparent power	1.0.9.7.0.255	kVA	READING	INSTANTANEOUS

3.3 Identification registers — `CUSTOMER` profile

Field	OBIS	Goes in
Meter serial number	0.0.96.1.0.255	CUSTOMER
Manufacturer name	0.0.96.1.1.255	CUSTOMER
Device ID	0.0.96.1.2.255	CUSTOMER (D1 / D2 meter categories)
Firmware version	1.0.0.2.0.255	CUSTOMER
Clock (real-time)	0.0.1.0.0.255	Implicitly handled by intervalPeriod / timestamp

---

4. IS 15959 Event IDs -> `EVENT` / `ALARM`

Events carry `eventId` (numeric, IS 15959) and a human-readable `eventName` directly. Examples are in `schemas/MeterData/v0.6/examples/EventProfile.json`.

Event ID (occur)	Event ID (restore)	Description	Category	Profile
1	2	Phase-to-neutral under-voltage	Voltage	`EVENT`
3	4	Phase-to-neutral over-voltage	Voltage	`EVENT`
11	12	Phase missing	Voltage	`EVENT`
51	52	Phase current unbalance	Current	`EVENT`
61	62	CT reverse	Current	`EVENT`
69	70	Neutral disturbance	Current / others	`EVENT`
101	102	Power failure / power up	Power	`EVENT`
151	—	Parameters programming (config change)	Transaction	`EVENT`
201	202	Magnetic interference	Tamper	`ALARM` (active) / `EVENT` (logged)
205	—	Meter cover open (non-restorable)	Tamper	`ALARM`
301	302	Relay disconnect / connect (manual)	Control	`EVENT`
303	—	Relay disconnect (overload)	Control	`EVENT` / `ALARM`
304	—	Relay disconnect (low credit)	Control	`ALARM`
305	—	Relay disconnect (tamper auto-cutoff)	Control	`ALARM`
306	307	Token acceptance / rejection (prepaid)	Control	`EVENT`

`EVENT` profiles can carry the IS 15959 magnitude and duration directly (`magnitude`, `duration`); `ALARM` profiles are state-based (active / cleared) and intended for real-time pushes.

---

5. Data quality — `validationStatus`

v0.6 carries per-reading validation via the `overrides[]` block on a compact interval entry (explicit `readings[]` carry `validationStatus` and `source` inline). Each override pins the `descriptorIndex` it qualifies, the `validationStatus`, the `source`, an optional `changeMethod`, and a `failCode`.

`validationStatus`	Meaning
`VALID`	Read direct from the meter; no estimation
`ESTIMATED`	Filled by VEE / interpolation
`MANUAL`	Manually entered or corrected by an operator
`SUSPECT`	Value present but flagged as out-of-range / questionable
`REJECTED`	Failed validation; not usable for billing or settlement

Example fragment from `IntervalProfile.json`:

```
{ "id": 2, "payloads": [ 5.21, 0.12 ],
  "overrides": [{ "descriptorIndex": 0, "validationStatus": "ESTIMATED",
                  "source": "ESTIMATED", "changeMethod": "linear-interpolation",
                  "failCode": "VEE-MISSING-INTERVAL" }] }
```

---

6. CIM master data -> `CUSTOMER` profile

The CIM (IEC 61968-1 / 61968-9) master data the AMI deployment requires lands in the `CUSTOMER` profile. Use `[BaseProfile].meterRefs[]`, `serviceDeliveryPointRefs[]`, `customerRefs[]` for the relational keys, and `attributes` for the descriptive payload.

CIM area	Key fields	Source system	Reference in `CUSTOMER`
Asset master	Meter Serial, Make, Model, Hardware / Firmware, CT / PT ratios	Inventory / ERP	`meterRefs[]` + `attributes`
Consumer master	CA number, name, address, category (DS / NDS), sanctioned load	CIS / billing	`customerRefs[]` + linked ElectricityCredential v1.2 `customerProfile`
Network master	Substation, Feeder, DT, Pole, Phase	GIS	`attributes` (FEEDER_ID, DT_ID, SUBSTATION_ID, POLE_ID, PHASE)
Tariff master	ToU slots, slabs, fixed charges	Billing system	Out of band of MeterData; carried by `MeterServiceProfile` inside ElectricityCredential v1.2

When an OBIS reading arrives, the receiving MDM matches it via `meterRefs[]` -> asset -> consumer -> network -> tariff. The `serviceDeliveryPointRefs[]` value is the "service point" object that ties the three together (CIM IEC 61968-9 `UsagePoint`).

---

7. Worked example — 30-minute Load Survey

The full schema-validating example is at `schemas/MeterData/v0.6/examples/IntervalProfile.json`. The shape:

1. One `PayloadDescriptorProfile` declares `IntervalLoadSurveySet` with `kWh imp block` (OBIS `1.0.1.29.0.255`) and `kWh exp block` (OBIS `1.0.2.29.0.255`).
2. One `IntervalProfile` references the descriptor set, carries `intervalPeriod` (`start`, `duration: PT30M`), and packs each interval into a tight `payloads: [imp, exp]` array.
3. Quality overrides ride alongside, addressing the descriptor by index.

This is the minimal pattern. Daily / Monthly / Instantaneous / Event / Alarm follow the same descriptor-then-profile shape; their examples sit alongside in `schemas/MeterData/v0.6/examples/`.

---

8. References

- `MeterData` — schema, examples, user guide, reference guide (mappings on this page are pinned to `v0.6`)
- `IES codes.json` — canonical OBIS catalogue (machine-readable). Source on GitHub.
- Smart Meter Data Exchange — the use case that carries this payload
- Data Exchange — the wire that carries it
- `MeterDataRequest` — the matching request schema (current: `v0.6`)
- `MeterDataRequestCredential` — optional authorisation VC (current: `v0.1`)
- `ElectricityCredential` — sits alongside `MeterData` for the slow-changing customer / asset / service-connection data (current: `v1.2`)

Appendix — IS 15959 deep reference

This appendix is the full IS 15959 source-of-truth survey (formerly published as a standalone page). Use it to verify the mappings above against the underlying Indian standard.

A1. Profile buffers

Profile type	OBIS	Integration period	Mandatory capture sequence (typical)
Load Profile	`1.0.99.1.0.255`	15 / 30 min	Clock, kWh Imp, kVAh Imp, Avg Voltage, Avg Current
Billing Profile	`0.0.98.1.0.255`	Monthly	Billing Date, Total kWh, Total kVAh, MD kW, MD kVA, ToU registers
Daily Profile	`1.0.94.91.0.255`	Daily (24 h)	Clock (midnight), Cumulative kWh, Cumulative kVAh

A2. Event log buffers

Category	OBIS profile	Event types
Voltage	`0.0.99.98.0.255`	Under-voltage, over-voltage, phase missing
Current	`0.0.99.98.1.255`	Unbalance, CT reverse, neutral disturbance
Power failure	`0.0.99.98.2.255`	Power ON / OFF logs with timestamps
Transaction	`0.0.99.98.3.255`	Programming, date change, relay operation
Others / tamper	`0.0.99.98.4.255`	Magnetic interference, cover open
Control	`0.0.99.98.5.255`	Load-limit changes, relay triggers (prepaid)

A3. Communication flow **Polling (HES -> meter):** Association Request (AARQ) -> HLS authentication (AES-GCM-128, Security Suite 0) -> Selective Read of `profile_generic` by range -> meter responds with `compact-array` -> Release (RLRQ).

Push (meter -> HES -> MDM): Critical events (tamper / power-fail) are pushed via the PUSH Setup Object (`0.0.25.9.0.255`). Routine logs are fetched during the daily scheduled read. HES converts raw OBIS to an IEC 61968-9 message for the MDM.

Control (MDM -> HES -> meter): MDM issues `Disconnect` request via Northbound REST / SOAP -> HES authenticates with meter, sends `ACTION` to Relay Control Object (`0.0.96.3.10.255`) -> HES reads Relay Status to confirm -> Feedback relayed to MDM.

A4. Network topology assumed by AT&C-loss accounting

- **Substation / Feeder level** — high-accuracy meters (Class 0.2s / 0.5s) on the outgoing 11 kV / 33 kV lines.
- **DT (LV side) level** — meters at the DTR for energy accounting.
- **Consumer level** — 1-phase or 3-phase end-user meters.

GIS holds Feeder <-> DT and DT <-> Consumer mappings; MDM stores them. The `CUSTOMER` profile carries the relevant identifiers via its `attributes` block (FEEDER_ID, DT_ID, SUBSTATION_ID, POLE_ID, PHASE).

A5. IS 15959 annexure summary

Annexure	Subject	Detail
A	Data types	Mandates `double-long-unsigned` for energy to avoid overflow
B	OBIS mapping	Master list for 1-phase, 3-phase, CT / PT meters
C	Load profile	Standardises the sequence of capture objects
D	Event IDs	Canonical 8-bit list (1–255)
E	Billing logic	Auto-reset (midnight of 1st) and MD reset behaviour
F	Security	Security Suite 0 requirements for AMI
G	Smart features	Protocols for firmware image transfer and PUSH parameters

DER Visibility

In a hurry? Jump to the Checklist. For the standards basis and full field schedule, see the **Overview**.

A DISCOM's future, illustrative per-feeder view of every distributed energy resource (rooftop solar, battery, EV charger) behind its meters — PII-free, conceptually reusing the same `EnergyResource` and `ConsumptionProfile` building blocks that the consumer's Energy Passport (ElectricityCredential v1.2) composes. The only currently executable EC v1.2 path is the per-consumer Energy Passport itself.

Scenario

As rooftop solar, batteries and EV charging spread, a DISCOM often cannot answer the basic question: what is connected on feeder F-02, at what capacity, where on the network, and is it controllable?

What is executable today. The only currently executable ElectricityCredential v1.2 path is the per-consumer credential — the Consumer Energy Passport — whose `credentialSubject.customerProfile.customerNumber` is a required field. See the validated Schedule I example. A grid operator or aggregator wanting per-connection detail today would consume that credential, with consumer consent, via the same Setup Exchange flow.

The live half — DER telemetry. The asset register above is only Schedule I of this use case. Schedule II — what each inverter, PV array or battery is doing now — is also executable today, per DER: an electrical subset mapped to MeterData v0.6, delivered inverter -> MQTT -> the national MNRE M2M platform rather than over Beckn, with a validated fixture at `schedule-ii-example.json`. The field mapping, the informative extensions and the transport requirements are in Overview §9 — Schedule II.

A note on the aggregate (illustrative, future). ElectricityCredential is issued per consumer connection — one `customerProfile`, one customer number — so it cannot combine multiple consumers into a feeder- or substation-level record. Each consumer keeps their own Energy Passport. A PII-free, per-locus view — an array of `EnergyResource` and `ConsumptionProfile` entries for the locus, subject to the feeder / substation / licensee DID — is described below as an **illustrative future profile**. It may conceptually reuse the same building blocks the Passport composes, but it **does not validate as EC v1.2** (which requires a single `customerProfile` / `customerNumber`), is **not a signed EC v1.2 credential**, and has **no canonical executable example** today — see Open Items.

How it differs from the Consumer Energy Passport

The "DER Visibility" column below describes the illustrative future aggregate (see Scenario, above) — not an executable credential today.

Concern	Consumer Energy Passport	DER Visibility
Shape	Full `ElectricityCredential v1.2`, one consumer per credential	PII-free arrays of `EnergyResource` + `ConsumptionProfile` entries, one locus per record
Subject	Consumer's wallet DID	The feeder / substation / licensee DID
`customerProfile`, `customerDetails`	Populated	**Omitted entirely** — no consumer PII, and consumers are never merged together

Actors and Roles

Role	Who	What they do
Issuer	DISCOM	Publishes the aggregated record per locus (feeder / substation / licensee-wide), on a refresh cadence or on material change
Consumer (BAP)	Grid operator	Ingests for forecasting, planning, dispatch, outage analysis
Consumer (BAP)	Aggregator	Ingests as a discovery surface for controllable resources

Unlike the Passport, the DER Visibility record is **published, not held** — consumed from the DISCOM's BPP catalogue, not carried in a wallet.

Building Blocks Used

Block	Role in this use case
Identifiers	Same `did:web` as the Passport; subject set to a feeder / substation / licensee DID instead of a consumer DID
Energy Credentials	Same signing / verification / revocation pipeline used for any ElectricityCredential v1.2 building block
Data Exchange	BPP catalogue entries published per locus; grid operators and aggregators consume as BAPs

---

The Dataset — `EnergyResource[]` + `ConsumptionProfile[]` (grid-side publication)

Illustrative / non-normative — describes the future per-locus aggregate (see Scenario, above), not an executable credential today.

`energyResources[]` would be populated per locus — solar, battery, EV charger, inverter, controllable load — with capacity, inspection status, equipment details, and the parent/sub-resource topology (PV and BESS -> Inverter -> Meter -> DT). `consumptionProfiles[]` would be included where sanctioned-load / export-limit context is needed. No `customerProfile`, no `customerDetails` — these fields would never be populated for this issuance.

```
Feeder F-02
 ├── DT F02-DT-15 ── 14 consumers ──┐
 ├── DT F02-DT-16 ── 22 consumers ──┤ aggregated into one
 └── DT F02-DT-17 ── 31 consumers ──┘ future PII-free aggregate record
                                        (illustrative — energyResources[] + consumptionProfiles[], no PII)
```

The subject scopes by locus:

Scope	Subject DID example
Per feeder	`did:web:ies.discom.example:assets:feeder:F02`
Per substation	`did:web:ies.discom.example:assets:substation:SS-11KV-12`
Licensee-wide	`did:web:ies.discom.example`

Setup: Register -> Discover -> Exchange

The steps below describe the future implementation path for the illustrative per-locus aggregate, once its credential-subject shape is formalised (see Open Items). They are not executable today; for the currently executable path, follow Issue Credentials for the per-consumer Energy Passport.

1. Register — identity reused

Same `did:web` as the Energy Passport — no new identity setup if you already issue Passports -> **Setup Register**.

2. Discover — grid-side catalogue

Publish a BPP catalogue entry per locus (feeder / substation / licensee-wide), `accessMethod: INLINE` -> **Setup Exchange**. Test grid-operator and aggregator consumers as BAPs.

3. Exchange — issuance shape

- Exclude PII at issuance time — no `customerDetails`, no customer numbers; one consumer's data is never merged with another's
- Populate `energyResources[]` from CIS / DERMS / inspection register
- Populate `consumptionProfiles[]` only where sanctioned-load / export-limit context is needed
- Populate topology links (`parentResources`, `subResources`) where the DT mapping is known
- Validate against the aggregate's own credential-subject schema once formalised (not ElectricityCredential v1.2 — see Open Items)
- Agree and schedule a refresh cadence (typical: weekly for an active growth area, monthly otherwise, or on material change)

Checklist

Step 0 — Prerequisites

- [ ] Register your organisation in the DeDi directory and create your `did:web` identity — see Setting up Register
- [ ] Stand up your Beckn ONIX adapter and connect it to the network — see Setting up Discover & Exchange
- [ ] Pass the basic conformance check — see Conformance Checklist

The items below track the future implementation path for the illustrative aggregate (see Scenario, above) — not applicable until its credential-subject shape is formalised.

- [ ] Same `did:web` as the Energy Passport confirmed working
- [ ] Feeder / substation identifier convention confirmed (existing IDs wrapped in `did:web`)
- [ ] BPP catalogue entries published per locus
- [ ] Grid-operator and aggregator consumers tested as BAPs
- [ ] PII excluded at issuance — no `customerDetails`, no customer numbers; no merging across consumers
- [ ] `energyResources[]` populated from CIS / DERMS / inspection register
- [ ] `consumptionProfiles[]` included only where sanctioned-load / export-limit context is needed
- [ ] Topology links populated where known
- [ ] Schema validation passes against the aggregate's own credential-subject schema (once formalised)

- [ ] Refresh cadence agreed and scheduled
- [ ] Revocation tested

Team. [ ] Network / planning SPOC · [ ] IT SPOC · [ ] Inspection / commissioning SPOC

Open Items

- **Refresh cadence per locus** — weekly vs monthly vs on-material-change — to be tuned per deployment.
- **Aggregator binding** — the exact field and credential proof an aggregator presents to claim a resource.
- **Privacy review** — confirmation that the aggregated, PII-free issuance meets DPDP grid-side disclosure norms; `consumptionProfiles[]` entries are keyed by meter id (pseudonymous, not anonymous), so their inclusion belongs behind the authenticated tier where required.
- **Aggregate record shape** — ElectricityCredential requires a single `customerProfile` with one customer number, so it cannot represent a PII-free, multi-consumer aggregate. The aggregate needs its own credential-subject shape, formalised upstream through separate governance; until then it remains an illustrative future profile with no canonical executable example.

References

- ElectricityCredential v1.2 schema
- Overview — DER Visibility — standards basis, definitions, full field schedule
- Consumer Energy Passport — the currently executable per-consumer EC v1.2 issuance
- Consumer Energy Passport Schedule I example — the validated, executable EC v1.2 payload
- Energy Credentials

P2P Energy Transaction

In a hurry? Jump to the Checklist, or See it run in one `docker compose up`. For the standards basis, the full field schedule and definitions, see the **Overview**.

Two prosumers on different DISCOMs execute a direct, signed energy trade over the same Beckn wire that carries dataset exchanges. Each DISCOM is represented in the protocol by a regulated Ledger Provider; allocation and settlement are computed by signed Rego policy, with no central exchange.

> **Current deployment.** The architecture supports one Ledger Provider per DISCOM, but to start with **all trading platforms connect to a single Ledger Provider**, hosted at `ies-p2p-energy-ledger.beckn.io` — the two LPs collapse into one (the intra-DISCOM topology; same protocol, fewer hops). The network namespaces are `indiaenergystack.in/test-ies-p2p-trading-network` (test) and `indiaenergystack.in/ies-p2p-trading-network` (production) — use them as the `networkId` in your ONIX adapter's configuration, described in How you implement IES -> Setup Exchange (ONIX).

If you are a…	Start here
Trading platform (BAP / BPP)	What each actor does, per phase — the unshaded arrows are yours · Payload snapshots · Setup
Ledger Provider (LP)	What each actor does, per phase · Auto-routing of contracts and allocations · Ledger interfaces · Setup
DISCOM (utility)	What each actor does, per phase — the daily meter-data pull is yours · Payload snapshots — the allocation columns are yours · the DISCOM rows in Setup -> Exchange

---

Scenario

A buyer prosumer and a seller prosumer — potentially on different DISCOMs — want a direct energy trade with cryptographic settlement and no central exchange. Each is represented by a trading platform (TP); each DISCOM contracts one regulated Ledger Provider (LP) that holds its slice of the record. The contract is a `DEGContract` with a `P2PTrade` body; per-interval negotiation, allocation and reconciliation ride as `BecknTimeSeries` inside the same envelope. Network rules and settlement terms are signed Rego bundles, evaluated locally by every participant.

The same integration works **inter-DISCOM** (two LPs, one peer leg) and **intra-DISCOM** (buyer and seller behind the same DISCOM — the two LPs collapse into one). You integrate once.

Actors and Roles

Role	Who	What they run	Talks to
Trading platform (BAP)	Buyer's platform	Matching engine + ONIX BAP wiring	The peer TP (Beckn `/select`–`/status`) + its own LP (Beckn `/confirm`, `/status`)
Trading platform (BPP)	Seller's platform	Catalogue + matching engine + ONIX BPP wiring	The peer TP + its own LP; hosts the `contractpolicyenforcer` step
Ledger Provider (LP)	One per DISCOM (may be shared)	Ledger app behind a Beckn BPP **and** BAP	Both TPs it serves + its own DISCOM (meter-data pull)
DISCOM (utility)	Buyer's / Seller's DISCOM	A thin Beckn BPP that answers meter-data pulls	Its contracted LP only

Participants are identified by their plain **network subscriber IDs** (from the DeDi registry) — no `did:web` for participants. See Overview §3.

Building Blocks Used

Block	Role in this use case
Register	Subscriber IDs for all four actors; the LP<->DISCOM `utilityId` binding in `DiscomLedgerProvider`
Discover	Seller TP lists an `EnergyTradeOffer` catalog via `publish-catalog`; buyer TP queries with `discover`
Exchange	The `select` -> `init` -> `confirm` -> `status` lifecycle, plus the `degledgerrecorder` and `contractpolicyenforcer` ONIX steps

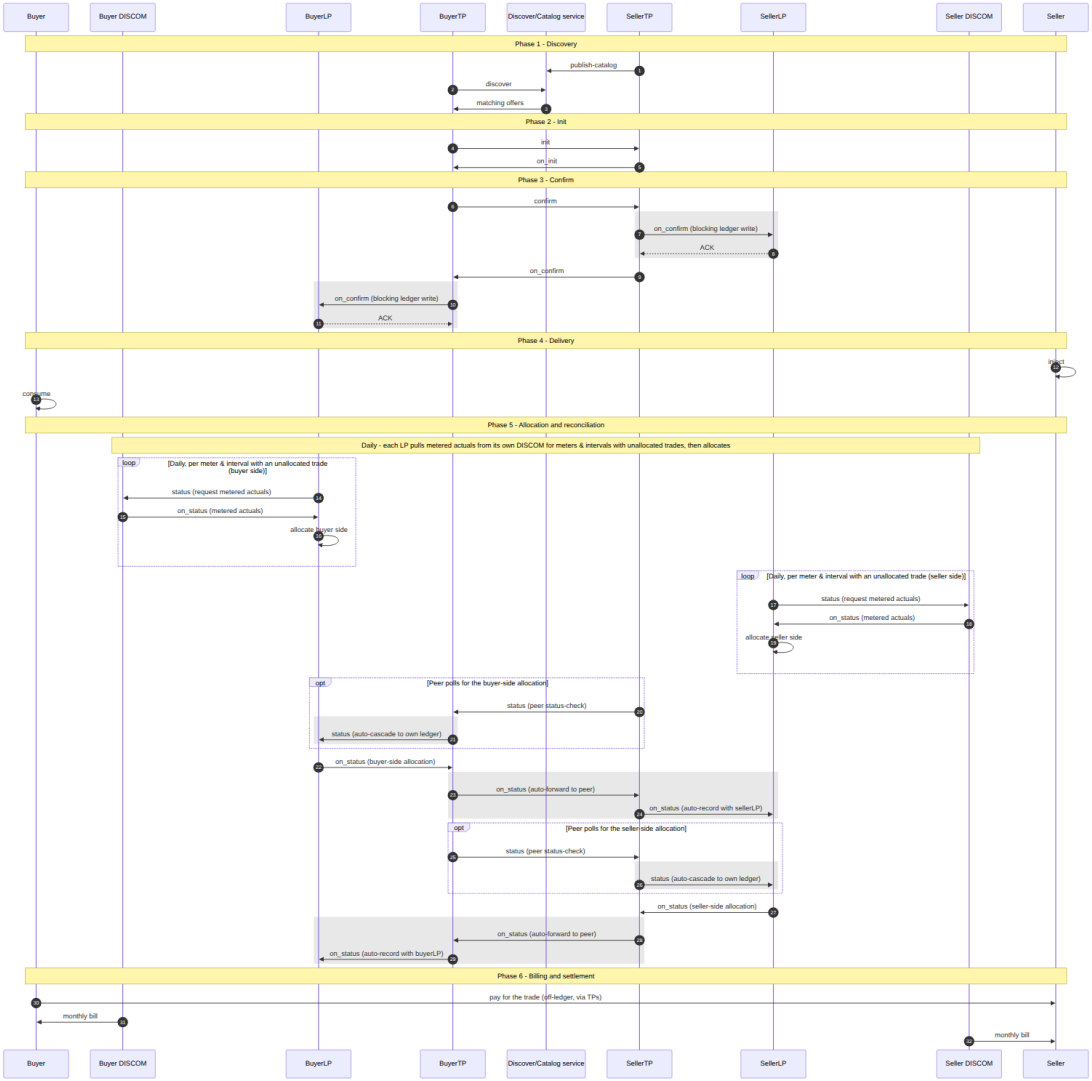
Topology



Everything buyer-side sits on the left, everything seller-side mirrors it on the right, and the two TPs meet in the middle over the `select / init / confirm / status` leg. Two regulated LPs in the protocol — one per DISCOM. No central exchange. The two LPs never speak to each other directly; the two TPs are the only liaison between them. Each LP pulls metered actuals daily from its **own** DISCOM (dashed lines) as the input to its allocation. Discovery (not drawn) goes through the network's Discovery service: the SellerTP lists offers via `publish-catalog`, the BuyerTP queries with `discover`.

What each actor does, per phase

The full wire sequence for the inter-DISCOM flow. **Shaded (grey) bands are automated by ONIX plugins** (`degledgerrecorder`, `contractpolicyenforcer`) — you configure them, you don't code them; **unshaded arrows are your application logic.** Intra-DISCOM (buyer and seller behind the same DISCOM) collapses Phase 2's optional limit check and Phase 5 into a single ledger.



The lanes are laid out **symmetrically** — `Buyer · Buyer DISCOM · BuyerLP · BuyerTP` on the left mirror `SellerTP · SellerLP · Seller DISCOM · Seller` on the right, with the Discover/Catalog service as the centre axis and the two trading platforms meeting in the middle. The flow is a mirror image side-to-side, with **two real seller-side specifics**: on the seller side, `SDL->STP` `on_status (seller-side allocation)` carries the `FINAL_ALLOC` settled quantity, and the `contractpolicyenforcer` step computes the revenue flows as that `on_status` passes SellerTP.

The table below maps each phase to what every actor does:

Phase	Buyer TP	Seller TP	Each LP	Each DISCOM
1 · Discovery	`discover` against the Discovery service with a JSONPath intent filter	`publish-catalog` an `EnergyTradeOffer`	—	—
2 · Init	`init` — refine quantity and price	Answer `on_init`; optional LP headroom pre-check	*(optional)* answer a headroom pre-check	—

Phase	Buyer TP	Seller TP	Each LP	Each DISCOM
3 · Confirm	`confirm`	Answer `on_confirm`	**(auto)** record the blocking `on_confirm` forward and sync-ACK it	—
4 · Delivery	Buyer consumes	Seller injects	—	—
5 · Allocation & reconciliation	*(optional)* poll the seller side with `status`	*(optional)* poll the buyer side with `status`; the `contractpolicyenforcer` step computes revenue flows	**Daily**, `status`-pull metered actuals from your own DISCOM for meters/intervals with unallocated trades, allocate, emit `on_status`; **(auto)** cascade	**Daily**, answer your LP's `status` pull with an `on_status` carrying the metered actuals
6 · Billing & settlement	Buyer pays seller off-ledger via TPs	—	Adjust the monthly bill (excl. traded volume, incl. wheeling)	—

The three things you must build:

1. **A trading platform** builds its matching engine and answers/drives the Beckn lifecycle. The cascade to ledgers is not your code — it is the `degledgerrecorder` plugin.
2. **A Ledger Provider** builds the **allocation function** (as simple as pro-rata across the customer's trades in the delivery window) and the **daily meter-data pull** from its DISCOM. Everything routing-related is the plugin.
3. **A DISCOM** builds a thin BPP that answers its LP's daily `status` pull with the metered actual injected / consumed quantities per interval — delivered as a `BecknTimeSeries` sub-transaction, **not** as a separate `MeterData` exchange.

Phase 5 supports multiple rounds — provisional allocation, final allocation after meter-data finalisation, deviation true-up — by repeating the `/status` round-trip with a fresh `BecknTimeSeries` payload. Iteration is a payload concern, not a protocol concern.

Auto-routing of contracts and allocations

The hard part of a four-actor topology is making sure every contract and every allocation update reaches both TPs **and** both LPs — without a central exchange, without the LPs talking to each other, and without loops. That cascaded routing is a handful of behaviours, implemented entirely by the `degledgerrecorder` ONIX plugin. **You configure it; you do not write it.**

Behaviour	Trigger	What the plugin does
Record the contract — blocking ledger write	The seller TP emits `on_confirm` (at `/bpp/caller`); the buyer TP receives it (at `/bap/receiver`)	Each TP forwards a context-rewritten `on_confirm` to its **own** DISCOM's LP and **blocks until the LP ACKs**: the `on_confirm` leaves the seller only after the seller-side ledger has recorded it, and reaches the buyer app only after the buyer-side ledger has. A confirmed trade is by definition a recorded trade. The LPs answer with a synchronous ACK — they do not emit an `on_confirm` of their own. There is no cascade on the `/confirm` request itself.
Record status with your own ledger	`/status` arrives at a TP's `/bpp/receiver` — including a **peer status-check poll**	Cascade a copy (async) to the TP's **own** DISCOM's LP. The LP endpoint is read from the payload itself — `participants[role=…Discom].participantAttributes.ledgerUrl` (`ledgerUriSource: payload`) — not from static config. This is how a peer's `status` poll reaches the polled side's ledger.
Forward your DISCOM's update to the peer	`/on_status` arrives at a TP's `/bap/receiver` **and** `context.bppId` equals the TP's own DISCOM's participantId	The TP's own LP has just computed its allocation. Forward the `on_status` to the **peer TP**, so the peer can record it and pass it down to its own LP.
Record the peer's update with your DISCOM	`/on_status` arrives at `/bap/receiver` from anyone **other than** the own DISCOM (i.e. the peer TP)	Push the payload to the TP's **own** LP so it records the full bilateral settlement. Skipped when the payload carries no performance data (a bare status-check ACK is not cascaded).

Chained together they produce one linear path per update, and the routing is **symmetric** — a seller-side update runs `SellerLP -> SellerTP -> BuyerTP -> BuyerLP`, and a buyer-side update runs the mirror chain `BuyerLP -> BuyerTP -> SellerTP -> SellerLP` — after which every party holds the same signed record. (The plugin source, devkit configs and workflows label the last three behaviours *Rule 1*, *Rule 2a* and *Rule 2b* — you'll meet those names when reading the config comments.)

Why it cannot loop. The chain always alternates *ledger -> platform -> ledger*; there is no ledger->ledger edge. LPs receive `on_status` at `/bap/receiver`, which routes to an ACK-only webhook and never re-cascades — every peer-forward terminates in a ledger write at an LP sink. Degenerate topologies collapse safely: if buyer and seller share one platform the self-forward is skipped and the cascade goes straight to the peer's DISCOM; if they share one DISCOM the single LP is written once.

What the rules enable:

- **No central exchange, full replication** — all four parties converge on the same contract and allocation state through pairwise Beckn legs only.
- **Zero routing code for implementers** — a TP or LP enables the plugin and edits the `participants` block; the routes are read from the payload's participants, so inter-DISCOM, intra-DISCOM and single-platform-prosumer topologies all work from the same config.
- **A per-leg audit trail** — every cascade leg rewrites `context.bapId` / `bapUri` / `bppId` / `bppUri` for the sub-transaction and is separately signed, so each hop is independently attributable end-to-end.

If a cascade leg exhausts its retries, the plugin returns a best-effort error `on_status` to the original sender with `error.code = "DEG_ASYNC_ACK_TIMEOUT"`. The full design and the loop-free argument live in the plugin README and the wave-2 devkit.

Ledger interfaces

Each LP runs both a BPP and a BAP face:

- `/bap/receiver` — accepts the blocking `on_confirm` forward (contract entry, answered with a synchronous ACK) and `on_status` callbacks (e.g. the daily meter-data actuals from the DISCOM).
- `/bpp/receiver` — accepts the `/status` cascades from the TPs (including peer status-check polls).
- `/bpp/caller` — emits `on_status` callbacks (allocations, settled quantities) toward the TPs.
- `/bap/caller` — emits the **daily** `/status` requests (meter-data pulls) toward the DISCOM actor's `/bpp/receiver`.

Authentication is the standard Beckn signing flow against the network registry. Nothing custom is required of the implementer beyond the ONIX config blocks the devkit ships.

Payload snapshots

One `BecknTimeSeries` envelope carries the whole trade; what changes between phases is only **which payloadType columns exist and who inserts them** (`insertedBy`). Three moments from the devkit's uc1 examples, trimmed to one interval:

1 — At confirm (trade negotiation). The TPs have inserted the negotiation columns (`objectType: EVENT_PAYLOAD_DESCRIPTOR`); no allocation data exists yet:

```
"commitmentAttributes": {
  "@context": "https://schema.nfh.global/BecknTimeSeries/v1.0/context.jsonld",
  "@type": "TimeSeries",
  "intervalPeriod": { "start": "2026-04-26T04:30:00Z", "duration": "PT1H" },
  "payloadDescriptors": [
    { "objectType": "EVENT_PAYLOAD_DESCRIPTOR", "payloadType": "PRICE_PER_KWH", "currency": "INR",
    ↪  "insertedBy": "sellerPlatform" },
    { "objectType": "EVENT_PAYLOAD_DESCRIPTOR", "payloadType": "REQUESTED_QTY", "units": "KWH",
    ↪  "insertedBy": "buyerPlatform" }
  ],
  "intervals": [
    { "id": 0, "payloads": [
      { "type": "PRICE_PER_KWH", "values": [12.5] },
      { "type": "REQUESTED_QTY", "values": [20.5] } ] }
  ]
}
```

2 — During reconciliation (DISCOM allocation). Same envelope, same intervals — the DISCOM sides have appended their allocation columns (`objectType: REPORT_PAYLOAD_DESCRIPTOR`), populated from the daily meter-data pull. Note `FINAL_ALLOC <= min(BUYER_DISCOM_ALLOC, SELLER_DISCOM_ALLOC)` — the network policy enforces it per interval:

```
"payloadDescriptors": [
  { "objectType": "EVENT_PAYLOAD_DESCRIPTOR",  "payloadType": "PRICE_PER_KWH", "currency": "INR",
  ↪  "insertedBy": "sellerPlatform" },
  { "objectType": "EVENT_PAYLOAD_DESCRIPTOR",  "payloadType": "REQUESTED_QTY", "units": "KWH",
  ↪  "insertedBy": "buyerPlatform" },
  { "objectType": "REPORT_PAYLOAD_DESCRIPTOR", "payloadType": "BUYER_DISCOM_ALLOC", "units": "KWH",
  ↪  "insertedBy": "buyerDiscom" },
  { "objectType": "REPORT_PAYLOAD_DESCRIPTOR", "payloadType": "SELLER_DISCOM_ALLOC", "units": "KWH",
  ↪  "insertedBy": "sellerDiscom" },
  { "objectType": "REPORT_PAYLOAD_DESCRIPTOR", "payloadType": "FINAL_ALLOC", "units": "KWH", "insertedBy":
  ↪  "sellerDiscom" }
],
```

```
"intervals": [
  { "id": 0, "payloads": [
    { "type": "PRICE_PER_KWH", "values": [12.5] },
    { "type": "REQUESTED_QTY", "values": [20.5] },
    { "type": "BUYER_DISCOM_ALLOC", "values": [18.5] },
    { "type": "SELLER_DISCOM_ALLOC", "values": [19.2] },
    { "type": "FINAL_ALLOC", "values": [18.5] } ] }
]
```

(The full example also carries `BUYER_DISCOM_STATUS` / `SELLER_DISCOM_STATUS` string columns — `COMPLETED` per interval.)

3 — Revenue flows. As the settled `on_status` passes the seller TP, the `contractpolicyenforcer` step resolves the DeDi-published contract policy, evaluates it locally and injects the result into the contract's `consideration` block — no application wrote this (illustrative values):

```
"consideration": [
  { "id": "auto-settlement-flows",
    "considerationAttributes": {
      "@context": "https://schema.nfh.global/RevenueFlow/v2.0/context.jsonld",
      "@type": "RevenueFlow",
      "revenueFlows": [
        { "role": "buyerPlatform",  "value": -437.15, "currency": "INR", "description": "Energy purchase
        ↪  cost (18.5 kWh × ₹12.5 + 14.2 kWh × ₹14.5)" },
        { "role": "sellerPlatform", "value": 437.15,  "currency": "INR", "description": "Energy sale
        ↪  proceeds" },
        { "role": "buyerDiscom",    "value": 0, "currency": "INR", "description": "Buyer-side wheeling
        ↪  charge (rate per the published policy)" },
        { "role": "sellerDiscom",   "value": 0, "currency": "INR", "description": "Seller-side wheeling +
        ↪  shortfall penalty (rates per the published policy)" }
      ] } }
]
```

The energy legs net to zero between the two platforms; the DISCOM rows itemize whatever wheeling and shortfall-penalty charges the seller-DISCOM's published policy defines, alongside a disclosed platform-charge cap. A DISCOM sets its own rates by publishing a new policy version — no payload or code change needed.

Policy-as-code (Rego / OPA)

Every `DEGContract` carries a **`policyUrl`**. That URL points to a Rego policy bundle hosted on a DeDi runtime and digitally signed. **Two distinct rego bundles** apply, both enforceable offline by any participant.

The IES network mandates which policy bundles are in force on a given network and **publishes them as policy-as-code records on a DeDi runtime**. DeDi serves the same role for policy that it does for keys: a trusted, verifiable, version-controlled source. A participant fetches the bundle, evaluates locally with OPA, and the answer is cryptographically attributable to the published version. There is no central policy server to call out to at trade time.

Network policy

Enforces "is this a valid trade on this network at all?" Examples drawn from `p2p-trading-ies-wave2-networkpolicy.rego`:

Rule	What it checks
Roles	The four required roles (`buyer`, `seller`, `buyerDiscom`, `sellerDiscom`) are present and each maps to a known `utilityId`; buyer's DISCOM seller's DISCOM for inter-DISCOM trades.
No self-trade	Buyer and seller meter IDs are different.

Rule	What it checks
Generation source	Offer's `sourceType` is not `GRID` (network admits only DER-sourced energy).
TimeSeries shape	`PRICE_PER_KWH` is denominated in INR; `AVAILABLE_QTY` / `REQUESTED_QTY` in kWh; every `payloadType` used in `payloads[]` is declared in `payloadDescriptors`.
Context alignment	`context.bppId` / `context.bapId` match the participantIds the payload claims.
Performance integrity	`FINAL_ALLOC <= min(BUYER_DISCOM_ALLOC, SELLER_DISCOM_ALLOC)` per interval.
TEST / PROD separation	Test-network identifiers carry the `TEST_` prefix consistently; production networks check approved `utilityId`s only.

A network operator can change the rule set without recompiling code — publish a new bundle on DeDi, bump the `policyUrl` on the next contract, every participant resolves the new bundle on first use.

Contract policy (seller-DISCOM policy)

A second rego bundle (`p2p-trading-ies-wave2-contractpolicy.rego`, published on DeDi as `ies-p2p-network-settlement-rego-policy-v1`) is the policy of the *seller's* DISCOM, linked by the catalog publisher into every trade involving that DISCOM's prosumers. It computes the **revenue flows** on each contract from the final allocation:

- Buyer pays `FINAL_ALLOC × PRICE_PER_KWH` (signed negative); seller receives the same amount.
- BuyerDiscom and SellerDiscom collect wheeling charges and any delivery-shortfall penalty, with a disclosed platform-charge cap — all rates defined by the published policy version in force.

On the seller TP, the `contractpolicyenforcer` ONIX pipeline step resolves the DeDi record on `on_status` (accepting only URLs under the configured `allowedPolicyUrlPrefixes`), verifies its checksum, evaluates the rego locally with a >=1-day cache, and writes the result to `message.contract.consideration[id=auto-settlement-flows].considerationAttributes` as a `RevenueFlow` JSON-LD object (wire key `revenueFlows`). Settlement reconciliation reads from there.

Mandating the contract policy the same way as the network policy keeps every participant computing the same answer from the same inputs. The wheeling charge a DISCOM collects is no longer a bilateral spreadsheet — it is the output of a signed rego function over a signed contract. DISCOMs author and publish their own via the discom policy guide.

Setup: Register -> Discover -> Exchange

Built on the four implementation steps in **Concepts — Before you build**. Prerequisites: Git, Docker + Docker Compose, and Postman — nothing else. If you have never run an IES exchange before, do the Data Exchange quick start first; this use case reuses that stack unchanged.

Step 0 — See it run before you build (local devkit)

Prove the four-actor topology and the cascade on your laptop before touching real systems:

```
git clone https://github.com/beckn/DEG.git
cd DEG/devkits/p2p-trading-ies-wave2/install
docker compose up -d
```

This starts the buyer side (`onix-buyerapp`, `sandbox-buyerapp`, `onix-ledger-buyerdiscom`, `onix-buyerdiscom`), the seller side (`onix-sellerapp`, `sandbox-sellerapp`, `onix-ledger-sellerdiscom`, `onix-sellerdiscom`), and one `beckn-router` (Caddy) on `:9000` resolving each actor by per-node hostname (e.g. `seller-discom-ledger.example.com`). The sandbox containers stand in for your application; the ONIX containers are the adapters you will keep.

Drive the flow from Postman. Four collections sit under `uc1/postman/` — one per role (buyer TP, seller TP, buyer DISCOM ledger, seller DISCOM ledger). Import the role you are integrating, leave the defaults in place, and fire `publish-catalog` / `discover` (buyer TP) then `/select` -> `/init` -> `/confirm` -> `/status`. The full Phase 1–5 lifecycle is covered by the role-specific requests.

Register — four-actor network identity

- [ ] Identity setup complete for your role (TP, LP, or DISCOM)
- [ ] DeDi subscriber record under the correct network namespace — `indiaenergystack.in/test-ies-p2p-trading-network` (test), `indiaenergystack.in/ies-p2p-trading-network` (production)
- [ ] Signing key in a secrets manager
- [ ] (LP) `DiscomLedgerProvider` entry registered with the LP<->DISCOM `utilityId` binding

Discover — catalog and offers

- [ ] Discovery setup complete
- [ ] (Seller TP) Catalogue entry published via `publish-catalog` — `EnergyTradeOffer` with `BecknTimeSeries` descriptors per `payloadType`
- [ ] (Buyer TP) `discover` against the network's Discovery service returns your counterparty's offers

Exchange — adapter, cascade, policy

- [ ] Adapter built mapping your application logic to your role (see role matrix below)
- [ ] `degledgerrecorder` ONIX plugin enabled (`ledgerUriSource: payload`, `ledgerApi: beckn`); auto-routing verified on the devkit, no loops
- [ ] Network policy bundle URL resolves and signature verifies; local OPA eval rejects rule-violating payloads
- [ ] Contract policy record URL resolves; `contractpolicyenforcer` step computes revenue flows on `on_status`
- [ ] (DISCOM) daily meter-data pull answered — actual injected / consumed quantities published to your LP as `BecknTimeSeries`, **not** as `MeterData` / `DatasetItem`
- [ ] (LP) meter-quantity payloadTypes wired into the allocation function; daily pull over meters/intervals with unallocated trades scheduled
- [ ] One real trade completed end-to-end and reconciled
- [ ] Runbook in place (key rotation, bundle upgrades, disputes)

If you are a …	You implement	Talks to
Trading-platform vendor (BAP / BPP)	Your matching engine + the ONIX BAP/BPP wiring	The peer TP (Beckn `/select`–`/status`) + your own LP (Beckn `/confirm`, `/status`)
LP for one or more DISCOMs	Your ledger app (allocation function + daily meter-data pull) behind a Beckn BPP+BAP	Both TPs you serve + the DISCOM actor (meter-data sub-tx)
DISCOM (utility)	A thin Beckn BPP that answers its LP's daily meter-data pull	Your contracted LP only

The devkit ships sandbox implementations of all four roles — replace one at a time as your real component matures: swap the sandbox container for your application, then point the ONIX adapter's `allowedNetworkIDs`, `networkParticipant` and `keyId` at your real identity.

Go live — join the production network

The production fabric is operated on NFH (Networks for Humanity). Its Join the network instructions are the final mile, and map one-to-one onto what you just proved locally:

1. **Register an identity** — register your subscriber ID and publish your public key via the Registry (the same DeDi machinery as your devkit subscriber record).
2. **Stand up a network adapter** — run ONIX inside your own infrastructure, with the signing, schema-validation, policy and audit plug-ins configured (your devkit ONIX config carries over).

3. **Publish, discover, or both** — list your offers via the Catalog service; query via the Discovery service — the production endpoints for the `publish-catalog` / `discover` calls you tested in Step 0.
4. **Transact** — the `select -> init -> confirm -> status` lifecycle, now against real counterparties under the production policy bundles.

None of the four steps require permission from a platform owner. For a guided first run on the fabric, see Getting started with Fabric.

Checklist

Step 0 — Prerequisites

- [ ] Register your organisation in the DeDi directory and create your `did:web` identity — see Setting up Register
- [ ] Stand up your Beckn ONIX adapter and connect it to the network — see Setting up Discover & Exchange
- [ ] Pass the basic conformance check — see Conformance Checklist

Role-specific rollout

- [ ] Subscriber record under the correct network namespace (test / production)
- [ ] Signing key in a secrets manager
- [ ] (LP) `DiscomLedgerProvider` entry registered with the `utilityId` binding
- [ ] (Seller TP) `EnergyTradeOffer` catalogue published via `publish-catalog`
- [ ] (Buyer TP) `discover` returns your counterparty's offers
- [ ] `degledgerrecorder` enabled (`ledgerUriSource: payload`, `ledgerApi: beckn`); cascade verified on the devkit, no loops
- [ ] Network policy bundle resolves and verifies; local OPA rejects rule-violating payloads
- [ ] Contract policy resolves; `contractpolicyenforcer` computes revenue flows on `on_status`
- [ ] (DISCOM) daily meter-data pull answered with `BecknTimeSeries` actuals — **not** a `MeterData` exchange
- [ ] (LP) allocation function wired to the meter-quantity payloadTypes; daily pull scheduled
- [ ] One real trade completed end-to-end and reconciled
- [ ] Runbook in place (key rotation, bundle upgrades, disputes)

Team. [ ] IT / data SPOC · [ ] Commercial / settlement SPOC · [ ] Authorised Signatory

---

Open Items

1. **Wheeling / penalty tariff values** — the rates in force are whatever the seller-DISCOM's published contract policy defines; each DISCOM sets production values per its tariff order by publishing its own policy version.
2. **`contractpolicyenforcer` ONIX step** — ships in the wave-2 seller-TP config; being aligned across LP implementations.
3. **TEST -> PROD `utilityId` allow-list** — production allow-list is governance-pending.
4. **CERC sandbox graduation** — production-grade network policy bundle awaits CERC sign-off post-sandbox.
5. **Intra-DISCOM topology** — collapsing the two LPs into one is supported and lighter; the configuration convention is in the wave-2 devkit, being formalised.
6. **Daily meter-data pull cadence** — the LP->DISCOM actuals pull is modelled as a daily job over meters/intervals with unallocated trades; the exact cadence and retry/true-up window are per-DISCOM and being formalised.

Dev kits and code

- **Devkit** — `devkits/p2p-trading-ies-wave2` (code, examples, four role-specific Postman collections)
- **Scripted lifecycle** — `uc1/workflows/p2p-trading-ies-wave2.arazzo.yaml`
- **Cascade plugin** — `plugins/degledgerrecorder`
- **Contract-policy plugin** — `plugins/contractpolicyenforcer`
- **Network policy** — `p2p-trading-ies-wave2-networkpolicy.rego`

- **Contract policy** — `p2p-trading-ies-wave2-contractpolicy.rego`
- **Inter-DISCOM specification** — Beckn DEG full spec
- **IES architecture note** — "Inter-DISCOM P2P trading" in the ies-docs repository (repository has since been archived and made private; ask the IES Secretariat for access)
- **NFH fabric onboarding** — Join the network · Getting started with Fabric
- **Sample bill worksheet** — Google Sheet

References

- **Overview — P2P Energy Transaction** — standards basis, definitions, full field schedule, the six-phase walkthrough, and Points for Confirmation
- **External Schemas — Energy Trading** — consolidated field reference
- Canonical schemas at **schema.nfh.global** — P2PTrade/v2.0 · DEGContract/v2.0 · EnergyTradeOffer/v2.0 · EnergyTradeDelivery/v2.0 · DiscomLedgerProvider/v2.0 · BecknTimeSeries/v1.0
- **ElectricityCredential v1.2** *(optional)* — seller's attestation backing the offer

Concepts

Before you build

Every IES participant completes the same setup once, before any use case ships. It is deliberately small — a domain, a few keys, Docker containers, one sign-off — and everything you build afterwards reuses it. The sequence:

1. **Register your organisation.** Claim and domain-verify a namespace in the DeDi directory, and publish a `did:web` identity — the cryptographic name counterparties and verifiers resolve to check your signatures. Beckn participants also publish a subscriber record. -> **Setting up Register**
2. **Stand up the Beckn ONIX adapter.** Deploy the ready-made adapter that signs, verifies, and routes messages, and complete a signed `confirm` -> `on_confirm` round-trip on the IES network. -> **Setting up Discover & Exchange**
3. **Pass the conformance check.** Run the conformance checks end-to-end and sign off — that page states exactly what a self-run sign-off does and doesn't prove. -> **Conformance Checklist**

Two more pieces slot in between, depending on what you're building:

- **Issuing Credentials** — for consumer-facing (B2C) credential use cases: run OpenCred and issue, verify, and revoke W3C Verifiable Credentials. A credential is signed once by you and verified by *any* third party against your published `did:web`, over any delivery channel — no network membership involved. A credentials-only participant can defer the ONIX step until a data-exchange use case calls for it; a pure Beckn participant (e.g. a trading platform) skips this one.
- **Build your Internal-facing Adapter** — the only place you write code: a small mapping (typically 200–1,000 lines per use case) between your internal systems and the IES schemas, feeding OpenCred and/or ONIX. Your existing CIS / MDM / billing / DERMS / ERP stays exactly as it is.

How this section is organised. The "Setting up …" pages (and Issuing Credentials) are do-guides — prerequisites first, then numbered copy-pasteable steps, then a checklist. Nested under each is an *in depth* page — Register & Identifiers, Discover, Exchange, Verifiable Credentials — that explains the concepts behind the steps: what a DID is, how the DeDi directory is structured, why Beckn, the credential trust model. Work through the setup page; open the in-depth page when you want to know why a step is what it is.

Setup	Page	Time	What you get
Register *(everyone)*	Setting up Register	1–2 days	A verifiable `did:web`, a verified DeDi namespace — plus, for Beckn participants, a published subscriber record and an IES network reference
Exchange *(B2B use cases)*	Setting up Discover & Exchange	1–2 days	A running Beckn adapter (ONIX) exchanging signed messages on the IES network
Credentials *(B2C use cases)*	Issuing Credentials	½ day	A running OpenCred signing service: issue, verify, revoke W3C Verifiable Credentials

Setup	Page	Time	What you get
Adapter	Build your Internal-facing Adapter	1–3 weeks	Small mapping layers between your internal systems and the IES schemas
Conformance	Conformance Checklist	1 day	A self-run conformance sign-off, end-to-end — scope explained on that page

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What you need

- **A domain you control.** Most DISCOMs use a dedicated subdomain like `ies.<discom>.in`. A bare apex domain works too.
- **DNS access.** To add a TXT record once (DeDi namespace verification).
- **One Linux host** that can run Docker (for OpenCred and/or ONIX) and serve HTTPS. Cloud VM is fine. Air-gapped works too.
- **One engineer.** Total effort is small; you do not need a team.
- **A vendor or in-house developer** to write the adapter mapping. The same person can do everything above.

What you do NOT need

- A new database. Your existing CIS / MDM / billing / DERMS / ERP stays exactly as is.
- A new compliance filing.
- A new procurement contract.
- A licence fee.
- Approval from anyone before starting on the sandbox. Get something working, then talk to the IES Secretariat about going to production.

Who needs to be involved

Role	Why	Time commitment
DNS / web admin	Publish `did.json` on your domain; add DeDi TXT record	15 minutes, once
IT / security admin	Generate signing keys (KMS preferred); deploy Docker containers	A few hours, once
Application developer / vendor	Write the mapping (your data -> IES schema)	The bulk of the work
Authorised signatory	Submit your subscriber record for IES network whitelisting	Email/form, once
Customer-ops / compliance	(For consumer credentials only) Identity-proofing procedure and privacy review	One review pass

---

How long the whole thing takes

The four pilot DISCOMs each went from cold start to four demonstrated use cases in **30 days** during the (now completed) DISCOM Challenge. The bulk of that time went into the adapter mapping and the customer-ops procedures for consumer-facing credentials.

Phase	Calendar time (typical)
Register (identity + directory)	1–2 days
Engines (OpenCred and/or ONIX)	1–3 days
Adapter for the first use case	1–3 weeks
Conformance	1–2 days
Each subsequent use case	A few days to a week

After you're set up

You only do the setup once. Adding new use cases on top only adds a small bit to the adapter mapping — the identity, the engines, the network membership are all reused. Pick your first from the Use Case Implementation Guides: Consumer Energy Passport, Consumer Meter Digest, Smart Meter Data Exchange, DER Visibility, or P2P Energy Transaction.

Setting up Register

> **The first setup every participant completes.** Get your organisation a verifiable digital identity, list it in the shared directory, and — if your use cases need a Beckn network — publish your network identity too. *Done once.* Hands-on work is under 30 minutes; allow 1–2 days elapsed for DNS verification and (for Beckn participants) the network reference.

The concepts behind every artefact on this page — DIDs, DeDi, namespaces, subscriber records — are in **Register & Identifiers in depth**. This page is the do-guide.

Who does what

Registration is role-based. Everyone does §1.1–1.4; the rest depends on what you plan to run.

You are…	Do	Then go to
A DISCOM / AMISP issuing credentials only (Consumer Energy Passport, Meter Digest)	§1.1 – §1.4	Issuing Credentials
Any organisation joining a Beckn network (data exchange, P2P trading — DISCOMs, AMISPs, trading platforms, aggregators)	§1.1 – §1.4, then §1.5 – §1.7	Setting up Discover & Exchange
A network operator (NFO) running your own Beckn network	§1.1 – §1.4, then §1.8	NFH network-operator docs (linked in §1.8)

Trading platforms and other pure Beckn participants who never issue credentials still need §1.1–1.4: the domain, `did.json`, and the verified DeDi namespace are the root of trust that the Beckn subscriber record hangs off.

What you'll have at the end

- A resolvable `did:web` for your organisation.
- Your public signing key published at a well-known URL under your domain.
- A verified DeDi namespace anchored to your domain, plus a DeDi API key.
- *(Beckn participants)* An Ed25519 message-signing keypair, a published Beckn subscriber record, and a reference entry in an IES network registry.

Before you start

- **A domain or subdomain you control**, with the ability to host one small static file under it.
- **DNS access** — to add one TXT record (DeDi domain verification).
- **`openssl`, `python3`, `curl`, `jq`** on the machine where you generate keys.
- *(Beckn participants, §1.5 only)* **Go 1.21+** — see the official Go install guide.

1.1 — Pick a domain (or subdomain) you control

Either works. Most DISCOMs use a dedicated subdomain so the credential-issuing identity is cleanly separated from the marketing site:

- `ies.discom.example` (subdomain)
- `discom.example` (apex domain — equally valid)

The host you pick becomes the host portion of your `did:web`. Your DID document will live at `https://<your-host>/.well-known/did.json`.

Once picked, record it in an environment variable — the copy-pasteable commands in this guide and in Issuing Credentials all reference `$OPENCRED_ISSUER_DOMAIN`, so you set your domain once here and never hand-edit a command:

```
export OPENCRED_ISSUER_DOMAIN="ies.yourdiscom.in"
```

> **Hosting `did.json` in a subfolder instead of the domain root?** `did:web` encodes sub-paths with colons, and `OPENCRED_ISSUER_DOMAIN` accepts the same form: set `OPENCRED_ISSUER_DOMAIN="<discom-domain>:subfolder"` and your DID becomes `did:web:<discom-domain>:subfolder`. The generator in §1.3 works unchanged, but the hosting path changes: a path-form DID resolves to `https://<discom-domain>/subfolder/did.json` — **no `.well-known/`** — so in §1.3 replace the colon with a slash and drop `.well-known/` when uploading and verifying. Variant table in Register & Identifiers — DID and DID document.

1.2 — Generate your credential-signing keypair

You will sign every credential with this key. Treat it as a production credential — KMS / HSM where possible, otherwise a securely backed-up file on a hardened host. The key is **EC P-256** (matches W3C VC Data Model 2.0 defaults):

```
mkdir -p ~/opencred/keys
cd ~/opencred

openssl genpkey -algorithm EC -pkeyopt ec_paramgen_curve:P-256 \
  -out keys/issuer-key.pem
chmod 600 keys/issuer-key.pem
```

Keep `issuer-key.pem` in your KMS in production. Treat it like a TLS private key. Issuing Credentials mounts this exact file into the OpenCred signing container, so keep the `~/opencred/keys/` path (or adjust consistently later).

> This key signs **credentials**. Beckn messages use a different key (Ed25519, §1.5). The two identities intentionally use separate keys, live in different registries, and serve different verifiers, so a compromise of one does not compromise the other.

1.3 — Generate and publish `did.json`

The DID document is one small JSON file: your DID string plus the public half of the key from §1.2. The block below is a complete generator — paste it as-is and it writes a finished `did.json`, with the DID `id`, `controller`, and JWK `x`/`y` coordinates all filled in from `$OPENCRED_ISSUER_DOMAIN` and `~/opencred/keys/issuer-key.pem`. Nothing to hand-edit.

```
python3 - > did.json <<'PY'
import subprocess, base64, json, os, sys
domain = os.environ.get("OPENCRED_ISSUER_DOMAIN")
if not domain:
    sys.exit("OPENCRED_ISSUER_DOMAIN is not set - run the export in 1.1 first")
key = os.path.expanduser("~/opencred/keys/issuer-key.pem")
der = subprocess.run(["openssl", "pkey", "-in", key, "-pubout", "-outform", "DER"],
                     capture_output=True, check=True).stdout
pt = der[-65:]  # uncompressed EC point: 0x04 || X(32) || Y(32)
b64u = lambda b: base64.urlsafe_b64encode(b).rstrip(b"=").decode()
did = f"did:web:{domain}"
print(json.dumps({
    "@context": [
        "https://www.w3.org/ns/did/v1",
        "https://w3id.org/security/suites/jws-2020/v1"
    ],
    "id": did,
    "verificationMethod": [{
        "id": f"{did}#key-0",
        "type": "JsonWebKey2020",
        "controller": did,
        "publicKeyJwk": {"kty": "EC", "crv": "P-256",
                         "x": b64u(pt[1:33]), "y": b64u(pt[33:65])}
    }],
    "authentication":  [f"{did}#key-0"],
    "assertionMethod": [f"{did}#key-0"],
}, indent=2))
PY
cat did.json
```

Expected: `cat did.json` prints a complete DID document with your domain in `id` and real `x`/`y` coordinates in `publicKeyJwk`.

> **On Windows?** The Python itself is cross-platform, but this block is written for a POSIX shell — the `python3 - <<'PY' … PY` heredoc, the `python3` command name, and `cat` don't work in native `cmd.exe`/PowerShell. Run it from **WSL** or **Git Bash**, where it works as-is (and where `openssl`, `curl`, and `jq` from §1.1 are also available). If you must stay in PowerShell, save the Python between `<<'PY'` and `PY` to a file (e.g. `gen-did.py`), then run `python gen-did.py > did.json` (use `python` or `py`, not `python3`).

Three fields matter, and you can ignore the rest until later:

- **`verificationMethod`** is the public key. Verifiers use it to check your signatures.
- **`assertionMethod`** says which key is allowed to issue credentials.
- **`authentication`** says which key can sign requests on behalf of the DID.

> **Keep the verification-method `type` as `JsonWebKey2020`.** It is the type defined by the `jws-2020` context already listed in `@context`, and it is the type that mainstream verifier libraries (did-jwt / Veramo, Spruce, `jose`-based stacks) map to your P-256 key for `ES256`. The plain `JsonWebKey` name is valid did-core, but several widely used verifiers do **not** resolve it to an `ES256` key and reject the signature with a "no suitable keys" error — a genuine, correctly signed credential then fails to verify. Publish `JsonWebKey2020` so any verifier can find your key.

You can add a `service` array later when your Beckn and OpenCred endpoints are publicly addressable; the DID is valid without it.

Publish it. Upload the file so this URL returns the JSON:

```
https://$OPENCRED_ISSUER_DOMAIN/.well-known/did.json
```

The `.well-known/` path is a standard convention; verifiers know to look there. A normal TLS cert is enough — the same one that already terminates your subdomain — and there must be **no redirect** on that URL.

Verify it from outside your network:

```
curl -s "https://$OPENCRED_ISSUER_DOMAIN/.well-known/did.json" | jq .id
```

Expected: your DID, e.g. `"did:web:ies.yourdiscom.in"`. If that prints, identity setup is done from the wire-protocol side — any participant can now resolve your `did:web` to your public key and verify anything you sign.

1.4 — Claim a DeDi namespace and verify your domain

DeDi is the public registry mechanism IES uses for namespaces, credential revocation, and Beckn subscriber membership. Claiming a namespace ties your organisation's records to your domain in a way anyone can verify.

1. **Sign up at publish.dedi.global** using an organisational role mailbox (`registry-admin@discom.example`).
2. **Create a namespace** — use a short name that maps to your organisation (`discom`, `np.example.com`).
3. **Verify your domain** — DeDi issues a DNS TXT record; add it to your DNS zone and click *Verify*. Wait for DNS propagation (usually 15 minutes; can take up to 48 hours). Once verification completes, a green **verified** label appears on your namespace in publish.dedi.global, and the namespace becomes publicly visible on explore.dedi.global — only verified namespaces are listed there, so appearing in explore results is itself the public verification signal.
4. **Create an API key** — in the DeDi UI, click your avatar in the top-right corner, then **Manage API key**. Tools that write into your namespace (OpenCred for revocation entries, your registry publisher) authenticate with this key. Store it alongside your other secrets.

You create the empty namespace here. What goes inside it depends on your role:

- **For credential issuance**, you do **not** need to create the registries by hand — when you run OpenCred in Issuing Credentials, it auto-creates the four it needs on first boot (`vc-revocation-registry`, `opencred-key-registry`, `schema_registry`, `context_registry`).
- **As a participant on a Beckn network**, you create **subscriber registries** by hand (§1.6) — one record per role/environment declaring your callback URL and message-signing key.
- **As a network operator (NFO)**, you create a **directory of participants** — a Beckn *subscriber-reference* registry (§1.8) that curates which subscribers belong to your network.

DeDi itself is documented at **docs.nfh.global**; the IES-specific registry map is in **Register & Identifiers — The directory: DeDi**.

> **Checkpoint — issuing credentials only?** You are done with registration. Continue to **Issuing Credentials**, which runs the OpenCred signing container against the domain, key, namespace and API key you just set up (and auto-creates those four registries). The remaining sections of this page are for Beckn-network participants.

1.5 — *(Beckn participants)* Generate your Beckn signing keypair

Prerequisite: §1.1–1.4 complete; Go installed.

Beckn message signing uses **Ed25519** keys — a different key from the P-256 credential key in §1.2. Beckn ONIX ships a small generator that prints the two Base64 values the subscriber record and your ONIX config need:

```
git clone https://github.com/beckn/beckn-onix.git
cd beckn-onix
go run install/generate-ed25519-keys.go
```

Expected output — two lines:

```
signingPrivateKey: <Base64, 32-byte seed>
signingPublicKey:  <Base64, 32-byte public key>
```

- **`signingPrivateKey`** — store in your secret manager; ONIX loads it to sign outgoing messages (Setting up Discover & Exchange §2.3). Never commit it.

- **signingPublicKey** — publish in your subscriber record (§1.6). Other Beckn nodes fetch it to verify your message signatures.

The generator source is `install/generate-ed25519-keys.go`; any other Ed25519 keypair generator works as long as it produces the same two Base64 artefacts (raw 32-byte seed and raw public key, no PEM headers).

1.6 — *(Beckn participants)* Publish your Beckn subscriber record

Prerequisite: verified namespace (§1.4), Ed25519 public key (§1.5), and the public HTTPS URL where your Beckn adapter will receive messages (you can publish first and deploy the adapter in Setting up Discover & Exchange; the URL just has to be the one you will use).

1. **Create a subscriber registry** under your verified namespace in publish.dedi.global, with the built-in **beckn_subscriber** tag. Create **one registry per environment** — `subscribers-test` and `subscribers-prod` — so a misconfigured test run can never pollute a production lookup.
2. **Add a record** with these fields:

Field	What to fill	Example
`subscriber_id`	Your **did:web** — the same org identity you published in §1.3. Using the DID (not a bare hostname) keeps your Beckn identity and your credential identity the same string.	`did:web:ies.discom.example`
`subscriber_url`	Your Beckn ONIX receiver endpoint	`https://ies.discom.example/bpp/receiver`
`type`	Your role on the network	`BAP` (consumer) or `BPP` (provider)
`signing_public_key`	The Ed25519 public key from §1.5, Base64, no header/footer	`eyAeqGFtAuks...`
`encryption_public_key`	*(Optional)* encryption public key	`lCI84I0Q0U0w...`
`countries`	Countries where you operate	`["IND"]`

   If you operate in both roles (BAP *and* BPP), publish a separate record per role. That `did:web subscriber_id` is what you set as `networkParticipant` in your ONIX config (Setting up Discover & Exchange §2.3).
3. **Note the record ID** DeDi assigns — you will configure it into ONIX as the `keyId` in Setting up Discover & Exchange §2.3.
4. **Verify the lookup.** Other nodes resolve your record through the Beckn fabric lookup URL. Substitute `<your-subscriber-id>` (your DeDi namespace from §1.4 — the path is addressed by this subscriber id, not by the `did:web` value of the `subscriber_id` record field) and `<your_record_id>` (from step 3); `subscribers.beckn.one` is the fixed fabric schema keyword — leave it literal, it is not your registry name. Allow 5–10 minutes for the cache, then:

   ```
   curl -s
    ↪  "https://fabric.nfh.global/registry/dedi/lookup/<your-subscriber-id>/subscribers.beckn.one/<your_record_id>"
    ↪  | jq
   ```

   Expected: the record you just published, including your `signing_public_key`. At this point `network_memberships` is empty — the NFO fills it in §1.7.

1.7 — *(Beckn participants)* Get referenced into an IES network

Prerequisite: subscriber record published and resolvable (§1.6).

A subscriber record alone does not put you on a network. Each IES network is a curated registry operated by the IES network operator (the Network Facilitator Organisation, NFO — currently REC / IES Secretariat under the

`indiaenergystack.in` namespace). To join, the NFO writes a **reference entry** pointing at your subscriber record; your identity stays self-owned, nothing is copied.

This section describes the documented application process. Whether the Secretariat and these network registries are currently staffed and reachable is not something this repository can verify on its own — see where things stand on the Getting Started page before treating it as a live pipeline.

The networks you can apply to:

Network registry	Env	Purpose
`ies-data-sharing-network` / `test-…`	prod / test	Energy data sharing (DISCOM <-> DISCOM, regulators, aggregators)
`ies-p2p-trading-network` / `test-…`	prod / test	P2P energy trading between consumers, prosumers, aggregators
`ies-der-integration-network` / `test-…`	prod / test	DER (rooftop solar, BESS, EV) integration with DISCOMs

Apply by email to the IES Secretariat — IES.Secretariat@fsrglobal.org (alternate: ies@recindia.com; web: ies.fsrglobal.org) — with:

- Your short identifier (`<discom-short-code>` for DISCOMs, FQDN for other participants).
- Your verified DeDi namespace (e.g. `discom`, `np.example.com`).
- The DeDi lookup URL of either your **whole subscriber registry** (typically `subscribers-prod`) or a **single record** inside it.
- Whether that URL is a `Registry` or a `Record` (one of those two values).
- Which network(s) you want to join, including the `test-` variant if you want sandbox first (recommended — you must certify on test before prod; see Conformance).
- Role: `BAP`, `BPP`, or both.
- Service areas / countries (`IND` for India).
- A point of contact (name, email, phone).

Before approving, the Secretariat validates that your namespace is domain-verified, your callback URL is reachable, your signing public key is present and valid, and your declared role matches the network's expectations. Once the reference entry is written, your record is queryable as in-network by every counterparty — no separate bilateral handshake.

Confirm the reference landed. Re-run the same lookup from §1.6 and check the `network_memberships` array — it should now list the network(s) you were added to:

```
curl -s
 ↪  "https://fabric.nfh.global/registry/dedi/lookup/<your-subscriber-id>/subscribers.beckn.one/<your_record_id>"
 ↪  | jq '.data.network_memberships'
```

Expected: the parent network IDs appear, e.g. `["indiaenergystack.in/test-ies-data-sharing-network"]`. Empty or missing means the NFO hasn't written the reference yet (or wrote it against a different record) — this is exactly the value your ONIX checks against `allowedNetworkIDs` in Setting up Discover & Exchange §2.3.

Membership in the test network does **not** imply membership in prod; each is referenced separately.

1.8 — *(NFOs)* Run your own Beckn network

Prerequisite: §1.1–1.4 complete for your own organisation.

If you operate a network rather than joining one, create a registry under your namespace with the built-in **`beckn_subscriber_reference`** tag — one registry per network you facilitate (typically separate `test-` and prod variants). The `network_id` becomes `<your-domain>/<registry-name>`. You curate membership by writing reference records that point at participant-owned subscriber records — you never copy participant data.

The NFO operational details (network manifest, policies, key rotation) are in the NFH docs: setting up the network environment and configuring network policies.

Troubleshooting

Symptom	Likely cause	Action
`curl` of `did.json` fails or redirects	File not at the exact path, or a `www.`/HTTPS redirect in front of it	The DID document URL must return the JSON directly, status 200, no redirect
DNS TXT verification stuck on "pending" for > 1 hour	Propagation delay (normal up to 48 h) or wrong record in zone	Check with `dig +short TXT <your-domain>`; ensure the record matches what DeDi issued
`404` on `GET /dedi/lookup/<namespace>/<registry>/<record-id>`	Wrong record-id (case-sensitive), or the record was never published (left as `draft`)	Verify in publish.dedi.global; re-publish with the correct id
`403 / signature invalid` on a DeDi write	Namespace controller key changed, or wrong API key	Confirm the API key belongs to the account controlling the namespace
Beckn counterparty cannot find your subscriber	Network reference not yet written (§1.7), or you're looking at the wrong environment	Verify the lookup URL the NFO wrote; check that `test-` vs prod matches the network you're targeting

Checklist

Everyone:

- [ ] Domain picked (typically a dedicated subdomain like `ies.<discom>.in`); `OPENCRED_ISSUER_DOMAIN` exported
- [ ] EC P-256 signing keypair generated, stored in KMS / HSM where available
- [ ] `did.json` generated and published at `https://<domain>/.well-known/did.json`
- [ ] `curl` of the DID document from outside returns your DID
- [ ] DeDi namespace claimed at `publish.dedi.global` and verified via DNS TXT (green **verified** label; listed on `explore.dedi.global`)
- [ ] DeDi API key created (avatar menu -> **Manage API key**) and stored alongside your other secrets

Beckn participants additionally:

- [ ] Ed25519 keypair generated (`go run install/generate-ed25519-keys.go`); private key in secret manager
- [ ] `subscribers-test` / `subscribers-prod` registries created; subscriber record(s) published with public key, callback URL, role
- [ ] Fabric lookup URL returns your record
- [ ] IES Secretariat emailed; reference entry written into the network registry you applied for

When your role's boxes are ticked, registration is done. Beckn participants: continue to **Setting up Discover & Exchange**. Credential issuers: continue to **Issuing Credentials**.

Register & Identifiers in depth

> **The first of the three IES steps — Register, Discover, Exchange.** Verifiable digital identity. Every participant gets a digital identity and is listed in a shared directory, and every meter, asset, connection and dataset gets a stable identifier under the same root. *Done once.*

Register is the foundation. Before any two systems can exchange data — and before any third party can trust a credential — both sides must be able to **name** each other and **prove** they are who they claim to be. IES does this with two cooperating pieces:

Piece	What it is	Standard
Identifier	A cryptographic name for an organisation, a regulator, a meter, a consumer, an asset, a credential or a dataset.	W3C Decentralised Identifiers (DIDs)
Directory	A public registry that lists records that go with those identifiers — signing keys, callback URLs, revocation status, network membership.	DeDi (Decentralised Directory Protocol)

Anyone can resolve an IES identifier in milliseconds, with no callback to the publisher, and check that a signature matches the registered key.

Hands-on setup: Setting up Register — domain, keys, `did.json`, DeDi namespace, and (for Beckn participants) subscriber records, as numbered copy-pasteable steps.

Why a verifiable identity, not a username

When a DISCOM hands a consumer a digital electricity credential, or shares meter data with a regulator, the recipient needs to answer one question on their own: *"Is this really from the DISCOM?"* If they have to call you, email your IT team, or trust a screenshot, the system does not scale and it is not really verifiable.

A username on a central platform is only as trustworthy as the platform — and IES has no central platform. Instead, every IES participant publishes a small JSON file on a web address it controls, listing the public key it signs things with. Anyone — a wallet, another DISCOM, a regulator, a bank — can fetch that file over plain HTTPS and check signatures on their own. No central authority, no API key, no special middleware. You need two things: **a domain you own** and **a key your IT team can generate**.

The same model covers the organisation (`did:web` on its domain), the consumer (`did:key` in a wallet), the asset (a `did:web` path that reuses the existing meter serial), and the dataset. One method family, one resolution flow, every actor.

Two identities you'll set up (and why)

An organisation on IES needs up to **two** identifier setups. They exist for different reasons, use **different keys** generated by different procedures, live in different registries, and serve different verifiers. They share only the organisational root — your domain.

	(a) Org identity	(b) Beckn network identity
Proves	*"This credential / payload was issued by us"* — content-level trust	*"This Beckn message was sent by us, and we belong to this network"* — transport-level trust
Identifier	`did:web:<your-domain>`	Your `subscriber_id` (your `did:web` — the same string as the org identity) in a DeDi subscriber record
Key	EC P-256 (credential signing)	Ed25519 (Beckn message signing)
Published in	`did.json` at `https://<your-domain>/.well-known/did.json`	Your DeDi namespace (`beckn_subscriber` registry), plus an NFO-side network reference
Verified by	Credential verifiers — banks, regulators, marketplaces, wallets	Other Beckn nodes on the network
Who needs it	Anyone issuing credentials or signing payloads	Anyone joining a Beckn network (data exchange, P2P trading)

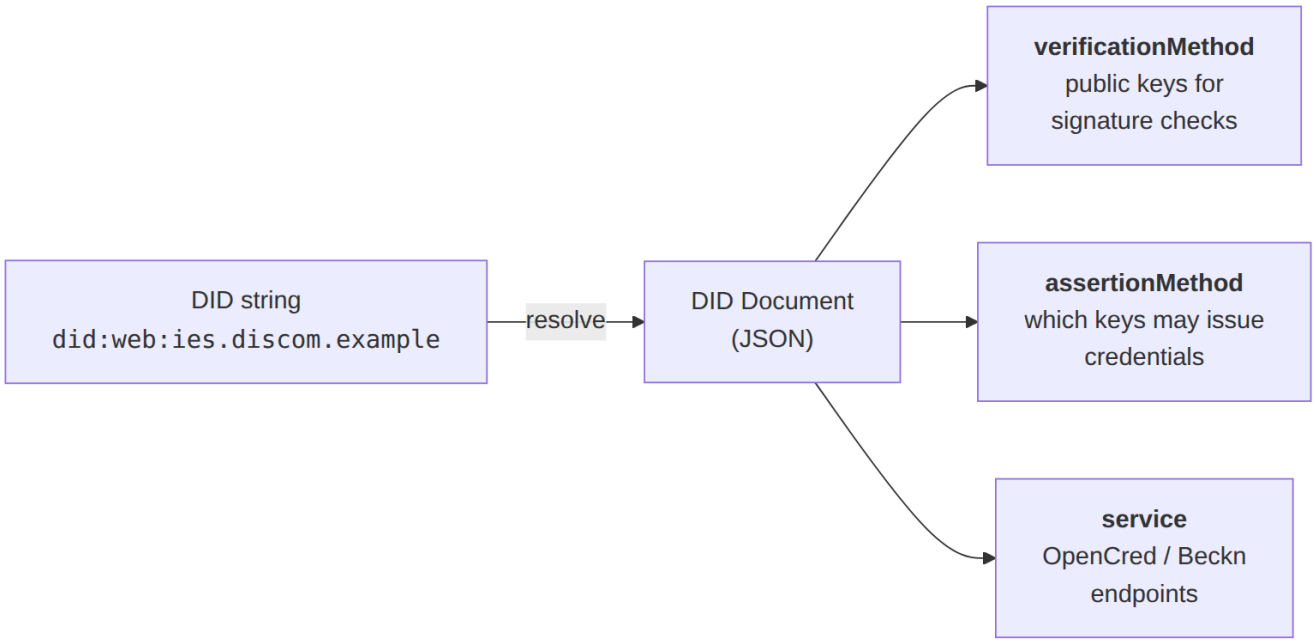
The (a) identity is the issuer string on every credential you sign, and the root that all the IDs you reference inside payloads — meter DIDs, transformer DIDs, connection DIDs — extend by adding path segments. Verifiers resolve only this one DID to check your signature; the asset DIDs ride inside the signed payload as stable references. **You do not need to be listed in any IES-side registry to issue credentials** — verifiers fetch your `did.json` directly; that is the only mandatory leg of the trust chain. If a regulator can vouch for your licence, cite them in the optional `issuer.idRef` and verifiers gain a second, licence-anchored leg.

The (b) identity exists because Beckn is a **trust-bounded network**: a Network Facilitator Organisation (NFO) curates who is on the network, and counterparties verify every message against that membership boundary — your own subscriber record (callback URL, role, Ed25519 public key) plus the NFO's reference entry that marks you in-network.

Separate keys mean a compromise of one identity does not automatically compromise the other.

The DID methods IES uses

A DID string by itself is just a name. The useful part is the **DID document** it resolves to — a small JSON object carrying the public keys a verifier needs, and optionally the service endpoints where the network can reach you.



IES uses three standard W3C DID methods, all listed in the W3C DID Spec Registries, so any standards-compliant verifier already knows how to resolve them. **There is no separate `did:dedi` method** — where DeDi appears, it is a key-discovery and registry layer over `did:web`, not a new method.

Subject	Method	Why
DISCOM / regulator / AMISP (Advanced Metering Infrastructure Service Provider) — any issuer	`did:web`	Trust roots in the domain and its existing TLS certificate; key rotation is one file replace; no new infrastructure beyond a static-file host
Consumer (holder)	`did:key` (or `did:jwk`)	Wallet-generated, no domain needed, verifies offline — the public key *is* the identifier

`did:web` is a URL in disguise: `did:web:ies.discom.example` resolves to `https://ies.discom.example/.well-known/did.json`, and path segments encode with colons:

DID string	DID document URL
`did:web:ies.discom.example`	`https://ies.discom.example/.well-known/did.json`
`did:web:discom.example:ies`	`https://discom.example/ies/did.json`
`did:web:discom.example:ies:issuer`	`https://discom.example/ies/issuer/did.json`
`did:web:discom.example%3A8443`	`https://discom.example:8443/.well-known/did.json` (port encoded as `%3A`)

Its main limitation: domain hijack would mean issuer-key hijack. Mitigations layer on top — the regulator's `issuer.idRef` licensing assertion on credentials (an attacker cannot forge the regulator's vouching record), and, on the Beckn side, the NFO's curated network registry. In addition to self-hosting `did.json`, an issuer can publish its key (and optionally a frozen `did.json` snapshot) into DeDi's key registry under its verified namespace — a second discovery path for the same key, useful as a fallback when the domain is unreachable. The method is still `did:web`.

`did:key` encodes the public key in the DID string itself — no network call, no website. Exactly right for consumers, who don't own domains; also handy for dev/test before a real subdomain exists. The price: the key cannot rotate (rotating means a new DID), so it is inappropriate for long-lived institutional issuers. **`did:jwk`** is the same idea with JWK encoding — IES accepts it for holders; default to `did:key` unless you have a reason.

Identifier vs. record Think of a DID like a **vehicle's licence plate**. The plate stays the same for the life of the registration; the RTO record it resolves to — owner, insurance, address — changes over time. A cop reads the plate and queries the RTO for the *current* record; they never try to guess the owner from the digits. Likewise:

- **Don't parse a DID for business logic.** Resolve it and read the record's fields. Code that routes on substrings of a DID breaks the day a character needs percent-encoding or an asset hierarchy is restructured.
- **Records update without re-issuing identifiers.** Key rotation, meter replacement, address correction — the record changes, the identifier stays, and everything ever signed against that identifier keeps verifying (verifiers fetch the current document). DeDi additionally keeps full version history, so `?as_on=<date>` answers "what was the public key when this was signed?".

Identifier patterns

Internal numbering — CIS account numbers, meter SLNOs, SAP codes — is preserved verbatim as the tail of the DID. Nothing renames.

Subject	Pattern	Example
Your organisation (issuer)	`did:web:<domain>`	`did:web:ies.discom.example`
A regulator	`did:web:<their-domain>`	`did:web:ies.serc.example`
A consumer (holder, optional)	`did:key:…` (wallet-generated) or `tel:+91…` (RFC 3966)	`did:key:z6MkjVQ8r4f3rPuY…`
A consumer (DISCOM-assigned stable DID, when no wallet)	`did:web:<domain>:consumers:<consumer-number>`	`did:web:ies.discom.example:consumers:DISCOM-2025-00987654`
The consumer's CIS account number	Plain string, kept verbatim in the credential	`DISCOM-2025-00987654`
Meter	`did:web:<domain>:assets:meter:<slno>`	`did:web:ies.discom.example:assets:meter:MET-IMPORT-001`
Other asset (transformer, feeder, substation, solar-plant, wind-farm, bess, ev-charger)	`did:web:<domain>:assets:<class>:<id>` (kebab-case class)	`did:web:ies.discom.example:assets:feeder:FDR-11KV-NDL-072`
Service connection	`did:web:<domain>:connections:<id>`	`did:web:ies.discom.example:connections:CONN-2025-001234567`
Dataset (Beckn `DatasetItem`)	`did:web:<domain>:datasets:<class>:<id>`	`did:web:ies.discom.example:datasets:meter-telemetry:2026-01`

Three notes that head off most confusion:

- **You do not assign wallet DIDs.** The consumer's wallet (or DigiLocker) generates the `did:key` and sends it to you; you verify the consumer controls it, then include it verbatim. Binding patterns and the identity-proofing that must precede them: Issuing Credentials — Holder binding.
- **Asset DIDs are stable identifiers, not necessarily resolvable documents.** Trust on an asset DID inside a credential comes from the *issuer's* signature; the asset DID rides inside the signed payload as a stable reference. Three hosting patterns exist, in increasing effort: **pragmatic** (host only your top-level `did.json` — right for most internal assets), **programmatic** (one small service synthesises DID documents for any `assets/<class>/<id>` path — strict `did:web` compliance without per-asset files), **per-asset documents** (for high-value public assets other networks resolve independently). Start pragmatic.
- **Replace any characters outside `[A-Za-z0-9._-]`** in path segments with `%xx` percent-encoding.

Where each ID goes in a credential One filled-in ElectricityCredential v1.2 showing every identifier in place:

```
{
  "@context": [
    "https://www.w3.org/ns/credentials/v2",

      ↪  "https://india-energy-stack.github.io/ies-accelerator/schemas/ElectricityCredential/v1.2/context.jsonld"
  ],
  "id": "urn:uuid:b2c3d4e5-0000-0000-0000-aabbccdd0001",
  "type": ["VerifiableCredential", "ElectricityCredential"],

  "issuer": {
    "id":   "did:web:ies.discom.example",
    "name": "Example State Distribution Company Limited",
    "idRef": {
      "_comment":  "optional — include only when citing a regulator",
      "issuedBy":  "did:web:ies.serc.example",
      "subjectId": "serc.example:DISCOM-REG-0042"
    }
  },

  "validFrom":  "2025-03-01T00:00:00+05:30",
  "validUntil": "2026-03-01T00:00:00+05:30",

  "credentialSubject": {
    "customerProfile": {
      "customerNumber": "DISCOM-2025-00987654",
      "energyResources": [{
        "id":   "did:web:ies.discom.example:assets:meter:MET-IMPORT-001",
        "type": "METER",
        "attributes": {"meterCapability": "AMI", "energyDirection": "Forward"}
      }]
    },
    "customerDetails": {
      "fullName": "Arjun Mehra"
    }
  },

  "proof": {
    "type": "JsonWebSignature2020",
    "verificationMethod": "did:web:ies.discom.example#key-0",
    "proofPurpose": "assertionMethod",
    "jws": "eyJhbGciOiJFUzI1NiJ9..UAF...g"
  }
}
```

Field	What it carries	Set by
`issuer.id`	Your `did:web`	You
`issuer.idRef` *(optional)*	A pointer the regulator gave you confirming you are a licensed DISCOM in their area. Optional per both the v1.2 schema and W3C VC 2.0.	Regulator + you
`customerProfile.customerNumber`	Your existing CIS number, unchanged	You
`energyResources[].id`	A `did:web` per meter / asset, built from your domain plus a path segment	You
`proof.verificationMethod`	A pointer back into your `did.json` saying which key did the signing	Your signing pipeline
`credentialSubject.id` *(optional)*	A holder identifier (wallet `did:key` or `tel:` URI) — set when you want presentation-time proof that the presenter is the legitimate subject	You, after verifying the consumer controls it

Notice what is *not* required: no central registry of consumers, no national ID, no identifier system competing with your CIS. Everything cross-references your DID document plus (optionally) the regulator's — static files on websites you each control.

The directory: DeDi

DeDi is the open registry runtime IES uses to anchor namespaces, publish subscriber records, and check credential revocation. When two organisations exchange data without a central middleman, the receiver needs to answer three questions — and DeDi answers all three through cryptography rather than calls to the publisher:

Question	What DeDi gives you
Integrity — has this data changed since publication?	Every record carries a BLAKE2 H256 digest anchored on the CORD blockchain. Recompute, compare, done — tampering is detectable.
Validity — is this data still current?	Version history per record, optional `valid_till`, a `state` field (`live` / `inactive` / `draft`), and revocation registries with `as_on=<date>` time-travel.
Authenticity — is the publisher who they claim to be?	A namespace is bound to a domain via a DNS TXT-record proof. Only the namespace controller can write under it.

DeDi is open-source (Linux Foundation Decentralized Trust labs); the hosted runtime IES uses is dedi.global — you publish at publish.dedi.global, the public read API is `api.dedi.global`, and verified namespaces are browsable at explore.dedi.global. Upstream docs: docs.nfh.global.

The three-coordinate model

```
api.dedi.global
└── <namespace>              <- owned by one verified domain; only the controller can write
    └── <registry>           <- a typed collection of records (schema = built-in tag or custom)
        └── <record-id>      <- a specific record — the resolvable end-point
```

Namespace + registry + record-id uniquely address any record. The namespace is the unit of governance; the registry is the unit of schema; the record is the unit of data. Any record resolves over a public read API, and records

never disappear silently — every update creates a version (queryable at a point in time), and registries and records carry explicit `live` / `inactive` / `draft` states. The exact endpoints and how to publish are in Setting up Register.

The registries IES uses, by role

Role	Registries you operate	Registries operated for you
DISCOM / issuer running OpenCred *(the reference credential-issuance service)*	`vc-revocation-registry`, `opencred-key-registry`, `schema_registry`, `context_registry` — all four **auto-created by OpenCred on first boot** (tags `revocation`, `public_key`, custom ×2). That's everything credential issuance needs.	—
Beckn network participant (BAP / BPP, aggregator, AMISP, trading platform)	`subscribers-test` and `subscribers-prod` (tag `beckn_subscriber`) — your identity + Ed25519 key + callback URL per environment	The NFO writes a **reference entry** (tag `beckn_subscriber_reference`) marking you in-network
NFO (network operator)	One `beckn_subscriber_reference` registry per network you facilitate — reference records pointing at participant-owned subscriber records; nothing copied	—
Verifier / wallet	Nothing — read-only. Resolve the issuer's `did:web` for the key; check the issuer's `vc-revocation-registry` for revocation.	—

You do not create these four by hand — Issuing Credentials covers running OpenCred and confirming the registries appeared.

The IES networks today The IES network operator publishes its registries under the namespace `indiaenergystack.in` — anyone can list them through the public DeDi read API. They fall into two groups:

Group	Registries	Purpose
Beckn networks (NFO-curated, prod + `test-` variants)	`ies-data-sharing-network`, `ies-p2p-trading-network`, `ies-der-integration-network`	The membership boundary for each IES Beckn network — energy data sharing, P2P trading, DER integration
Reference allow-lists (industry coordination)	`ies-discoms-reference-registry`, `ies-regulators-reference-registry`, `ies-service-providers-reference-registry`	Curated lists of recognised DISCOMs, regulators (SERCs, CERC), and service providers, referenced into network registries to enforce a trust boundary

Membership in a test network does not imply membership in prod — each is referenced separately. Credential issuance requires **no entry in any of these** — they are the Beckn-side trust boundary only. How to apply for a listing (what to email the IES Secretariat, what they validate): **Setting up Register §1.7**.

Setup

Everything on this page turns into action in **Setting up Register**: domain -> keypair -> `did.json` -> DeDi namespace (§1.1–1.4 for everyone), then subscriber records and network references for Beckn participants (§1.5–1.7)

and network registries for NFOs (§1.8). From there: **Issuing Credentials** (B2C rail) or **Setting up Discover & Exchange** (B2B rail).

Where this fits

Step	Page
Step 1 — Register *(this page)*	—
Step 2 — Discover	Beckn-protocol interaction — look up, negotiate, contract
Step 3 — Exchange	Signed Beckn payload delivery — schemas + verifiable credentials

Setting up Discover & Exchange

> **The B2B rail.** Stand up the Beckn adapter — the ready-made "ONIX" reference software — run a local end-to-end exchange, then swap in your real identity and go live on the IES network. Structured datasets move between registered organisations over a trust-bounded open network. About 1–2 days.

Registering on Beckn and being discoverable — your subscriber record, the network reference — is already done in **Setting up Register §1.5–1.7**. This page is what comes after: standing up the adapter that actually runs the exchange. The concepts — why Beckn, what the network provides, the protocol lifecycle — are in **Discover in depth** and **Exchange in depth**; the wire-level reference (message envelope, pagination, architecture) is in the appendices below.

> **About the walkthrough.** The commands use the DEG Data Exchange devkit — a ready-to-run Docker stack that bundles the **ONIX** Beckn protocol adapter (signing, verification, registry lookup), a sandbox BAP/BPP pair, and a Caddy router. ONIX is the recommended adapter for IES; if you already run one, swap it in — the wire format and registry contracts are the same.

What you'll have at the end

- A running ONIX container that speaks Beckn (`confirm` / `status` / `discover` / `on_*`).
- A verified local `confirm` -> `on_confirm` round-trip, then the same over the public internet.
- ONIX configured with your own subscriber identity and network boundary (`allowedNetworkIDs`).
- A clear picture of the two places your own application plugs in.

After this you can issue and respond to requests on the IES network. You still need **Build your Internal-facing Adapter** before you can return real data from your internal systems.

Before you start

From **Setting up Register** you need, for the real-network sections (§2.3 onward; the local sandbox §2.1–3.2 needs none of it):

- A verified DeDi namespace (§1.4).
- Your Ed25519 keypair (§1.5), published subscriber record with its **record ID** (§1.6), and the IES network reference written by the Secretariat (§1.7).

On this machine:

- **Docker 24+**, **Docker Compose**, **Git**, **Python 3**, and **Postman** *(or `curl`)*. ~2 GB free disk.

Domains to whitelist — if your organisation restricts outbound traffic, allow these before starting:

Purpose	Domain
Devkit and related repos	`https://github.com/beckn`
Discovery service (DS) — where ONIX routes `discover`	`https://34.93.165.42.sslip.io`
Catalog service (CS) + DeDi registry lookups	`https://fabric.nfh.global`
DeDi publishing	`https://publish.dedi.global`
Schema contexts	`https://schema.nfh.global`, `https://schema.nfh.global`, `https://raw.githubusercontent.com`
IES docs and hosted schemas	`https://india-energy-stack.gitbook.io/docs`, `https://india-energy-stack.github.io`

The devkit ships pre-configured sandbox identities (`bap.example.com` / `bpp.example.com`) — no real credentials needed for the local walkthrough.

2.1 — Deploy the local sandbox

```
git clone https://github.com/beckn/DEG.git
cd DEG/devkits/data-exchange/install
docker compose up -d
```

This brings up two ONIX adapters (`onix-bap`, `onix-bpp`), two sandbox stand-in apps (`sandbox-bap`, `sandbox-bpp`), a Caddy router, and a Redis per side — pre-wired with placeholder identities so you can test end-to-end on your laptop before plugging in your own. What each container does is in Appendix C — Architecture.

Verify the services:

```
docker compose ps
curl -s http://localhost:8081/health
curl -s http://localhost:8082/health
curl -s http://localhost:9000/
```

Expected: `docker compose ps` shows all 7 containers running (redis/sandbox: healthy); the two `/health` endpoints return `{"status":"ok"}` (onix-bap on 8081, onix-bpp on 8082); port 9000 returns `beckn-router ok`.

If checks fail, give ONIX ~15 s after `docker compose up` to initialise, then re-curl. Persistent `connection refused` usually means a container isn't running — `docker compose logs onix-bap` / `onix-bpp` shows why.

2.2 — Run your first exchange

Import the Postman collection for your use case. Each use case ships matching BAP and BPP collections; import the **BAP** one:

```
devkits/data-exchange/uc1-meter-data/postman/data-exchange-uc1-meter-data.BAP-DEG.postman_collection.json
devkits/data-exchange/uc2-regulatory-data/postman/
↪ data-exchange-uc2-regulatory-data.BAP-DEG.postman_collection.json
devkits/data-exchange/uc3-tariff-policy/postman/
↪ data-exchange-uc3-tariff-policy.BAP-DEG.postman_collection.json
```

Defaults are correct for local runs — don't change `beckn_adapter_url` (`http://localhost:8081/bap/caller`) or the `bap_host_root` / `bpp_host_root` docker hostnames. The two layers of URL are explained in Appendix E — Hostnames.

Send `confirm`. Open the `confirm` request in Postman and click **Send**.

Expected: the synchronous response is the **ACK envelope** — `{"message":{"status":"ACK","messageId":<id>}}` — meaning *"accepted, async callback pending"*. The ACK originates at the provider's webhook and cascades back through the adapters to Postman.

Watch `on_confirm` land. Tail the BAP sandbox logs:

```
docker logs sandbox-bap 2>&1 | grep on_confirm | tail -5
```

Expected: an `on_confirm` entry. Its body carries the dataset (inline) or a handle pointing to a later `on_status`. Either way, seeing `on_confirm` in `sandbox-bap` is your end-to-end signal.

If nothing arrives: (a) check `bap_host_root` / `bpp_host_root` are still the docker hostnames, (b) `docker logs onix-bap | grep -i error` and the same for `onix-bpp`, (c) host-alias DNS — see beckn/DEG#319.

BPP path. Same sanity check in reverse — import the **BPP** collection, trigger `on_confirm`, and tail `docker logs sandbox-bpp 2>&1 | grep -E 'confirm|status' | tail -5`. To replace the sandbox with your own provider, see §2.6.

Stop the stack when done:

```
docker compose down
```

2.3 — Swap in your real identity

Prerequisite: Setting up Register §1.5–1.7 — your Ed25519 keypair, subscriber record + record ID, and network reference.

The walkthrough so far ran on the shipped sandbox identities. To join the IES networks, replace them with yours in `config/local-simple-bap.yaml` (and the matching `local-simple-bpp.yaml`):

```yaml
modules:
  - name: bapTxnReceiver
    handler:
      plugins:
        registry:
          id: dediregistry
          config:
            url: "https://fabric.nfh.global/registry/dedi"
            allowedNetworkIDs: "indiaenergystack.in/test-ies-data-sharing-network"
            timeout: 30
            retry_max: 3
        keyManager:
          config:
            networkParticipant: your.subscriber.id
            keyId: <recordId-from-DeDi>
            signingPrivateKey: <base64-ed25519-private>
            signingPublicKey:  <base64-ed25519-public>
```

Value from Setting up Register	ONIX YAML path
Your `subscriber_id` (§1.6)	`plugins.keyManager.config.networkParticipant`
Your record ID (§1.6)	`plugins.keyManager.config.keyId`
Your Ed25519 **private** key (§1.5)	`plugins.keyManager.config.signingPrivateKey` *(Base64, 32-byte seed)*
Your Ed25519 **public** key (§1.5)	`plugins.keyManager.config.signingPublicKey` *(matches what you published in DeDi)*
Network you were referenced into (§1.7)	`plugins.registry.config.allowedNetworkIDs` *(comma-separated string)*

`allowedNetworkIDs` is your trust boundary. A subscriber record on DeDi carries a `network_memberships[]` field listing every network the subscriber has been referenced into. On each inbound message, ONIX's `dediregistry` plugin looks the sender up and intersects their memberships with this list — a sender outside it is rejected with `registry entry with subscriber_id '...' does not belong to any configured networks`. Rules of thumb:

- **Use full network IDs**, not bare registry names — `indiaenergystack.in/ies-data-sharing-network`, not `ies-data-sharing-network`.
- **Leaving `allowedNetworkIDs` unset accepts everything.** Useful for a discovery / catalogue node; never the right setting for a transactional BAP / BPP.

- **One ONIX instance per environment** — see §2.5. Configure the prod instance with prod-network IDs and the test instance with `test-` IDs only.
- The older config key `allowedParentNamespaces` is **deprecated**; the plugin errors until you migrate to `allowedNetworkIDs`.

After the config swap, restart the stack and re-import the same Postman collection — only the identity ONIX signs with and the network it claims change. Use the same network ID in every payload's `context.networkId`: ONIX rejects messages whose `networkId` isn't in its `allowedNetworkIDs`. Stay on the **test network** (`indiaenergystack.in/test-ies-data-sharing-network`) until certified — see Conformance.

2.4 — Go over the public internet (ngrok)

Once the local exchange is green, prove the same stack interoperates through a real public tunnel — typically with another participant's laptop or a cloud host:

1. Create a free ngrok account; copy the devkit's tunnel config and add your authtoken:

   ```
   cd DEG/devkits/data-exchange/install
   cp ngrok.yml.example ngrok.yml
   ngrok start --all --config ngrok.yml
   ```

   (Edit `ngrok.yml` to paste your authtoken before starting.)

2. In Postman, set `bap_host_root` / `bpp_host_root` to the HTTPS URL ngrok prints (e.g. `https://abc123.ngrok-free.dev`) — docker-only dummy hostnames aren't reachable from outside, so the payload URIs must carry your public tunnel URL.
3. Fire `confirm` again; watch the hops in the ngrok inspector at http://localhost:4040.
4. For two-party interop, each side runs its own tunnel and points `bpp_host_root` (or `bap_host_root`) at the *other side's* URL. Revert the host variables to `http://beckn-router:9000` to return to local-only.

Full recipe: devkit README — Over-the-internet notes.

2.5 — Two registries, two ONIX deployments

The recommended production pattern — keep test and prod cleanly separated:

1. **Run two ONIX deployments — test and prod — on separate hostnames** (e.g. `test.example-np.com` and `beckn.example-np.com`). Each has its own TLS cert, its own keypair, its own DeDi subscriber record under your namespace, in two separate subscriber registries (`subscribers-test` / `subscribers-prod` from Setting up Register §1.6) so they cannot be confused.
2. **`allowedNetworkIDs` is set narrowly on each side.** Test ONIX accepts only `indiaenergystack.in/test-ies-data-sharing-network`; prod ONIX accepts only `indiaenergystack.in/ies-data-sharing-network`. Cross-network traffic is rejected at the adapter — no stray test message can reach a prod counterparty.
3. **Phased onboarding:** start in test -> certification (Conformance) -> IES references your *prod* subscriber DeDi URL into the prod network -> from that moment your prod ONIX is allowed to transact with production NPs. Test stays useful for staging upgrades and certifying new use cases.

To run two devkit stacks side-by-side on the same host: copy `install/` to `install-prod/`, edit the published ports (`8081->8181`, `8082->8182`, `9000->9100`), adjust the `Caddyfile` listener, and use separate compose project names:

```
docker compose -p ies-test -f install/docker-compose.yml      up -d
docker compose -p ies-prod -f install-prod/docker-compose.yml up -d
```

In real production each stack would typically run on a separate host or VM with its own TLS termination. See devkit README — Hosting the site for the patterns.

2.6 — Make the sandbox your own

When you outgrow the sandbox, your app connects to ONIX in exactly two places — `bapUri` / `bppUri` are *not* one of them (those always point at ONIX receivers):

- **Inbound** — after verifying a message, ONIX forwards it to the **webhook URL set in the routing config** (`config/local-simple-routing-BAPReceiver.yaml` / `…-BPPReceiver.yaml`). Take over that URL and you've replaced the sandbox.
- **Outbound** — your app POSTs messages to ONIX's `caller` endpoint.

BAP. Stand up a service with a webhook endpoint for the `on_*` callbacks your flow needs, reachable from `onix-bap`. Change `target.url` in `local-simple-routing-BAPReceiver.yaml` from `http://sandbox-bap:3001/api/bap-webhook` to your service, restart `onix-bap`, then stop `sandbox-bap`.

BPP. Your provider receives requests (`confirm`, `status`, …) on the webhook set in `local-simple-routing-BPPReceiver.yaml` (devkit default: `http://sandbox-bpp:3002/api/webhook`). Acknowledge each with the ACK envelope, then respond asynchronously by POSTing the matching callback to `http://onix-bpp:8082/bpp/caller/on_<action>` — copy the wire shape from `uc*/examples/on-*-response*.json` and swap in your real data.

You can also build the bundled beckn/sandbox image yourself, modify the canned fixtures and handlers in `src/`, and use `sandbox-2.0:local` as a stepping stone before swapping in your own service.

This is where **Build your Internal-facing Adapter** picks up — the webhook service you wire in here is the Part-2 mapping you build there.

2.7 — Test the end-to-end loop

With your real identity configured, run a `discover` (or `confirm`) from your BAP to a sandbox BPP or a peer's BPP on the test network. Verify:

- Your message signature is accepted (your Ed25519 key resolves through DeDi).
- The counterparty's response signature verifies (their key resolves through DeDi).
- Discovery returns a catalogue / `on_confirm` returns data.
- ONIX's network-membership check passes (both subscribers are referenced into the same IES network registry).

If all four pass, this setup is done.

Checklist

- [ ] Local sandbox up; `confirm` -> `on_confirm` round-trip observed in `sandbox-bap` logs
- [ ] Same round-trip over a public tunnel (ngrok) with a second party
- [ ] ONIX configured with your subscriber ID, record ID, Ed25519 keypair, and `allowedNetworkIDs`
- [ ] End-to-end loop on the **test** network succeeds against a counterparty (all four §2.7 checks)
- [ ] Test/prod separation planned: two ONIX deployments, two subscriber registries, narrow `allowedNetworkIDs` each

When all five are ticked, move on to **Build your Internal-facing Adapter**.

Appendices

Wire-level reference for implementers. Skip on first read; come back when you're building the adapter or debugging.

Appendix A — Message envelope and correlation rules

Every Beckn message shares the same envelope (v2.0 wire format, camelCase):

```
{
  "context": {
    "networkId": "indiaenergystack.in/test-ies-data-sharing-network",
    "version": "2.0.0",
    "action": "confirm",
    "bapId": "bap.example.com",
    "bapUri": "https://bap.example.com/bap/receiver",
    "bppId": "bpp.example.com",
```

```
    "bppUri": "https://bpp.example.com/bpp/receiver",
    "transactionId": "b2c3d4e5-f6a7-8901-bcde-f12345678901",
    "messageId": "1e2f3a4b-5c6d-7890-4567-890123456789",
    "timestamp": "2026-04-01T09:25:00Z",
    "schemaContext": ["https://schema.nfh.global/DatasetItem/v1.1/context.jsonld"]
  },
  "message": { /* action-specific payload */ }
}
```

Two correlation rules from the Beckn v2.0 spec — every implementation honours them so participants and audit trails can stitch related messages together:

- **`transactionId` is constant** across every message in one exchange. From the first `discover` / `select` / `confirm` to the final `on_status` / `on_update`, the same UUID flows through.
- **`messageId` is shared** by a request and its paired `on_*` callback; each new request gets a new one. Treat the pair as one logical message with two hops.

Authoritative reference: beckn/protocol-specifications-v2 — `api/v2.0.0`.

`DatasetItem` and `accessMethod` The resource a data exchange agrees on is a **dataset**. **`DatasetItem`** (from DDM, Beckn's Decentralized Data Marketplace schema family) is the per-dataset record shape that rides inside `message.contract.commitments[].resources[]`, qualified by `resourceAttributes`:

```
"resources": [{
  "id": "ds-ami-meter-data-blr-zone-a-q1-2026",
  "descriptor": { "name": "IntelliGrid AMI Meter Data — Bengaluru Zone A — Q1 2026" },
  "resourceAttributes": {
    "@context": "https://schema.nfh.global/DatasetItem/v1.1/context.jsonld",
    "@type": "DatasetItem",
    "accessMethod": "INLINE",
    "dataPayload": { /* inline MeterData, ArrFiling, ... */ }
  }
}]
```

`accessMethod` values:

Mode	Description
`INLINE`	Embedded in the Beckn callback message (typically `on_confirm` or paged `on_status`)
`DOWNLOAD`	BPP provides a signed download URL + checksum
`MQTT` / `KAFKA` / `API` / `DATA_LAKE`	Streaming transports — connection credentials in `on_confirm.performance[].performanceAttributes`
`DATA_ENCLAVE`	Data accessible only within a secure compute environment
`OFF_CHANNEL`	Delivery through an agreed out-of-band channel

IES Data Exchange today primarily uses `INLINE` (often paged via Appendix B) — data arrives as part of the protocol message, making the exchange end-to-end verifiable.

Appendix B — Pagination: large datasets across multiple `status` / `on_status` messages

When a dataset is too large to fit in a single `on_confirm` (e.g. a quarter of AMI interval reads for a 500-meter feeder), the BAP and BPP exchange the payload across multiple `on_status` messages — one **page** per message. Two complementary patterns are in active use in the devkit:

BAP-PULL — the BAP drives the cadence

1. **First page.** The BAP sends `status` **without** a `pageCursor` (or omits the `pagination` tag entirely). The BPP treats a missing cursor as page 0.

2. **Subsequent pages.** The BPP's `on_status` carries `pageInfo.nextCursor` (and `pageInfo.isLast: false`). The BAP issues a new `status` with that cursor.
3. **Done.** The BAP keeps pulling until `pageInfo.nextCursor` is null **and** `pageInfo.isLast` is `true`. It then assembles all `dataPayload` slices and runs downstream processing.

Where the cursor lives on the request. Beckn v2 `Contract` is a closed schema and cannot host tags directly; `context.tags` is reserved for routing / transaction state. The pagination cursor therefore rides on `message.tags`:

```
{
  "context": {"action": "status", "transactionId": "...", "messageId": "...", "timestamp": "..."},
  "message": {
    "tags": [{
      "descriptor": {"code": "pagination"},
      "list": [
        {"descriptor": {"code": "pageCursor"},   "value": "ami-meter-blr-zone-a-q1-p2"},
        {"descriptor": {"code": "collectionId"}, "value": "ami-meter-blr-zone-a-q1-2026"}
      ]
    }],
    "contract": { "id": "<contract-id>" }
  }
}
```

Where `pageInfo` lives on the response. The BPP's `on_status` carries the slice in `message.contract.performance[].performanceAttributes`, alongside a `pageInfo` block:

```
"performanceAttributes": {
  "@context":  "https://schema.nfh.global/DatasetItem/v1.1/context.jsonld",
  "@type":     "DatasetItem",
  "dataPayload": [ /* this page's slice of CustomerProfile / IntervalProfile entries */ ],
  "pageInfo": {
    "@type":        "PageInfo",
    "sequence":     1,
    "pageSize":     167,
    "total":        500,
    "isLast":       false,
    "cursor":       "ami-meter-blr-zone-a-q1-p2",
    "nextCursor":   "ami-meter-blr-zone-a-q1-p3",
    "collectionId": "ami-meter-blr-zone-a-q1-2026"
  }
}
```

The cursor is **opaque** to the BAP — it's the BPP's internal identifier for the next slice. Echo it back unchanged.

BPP-PUSH — the BPP drives the cadence

1. **The BPP fires several back-to-back `on_status` messages**, each with its own slice of `dataPayload` and a `pageInfo` carrying `sequence`, `isLast`, and `collectionId`. The BAP accumulates entries across all messages with the same `(transactionId, collectionId)`.
2. **Done.** The BAP only triggers downstream processing once it sees `pageInfo.isLast: true`.

`pageInfo.cursor` / `nextCursor` are unused in push mode (the BAP isn't requesting anything); `sequence` and `isLast` are what matter.

Worked examples in the devkit The on-the-wire payloads (with sample interval reads and the full `pageInfo` block) live next to each use case:

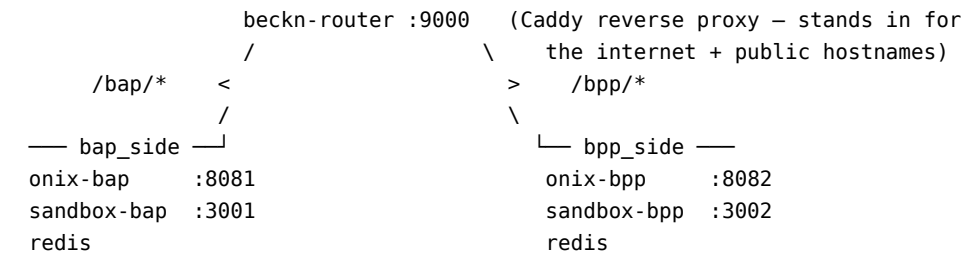
- BAP-PULL: `uc1-meter-data/.../status-request-paged-pull.json` + matching `on-status-response-paged-pull.json`.
- BPP-PUSH: `on-status-response-paged-push-p1.json`, `…-p2.json`, `…-p3.json` (with `isLast: true` on the last one).

Both patterns share the same `transactionId` across every message in the exchange — see Appendix A.

Appendix C — Architecture at a glance

Your application never speaks Beckn directly — it talks to **ONIX**, which exposes a `caller` endpoint (where your app hands it outbound messages to sign and dispatch) and a `receiver` endpoint (where the network delivers inbound messages, which ONIX verifies and forwards to your app's webhook). One caller / receiver pair per role — `/bap/caller`, `/bap/receiver`, `/bpp/caller`, `/bpp/receiver` — and one ONIX deployment can host any combination under one identity. **Roles are configuration, not separate software.**

The devkit simulates **two participants** (`bap.example.com` and `bpp.example.com`), so it runs two ONIX containers on mutually isolated docker networks:



Component	Role
`onix-bap` / `onix-bpp`	Beckn protocol adapter — signs, verifies, validates `dataPayload`, dispatches, forwards inbound to the app webhook
`sandbox-bap`	Consumer-side stand-in app — receives `on_*` callbacks on its webhook and logs them
`sandbox-bpp`	Provider-side stand-in app — receives requests on its webhook and auto-responds with example payloads via `/bpp/caller`
`beckn-router`	Plain Caddy reverse proxy — routes by path (`/bap/*` -> `onix-bap`, `/bpp/*` -> `onix-bpp`) and carries the dummy participant hostnames as docker aliases. **Not a Beckn entity** — deployment plumbing
`redis`	Per-side message cache used by ONIX

The four ONIX endpoints

Endpoint	Role	Direction	Who calls it
`/bap/caller`	BAP	outbound	**Your consumer app** hands ONIX a request (`confirm`, `discover`, …) to sign and dispatch
`/bap/receiver`	BAP	inbound	**The network** (the counterparty's ONIX) delivers `on_*` callbacks here; ONIX verifies and forwards them to your app's webhook
`/bpp/caller`	BPP	outbound	**Your provider app** hands ONIX an `on_*` response to sign and dispatch

Endpoint	Role	Direction	Who calls it
`/bpp/receiver`	BPP	inbound	**The network** delivers requests (`confirm`, `status`, …) here; ONIX verifies and forwards them to your provider's webhook

What happens after a `receiver` verifies a message is defined in the routing configs (`local-simple-routing-{BAPReceiver,BPPReceiver,BAPCaller,BPPCaller}.yaml`). The receiver routing config controls the webhook URL inbound messages are forwarded to — that's where you wire in your own app (§2.6).

Identity resolution When a message arrives, ONIX:

1. Reads the sender from the message context (`bapId` on a forward request, `bppId` on a callback).
2. Looks the sender up in the **DeDi registry**: `GET https://fabric.nfh.global/registry/dedi/lookup/<subscriber_id>/subscribers.beckn.one/<record_id>`. The response carries the sender's callback URL, signing public key, and network memberships.
3. Cross-checks those memberships against the local `allowedNetworkIDs` config (§2.3). A sender outside the boundary of trust is rejected.
4. Verifies the signature.

Two participants on different (or non-overlapping) networks cannot reach each other through their ONIX adapters.

Appendix D — Schema validation on the wire

The wire envelope accepts arbitrary JSON inside `dataPayload`; validation is **opt-in per object**, driven by JSON-LD self-description:

- **Payload declares `@context` + `@type` -> ONIX validates it.** ONIX fetches the OpenAPI 3.x **`attributes.yaml` that sits next to the `context.jsonld`** named in `@context` (same URL, different filename — every IES / Beckn schema family publishes the pair side by side), and validates the object against the `components.schemas` entry whose name matches `@type` exactly (`DatasetItem`, `MeterData`, `ArrFiling`, …). A failed validation rejects the message.
- **Payload omits `@context` -> ONIX passes it through.** Useful for prototyping a new dataset shape.

ONIX only fetches `@context` URLs from **allow-listed hosts** (default: `raw.githubusercontent.com`, `schema.nfh.global`). To validate payloads from your own schema repo, add the host to `extendedSchema` in your ONIX config.

Failure modes

- `no schema found for @type: XYZ` — the `attributes.yaml` doesn't define a schema with that name. Either `@type` is wrong, or the schema file at the resolved URL doesn't include it.
- Schema validation error — the object is missing a required field, has the wrong type, or otherwise doesn't match the OpenAPI definition. ONIX includes the JSON path of the failure in the error.

Publishing your own schema for ONIX to validate

1. Author an OpenAPI 3.x `attributes.yaml` with your schemas in `components.schemas`.
2. Author a JSON-LD `context.jsonld` mapping your field names to URIs.
3. Host both at a URL ONIX is allowed to reach — they must sit at the same path (`<base>/context.jsonld` and `<base>/attributes.yaml`).
4. In your payload, set `@context` to the context URL and `@type` to the matching schema name.

No code change in ONIX — the dispatch is purely URL-driven. The families under IES Schemas and beckn/DDM are working references for how to lay out the pair.

Appendix E — Hostnames: sandbox vs production

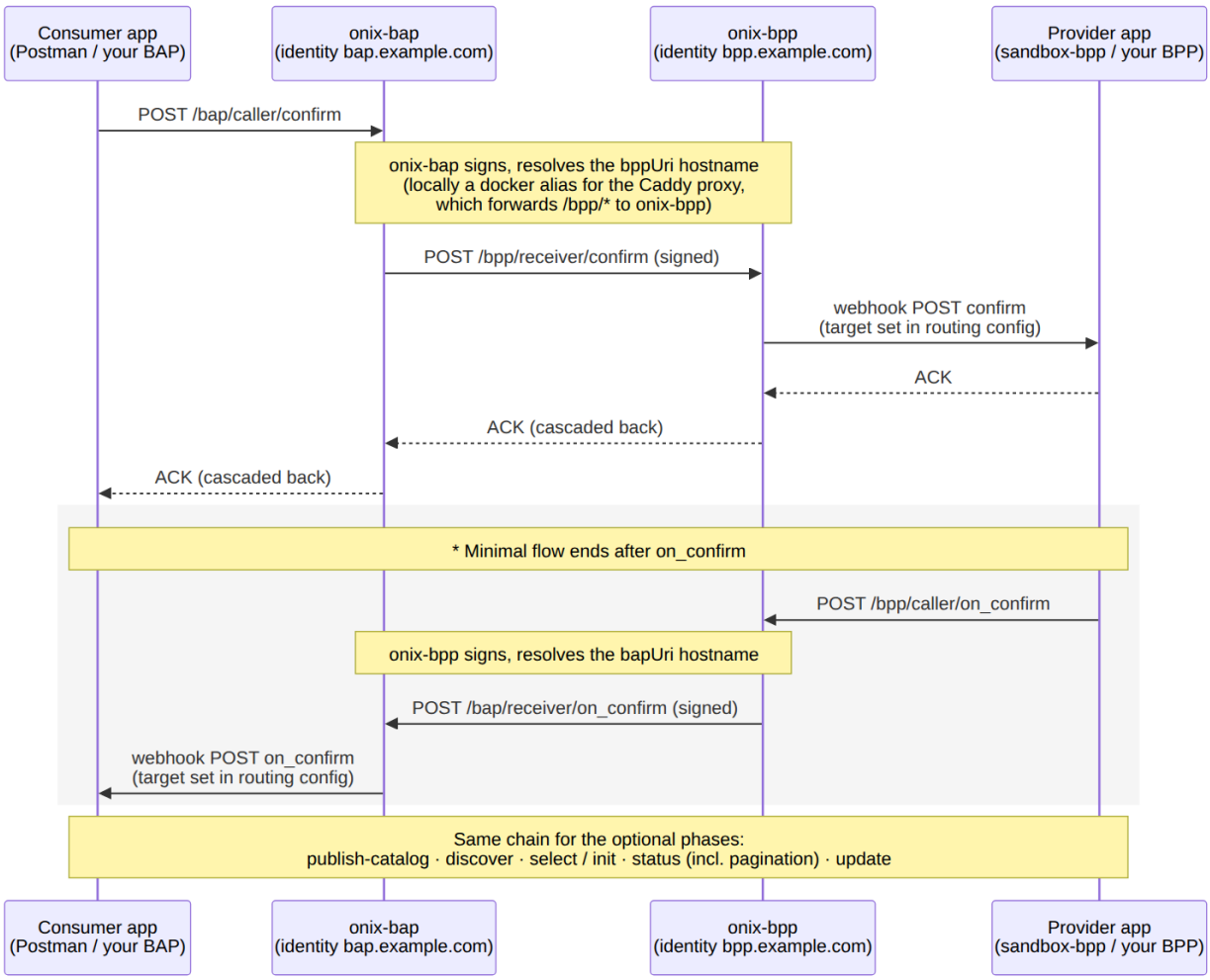
The hostnames in `bapUri` / `bppUri` represent **participant identities**. In production they are your real public URLs, published in your DeDi subscriber record. In the devkit they are **dummy aliases that only resolve inside the docker network**: compose attaches `bap.example.com` and `bpp.example.com` as aliases of `beckn-router`, so dispatch lands on the Caddy proxy, which path-routes to the right ONIX `receiver`.

That gives two kinds of URL at different layers:

Concern	Devkit sandbox	Real network
HTTP target — where your client (Postman, your app) POSTs `confirm`, `discover`, etc.	`http://localhost:8081/bap/caller` *(port-mapped to `onix-bap`)*	Your ONIX deployment URL behind TLS
HTTP target — where your provider POSTs `on_confirm` / `on_status`	`http://localhost:8082/bpp/caller` *(port-mapped to `onix-bpp`)*	Your ONIX deployment URL behind TLS
Payload hostnames — `bapUri` / `bppUri` in each message's `context`; resolved by the *sending* ONIX to reach the counterparty's `receiver`	`http://beckn-router:9000/{bap, bpp}/receiver` (or the docker aliases — both resolve to the proxy)	Your public callback URLs (matching what you published in your DeDi subscriber record)
`networkId`, `bapId` / `bppId`, `allowedNetworkIDs`	placeholder values shipped in `config/`	Your values (§2.3)

Your Postman client never connects to `beckn-router` — it POSTs to the port-mapped `caller` endpoints (`:8081` / `:8082`); the payload variables substitute the docker-resolvable hostnames into the message body for ONIX to use.

Appendix F — Generic Beckn flow (sequence diagram)



`beckn-router` is intentionally not a participant in this diagram — it's a path-only Caddy reverse proxy, not a Beckn entity. In production it's replaced by your own TLS-terminating reverse proxy.

References

- DEG Data Exchange devkit — the source for the walkthrough above (Postman collections, fixtures, configs)
- Beckn ONIX — the protocol adapter; `install/generate-ed25519-keys.go` for the Ed25519 keypair
- Beckn Protocol v2.0.0 spec
- Discover in depth — the concepts this page implements
- Smart Meter Data Exchange and P2P Energy Transaction — end-to-end use-case flows that exercise this protocol

Discover in depth

> **The second of the three IES steps — Register, Discover, Exchange.** Interaction protocol. Before every exchange, both systems look each other up, confirm the other is genuine, and agree on what will be exchanged and on what terms. *No bilateral arrangement is needed.*

Once participants are Registered (Step 1), they need a way to **find each other, negotiate terms, and produce a signed audit trail** of what was agreed — without anyone phoning anyone. IES uses the open **Beckn Protocol v2** for this — and the same signed Beckn channel then carries the Exchange payloads, so Beckn is the rail for both steps, with Discover as its look-up-and-contract stage.

Discover belongs to the **data exchange** capability — B2B exchange of structured datasets between registered organisations. The other IES capability, **Verifiable Credentials**, does not need this step: a credential is issued and verified against the issuer's published key, with no Beckn network involved.

Two rails, two kinds of trust

	B2B data exchange (this page + Exchange)	**B2C credentials** (Issuing Credentials)
What moves	Structured datasets — meter telemetry, regulatory filings, tariff data, trade offers	Signed attestations — a consumer's connection record, a meter digest
Between whom	Two **registered organisations** that know who they are transacting with	An issuer and **any third party** — a bank, a marketplace, a housing society — that the issuer has never met
Trust layer	**Bilateral, anchored in DeDi.** Each side resolves the other's subscriber record, verifies its message signature, and checks that both belong to the same curated network. The network operator (NFO) maintains that membership boundary.	**Unilateral, anchored in `did:web`.** The issuer signs once; any verifier, anywhere, fetches the issuer's published key over HTTPS and checks the signature — no membership, no prior relationship, no callback to the issuer.
Channel	Must ride a **trust-bounded open network** — IES uses Beckn Protocol v2 — because discovery, negotiation, consent and the signed audit trail are part of the exchange itself	**Any channel** — DigiLocker, a web portal, email, SMS, chat. The trust travels inside the credential, not the pipe.
Typical use cases	Smart Meter Data Exchange, Regulatory Filing, P2P Trading	Consumer Energy Passport, Consumer Meter Digest

The rest of this page covers how the B2B rail's two registered parties find each other, agree terms, and sign — the *Discover* step. What they actually exchange once agreed — the schemas and, where needed, verifiable credentials — is *Exchange*, covered on its own page.

Find, agree, and sign without a pre-negotiated integration

Energy data in India moves through bespoke point-to-point channels today — PDF filings, vendor-locked exports, one-off API integrations. A bespoke REST API per counterparty is exactly the n×m problem IES is solving. (Note: the *trust* here is still bilateral — each side verifies the other — but the wiring is not; you build to the protocol once, not to each partner.) Beckn is a single, open, **asynchronous and peer-to-peer** interaction protocol that any party builds to once and uses against every counterparty:

Concern	What Beckn provides
Discovery	Either party can publish a catalogue of what it offers (datasets, credentials, dispatch programmes) and other parties can query it.
Negotiation	A standard `select` / `init` round-trip lets the two sides agree on quantity, price, SLA, delivery method.

Concern	What Beckn provides
Commitment	A signed `confirm` / `on_confirm` round-trip **is the contract**. The signed envelope is the non-repudiable record of agreement.
Status and delivery	`status` / `on_status` carry asynchronous fulfilment — including payload delivery, pagination, settlement updates.
Consent and audit	Every message is signed; every leg leaves a tamper-evident receipt.

There is no central broker: the network operator curates the membership list, but every message flows directly between participants. The wire and the lifecycle are the same whether the payload is meter telemetry, a regulatory filing, a tariff order, or a peer-to-peer energy trade.

The lifecycle at a glance

Phase	BAP — the consumer side (Beckn Application Platform) calls	BPP — the provider side (Beckn Provider Platform) responds	When you need it
Discovery	`discover`	`on_discover`	Consumer doesn't yet know which provider to contract with
Negotiation	`select` / `init`	`on_select` / `on_init`	Terms need agreeing before commitment
Commitment	`confirm`	`on_confirm`	**The minimal flow** — payload can be delivered inline here
Delivery	`status`	`on_status`	Payload prepared asynchronously, or paged across messages
Post-fulfilment	`update`	`on_update`	Amendments, credential rotation

Minimal viable exchange: `confirm` -> `on_confirm`, with the dataset embedded in the callback. Everything else is optional; each use-case implementation guide lists which actions it actually exercises. Small datasets ride **inline** in the message (the IES default — end-to-end verifiable); large ones hand off to an established channel (signed URL, MQTT, Kafka, SFTP) agreed inside the same contract, so every exchange gets the same signed-audit story. Wire-level detail — message envelope, correlation rules, pagination, `accessMethod` values — is in the Setting up Discover & Exchange appendices.

The IES networks

IES currently operates three Beckn networks (each with a `test-` twin for onboarding and certification) under the `indiaenergystack.in` namespace: **data sharing**, **P2P trading**, and **DER integration**. Membership is per-network and per-environment; the adapter rejects messages from outside the configured boundary. The registry mechanics are in Register — The directory: DeDi; how to get listed is Setting up Register §1.7.

What you set up under Discover

For an IES participant, Discover means registering on Beckn and becoming findable — before anyone can exchange anything with you, they need to be able to look you up and verify who you are:

1. **Beckn signing keypair** — an Ed25519 keypair used to sign and verify messages (separate from the EC P-256 keypair used for credentials).

2. **Beckn subscriber record** — a small entry in your DeDi namespace declaring your callback URL, role (`BAP` / `BPP`), and Ed25519 message-signing public key. Other Beckn nodes look this up to verify your signatures.
3. **Network reference** — the IES network operator (acting as Network Facilitator Organisation, NFO) writes a *reference* into the network registry pointing at your subscriber record. This is the membership boundary.

These three are covered by **Setting up Register §1.5–1.7**. Standing up the adapter that actually runs an exchange against this registration — ONIX, the sandbox walkthrough, wire mechanics — is **Setting up Discover & Exchange**.

Where this fits

Step	Page
Step 1 — Register	Identity + directory
Step 2 — Discover *(this page)*	—
Step 3 — Exchange	Signed Beckn payload delivery — schemas + verifiable credentials

To register on Beckn hands-on: **Setting up Register §1.5–1.7**. To run the adapter and exchange data: **Setting up Discover & Exchange**.

Exchange in depth

> **The third of the three IES steps — Register, Discover, Exchange.** The schemas and verifiable credentials. Data moves using agreed field names and structure, following the public standard for that domain. Where the use case needs a durable record, a **verifiable credential** is issued that the holder keeps and can re-present anywhere.

Once two parties have Registered and Discovered each other (or, for credentials, once the issuer alone is registered), the data itself moves using agreed field names and structure — the **schemas**. Exchange has two parts:

Part	Purpose	IES choice
Schemas	The master vocabulary of IES — every domain object, what it is for, which use cases combine it, how it evolves and how new ones are proposed. The shape of each object is described by its **schema**: field names, types, units, optionality — one canonical schema per object.	IES-published JSON Schema + JSON-LD context, built on public standards (DLMS/COSEM, IEEE 2030.5, OpenADR, CIM, etc.)
Verifiable Credentials	Where the use case needs a durable record — a Passport, a Digest — a W3C Verifiable Credential is issued that the holder keeps in DigiLocker or an EntityLocker.	W3C VC Data Model 2.0; W3C DID Core

IES does not write new standards. It picks the right open standard for each domain — DLMS/COSEM for meter data, IEEE 2030.5 for solar and storage, OpenADR for demand response — and publishes a faithful schema on top. A schema is valid whether or not the payload ever travels over Beckn: the same `MeterData` object can ride a Beckn `on_confirm`, sit inside a signed `MeterDataCredential`, or be validated standalone.

Exchange happens over two kinds of rail, and a use case picks whichever fits:

- **B2B data exchange** — structured payloads moving between organisations, after the parties are **Registered** and have **Discovered** each other. IES uses the Beckn protocol here — the same signed channel carries discovery, consent, contract, payload delivery and the audit trail.

- **Credential flows** — issued with OpenCred (the reference credential-signing service) and verified offline against the issuer's published key, credentials stand on their own and **do not require a Beckn network**. Consumer-facing (B2C) delivery happens over DigiLocker, a web portal, or any channel the issuer already runs.

Don't hand-map schemas to use cases here

That lives in one place: the **IES Schemas** developer catalog (every schema, its purpose, current version, field reference) and the plain-language **Schemas Overview** pages (the *why* before the field-level reference). The wire envelope itself accepts any JSON payload; validation is opt-in per object, driven by the payload's own `@context` / `@type` declaration.

Verifiable Credentials

A subset of Exchange payloads are issued as W3C Verifiable Credentials — durable, holder-bound records the consumer or another stakeholder keeps independently of IES and can re-present anywhere. A credential is verified against the issuer's published key, so issuing, holding and verifying one involves no Beckn network at all. One credential, `MeterDataRequestCredential`, is the exception that rides *inside* a Beckn message (a seeker's proof-of-right-to-ask). The three credential schemas today:

Credential	What it carries
ElectricityCredential v1.2	Static facts of a consumer's connection — sanctioned load, tariff, assets behind the meter
MeterDataCredential v0.6	A signed envelope around a MeterData payload — consumer's own readings for a window
MeterDataRequestCredential v0.1	A signed proof-of-right-to-ask that a seeker presents to a meter-data provider

For the concept, trust model and issuance operations, see **Verifiable Credentials**; to issue them, **Issuing Credentials**.

What you set up under Exchange

For an IES participant, Exchange means standing up the adapter that runs the exchange and then wiring your own data into it:

1. **Deploy ONIX** (the reference Beckn protocol adapter) and run a sandbox `confirm -> on_confirm` round-trip end to end, then over the public internet — the Beckn stack that actually moves a payload once Discover's registration is in place.
2. **Write the Part-2 mapping** — the small adapter layer that translates between your internal systems and the IES schemas:
    1. Pick the use case you want to ship first.
    2. Map each IES field to the corresponding column / API field / file format in your internal systems (CIS, MDM, billing, DERMS, etc.).
    3. Hook the mapping into your ONIX as a BPP handler (or BAP callback), and/or into OpenCred as an issuance feed if your use case issues credentials.
    4. Validate the output against the canonical JSON Schema.

For the step-by-step setup, follow **Setting up Discover & Exchange** first, then **Build your Internal-facing Adapter**. If your use case issues credentials, also see **Issuing Credentials**.

The schema canonical references — JSON Schema, JSON-LD context, RDF vocabulary, example payloads — are in **IES Schemas**.

Detail pages

- **Verifiable Credentials** — issuance / verification / revocation operations using OpenCred; credential variants; trust model; DigiLocker delivery.
- **IES Schemas** — the developer catalog: every schema's purpose, versions, field reference and example payloads.
- **External Schemas** — DEG-published schemas (P2PTrade, DemandFlex*) referenced by IES use cases.

Where this fits

Step	Page
Step 1 — Register	Identity + directory
Step 2 — Discover	Beckn-protocol interaction
Step 3 — Exchange *(this page)*	Signed Beckn payload delivery — schemas + credentials

To set up the Part-2 mapping hands-on: **Build your Internal-facing Adapter**.

For how schemas evolve and how to propose a new one: **Propose a Schema**.

Issuing Credentials

> **The B2C rail — set up + operate.** Run the OpenCred signing container and issue, verify, and revoke W3C Verifiable Credentials. A credential is signed once by you and can then be trusted by *any* third party — a bank, a marketplace, a housing society — that resolves your `did:web`, and delivered over any channel (DigiLocker, web portal, email, SMS, chat). **No Beckn network is involved.** About half a day.

The concepts — what a credential is, the trust model, the credential variants and when to use each — are in **Verifiable Credentials in depth**. This page is the do-guide.

> **About the walkthrough.** The commands use **OpenCred**, the reference credential-signing service — W3C VC 2.0 compliant, with built-in DeDi integration. Any W3C-compliant signing pipeline that publishes the same `did.json` and VC-2.0 proofs is a drop-in replacement.

Before you start

From **Setting up Register** you need §1.1–1.4 complete:

- `OPENCRED_ISSUER_DOMAIN` exported, and `did.json` published and resolvable from outside (§1.1–1.3).
- The EC P-256 signing key at `~/opencred/keys/issuer-key.pem` (§1.2).
- A **verified DeDi namespace** and a **DeDi API key** (§1.4) — these enable revocation.

On this machine:

- **Docker 24+**, plus `curl`, `jq`, `openssl`, and ~2 GB free disk.
- *(Optional, recommended for licensed utilities)* your regulator's `did:web` and your licence identifier, to quote in `issuer.idRef`. Omit for pilots / non-regulated issuers.
- **A payload schema in mind.** Default: ElectricityCredential v1.2. For telemetry, MeterDataCredential v0.6.

> **No IES-side DISCOM-registry entry is required to issue credentials.** Verifiers fetch your `did.json` directly to check signatures. The IES network registries are the Beckn-side trust boundary — a separate concern (Setting up Register §1.7).

3.1 — Pull the OpenCred image

```
docker pull ghcr.io/nfh-trust-labs/opencred/opencred-server:latest
docker tag  ghcr.io/nfh-trust-labs/opencred/opencred-server:latest opencred:bootcamp
```

3.2 — Generate the OpenCred API token

This token is the only auth on OpenCred's HTTP API — whoever holds it can issue credentials in your name. Generate and save it:

```
export OPENCRED_API_KEY="$(openssl rand -base64 32)"
echo "Save this: $OPENCRED_API_KEY"
```

3.3 — Run OpenCred in `did:web` mode

First set the two DeDi variables (in addition to `OPENCRED_ISSUER_DOMAIN` from Setting up Register §1.1 — re-export it if this is a fresh shell):

```
export OPENCRED_DEDI_NAMESPACE="discom.example"
export OPENCRED_DEDI_API_KEY="paste-your-dedi-api-key-here"
```

- **`OPENCRED_DEDI_NAMESPACE`** — the **domain** your DeDi namespace is linked to and verified with (the same domain you domain-verified in Setting up Register §1.4) — *not* a free-text namespace name. Set it to that domain, so that `https://api.dedi.global/dedi/lookup/<namespace-domain>` resolves.
- **`OPENCRED_DEDI_API_KEY`** — the key you created in the DeDi UI at publish.dedi.global (avatar in the top-right corner -> **Manage API key**).

The `OPENCRED_DEDI_*` variables connect the container to DeDi at startup. That is what enables **revocation** (§3.8) and lets OpenCred auto-create the four registries it needs in your namespace on first boot.

Now start the container:

```
docker run -d \
  --name opencred \
  -p 3100:3100 \
  -e OPENCRED_API_KEY="$OPENCRED_API_KEY" \
  -e OPENCRED_KEY_PATH=/secrets/issuer-key.pem \
  -e OPENCRED_ISSUER_DID_METHOD=web \
  -e OPENCRED_ISSUER_DOMAIN="$OPENCRED_ISSUER_DOMAIN" \
  -e OPENCRED_DEDI_BASE_URL=https://api.dedi.global \
  -e OPENCRED_DEDI_AUTH_TYPE=api-key \
  -e OPENCRED_DEDI_API_KEY="$OPENCRED_DEDI_API_KEY" \
  -e OPENCRED_DEDI_NAMESPACE="$OPENCRED_DEDI_NAMESPACE" \
  -v "$HOME/opencred/keys/issuer-key.pem:/secrets/issuer-key.pem:ro" \
  --read-only --cap-drop ALL \
  opencred:bootcamp
```

Check it came up healthy:

```
curl -s http://localhost:3100/v1/health | jq
```

Expected: the JSON includes `"signingKeyLoaded": true`. If it is `false`, see Troubleshooting.

> **Want offline-verifiable identity instead?** Drop `OPENCRED_ISSUER_DID_METHOD` and `OPENCRED_ISSUER_DOMAIN` and OpenCred runs in `did:key` mode by default. The same key, the same API — only the `did:` string the container reports changes. This is the right choice for first-deploy testing, demos, and consumer wallets. You can also drop the four `OPENCRED_DEDI_*` variables while testing — the container still issues and verifies, but revocation stays disabled until you add them back.

3.4 — Confirm your DeDi namespace is live

Credential **revocation** rides on your DeDi namespace, so before issuing anything, confirm the namespace side is in order. Two checks, both in a browser:

1. **Is the namespace verified?** Open explore.dedi.global and search for your namespace. Only verified namespaces show up in explore results — if yours appears, it is verified. (In publish.dedi.global, where you log in and manage the namespace, verification shows as a green **verified** label.) Namespace not in explore? The DNS TXT record hasn't propagated or wasn't added; revisit Setting up Register §1.4.
2. **Did OpenCred create its registries?** Log in at publish.dedi.global and open your namespace. Can you see the four registries OpenCred auto-created on first boot — `vc-revocation-registry`, `opencred-key-registry`, `schema_registry`, `context_registry`? If they are there, revocation is wired up and you're ready to issue. If not, the container couldn't reach DeDi — check `docker logs opencred` and re-check the `OPENCRED_DEDI_*` values from §3.3 (a wrong API key or namespace name is the usual cause).

3.5 — Confirm the issuer DID OpenCred reports

```
export ISSUER_DID="$(curl -s http://localhost:3100/v1/keys \
  -H "Authorization: Bearer $OPENCRED_API_KEY" | jq -r '.keys[0].id | split("#")[0]')"
echo "$ISSUER_DID"
```

Expected: your DID, e.g. `did:web:ies.yourdiscom.in`. If you ran the container in `did:key` mode (for early dev), this prints `did:key:z…` instead — the rest of the flow is identical, only the issuer string changes.

3.6 — Issue your first credential

Default: **bearer-style** (no `credentialSubject.id`) — anyone holding the JSON is treated as the subject. This mirrors the OpenCred bootcamp. For consumer-facing flows where the presenter must be the legitimate subject, bind to a holder identifier — see Appendix — Holder binding.

```
curl -s http://localhost:3100/v1/credentials/issue \
  -H "Authorization: Bearer $OPENCRED_API_KEY" \
  -H "Content-Type: application/json" \
  -d "{
    \"schemaId\":    \"ies/electricity-credential/v1.2\",
    \"issuerDid\":   \"$ISSUER_DID\",
    \"proofFormat\": \"vc-jwt\",
    \"validFrom\":   \"2026-04-01T00:00:00+05:30\",
    \"validUntil\":  \"2031-04-01T00:00:00+05:30\",
    \"revocationRegistryUrl\":
↪   \"https://api.dedi.global/dedi/query/$OPENCRED_DEDI_NAMESPACE/vc-revocation-registry\",
    \"credentialSubject\": {
      \"customerProfile\": {
        \"customerNumber\": \"DISCOM-2025-00987654\",
        \"energyResources\": [{
          \"id\":   \"did:web:${OPENCRED_ISSUER_DOMAIN}:assets:meter:MET-IMPORT-001\",
          \"type\": \"METER\",
          \"attributes\": {\"meterCapability\": \"AMI\", \"energyDirection\": \"Forward\"}
        }],
        \"consumptionProfiles\": [{
          \"meterId\":            \"did:web:${OPENCRED_ISSUER_DOMAIN}:assets:meter:MET-IMPORT-001\",
          \"sanctionedLoad\":     {\"value\": 10, \"unit\": \"kW\"},
          \"tariffCategoryCode\": \"DS-I\",
          \"premisesType\":       \"Residential\",
          \"connectionType\":     \"Single-phase\"
        }]
      },
      \"customerDetails\": {
        \"fullName\":              \"Arjun Mehra\",
        \"installationAddress\": {
          \"geo\": {
            \"type\": \"Point\",
            \"coordinates\": [77.5946, 12.9716]
```

```
        },
        \"address\": {
          \"streetAddress\": \"12, 4th Cross, Indiranagar\",
          \"addressLocality\": \"Bengaluru\",
          \"addressRegion\": \"Karnataka\",
          \"postalCode\": \"560038\",
          \"addressCountry\": \"IN\"
        }
      },
      \"serviceConnectionDate\": \"2018-07-15T00:00:00+05:30\"
    }
  }
}" | tee credential.json | jq .credential
```

Four things worth noting:

- **Asset IDs are `did:web` under your own domain**, with colon-path segments (`did:web:<your-domain>:assets:meter:<slno>` — the command above builds them from `$OPENCRED_ISSUER_DOMAIN`). Same pattern for transformers, feeders, substations — see Register & Identifiers — Identifier patterns. No per-asset `did.json` hosting required for the pragmatic case.
- **`issuer.idRef` is optional.** OpenCred fills `issuer` with the DID string only. Your integration service appends `name` and (if you have a regulator to cite) `idRef` on egress, then re-signs if your flow requires a single signed artefact.
- **`credentialSubject.id` is absent here.** That's the bearer-style default. Set it to a wallet `did:key` or `tel:+91...` URI for holder-bound issuance — see Appendix — Holder binding.
- **This credential is revocable — it carries a `credentialStatus`.** The `revocationRegistryUrl` line points OpenCred at your DeDi revocation registry (built from `$OPENCRED_DEDI_NAMESPACE`), so it stamps a `credentialStatus` block onto both the credential and its JWT payload:

  ```
  "credentialStatus": {
    "id": "https://api.dedi.global/dedi/lookup/<namespace>/vc-revocation-registry/<credential-hash>",
    "type": "dedi",
    "statusPurpose": "revocation",
    "statusListCredential": "https://api.dedi.global/dedi/lookup/<namespace>/vc-revocation-registry"
  }
  ```

  You pass the **`query`** (whole-registry) URL at issue; OpenCred writes back the **`lookup`** (per-credential record) URL into `credentialStatus.id` — the two URL shapes are deliberate. The stamping happens at issue and needs no live DeDi connection; the actual revocation write in §3.8 does, so make sure the `OPENCRED_DEDI_*` variables from §3.3 are set before you rely on it.

  To issue *without* revocation — bearer-style, no `credentialStatus`, e.g. for `did:key` demos or throwaway testing — drop the `revocationRegistryUrl` line from the request body. The credential then can't be revoked, which is fine for those cases but not for anything a verifier will trust in production.

  The credential is W3C VC Data Model 2.0 — its `@context` is `https://www.w3.org/ns/credentials/v2` (single context, no `credentials/v1`). Point third parties at a **VC-2.0-aware verifier**. Libraries still pinned to VC Data Model 1.1 reject the `v2` context outright (e.g. an error like `@context is missing default context "https://www.w3.org/2018/credentials/v1"`) even though the signature is valid. This is a verifier-version mismatch, not a defect in the credential: do **not** add the `credentials/v1` context to "fix" it — a v2 credential must not carry the v1 context. If you must interoperate with a v1.1-only consumer, issue that consumer a separate v1.1 credential rather than downgrading the v2 one.

3.7 — Verify

```
jq -n --arg c "$(jq -r '.credential.proof.jwt' credential.json)" '{credential: $c}' | \
  curl -s http://localhost:3100/v1/credentials/verify \
    -H "Authorization: Bearer $OPENCRED_API_KEY" \
```

```
    -H "Content-Type: application/json" \
    -d @- | jq
```

Expected: the response includes `"valid": true`.

For `vc-jwt`, the verify endpoint takes the compact JWS string (`.credential.proof.jwt`), not the JSON envelope. For `data-integrity` proofs, send the full credential JSON. What a *third-party* verifier checks — signature, licensing assertion, revocation, validity window — is walked through in Verify a credential you received below.

3.8 — Revoke

OpenCred publishes revocation as a hash entry into your DeDi revocation registry; the verifier reads `credentialStatus` and looks up the hash. DeDi stores **only revoked** hashes — the per-credential lookup returns `200` when revoked and `404` when not (record existence is the signal). For the credential to carry that `credentialStatus`, pass `revocationRegistryUrl` at issue — the §3.6 issue and the variant examples below all do; drop that line and the credential cannot be revoked.

> **Two URL shapes, on purpose.** `revocationRegistryUrl` at issue points at the *registry* (`…/dedi/query/<namespace>/vc-revocation-registry`); a verifier checking one credential reads the *record* (`…/dedi/lookup/<namespace>/vc-revocation-registry/<hash>`). `query` addresses the whole registry, `lookup` a single record inside it — not a typo.

Compute the revocation hash of the credential you issued:

```
HASH=$(jq '{credential: .credential}' credential.json | \
  curl -s http://localhost:3100/v1/credentials/revocation-hash \
    -H "Authorization: Bearer $OPENCRED_API_KEY" \
    -H "Content-Type: application/json" -d @- | jq -r .revocationHash)
echo "$HASH"
```

Revoke it:

```
curl -s http://localhost:3100/v1/credentials/revoke \
  -H "Authorization: Bearer $OPENCRED_API_KEY" \
  -H "Content-Type: application/json" \
  -d "{\"hash\": \"$HASH\", \"reason\": \"connection-terminated\"}" | jq
```

Check its status:

```
curl -s http://localhost:3100/v1/credentials/revocation-status \
  -H "Authorization: Bearer $OPENCRED_API_KEY" \
  -H "Content-Type: application/json" \
  -d "{\"hash\": \"$HASH\"}" | jq
```

Expected: the status response reports the credential as revoked, with the reason you supplied.

Revocation works here because the container was started with the `OPENCRED_DEDI_*` variables in §3.3 — if you dropped them for early testing, add them back and restart before trying this section. See OpenCred Revocation for the conceptual model.

3.9 — Smoke test

A passing integration test should:

1. Issue a credential.
2. Resolve `issuer.id` (`did.json` over HTTPS) and confirm the public key matches the one that signed `proof`.
3. *(If `issuer.idRef` is present)* Resolve `issuer.idRef.issuedBy` and confirm the regulator vouches for your DISCOM.
4. `POST /v1/credentials/verify` — expect `valid: true`.
5. Check `revocation-status` — expect not revoked.
6. Revoke.
7. Re-check `revocation-status` — expect revoked.

Run on every release. It exercises every leg of the trust chain.

3.10 — Package the credential as a PDF / QR code

A signed VC is a JSON blob — fine for machine-to-machine, awkward to hand to a consumer. OpenCred renders the same credential into a printable **PDF certificate** with an embedded **QR code**, so a DISCOM can email or DigiLocker-share a document a person can actually read, and a field verifier can scan the QR to recover the credential verbatim. This mirrors the OpenCred bootcamp — Package the credential as a PDF / QR code.

Packaging is a **rendering** step, not a signing step — no signature is added or checked. The integrity guarantee stays in the credential you issued in §3.6, which is preserved byte-for-byte inside the QR and the JSON output.

```
curl -s http://localhost:3100/v1/credentials/package \
  -H "Authorization: Bearer $OPENCRED_API_KEY" \
  -H "Content-Type: application/json" \
  -d "{
    \"credential\": $(jq '.credential' credential.json),
    \"formats\": [\"pdf\", \"qr-png\", \"json\"],
    \"customization\": {
      \"primaryColor\":      \"#003366\",
      \"issuerDisplayName\": \"Example DISCOM\",
      \"footerText\":        \"Issued under IES ElectricityCredential v1.2\"
    }
  }" | tee packaged.json | jq '{outputs: [.outputs[] | {format, mimeType, suggestedFileName}], errors:
↪  .errors}'
```

- **`credential`** takes either the signed VC JSON object (as above — `data-integrity` and `vc-jwt` both work) or the bare compact token string (the `.credential.proof.jwt` from a `vc-jwt` issue, or an `sd-jwt-vc` token). For `vc-jwt` the QR embeds the VC with the JWT in `proof.jwt`; the PDF layout is driven by the decoded payload.
- **`formats`** is the field the endpoint reads — `qr-png`, `qr-svg`, `pdf`, `json`, `json-compact`. It defaults to `["json"]`, so **a misspelled or omitted key silently gives you JSON only** (there is no error — an unknown top-level field like `packageFormats` is ignored and the default applies).
- **`customization`** is entirely optional. All fields: `primaryColor`, `secondaryColor`, `textColor`, `labelColor`, `backgroundColor`, `issuerDisplayName` (<=200 chars), `logoDataUri` + `logoWidth`/`logoHeight` (10–200), `sealDataUri`, `footerText` (<=500 chars). Colours are `#rrggbb`; logo/seal are `data:` URIs.

Per-format failures land in `errors[]` without failing the whole call — the response is still `200` with whatever rendered.

Extract the files. **PDF** data is pure base64; **QR PNG** data is a `data:` URL (strip the prefix first); **SVG** and **JSON** are UTF-8 text:

```
jq -r '.outputs[] | select(.format=="pdf") | .data' packaged.json | base64 -d > certificate.pdf

jq -r '.outputs[] | select(.format=="qr-png") | .data | sub("^"; "")' packaged.json |
↪  base64 -d > qr.png

jq -r '.outputs[] | select(.format=="json") | .data' packaged.json > credential-packaged.json
```

Expected: `certificate.pdf` is a two-page PDF — page 1 is the credential rendered as a certificate with the "Scan to verify" QR, page 2 is the digital-signature block (proof type, algorithm, verification method). `qr.png` is a 400×400 PNG that decodes back to the credential you issued.

Issue the credential variants

The walkthrough above issued an ElectricityCredential. The other two IES credentials use the same `POST /v1/credentials/issue` flow with different schemas — what each variant is *for* is described in Verifiable Credentials — Credential variants.

MeterDataCredential v0.6 — telemetry signing

The `schemaId` is the OpenCred registry id **`ies/meter-data-credential/v0.6`**, and `credentialSubject.meterData` carries the `MeterData` payload (a profile object or array — see the v0.6 examples). Pass `revocationRegistryUrl` so the

credential carries a `credentialStatus` verifiers can check (§3.8):

```
curl -s http://localhost:3100/v1/credentials/issue \
  -H "Authorization: Bearer $OPENCRED_API_KEY" \
  -H "Content-Type: application/json" \
  -d "{
    \"schemaId\":     \"ies/meter-data-credential/v0.6\",
    \"issuerDid\":    \"$ISSUER_DID\",
    \"proofFormat\":  \"vc-jwt\",
    \"validFrom\":    \"2026-06-28T00:00:00+05:30\",
    \"validUntil\":   \"2026-07-05T00:00:00+05:30\",
    \"revocationRegistryUrl\":
↪   \"https://api.dedi.global/dedi/query/$OPENCRED_DEDI_NAMESPACE/vc-revocation-registry\",
    \"credentialSubject\": {
      \"meterData\": { \"@type\": \"DailyProfile\", \"profileType\": \"DAILY\", \"…\": \"a MeterData v0.6
↪   profile or array\" }
    }
  }" | tee credential.json | jq .credential
```

MeterDataRequestCredential v0.1 — proof of right-to-ask

This schema is **not** in OpenCred's built-in registry, so issue it with an **`inlineSchema`** rather than a `schemaId`: pass the JSON Schema in the request and OpenCred validates `credentialSubject` against it, writes the schema `$id`, and signs. Here we reuse the published MeterDataRequest `$defs` so the inline schema stays canonical — the first command pulls them, the second wraps them as the `credentialSubject` schema and issues:

```
REQ_DEFS=$(curl -s
↪   https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataRequest/v0.6/schema.json | jq
↪   '.["$defs"]')

jq -n --argjson defs "$REQ_DEFS" --arg iss "$ISSUER_DID" \
  --arg reg "https://api.dedi.global/dedi/query/$OPENCRED_DEDI_NAMESPACE/vc-revocation-registry" '{
  inlineSchema: {
    "$id":
↪   "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataRequestCredential/v0.1/schema.json",
    type: "object", required: ["meterDataRequest"],
    properties: {
      id: { type: "string", format: "uri" },
      meterDataRequest: { "$ref": "#/$defs/MeterDataRequestObject" }
    },
    "$defs": $defs
  },
  issuerDid: $iss,
  proofFormat: "vc-jwt",
  validFrom: "2026-06-28T00:00:00+05:30",
  validUntil: "2026-12-28T00:00:00+05:30",
  revocationRegistryUrl: $reg,
  credentialSubject: {
    meterDataRequest: {
      "@context":
↪   "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataRequest/v0.6/context.jsonld",
      "@type": "MeterDataRequest",
      scope: "ResourceOnly",
      from: "2026-06-04T00:00:00Z",
      duration: "P6M",
      maxRecordsShared: 10000,
      capabilitiesRequested: { profiles: [ { profileType: "CustomerProfile" }, { profileType:
↪   "IntervalProfile" } ] }
```

```
    }
  }
}' | curl -s http://localhost:3100/v1/credentials/issue \
  -H "Authorization: Bearer $OPENCRED_API_KEY" -H "Content-Type: application/json" -d @- | jq .credential
```

`scope` must be one of the `ScopeType` enum (`ResourceOnly`, `ResourceAndChildren`, `ChildrenOnly`). The seeker embeds the resulting credential in the Beckn `confirm` message's receiver participant; the provider's adapter verifies it before fulfilling. Ready-to-send `confirm` payloads live in the DEG data-exchange devkit.

Verify a credential you received (the verifier's walkthrough)

The flip side: a wallet or a counterparty hands you a signed credential. What to check, in order:

1. **Parse** the credential JSON. Read `issuer.id` (e.g. `did:web:ies.discom.example`).
2. **Resolve `issuer.id`** over HTTPS — fetch `https://<issuer-domain>/.well-known/did.json` and extract the `verificationMethod` public key.
3. **Verify the credential's `proof`** signature against that key. If it fails, stop — the credential is forged or corrupted.
4. **(Optional) Check `issuer.idRef`** if present. Resolve `issuer.idRef.issuedBy` (the regulator's `did:web`) and confirm the regulator vouches for this DISCOM under the cited `subjectId`. This is the licensing leg of the trust chain; skip when `idRef` is absent.
5. **Check revocation.** GET the URL in `credentialStatus.id` — typically `https://api.dedi.global/dedi/lookup/<discom>/vc-revocation-registry/<credential-hash>`. A `404` (or `status: not_revoked`) means valid; a record with `status: revoked` means rejected.
6. **Check the validity window.** Confirm `validFrom <= now <= validUntil`.
7. **(Holder-bound credentials only)** Challenge the presenter to prove control of `credentialSubject.id` — see Appendix — Holder binding.

A verifier that caches the issuer's `did.json` and the regulator's `did.json` (e.g. refresh every 10 minutes) can complete steps 2–4 entirely offline; only step 5 needs a live DeDi lookup.

Operational notes

The bare minimum to run OpenCred in production.

Key rotation

1. Generate a new signing key in your KMS.
2. Publish the new key in `did.json` (keep the old key listed for a transition window).
3. Restart OpenCred pointing at the new key.
4. After the transition window, remove the old key from `did.json`.

Existing credentials signed by the old key keep verifying as long as the old key remains in `did.json`. Once you drop it, those credentials stop validating — schedule re-issuance before dropping.

Signing-key sources

OpenCred loads exactly one signing key from one of:

Source	Env var	When to use
Software file (PEM, JWK, PKCS#8, PFX)	`OPENCRED_KEY_PATH`	Dev, small DISCOMs
AWS KMS	`OPENCRED_KMS_PROVIDER=aws`, `OPENCRED_KMS_KEY_ARN`	Production on AWS
Azure Key Vault	`OPENCRED_KMS_PROVIDER=azure`, `OPENCRED_AZURE_*`	Production on Azure

Source	Env var	When to use
GCP Cloud KMS	`OPENCRED_KMS_PROVIDER=gcp`, `OPENCRED_GCP_KMS_KEY_NAME`	Production on GCP

The private key never leaves the container; in KMS modes it never leaves the HSM at all. There is no shared signing service and no key escrow.

Schema validation

Validate the body of every `POST /v1/credentials/issue` against the schema **before** sending it. OpenCred validates server-side too, but a client-side check catches integration bugs earlier and avoids logging PII into OpenCred's error trail. Use the JSON Schema at `schemas/ElectricityCredential/v1.2/schema.json` (or the matching version).

Batch issuance

For high-volume flows (annual re-issue, bulk Passport rollout):

- Process in batches of 100–1000; respect `Retry-After` if OpenCred returns 429.
- Sign in-process; do not externalise to a queue that buffers credentials in the clear.
- Persist `(credentialId, customerNumber, status)` after each successful issue so retries are idempotent.
- Run integration tests against `test-` networks before flipping the prod flag.

OpenCred's API reference covers concurrency, rate limits, and error shapes for production scale.

Reverse proxy + TLS

Never expose OpenCred's `:3100` directly. Terminate TLS at nginx / Envoy / your existing edge; forward to OpenCred over an internal network. The `OPENCRED_API_KEY` is the only auth — leaking it gives the bearer full issuance powers.

Troubleshooting

Symptom	Likely cause	Action
`/v1/health` returns `signingKeyLoaded: false`	Key path wrong, key file permission denied, or KMS creds missing	Check the container logs; verify the mount and the env vars
`POST /v1/credentials/issue` returns `400 schema_validation_error`	Body doesn't match the schema for the declared `schemaId`	Diff against `schemas/<Credential>/<version>/schema.json`
`POST /v1/credentials/verify` returns `valid: false` for a freshly issued credential	You sent the JSON envelope instead of the compact JWS for a `vc-jwt` proof	Send `.credential.proof.jwt`, not the whole envelope
`/v1/keys/publish` fails with a validation error	A pre-existing DeDi registry has an incompatible schema	Either let OpenCred recreate, or align the registry schema; see OpenCred Deployment
`revoke` succeeds but verifiers still see `not_revoked`	DeDi cache; OpenCred's local cache	Wait 5–10 minutes; verify the registry directly via `curl https://api.dedi.global/dedi/lookup/<ns>/vc-revocation-registry/<hash>`

Appendix — Binding the credential to a holder identity

A credential signed by you proves *who issued it*. It does **not** by itself prove *who is allowed to present it*. Without an explicit holder binding, the credential is a **bearer token** — whoever has the JSON is treated as the subject. That's fine for paper-style issuance, demos, and counter-issued attestations, but for consumer-facing flows (Consumer Energy Passport, Consumer Meter Digest, anything a bank or marketplace will read) you usually want presentation-time proof that the presenter is the same person the credential was issued to.

The W3C VC Data Model 2.0 § Presentations handles this through `credentialSubject.id`: set it to an identifier the subject controls, and require a proof-of-control at presentation time. The v1.2 schema makes `credentialSubject.id` optional, so you can pick the pattern that fits the consumer's situation.

Before you bind anything: identity-proofing at issuance

This is the easy-to-miss step. The holder identifier (a DID, a phone number, anything) lands inside `credentialSubject.id` because **you put it there**. If you copy it verbatim from an unauthenticated request, an attacker can submit their own DID or their own phone and you will happily bind a Passport to it. Holder binding only works if you verify, **at issue time**, that the consumer controls the identifier you're about to embed.

So before any `POST /v1/credentials/issue` call that sets `credentialSubject.id`:

Holder identifier	What to verify at issue time
`did:key:...` (wallet)	The wallet signs a fresh nonce you send it; you verify the signature against the public key in the DID. This proves the requester holds the wallet's private key.
`tel:+91XXXXXXXXXX`	You confirm the number is the one already on file for that `customerNumber` in your CIS, **and** you send a fresh OTP to that number and confirm the requester reads it back.
DigiLocker-mediated	DigiLocker has already done the Aadhaar-backed identity check before issuing the access grant; you can rely on that for the identity binding, and just record DigiLocker as the binding channel.

Without that check, holder binding adds zero security; with it, the wallet/phone/DigiLocker identifier becomes a real second factor at presentation time.

Pattern 1 — Wallet DID (cryptographic, recommended where a wallet exists)

Set `credentialSubject.id` to a DID the consumer's wallet controls — typically `did:key`, sometimes `did:jwk`. Both are offline-resolvable (the public key is encoded in the DID string), so the verifier needs no network call to fetch the holder's key.

```
"credentialSubject": {
  "id": "did:key:z6MkjVQ8r4f3rPuY7CG2D6Lf8WJxJBs5sjkR8d3v2Bv4nP4Z",
  "customerProfile": { "customerNumber": "DISCOM-2025-00987654", ... },
  "customerDetails": { ... }
}
```

The presentation-time flow. When the consumer presents the credential to a verifier (a bank, a marketplace, a housing society):

1. The verifier asks the wallet for the credential.
2. The wallet wraps the credential in a **Verifiable Presentation (VP)** — a small JSON envelope that can carry one or more credentials and add a fresh signature from the holder.
3. The verifier sends a **challenge** (a random nonce, often a UUID) and a **domain** (an identifier for the verifier itself, e.g. `bank.example.com`) — these go into `proof.challenge` and `proof.domain` on the VP.
4. The wallet signs the VP with the private key matching `credentialSubject.id`, embedding the challenge and domain inside the signature.
5. The verifier:
   - Resolves `credentialSubject.id`. For `did:key` this means decoding the multibase-encoded public key out of the DID string itself — no network call needed.
   - Verifies the VP's `proof.signature` against that public key.
   - Confirms `proof.challenge` matches the nonce the verifier just sent (rules out replays).
   - Confirms `proof.domain` matches the verifier's own identifier (rules out a presentation captured from another verifier).

- Separately verifies the embedded credential's `proof` against the issuer DISCOM's `did:web` key (this is the standard issuer-side check).

6. If all checks pass, the presenter has demonstrated control of the wallet's private key — they are the subject the credential was issued to.

A stolen credential JSON does not pass step 5: the attacker doesn't have the wallet's private key, so they cannot produce a VP signature that matches the public key embedded in `credentialSubject.id`.

This is the model OpenID for Verifiable Presentations (OID4VP), DIDComm, and most W3C wallet ecosystems implement.

Wallet rotation. A wallet `did:key` cannot rotate (rotating means a new DID). If a consumer's wallet is lost or compromised, they generate a new `did:key` and you re-issue the credential bound to it. The old credential is revoked through your DeDi revocation registry (§3.8).

Pattern 2 — Phone-number URI (out-of-band, when there is no wallet)

If the consumer doesn't have a wallet — and many Indian consumers won't, at least initially — you can still bind the credential to a channel they control: a phone number. Use the RFC 3966 `tel:` URI scheme so the identifier is a proper URI rather than a bare string:

```
"credentialSubject": {
  "id": "tel:+919876543210",
  "customerProfile": { "customerNumber": "DISCOM-2025-00987654", ... },
  "customerDetails": { ... }
}
```

The URI form is strict: **`tel:` + full E.164** (country code + national number, no spaces, hyphens, parentheses, or leading zeros). For India that is `tel:+91` followed by the ten-digit mobile number. This is the only form a verifier can compare unambiguously across systems.

The presentation-time flow.

1. The consumer presents the credential to a verifier (often via QR scan, email attachment, or a DigiLocker share).
2. The verifier reads `credentialSubject.id` and extracts the phone number.
3. The verifier sends a fresh OTP to that number through SMS, IVR, or a push notification on a known app.
4. The presenter reads the OTP back to the verifier.
5. If the OTP matches and is unexpired, the presenter has demonstrated control of the phone number.
6. The verifier independently verifies the credential's `proof` against the issuer's `did:web` key.

Compared to Pattern 1, this is weaker security: phone numbers can be ported, SIM-swapped, or temporarily shared, and the OTP channel itself can be phished. It is also slower (out-of-band OTP delivery). It is the right pragmatic choice when the consumer has no wallet and the verifier already runs phone-OTP plumbing.

Phone-number changes. Phone numbers change far more often than wallet DIDs. Either keep `validUntil` short (e.g. annual re-issue) so a stale number doesn't outlive its accuracy, or re-issue the credential whenever the consumer updates their number in your CIS.

Pattern 3 — DigiLocker-mediated (Indian-context shortcut)

For consumers who already use DigiLocker, the identity binding is largely solved before the credential reaches the consumer: DigiLocker performs an Aadhaar-backed identity check before granting access, and the credential is delivered into the consumer's DigiLocker vault. A verifier consuming a credential out of DigiLocker can rely on DigiLocker's own access-control rather than re-running a presentation-time challenge.

In this case you may issue the credential without a `credentialSubject.id` (bearer style) and document DigiLocker as the binding channel, **or** set `credentialSubject.id` to the consumer's DigiLocker-resident `did:key` if their wallet exposes one. The first is simpler; the second future-proofs the credential for re-presentation outside DigiLocker. Delivery mechanics: DigiLocker delivery.

Pattern 4 — DISCOM-assigned consumer DID (stable subject reference)

When the consumer has no wallet and you still want a **stable, resolvable subject identifier** on the Passport, mint one under your own domain from the consumer number:

```
"credentialSubject": {
  "id": "did:web:ies.discom.example:consumers:DISCOM-2025-00987654",
  "customerProfile": { "customerNumber": "DISCOM-2025-00987654", ... },
  "customerDetails": { ... }
}
```

This is the same `did:web` path grammar used for assets and connections (Register & Identifiers — Identifier patterns), scoped to `consumers:<consumer-number>`. Its value: every credential you issue for that consumer carries the **same subject id**, so a verifier (or the consumer's own history) can correlate the Passport, a Meter Digest, and a re-issued Passport as being about one subject — without exposing a wallet key or a phone number.

What it does and doesn't give you. The DID resolves under *your* domain, so it is a stable name you control — not proof the *presenter* controls it. It does **not** replace holder proof-of-control: for presentation-time proof, layer Pattern 1 (a wallet `did:key` the consumer signs a VP with) or Pattern 2 (phone OTP) on top, or deliver via DigiLocker (Pattern 3). Use Pattern 4 as the default subject id for bearer/counter-issued Passports and DigiLocker-delivered Passports where the channel supplies the identity binding and you still want a durable, correlatable subject reference. It carries no PII — the consumer number is already inside `customerProfile.customerNumber`.

Where does the contact identifier live in the schema?

The `customerDetails` block in ElectricityCredential v1.2 (which references the shared `CustomerDetails/v1.0` shape) currently carries **`fullName`, `installationAddress`, and `serviceConnectionDate`** — there is **no telephone field**. So if you choose Pattern 2 today, the `tel:` URI lives in `credentialSubject.id`, not in `customerDetails`. If a future schema revision adds a `telephone` field, the canonical URI form there should be `tel:+<E.164>` so that the two locations agree.

Picking a pattern

Consumer situation	Pattern	`credentialSubject.id` value
Has a digital wallet (DID-capable)	**Pattern 1** — wallet DID	`did:key:z6Mkj...`
Has only a phone number on record	**Pattern 2** — `tel:` URI	`tel:+919876543210`
Uses DigiLocker, no separate wallet	**Pattern 3** — DigiLocker-mediated	*Omit, or use DigiLocker's `did:key` if exposed*
No wallet, but you want a stable correlatable subject	**Pattern 4** — DISCOM-assigned DID	`did:web:ies.discom.example:consumers:DISCOM-2025-00987654`
Paper / counter issuance, presented in person	None — bearer	*Omit*

Patterns are not exclusive, and Pattern 4 composes with the others: a Passport can carry a `did:web:...:consumers:...` subject id for correlation **and** be presented with a wallet-signed VP (Pattern 1) or phone OTP (Pattern 2) for proof-of-control. Credentials sharing the same `customerProfile.customerNumber` (or the same Pattern-4 subject id) share the same revocation handle, so revoking one does the right thing operationally.

Quick consistency checklist for adopters

- [ ] Issuer-side identity proof in place before any `credentialSubject.id` is set (challenge-sign for wallet DIDs, OTP for phones, DigiLocker grant otherwise).
- [ ] `tel:` URIs use full E.164 — `tel:+91` + ten digits — no spaces, hyphens, parentheses, or leading zeros.
- [ ] Verifier flow documented: how the verifier will challenge the holder at presentation time (VP-with-challenge for Pattern 1, OTP for Pattern 2, DigiLocker grant for Pattern 3).
- [ ] Revocation tested: revoking a credential invalidates **all** presentations of it, regardless of which binding pattern is in use.
- [ ] `validUntil` set with the binding's churn rate in mind (years for wallet DIDs, months-to-a-year for phone bindings).

Checklist

- [ ] OpenCred running; `/v1/health` reports `signingKeyLoaded: true`
- [ ] Namespace visible on `explore.dedi.global`; four OpenCred registries visible in `publish.dedi.global`
- [ ] `ISSUER_DID` matches your published `did:web`
- [ ] One credential issued, verified (`valid: true`), revoked, and re-checked as revoked
- [ ] Smoke test (§3.9) scripted into your release pipeline
- [ ] For consumer-facing credentials: identity-proofing procedure in place before any holder binding

When all boxes are ticked, you can issue production credentials: each one is schema-valid and independently verifiable by anyone who resolves your `did:web` — there is no separate credential-issuer certification step, because verification is bilateral rather than centrally gated. That is a narrower claim than full JSON-LD or external-schema conformance; see Conformance — What this checklist proves for the boundary. To feed OpenCred from your CIS / MDM automatically, continue to **Build your Internal-facing Adapter**. To exchange data over a Beckn network as well, do **Setting up Register §1.5–1.7** then **Setting up Discover & Exchange**.

Verifiable Credentials in depth

The trust layer. W3C Verifiable Credentials, signed by a DISCOM's `did:web`, delivered through wallets (DigiLocker or DID-aware), a web portal, or any channel the issuer already runs. Credential flows stand on their own — issuing, holding and verifying a credential requires **no Beckn network**. This page is the **reference** for the credential lifecycle, the variants IES uses, and the trust model. The operational commands — run OpenCred, issue, verify, revoke — are in **Issuing Credentials**; identity prerequisites are in **Setting up Register**.

Why credentials

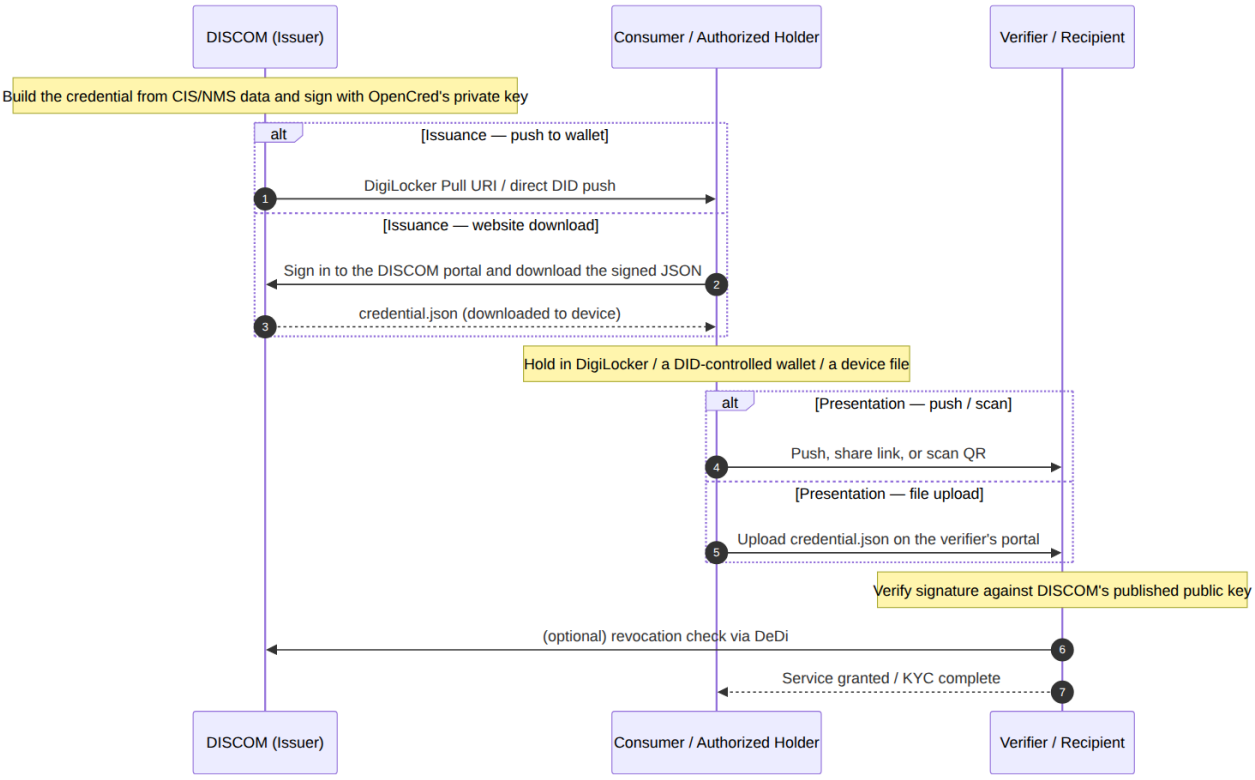
When a DISCOM hands a consumer a digital electricity attestation, or shares meter readings with a regulator or a marketplace, the receiver needs to answer one question on their own: *"Is this really from the DISCOM, intact, and still valid?"* If they have to call you, the system does not scale and is not really verifiable.

A **Verifiable Credential** is a small JSON object you sign with the private key behind your `did:web`. Anyone — a wallet, another DISCOM, a bank, a regulator — can fetch your `did.json` over HTTPS, check the signature, and consult a public revocation list. No callback to you required. That is what makes this the **B2C rail**: the verifier can be *anyone*, known to you or not, and the credential can travel over *any* channel — DigiLocker, a portal download, email, SMS, chat — because the trust is inside the object, not the pipe.

Three credentials cover almost everything IES does:

Credential	What it attests	Who signs	Typical receiver
ElectricityCredential v1.2	A service connection — customer number, sanctioned load, tariff, meter info, energy resources (rooftop solar, BESS, EV chargers)	DISCOM	The consumer's wallet, or a verifier (bank, marketplace, regulator) the consumer shares it with
MeterDataCredential v0.6	A signed meter-reading payload (raw `MeterData` profiles or derived summaries) for a specified period	AMISP, MDM, or DISCOM	DISCOM (B2B telemetry) or the consumer (their own readings)
MeterDataRequestCredential v0.1	A signed request for meter data — proves the requester has the right to ask	Seeker (typically a DISCOM)	Provider (typically an AMISP) at Beckn `confirm` time

Lifecycle at a glance



`Issued → Held / presented → Verified → (eventually) Revoked or expired`

- **Issued**: the issuer signs and emits the credential JSON.
- **Held**: a wallet or DigiLocker stores it.
- **Presented**: the holder shares it (raw or in a Verifiable Presentation) with a verifier.
- **Verified**: the verifier checks the issuer's signature, the regulator's `idRef` if present, revocation status, and the validity window.
- **Revoked**: the issuer publishes a hash in the DeDi revocation registry. Verifiers reject revoked credentials.
- **Expired**: `validUntil` passes. Issue a fresh credential on material change (rate revision, meter swap, ownership transfer) rather than relying on long expiry windows.

Pick your role

If you are…	Read	Then
A DISCOM / issuer (you sign and emit credentials)	Setting up Register §1.1–1.4 -> Issuing Credentials	Credential variants below, Build your Internal-facing Adapter
An AMISP / MDM / aggregator (you sign telemetry)	Same, issuing `MeterDataCredential`	Smart Meter Data Exchange use case
A holder / wallet (you hold credentials on behalf of a consumer)	Holder binding -> DigiLocker delivery	Issuing Credentials — Holder binding for binding patterns
A verifier (you receive and check credentials)	Trust model below	Issuing Credentials — Verify a credential you received for the step-by-step

---

Credential variants

The same schemas cover several use cases. **No new VC `type` values are introduced** — the variants are issuance configurations over the existing schemas. The issuance commands for each are in Issuing Credentials.

ElectricityCredential v1.2 — the default A DISCOM-signed attestation about a service connection. Carries `customerProfile` (non-PII: customer number, energy resources, consumption profile), `customerDetails` (optional PII: name, address, service-connection date), and the issuer block. Two common shapes:

- **Bearer / counter-issued** (no `credentialSubject.id`). Anyone holding the JSON is treated as the subject. Used for paper-style attestations, demos, or in-person verification.
- **Holder-bound, consumer-presentable** — the **Consumer Energy Passport** pattern. Same schema, but `credentialSubject.id` is the consumer's wallet `did:key` (or `did:jwk`), `customerProfile.idRef` carries a verifiable government-ID *reference* (never the raw number), and at presentation time the wallet proves control of the key via a challenge-signed Verifiable Presentation. The Consumer Energy Passport use case is about *who*, *when*, and *why* this shape is issued; the credential itself is an ElectricityCredential v1.2.

MeterDataCredential v0.6 — telemetry signing A signed VC wrapping a MeterData v0.6 payload (raw `INTERVAL`/`DAILY`/`MONTHLY` profiles or derived summaries) for a specified period. Two common shapes:

- **B2B**, typically without `credentialSubject.id`. The AMISP signs; the DISCOM consumes the payload.
- **Holder-bound, consumer-presentable** — the **Consumer Meter Digest** pattern. `credentialSubject.id` is the consumer's wallet DID, `validUntil` is short (hours to days), and the readings cover a period the consumer asked for. Delivered into the consumer's wallet / DigiLocker; verifiers (banks, marketplaces) check it without phoning the DISCOM. Use case: Consumer Meter Digest.

MeterDataRequestCredential v0.1 — proof of right-to-ask A signed VC carried at Beckn `confirm` time by a seeker (typically a DISCOM) when an AMISP's offer policy requires it — the one credential that rides *inside* a Beckn message. Proves the seeker has been authorised to request the data they're confirming. Schema: MeterDataRequestCredential v0.1; ready-to-send `confirm` payloads live in the DEG data-exchange devkit.

Summary

Pattern	Schema	`credentialSubject.id`	`validUntil`	Issued by
Bearer ElectricityCredential	`ElectricityCredential/v1.2`	absent	years	DISCOM
Consumer Energy Passport	`ElectricityCredential/v1.2`	wallet `did:key` (+ `customerProfile.idRef`)	years	DISCOM
B2B MeterDataCredential	`MeterDataCredential/v0.6`	absent	days to a year	AMISP / MDM
Consumer Meter Digest	`MeterDataCredential/v0.6`	wallet `did:key`	hours to days	DISCOM (on consumer demand)
Meter-data request	`MeterDataRequestCredential/v0.1`	absent	weeks to months (the authorised request window)	Seeker (typically DISCOM)

Holder binding

Holder binding turns a credential from a bearer token into something only the consumer's wallet can present. Choose a pattern (wallet DID, `tel:+91...` URI, or DigiLocker-mediated) based on the consumer's situation. **Identity-proofing at issuance is mandatory** — you must verify the consumer controls the identifier before embedding it.

Full patterns, presentation-time flows, and the adopter checklist: **Issuing Credentials — Binding the credential to a holder identity**.

DigiLocker delivery

DigiLocker is the dominant consumer wallet in India. Once issued, an ElectricityCredential or MeterDataCredential can be delivered into a consumer's DigiLocker via a Pull URI, and any verifier reading from DigiLocker inherits DigiLocker's Aadhaar-mediated identity binding.

Walkthrough (Pull URI endpoint, DocType registration, HMAC verification, issuing both DocTypes): **DigiLocker delivery**.

Trust model

A credential's trust chain has at most two legs, plus two freshness checks:

1. **Mandatory** — the issuer's `did:web` signature. A verifier resolves `issuer.id` over HTTPS to `did.json`, extracts the public key, and verifies `proof`. If this fails, stop — the credential is forged or corrupted.
2. **Optional** — the regulator's licensing assertion in `issuer.idRef`. When present, the verifier resolves `issuer.idRef.issuedBy` (the regulator's `did:web`) and confirms the regulator vouches for the DISCOM under the cited `subjectId`. When absent (pilots, non-regulated issuers), the verifier falls back to whatever out-of-band recognition they have of your `did:web`.
3. **Revocation status** — the URL in `credentialStatus.id`, resolved against the issuer's DeDi revocation registry.
4. **Validity window** — `validFrom <= now <= validUntil`.

Holder-bound variants add a fifth check at presentation time: the wallet signs a Verifiable Presentation with the key matching `credentialSubject.id`, embedding a fresh `challenge` and `domain`.

The verifier's step-by-step (with URLs and expected responses) is in Issuing Credentials — Verify a credential you received. No IES-curated registry sits between the credential and the verifier — the IES network registries are the Beckn-side (B2B) trust boundary only; see Register & Identifiers — Two identities.

Proof formats

Format	When to choose	Where it shines
`vc-jwt` (default)	Most flows	Compact wire form, easy to embed in headers, fast to verify
`data-integrity`	Custom-registered clean-context schemas	Linked-data-friendly, supports selective disclosure variants
`sd-jwt-vc`	Selective disclosure	The holder presents only chosen fields to each verifier

---

Core concepts

For readers new to verifiable credentials. Skip if you're mid-deployment.

What's a Verifiable Credential A **Verifiable Credential (VC)** is a JSON object with three properties:

- Who made the statement (`issuer`).
- The statement itself (`credentialSubject`).
- A cryptographic proof (`proof`) so anyone can verify the issuer signed it.

The IES profile uses the W3C VC Data Model 2.0. Required top-level fields: `@context`, `id`, `type`, `issuer`, `validFrom`, `credentialSubject`. Optional: `validUntil`, `credentialStatus`, `evidence`, `name`, `description`. The `proof` is added by the signing step.

What's a DID A **Decentralized Identifier (DID)** is a globally unique string that resolves to a DID document — a small JSON object listing the subject's current public keys. IES uses three standard W3C methods (`did:web` for institutions, `did:key` / `did:jwk` for consumer wallets); there is no `did:dedi` method — DeDi is a key-discovery and registry layer over `did:web`. Full treatment: Register & Identifiers — The DID methods IES uses.

References

- Register & Identifiers in depth — DIDs, identifier patterns, DeDi, holder identifiers
- Issuing Credentials — the operational walkthrough: run OpenCred, issue, verify, revoke
- IES Schemas — canonical schema mirrors for ElectricityCredential, MeterData(Credential), MeterDataRequest(Credential)
- DigiLocker delivery — Pull URI endpoint, HMAC verification, both DocTypes
- Use cases — Consumer Energy Passport, Consumer Meter Digest, Smart Meter Data Exchange
- OpenCred upstream documentation

DigiLocker delivery

This guide covers everything your DISCOM needs to build the DigiLocker Pull URI endpoint — the single piece of custom code required to make IES energy credentials available to consumers in DigiLocker.

Two document types flow into DigiLocker through the **same Pull URI mechanism**:

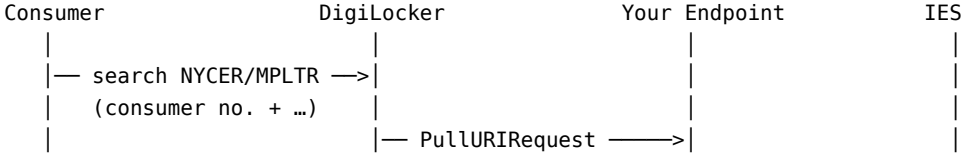
DocType	Credential	What it carries	Keyed on
`NYCER`	**Consumer Energy Passport** (`ElectricityCredential` v1.2)	Slow-changing facts — who holds the connection and every typed asset behind the meter (meter, solar, battery, EV, inverter, load, network), each power/capacity value a `{value, unit}` pair	Consumer number
`MPLTR`	**Consumer Meter Digest** (`MeterDataCredential` v0.6)	The consumer's own meter readings for a chosen period — a signed energy *statement*, not a bill	Consumer number + **period**

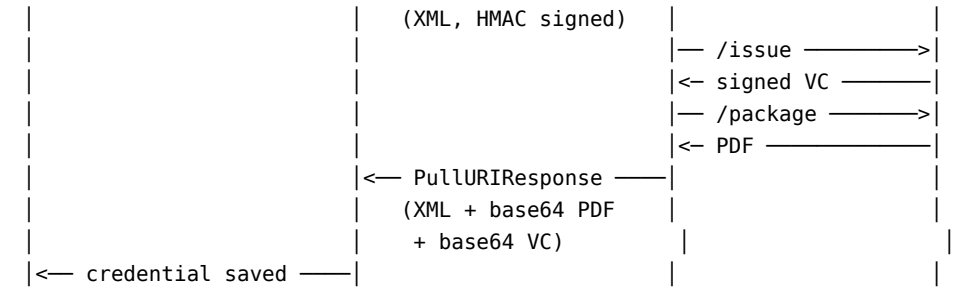
> **DocType confirmed.** NeGD has allotted `MPLTR` (Meter Document) as the DocType for the Consumer Meter Digest. `NYCER` stays unchanged for the Consumer Energy Passport — same DocType, same Pull URI flow; only the payload schema moves to v1.2.

The Digest reuses the connection-credential flow almost unchanged. The one structural difference is the document key: the Digest puts the **period into the URI** so every period coexists instead of overwriting — exactly the difference between a statement and a bill.

What DigiLocker Does

DigiLocker is India's national digital document wallet. When a consumer searches for one of these credentials in DigiLocker, DigiLocker calls your endpoint, your endpoint calls IES (OpenCred) to issue a signed credential, and DigiLocker stores it — all in real time.





A credential only has value once it reaches a use case — a lender assessing the consumer, an analytics app, a subsidy portal. DigiLocker is that bridge: it carries the consent and sharing rails (QR scan in person, OAuth consent pull for third-party apps) that make a credential useful beyond the wallet.

Flow Overview

The integration has three phases:

- **Phase 0 — One-time setup:** Register on API Setu, create issuer DID, register the DocTypes
- **Phase 1 — Pull URI endpoint:** Build and host the HTTPS endpoint DigiLocker calls (one handler, both DocTypes)
- **Phase 2 — Consumer sharing:** Consumers share the credential with verifiers via DigiLocker OAuth (no DISCOM involvement)

Phase 0 — One-Time Setup

Step 1 — Register on API Setu Go to https://apisetu.gov.in and register as a document issuer. You receive: - `issuer_id` (e.g. `in.gov.discom`) - `api_key` — store this securely; used for HMAC verification

> If your DISCOM already issues electricity bills (ELBIL) on DigiLocker, contact NeGD to add the `NYCER` and `MPLTR` document types to your existing account. Do not re-register.

Alongside the DocTypes, register your **credential issuer** with NeGD so DigiLocker can resolve and trust it. NeGD asks for a small block:

```
{
  "credential_type": "UtilityElectricityCredential",
  "issuer_did": "did:web:www.discom.example:energystack",
  "resource_schema": []
}
```

The `issuer_did` is the one from Step 2 — it may live on your own domain, or in a DigiLocker-hosted issuer namespace (e.g. `did:web:vc.dl6.in:issuers:in.gov.<state>electricity`) if NeGD hosts it for you.

> **Keep the org id consistent across registration and every URI.** The org segment of each Pull URI (`in.gov.<orgId>-<DocType>-<consumerNo>`, e.g. `in.gov.discom-NYCER-100000012345`) must equal the `issuer_id` / `orgId` you registered on API Setu. A URI whose org segment doesn't match the registered issuer fails at DigiLocker with an *org id / issuer id mismatch* — the response is accepted but its document data is rejected, which is easy to misread as a payload problem.

Step 2 — Stand Up OpenCred and Your Issuer DID Follow Setting up Register — keypair and did.json and Issuing Credentials — run OpenCred end-to-end. By the time you reach this page you should have:

- OpenCred running with `signingKeyLoaded: true`
- An issuer DID — `did:web:ies.discom.example` (recommended) or `did:key:…` (from an imported DSC)
- (Optional, recommended) DeDi configured for revocation

Save your issuer DID — it goes into every `POST /v1/credentials/issue` call.

Gotcha — the DID host must serve `did.json` *without* a redirect. A `did:web` DID resolves by fetching `https://<host>/<path>/did.json`. If that host 301-redirects — e.g. `https://discom.example/energystack/did.json -> https://www.discom.example/energystack/did.json` — lenient verifiers follow the redirect, but strict resolvers (OpenCred among them) reject it, and a consumer's credential then fails to verify. Name the **canonical host that answers directly** in your DID: if `www.` is where `did.json` is served, the issuer DID is `did:web:www.discom.example:energystack`, not `did:web:discom.example:energystack`. Confirm with `curl -sSL -o /dev/null -w '%{http_code} %{url_effective}\n' https://<host>/<path>/did.json` — a `200` with an unchanged URL, not a `301`.

Step 3 — Register the Pull URI Endpoints on API Setu Register both DocTypes against the same Pull URI endpoint. They differ only in the UDF fields and the URI key.

NYCER — Electricity Credential v1.2

Field	Value
Endpoint URL	`https://{your-domain}/digilocker/pulluri`
HTTP Method	`POST`
Content-Type	`application/xml`
DocType	`NYCER`
UDF1 Label	`Consumer Number`
UDF2 Label	`Registered Mobile Number`

MPLTR — Consumer Meter Digest

Field	Value
Endpoint URL	`https://{your-domain}/digilocker/pulluri`
HTTP Method	`POST`
Content-Type	`application/xml`
DocType	`MPLTR`
UDF1 Label	`Consumer Number`
UDF2 Label	`Statement Period` (e.g. `2026-05` or `last-12-months`)
UDF3 Label	`Registered Mobile Number`

MPLTR registers as an additional record on the same endpoint already registered for NYCER — one Pull URI handler, branched by `DocType`. One item is still pending **written** confirmation from DigiLocker even though it was agreed in principle on the 12 June 2026 call: treating each **period-keyed URI** as a distinct, independently listable/deletable document (see The Consumer Meter Digest). **Configurable rendering** of the statement card remains an open ask. Everything else reuses the NYCER flow unchanged.

Phase 1 — The Pull URI Endpoint

This is the **only endpoint your DISCOM builds**. DigiLocker calls it; you respond. A single handler serves both DocTypes — branch on `DocType` (see Routing by DocType).

Endpoint Specification

```
POST https://{your-domain}/digilocker/pulluri
Content-Type: application/xml
x-digilocker-hmac: <Base64(HMAC-SHA256(api_key, request_body))>
```

Step 1 — Verify the Inbound HMAC **Always verify the HMAC before doing anything else. Reject with HTTP 401 if invalid.**

```python
import hmac, hashlib, base64

def verify_digilocker_hmac(request_body: bytes, api_key: str, received_hmac: str) -> bool:
    expected = base64.b64encode(
        hmac.new(api_key.encode(), request_body, hashlib.sha256).digest()
    ).decode()
    return hmac.compare_digest(expected, received_hmac)  # constant-time comparison
```

Step 2 — Parse the Inbound Request DigiLocker sends a `PullURIRequest` XML v3.0. For NYCER:

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIRequest ver="3.0"
                ts="2026-04-01T10:30:00+05:30"
                txn="TXN-20260401-001"
                orgId="discom">
  <DocDetails>
    <DocType>NYCER</DocType>
    <DigiLockerId>7a3f9b2c-1e4d-4f8a-b6c2-0d5e8f1a2b3c</DigiLockerId>
    <UDF1>DISCOM-2025-001234567</UDF1>   <!-- consumer number -->
    <UDF2>9876543210</UDF2>             <!-- registered mobile number -->
    <FullName>Priya Sharma</FullName>    <!-- optional, from DigiLocker profile -->
  </DocDetails>
</PullURIRequest>
```

For MPLTR the period travels as a UDF, so the consumer can pull a chosen window:

```xml
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIRequest ver="3.0"
                ts="2026-05-15T10:00:00+05:30"
                txn="TXN-20260515-042"
                orgId="discom">
  <DocDetails>
    <DocType>MPLTR</DocType>
    <DigiLockerId>7a3f9b2c-1e4d-4f8a-b6c2-0d5e8f1a2b3c</DigiLockerId>
    <UDF1>DISCOM-2025-001234567</UDF1>   <!-- consumer number -->
    <UDF2>last-12-months</UDF2>          <!-- statement period: YYYY-MM, a range, or a keyword -->
    <UDF3>9876543210</UDF3>             <!-- registered mobile number -->
    <FullName>Priya Sharma</FullName>
  </DocDetails>
</PullURIRequest>
```

Field	Description
`ts`	Request timestamp — echo in response
`txn`	Transaction ID — echo in response
`DigiLockerId`	The consumer's DigiLocker-registered ID. Carry this into the issued credential's identity binding (`idRef`) for **both** DocTypes — see Identity Binding — the DigiLocker ID.
`UDF1`	Consumer number — primary CIS lookup key
`UDF2` (NYCER)	Registered mobile number — secondary verification
`UDF2` (MPLTR)	Statement period — `YYYY-MM`, a range, or a keyword like `last-12-months`. A recurring pull with the current month fetches the latest statement.
`UDF3` (MPLTR)	Registered mobile number — secondary verification

Field	Description
`FullName`	DigiLocker profile name — DigiLocker validates name match against your response

Step 3 — Look Up Consumer in CIS

```
consumer = cis.lookup(consumer_number=udf1, mobile=mobile)
if not consumer:
    return pulluri_error_response(ts, txn, status=0, message="Consumer not found")
```

Identity Binding — the DigiLocker ID NeGD's requirement, confirmed and accepted: the `DigiLockerId` received in the `PullURIRequest` must be carried inside the issued credential as part of the subject's identity binding, for **both** `NYCER` and `MPLTR`. Populate `customerProfile.idRef` (Passport) / the subject-level identity reference (Digest) with it:

```
holder_idref = {
    "issuedBy":  "did:web:digilocker.gov.in",
    "subjectId": f"digilocker.gov.in:{digilocker_id}",   # the <DigiLockerId> from the request
}
```

The rendered certificate (`DocContent`) should also show this in human-readable form — e.g. *"Identity Binding: binding method DIGILOCKER_PULL, binding date …"* — the pattern APEPDCL's pilot implementation followed.

> **Schema gap.** `customerProfile.idRef` exists on the ElectricityCredential v1.2 schema, so the Passport can carry this binding today. `MeterDataCredential` v0.6's `credentialSubject` does **not** yet define an `idRef` (or equivalent) property — until the schema adds one, the Digest cannot carry this binding as a structured VC field, only in the rendered `DocContent`. Track this as an open schema item alongside the MPLTR rollout.

Step 4 — Call OpenCred to Issue the Credential This step differs by DocType — see Issuing NYCER (v1.2) and Issuing the Digest (MPLTR) below. Both return a signed VC, then render the PDF with a **separate** `POST /v1/credentials/package` call (see Step 5).

Step 5 — Package the PDF and VC for the response Rendering the PDF is a **separate call** — `POST /v1/credentials/package` with `formats: ["pdf"]` (the DocType handlers in Step 4 make it). An inline `packageFormats` field on the *issue* body is silently ignored, so this call is not optional — see Issuing Credentials §3.10. The PDF carries the signed credential payload in its info dictionary and embeds a scannable QR — DigiLocker stores the PDF, and verifiers can later re-extract the credential from the PDF itself.

```
import base64, json

# pdf_bytes came back from the /v1/credentials/package call in Step 4
pdf_b64 = base64.b64encode(pdf_bytes).decode()
vc_b64  = base64.b64encode(json.dumps(vc_json).encode()).decode()
```

For a custom PDF design (branded letterhead, regional language, a consumption-ladder or time-of-day card), render server-side with your own template engine and embed the QR + credential JSON as you prefer. `VcContent` is what verifiers actually parse — the PDF is for human display only and can't be checked against the issuer's signature.

Step 6 — Return the PullURIResponse

```
def build_pulluri_response(ts, txn, doc_type, uri, issued_to, valid_from, valid_to, pdf_b64, vc_b64):
    return f"""<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIResponse ver="3.0" ts="{ts}" txn="{txn}" orgId="discom">
  <ResponseStatus Status="1" ts="{ts}" txn="{txn}"/>
  <DocDetails>
    <DocType>{doc_type}</DocType>
    <URI>{uri}</URI>
```

```
    <IssuedTo>{issued_to}</IssuedTo>
    <ValidFrom>{valid_from}</ValidFrom>
    <ValidTo>{valid_to}</ValidTo>
    <DocContent format="pdf">{pdf_b64}</DocContent>
    <VcContent format="json-ld">{vc_b64}</VcContent>
    <VcType>application/ld+json</VcType>
  </DocDetails>
</PullURIResponse>"""
```

Field	Description
`Status="1"`	Success. Use `Status="0"` for errors.
`URI`	Unique identifier for this document in DigiLocker — `{issuer_id}-{DocType}-{docId}`. The final **doc-id** segment is **required**: NeGD rejects a URI that ends at the DocType with no doc-id. Form it the same way your existing electricity-bill (`ELBIL`) integration forms its doc-id; for a connection-keyed credential the CA / consumer number serves this role. NYCER: `{issuer_id}-NYCER-{consumerNo}`. MPLTR: `{issuer_id}-MPLTR-{consumerNo}-{period}` — **the period makes each statement a distinct document** (see The document key).
`IssuedTo`	Consumer's full name — DigiLocker validates this against the user's profile
`DocContent`	Base64-encoded PDF (the rendered statement / certificate card)
`VcContent`	Base64-encoded W3C VC JSON-LD — the signed credential, proof intact
`VcType`	Media type of the `VcContent` payload — **`application/ld+json`**. Required by NeGD immediately after `VcContent`; a response without it is rejected.

Handler Logic Summary

```
1. Read raw request body (keep bytes for HMAC)
2. Verify x-digilocker-hmac -> HTTP 401 if invalid
3. Parse XML -> extract DocType, UDF1..n, ts, txn
4. Lookup consumer in CIS -> error response if not found
5. Resolve / generate the consumer's holder DID
6. Branch on DocType:
     NYCER -> build v1.2 credentialSubject (energyResources[])
     MPLTR -> build MeterData v0.6 payload for the requested period
7. POST /v1/credentials/issue -> signed VC; then POST /v1/credentials/package (formats=["pdf"]) -> PDF
8. Base64-encode PDF and VC
9. Return PullURIResponse XML with Status=1 and the DocType-appropriate URI
```

Routing by DocType

```python
doc_type = request_xml.find(".//DocType").text
if doc_type == "ELBIL":
    return handle_elbil(request_xml)   # existing electricity bill, if you serve it
elif doc_type == "NYCER":
    return handle_nycer(request_xml)   # Electricity Credential v1.2
elif doc_type == "MPLTR":
    return handle_mpltr(request_xml)   # Consumer Meter Digest
```

```
else:
    return pulluri_error_response(ts, txn, status=0, message="Unsupported DocType")
```

Issuing NYCER — the Consumer Energy Passport (Electricity Credential v1.2)

`NYCER` is the existing, unchanged DocType; only its payload schema is upgraded. Today's NYCER credential carries a flat set of customer-and-connection fields. The IES Electricity Credential, finalised at **v1.2** as the **Consumer Energy Passport**, carries far more: every physical asset behind the connection — meter, solar, battery, EV, inverter, controllable load — as one typed resource model with unit-bearing quantities.

What changed from the flat shape

	Today (flat)	v1.2 Electricity Credential
Customer & connection	Flat fields on `credentialSubject`	Non-PII `customerProfile` + PII `customerDetails`
Meter	`meterNumber` / `meterType` fields	`energyResources` entry, `type METER`, `attributes.meterCapability` + `energyDirection`
Generation, storage, EV, …	Not carried	Typed resources — `SOLAR_PV`, `BESS`, `EV_CHARGER`, `INVERTER`, …
Power & capacity values	n/a	`QuantitativeValue {value, unit}` — e.g. `maxExport {10, kW}`
Asset topology	Not carried	`parentResources` / `subResources` — submetering, parallel metering
Consumption / tariff	Not carried	`consumptionProfiles[]`, `sanctionedLoad` as `{value, unit}`

credentialSubject shape

```
credentialSubject
├─ id                       (optional)  customer DID
├─ customerProfile          (required)  non-PII
│  ├─ customerNumber        (required)  utility CA number
│  ├─ idRef                 (optional)  identity binding — the DigiLocker ID from the pull request
│  ├─ energyResources[]     (required)  typed assets, min 1 (at least one METER)
│  └─ consumptionProfiles[] (optional)  tariff / load per meter (QV units)
└─ customerDetails          (optional)  PII: fullName, address, connection date
```

Seven typed `EnergyResource` kinds, each aligned to CIM classes (IEC 61968 / 61970):

Kind	`type` values	CIM alignment
Meter	`METER`	`cim:Meter` / `EndDevice`
Generator	`SOLAR_PV`, `WIND`, `HYDRO`, `BIOGAS`, `CHP`, `FUEL_CELL`	`cim:GeneratingUnit`
Storage	`BESS`	`cim:BatteryUnit`
EV Charger	`EV_CHARGER`, `EV_V2G`	`cim:EVChargingStation`
Inverter	`INVERTER`	`cim:PowerElectronicsConnection`
Load	`SMART_HVAC`, `SMART_WATER_HEATER`, `CONTROLLABLE_LOAD`	`cim:EnergyConsumer`
Network	`DT`, `BUS`, `FEEDER`, `MICROGRID`	`cim:PowerTransformer` / `Feeder`

Common attributes (all kinds): `make`, `model`, `maxExport`, `maxImport`, `commissioningDate`, `location`. Per-kind adds, e.g. Storage -> `storageCapacity`, `storageType`, `stateOfHealthPct`; Meter -> `meterCapability`, `energyDirection`,

`functions[]`. Every power or capacity figure is a `QuantitativeValue` — a `{value, unit}` pair with short unit aliases (`kW`, `kWh`, `kVA`, `kVAR`, `kV`) mapped to QUDT IRIs in the JSON-LD context.

> Only the meter is mandatory — a consumer with no DERs has one `METER` entry. PII (the name) lives only in `customerDetails`, so a verifier needing asset facts but not identity can be given `customerProfile` alone. The v1.1 type aliases `SOLAR` and `BATTERY` remain valid for backward compatibility, but `SOLAR_PV` / `BESS` are preferred.

The OpenCred issue call

```python
import requests
from datetime import datetime, timezone, timedelta

IST = timezone(timedelta(hours=5, minutes=30))

credential_response = requests.post(
    "http://opencred:3100/v1/credentials/issue",
    headers={"Authorization": f"Bearer {OPENCRED_API_KEY}"},
    json={
        "issuerDid":  ISSUER_DID,               # e.g. "did:web:ies.discom.example"
        "schemaId":   "ies/electricity-credential/v1.2",
        "additionalTypes": ["ElectricityCredential"],
        "proofFormat": "vc-jwt",
        "validFrom":  datetime.now(IST).isoformat(),
        "validUntil": (datetime.now(IST) + timedelta(days=365 * 5)).isoformat(),
        "subjectDid": holder_did,
        "credentialSubject": {
            "id": holder_did,
            "customerProfile": {
                "customerNumber": consumer.consumer_number,
                "idRef": {
                    "issuedBy":  "did:web:digilocker.gov.in",
                    "subjectId": f"digilocker.gov.in:{digilocker_id}",  # <DigiLockerId> from Step 2 —
                     ↪ NeGD's identity-binding requirement
                },
                "energyResources": [
                    {
                        "id":   consumer.meter_id,
                        "type": "METER",
                        "attributes": {
                            "meterCapability":   consumer.meter_capability,   # e.g. "AMI"
                            "energyDirection":   consumer.energy_direction,    # e.g. "Bidirectional"
                            "functions":         consumer.meter_functions,     # ["NetMetering", "ToU"]
                            "applicationProtocol": "DLMS_COSEM",
                        },
                    },
                    # … append SOLAR_PV / BESS / EV_CHARGER / INVERTER entries as present,
                    #    each power/capacity field a {"value": …, "unit": "kW"|"kWh"|…} pair,
                    #    each linked to the meter via "parentResources": [meter_id]
                ],
                "consumptionProfiles": [
                    {
                        "meterId":           consumer.meter_id,
                        "sanctionedLoad":    {"value": consumer.sanctioned_load_kw, "unit": "kW"},
                        "tariffCategoryCode": consumer.tariff_code,
                        "premisesType":      consumer.premises_type,
                        "connectionType":    consumer.connection_type,
                        "paymentMode":       consumer.payment_mode,
```

```
                    }
                ],
            },
            "customerDetails": {
                "fullName": consumer.full_name,
                "installationAddress": {
                    "geo": {"type": "Point",
                            "coordinates": [consumer.longitude, consumer.latitude]},
                    "address": {
                        "streetAddress":   consumer.address_line,
                        "addressLocality": consumer.city,
                        "addressRegion":   consumer.state,
                        "postalCode":      consumer.pincode,
                        "addressCountry":  "IN",
                    },
                },
                "serviceConnectionDate": consumer.connection_date.isoformat(),
            },
        },
        "revocationRegistryUrl": DEDI_REVOCATION_URL,
    }
).json()

vc_json = credential_response["credential"]

# Render the PDF with a SEPARATE packaging call. Packaging is a distinct
# rendering step, not part of issue — an inline `packageFormats` field on the
# issue body is silently ignored (see Issuing Credentials §3.10). Package the
# credential as issued, proof intact, so the QR embedded in the PDF stays
# independently verifiable.
package_response = requests.post(
    "http://opencred:3100/v1/credentials/package",
    headers={"Authorization": f"Bearer {OPENCRED_API_KEY}"},
    json={"credential": vc_json, "formats": ["pdf"]},
).json()
pdf_bytes = base64.b64decode(
    next(o for o in package_response["outputs"] if o["format"] == "pdf")["data"]
)

# Inject the schema-required issuer object before delivery — OpenCred returns
# issuer as the DID string, but the ElectricityCredential schema requires the
# full object with name and (optional) idRef. This invalidates the original
# proof; re-sign downstream if you need a single signed-and-conformant artifact.
vc_json["issuer"] = {
    "id":   ISSUER_DID,
    "name": ISSUER_NAME,
    "idRef": {
        "issuedBy":  IES_REGISTRY_DID,         # DID of the indiaenergystack.in namespace on dedi.global
        "subjectId": IES_REGISTRY_SUBJECT_ID,  # e.g. indiaenergystack.in:discom
    },
}
```

The URI for NYCER stays keyed on the consumer number — a refresh overwrites the last, which is correct for a slow-changing connection credential:

```
uri = f"in.gov.discom-NYCER-{consumer.consumer_number}"   # trailing segment is the required doc-id (see
↪  URI note in Step 6)
```

What DigiLocker needs to accept for v1.2

1. **Accept the v1.2 schema for NYCER.** Validate against the published v1.2 context and JSON Schema — a `customerProfile` with `energyResources[]` (min one `METER`), optional `consumptionProfiles[]`, and `customerDetails` for PII. Power/capacity fields are `{value, unit}` objects, not bare numbers.
2. **Pass `VcContent` through unchanged.** The signed JSON-LD is the v1.2 credential as issued, proof intact — no reshaping. The proof is over the canonical form, and the QUDT unit context is part of it.
3. **Carry resource fields into the Certificate XML.** Extend the `DataContent` block so meter and DER resources (`type`, key attributes with value+unit) appear for DigiLocker-native parsers, alongside the existing connection and address fields.
4. **Render whatever resources are present.** Drive the card from `energyResources[]` as a list, showing each value with its unit. A meter-only consumer shows one row; a prosumer shows meter, solar, battery, EV. New kinds and unit types need no card change.
5. **Store and surface the DigiLocker identity binding.** `customerProfile.idRef` carries the `DigiLockerId` from the pull request (see Identity Binding) — reflect it on the rendered certificate alongside the binding date.

The Consumer Meter Digest (DocType `MPLTR`)

The **Consumer Meter Digest** is a DISCOM-issued, digitally signed credential carrying a consumer's own meter readings for a chosen window of time. Where NYCER attests to slow-changing facts — who holds the connection, what assets sit behind the meter — the Digest is the consumer's **energy statement**: the meter records a reading roughly every fifteen minutes (~96 blocks a day), and the Digest packages those readings for a requested period into a signed document the consumer can hold and present.

A statement, not a bill. A bill is the latest snapshot, overwriting the last. A statement is a series the consumer can pull by period ("April, May, last twelve months"), each one kept rather than replaced — the whole point of the `MPLTR` integration, and why the document key carries the period (see The one real gap).

Schema compliance — `MeterDataCredential` v0.6 The Digest is **not a new schema**. It is a consumer-facing issuance of the standard `MeterDataCredential` v0.6 — a W3C VC 2.0 subclassing `EnergyCredential` v2.0 and wrapping a `MeterData` v0.6 payload — the same credential AMISPs and MDMs issue provider-to-DISCOM in the Beckn data-exchange flow. Only the trigger differs: the consumer requests it for a period.

The envelope — `issuer` (with regulatory `licenseNumber`), `validFrom` / `validUntil`, DeDi `credentialStatus`, `proof` — is inherited from `EnergyCredential` v2.0. What the Digest defines is `credentialSubject.meterData`: a `MeterData` v0.6 payload of one or more typed profiles.

```
MeterDataCredential  (W3C VC 2.0  ·  subclass of EnergyCredential v2.0)
├─ @context        W3C credentials/v2 + EnergyCredential v2.0 + MeterDataCredential v0.6
├─ type            ["VerifiableCredential", "MeterDataCredential"]
├─ issuer          id, name, licenseNumber  (DISCOM / AMISP / MDM DID)
├─ validFrom / validUntil                  (inherited envelope)
├─ credentialStatus  DeDi revocation registry  (inherited)
├─ credentialSubject
│  ├─ id           consumer / asset DID
│  └─ meterData    MeterData v0.6 payload — single profile OR array of profiles
│                  profileType  in  CUSTOMER | INTERVAL | DAILY | MONTHLY |
│                  BILL_DETAILS | INSTANTANEOUS | EVENT | ALARM | DESCRIPTOR
└─ proof           Ed25519Signature2020
```

The **period and shape** the consumer chose are expressed by the profiles themselves:

- A **twelve-month statement** is an array of `MONTHLY` profiles — one `MonthlyProfile` object per month, each with a `timePeriod` (`P1M`) and its `readings`.
- **Raw analytics data** is an `INTERVAL` profile (`PT15M` / `PT30M` intervals, ~96 readings a day), preceded by a `DESCRIPTOR` profile that the interval profile references via `payloadDescriptorSetRef` / `compactSequenceRef`.

A monthly-bill view and a 15-minute analytics feed are the **same credential type** with different `meterData` profiles. The consumer's requested window simply bounds which profiles the DISCOM returns — within the scope authorised

by the originating request.

Example — a twelve-month statement (`MONTHLY` profile) The JSON-LD that travels in `VcContent`, per the canonical MeterDataCredential v0.6 monthly example — one `MonthlyProfile` per month, carried as an array (trimmed proof; first month shown in full):


```
{
  "@context": [
    "https://www.w3.org/ns/credentials/v2",
    "https://schema.nfh.global/EnergyCredential/v2.0/context.jsonld",
    "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataCredential/v0.6/context.jsonld"
  ],
  "id": "urn:uuid:9b3c1f0a-2d4e-4ba8-9f1c-1e7a90c8a5d2",
  "type": ["VerifiableCredential", "MeterDataCredential"],
  "issuer": {
    "id": "did:web:ies.discom.example",
    "name": "Example State Distribution Company Limited",
    "licenseNumber": "SERC-DISCOM-2025-007"
  },
  "validFrom": "2026-05-15T10:00:00+05:30",
  "validUntil": "2027-05-15T10:00:00+05:30",
  "credentialStatus": {
    "id": "https://dedi.global/dedi/query/did:web:ies.discom.example/vc-revocation-registry",
    "type": "dediregistry",
    "statusPurpose": "revocation",
    "statusListCredential":
     ↪  "https://dedi.global/dedi/lookup/did:web:ies.discom.example/vc-revocation-registry"
  },
  "credentialSubject": {
    "id": "did:key:z6MkjVQ8r4f3rPuY7CG2D6Lf8WJxJBs5sjkR8d3v2Bv4nP4Z",
    "meterData": [
      {
        "@context":
         ↪  "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/context.jsonld",
        "@type": "MonthlyProfile",
        "profileType": "MONTHLY",
        "meterRefs": [
          { "scheme": "DID", "value": "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }
        ],
        "serviceDeliveryPointRefs": [
          { "scheme": "DID", "value": "did:web:ies.discom.example:connections:SDP-A-12345" }
        ],
        "timePeriod": { "start": "2025-05-01T00:00:00+05:30", "duration": "P1M" },
        "readings": [
          { "readingType": "kWh imp", "value": 391.0, "validationStatus": "VALID", "source": "METER" }
        ]
      },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-06-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 412.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-07-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 478.0 }] },
```

```
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-08-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 463.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-09-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 421.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-10-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 384.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-11-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 352.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2025-12-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 339.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-01-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 367.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-02-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 358.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-03-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 405.0 }] },
      { "@type": "MonthlyProfile", "profileType": "MONTHLY", "meterRefs": [{ "scheme": "DID", "value":
       ↪  "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }], "timePeriod": { "start":
       ↪  "2026-04-01T00:00:00+05:30", "duration": "P1M" }, "readings": [{ "readingType": "kWh imp",
       ↪  "value": 437.0 }] }
    ]
  },
  "proof": { "type": "Ed25519Signature2020", "proofValue": "z5wQ…" }
}
```


Example — raw analytics (`INTERVAL` profile + `DESCRIPTOR`) For raw analytics data the same wrapper carries an `INTERVAL` profile — `PT15M` intervals, around 96 readings a day — preceded by a `DESCRIPTOR` profile the interval profile references via `payloadDescriptorSetRef` and `compactSequenceRef`. The `meterData` value is an **array of profiles** (fragment, per the canonical v0.6 interval example; envelope as in the statement example above):


```
{
  "credentialSubject": {
    "id": "did:key:z6MkjVQ8r4f3rPuY7CG2D6Lf8WJxJBs5sjkR8d3v2Bv4nP4Z",
    "meterData": [
      {
        "@context":
         ↪  "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/context.jsonld",
        "@type": "PayloadDescriptorProfile",
        "profileType": "DESCRIPTOR",
        "id": "DESC-INTERVAL-2026Q2",
        "payloadDescriptorSets": [
```

```
          {
            "name": "IntervalLoadSurveySet",
            "payloadDescriptors": [
              { "readingType": "kWh imp block", "name": "Active energy import – block incremental", "unit":
                ↪ "kWh", "flowDirection": "IMPORT", "reportedMode": "USAGE", "obis": "1.0.1.29.0.255" }
            ],
            "compactSequences": [
              { "name": "IntervalSeq", "sequenceItems": [{ "readingType": "kWh imp block" }] }
            ]
          }
        ]
      },
      {
        "@context":
         ↪ "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/context.jsonld",
        "@type": "IntervalProfile",
        "profileType": "INTERVAL",
        "meterRefs": [
          { "scheme": "DID", "value": "did:web:ies.discom.example:assets:meter:DISCOM-SM-2025-654321" }
        ],
        "payloadDescriptorSetRef": "DESC-INTERVAL-2026Q2",
        "compactSequenceRef": "IntervalSeq",
        "intervalPeriod": { "start": "2026-05-01T00:00:00+05:30", "duration": "PT15M" },
        "intervals": [
          { "id": 0, "payloads": [0.18] },
          { "id": 1, "payloads": [0.16] }
        ]
      }
    ]
  }
}
```


Mapping to DigiLocker as it works today DigiLocker already does everything the Digest needs — the **same Pull URI mechanism** the electricity-connection credential uses. The Digest is a new document type returned through the same response, with the signed `MeterDataCredential` in `VcContent` and a rendered statement in `DocContent` / `DataContent`. The one change that matters is the document key.

DigiLocker mechanism (today)	How the Digest uses it
Pull URI endpoint + DocType	`MPLTR` registered as an additional record for the Digest; DigiLocker calls the **same Pull URI** the connection credential uses.
URI — the document key	Put the period into the URI: `in.gov.{discom}-MPLTR-{consumerNo}-{YYYY-MM}`. This is the composite key that lets every period coexist instead of overwriting — **confirmed on the 12 June 2026 call**: URI uniqueness is managed on the issuer side, DigiLocker needs no change.
UDF request fields	Consumer number as `UDF1`; the period (or month) as a UDF — so the consumer pulls a chosen window, and a recurring pull fetches the current one.
`VcContent` (JSON-LD)	The signed `MeterDataCredential` as-is — proof intact — for verifiers to parse and check.
`DocContent` (PDF) / `DataContent` (XML)	A rendered statement card — consumption ladder or time-of-day — built from a configurable template, with a summary the consumer reads on screen.

DigiLocker mechanism (today)	How the Digest uses it
OAuth consent pull (`/xml/{uri}`)	A third-party app pulls a chosen period directly with consumer consent — no app-switching.

The one real gap — the document key The electricity bill and NYCER are keyed on the consumer number alone, so each refresh overwrites the last. The Digest must be keyed on **consumer number plus period**, so April's and May's statements sit side by side as distinct documents.

```
# NYCER — overwrite on refresh (correct for a connection credential)
uri = f"in.gov.discom-NYCER-{consumer.consumer_number}"   # trailing segment is the required doc-id (see
 ↪  URI note in Step 6)

# MPLTR — period in the key, so every statement coexists
uri = f"in.gov.discom-MPLTR-{consumer.consumer_number}-{period}"   # period e.g. "2026-05" or
 ↪  "2025-05_2026-04"
```

Issuing the Digest (`MeterDataCredential` v0.6) Inside `handle_mpltr`, resolve the requested window from `UDF2`, query MDMS/MDM for the matching readings, build the `MeterData` v0.6 payload, and call OpenCred. For a consumer-pulled Digest, `validUntil` may be set **shorter than the v0.6 default** where a verifier wants freshness (loan applications often want fresh; subsidy portals tolerate a week).

```
import requests
from datetime import datetime, timezone, timedelta

IST = timezone(timedelta(hours=5, minutes=30))

# 1. Resolve the requested window from UDF2 -> (start, end, profile_type, uri_period)
window = resolve_period(udf2)            # e.g. "last-12-months" -> MONTHLY, 2025-05..2026-04
readings = mdm.query(consumer.meter_id, window.start, window.end, window.granularity)

# 2. Build the MeterData v0.6 payload — an array of profile objects
#    (one MonthlyProfile per month for a statement; DESCRIPTOR + INTERVAL for analytics).
#    Every profile carries the required meterRefs; readings use the v0.6 Reading shape
#    (readingType / value / openingValue / closingValue / validationStatus / source).
meter_data = [
    {
        "@context":
         ↪  "https://india-energy-stack.github.io/ies-accelerator/schemas/MeterData/v0.6/context.jsonld",
        "@type": "MonthlyProfile",
        "profileType": "MONTHLY",
        "meterRefs": [{"scheme": "DID", "value": consumer.meter_did}],
        "serviceDeliveryPointRefs": [{"scheme": "DID", "value": consumer.sdp_did}],
        "timePeriod": {"start": month.start.isoformat(), "duration": "P1M"},
        "readings": [
            {"readingType": "kWh imp", "value": month.kwh_import,
             "validationStatus": "VALID", "source": "METER"}
        ],
    }
    for month in readings.by_month()
]

# 3. Issue via OpenCred — same instance, MeterDataCredential schema, short validUntil
credential_response = requests.post(
    "http://opencred:3100/v1/credentials/issue",
    headers={"Authorization": f"Bearer {OPENCRED_API_KEY}"},
    json={
```

```
        "issuerDid":  ISSUER_DID,
        "schemaId":   "ies/meter-data-credential/v0.6",
        "additionalTypes": ["MeterDataCredential"],
        "proofFormat": "vc-jwt",
        "validFrom":  datetime.now(IST).isoformat(),
        "validUntil": (datetime.now(IST) + timedelta(days=7)).isoformat(),   # freshness window
        "subjectDid": holder_did,
        "credentialSubject": {
            "id": holder_did,
            "meterData": meter_data,
            # NeGD's identity-binding requirement applies here too, but MeterDataCredential v0.6
            # has no idRef (or equivalent) field on credentialSubject yet — see the schema-gap
            # note under Identity Binding. Until the schema adds one, render the DigiLocker ID
            # binding into DocContent only (see Step 6 below).
        },
        "revocationRegistryUrl": DEDI_REVOCATION_URL,
    }
).json()

vc_json = credential_response["credential"]

# Render the PDF with a separate packaging call (same pattern as NYCER, and
# Issuing Credentials §3.10) — package the credential as issued, proof intact.
package_response = requests.post(
    "http://opencred:3100/v1/credentials/package",
    headers={"Authorization": f"Bearer {OPENCRED_API_KEY}"},
    json={"credential": vc_json, "formats": ["pdf"]},
).json()
pdf_bytes = base64.b64decode(
    next(o for o in package_response["outputs"] if o["format"] == "pdf")["data"]
)

# Inject the schema-required issuer object (with licenseNumber) — same pattern as NYCER.
vc_json["issuer"] = {
    "id":   ISSUER_DID,
    "name": ISSUER_NAME,
    "licenseNumber": DISCOM_LICENSE_NUMBER,      # e.g. "SERC-DISCOM-2025-007"
}

# 4. Period-keyed URI — this is the one structural difference from NYCER
uri = f"in.gov.discom-MPLTR-{consumer.consumer_number}-{window.uri_period}"
```

Revocations are rare in practice — Digests are usually short-lived enough to expire before any revoke would matter — but when a revoke is needed, pass an optional `reason` such as `"data-correction"` or `"holder-request"` on `POST /v1/credentials/revoke`; see Issuing Credentials -> Revoke.

What the consumer can do with it Three actions, all on DigiLocker's existing rails:

1. **View** a readable statement card (rendered from `DocContent` / `DataContent`) — a consumption ladder or time-of-day profile, with a summary the consumer reads on screen.
2. **Download** the signed document locally.
3. **Share** it — by **QR scan** in person, or by granting **OAuth consent** so a third-party app pulls the chosen period directly, with no app-switching.

In every mode the signed JSON moves as-is (see Step 5 above). Where DigiLocker adds the holder's own signature (e.g. on a QR-scan share), it must be an **envelope over the credential** — a verifiable-presentation pattern, not a reshaping of the credential body — so the issuer's proof stays intact and independently verifiable.

The ask to DigiLocker (MPLTR)

Capability	Status	What it means
Introduce the Digest document type	**Done** — NeGD has allotted `MPLTR` (Meter Document)	Register the Consumer Meter Digest (a `MeterDataCredential`) as `MPLTR` on API Setu, alongside the existing `NYCER` endpoint.
Key the URI on consumer + period	**Agreed in principle** on the 12 June 2026 call; written confirmation still owed	Put the period into the URI so multiple periods coexist rather than overwriting. URI uniqueness is managed issuer-side — DigiLocker needs no change, only to treat each unique URI as a distinct, independently listable/deletable document.
Accept period as a pull parameter	Open ask	Let the consumer pull a chosen historical window via `UDF2`, and let a recurring pull with the current period fetch the latest statement.
Render from a configurable template	Open ask	Drive the `DocContent` / `DataContent` card from an externalised, per-DocType renderer — not hardcoded fields — so the card need not change when a parameter does.
Position on optional `DocContent` for time-series types	Open ask — IES provides a simplified PDF meanwhile	Whether `DocContent` may be made optional for VC-only / time-series DocTypes in future, since the full ~96-blocks-a-day series lives only in `VcContent`.

Everything else reuses the connection-credential flow unchanged. Format is **W3C VC 2.0 JSON-LD** for both DocTypes (not SD-JWT) — selective disclosure via SD-JWT is a joint roadmap item once this initial integration is stable.

Error Response Format

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<PullURIResponse ver="3.0" ts="{ts}" txn="{txn}" orgId="discom">
  <ResponseStatus Status="0" ts="{ts}" txn="{txn}">Consumer not found</ResponseStatus>
</PullURIResponse>
```

Phase 2 — Consumer Shares Credential with a Verifier

Phase 2 requires no DISCOM involvement. Consumers share their NYCER credential or Meter Digest from DigiLocker using the standard DigiLocker OAuth flow. See Issuing Credentials -> Verify for the verifier-side implementation.

Security Checklist

- [ ] HMAC verified on every inbound request before any processing
- [ ] `hmac.compare_digest` used (not `==`) to prevent timing attacks
- [ ] Raw request bytes used for HMAC computation (not re-serialised XML)

- [ ] `customerProfile.idRef` carries the `DigiLockerId` from the request (identity binding, per NeGD) — logged only where necessary, never alongside the full Aadhaar / government ID
- [ ] Digest `meterData` carries readings only — no PII beyond the consumer/meter identifiers needed to bind the statement
- [ ] Period UDF validated and bounded to the scope authorised by the originating request (no arbitrary historical ranges)
- [ ] API key stored in environment variable / secrets manager, not in code
- [ ] Endpoint accessible only over HTTPS (TLS 1.2+)
- [ ] Rate limiting applied (DigiLocker can retry aggressively on timeout)

Reference

- Consumer Meter Digest — use-case guide
- Electricity Credential — schema reference
- OpenCred Documentation — credential service API reference
- API Setu — DigiLocker issuer registration and partner documentation
- DigiLocker Technical Specification v1.0.5 (the former partner-portal download, `partners.digitallocker.gov.in`, is offline; obtain the current spec through your API Setu / NeGD contact)

Build your Internal-facing Adapter

> Write the **Part-2 mapping** between your internal systems and the IES schemas. This is the only IES work where you write code — the rest is configuration. About 1–3 weeks for the first use case; subsequent use cases add only a few days each.

This is where the engines — OpenCred and/or ONIX — get fed from your internal systems. Per-use-case adapter shapes are in the individual Use Case Implementation Guides, linked from the table below.

Before you start

- Setting up Register complete — your `did:web` resolves and your DeDi namespace is verified.
- The engine your first use case needs is running: **OpenCred** for credential use cases (Issuing Credentials) and/or **ONIX** for data-exchange use cases (Setting up Discover & Exchange). You wire your mapping into whichever you set up.
- Read access to the internal system that holds the data (CIS, MDM, HES, DERMS, ERP) and a developer who can write ~200–1,000 lines of glue code.

What the adapter is

The IES adapter has two parts: a ready-made **engine** and your **mapping**. There is one engine per capability — **ONIX** for use cases that run over a Beckn network (Setting up Discover & Exchange), **OpenCred** for credential use cases (Issuing Credentials) — and you build an internal-facing mapping for each engine your use cases need.

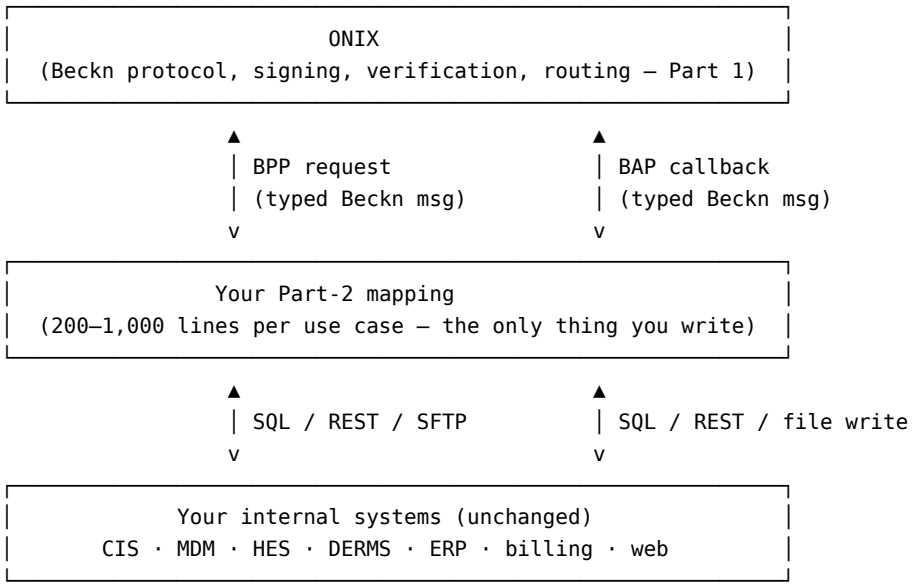
Part	What it does	You build it?
Part 1 — ONIX *(Beckn-network use cases)*	Finds other systems, exchanges messages, signs and verifies, routes callbacks	**No** — ready-made, the same for everyone
Part 1 — OpenCred *(credential-only use cases)*	Issues, verifies and revokes W3C Verifiable Credentials with your signing key	**No** — ready-made, the same for everyone

Part	What it does	You build it?
Part 2 — your mapping(s)	Translates between your data format and the IES specs, feeding the engine(s) above	**Yes** — specific to your organisation, set up once

The mapping is small. For a single use case it is typically 200–1,000 lines of code. It is the only piece that knows about your internal field names, your tariff codes, your meter SLNO format, your CIS schema.

Where the mapping lives

For Beckn-network use cases, the mapping plugs into ONIX as one or more **BPP handlers** (provider side) and **BAP callbacks** (buyer side). ONIX delivers a typed Beckn message to your handler; your handler talks to your CIS / MDM / DERMS / ERP and returns a typed Beckn response. For credential-only use cases the same pattern applies with **OpenCred** in place of ONIX: your mapping sits between your internal systems and OpenCred's issue API (see Task 4).



Pick your first use case

The pilot DISCOMs each picked one use case to ship first, then layered the rest on the same identity, network and adapter foundation. The natural first choice depends on what data is easiest for you to expose:

First use case	Internal system you'll touch	Schema	Why first
Consumer Energy Passport	CIS / billing master	ElectricityCredential v1.2	Smallest schema; consumer-facing value is immediate
Smart Meter Data Exchange	HES / MDM	MeterData v0.6	Solves a tangible AMISP-DISCOM pain; mostly DataOps work
Consumer Meter Digest	MDM (read path)	MeterDataCredential v0.6	Builds on the Passport adapter; small extra surface

First use case	Internal system you'll touch	Schema	Why first
DER Visibility	DER / inverter registration database	ElectricityCredential v1.2 energyResources	If rooftop / EV registrations are your priority
DISCOM Regulatory Filing	Regulatory affairs / finance	ArrFiling v0.5	If you have an ARR due

---

The five tasks for any use case

The shape is the same regardless of which use case you pick first.

Task 1 — Map fields

For each IES field in the schema, identify the corresponding column / API field / file format in your internal systems. Write it down as a table:

IES field	Your source	Notes
`customerProfile.customerNumber`	CIS `CA_NUMBER`	Plain string, no transformation
`consumptionProfiles[].sanctionedLoad.value`	CIS `SANC_LOAD_KW`	Multiplied by 1.0; unit always `kW` for residential
`consumptionProfiles[].sanctionedLoad.unit`	hardcoded `"kW"`	LT-Domestic always reported in kW
`consumptionProfiles[].tariffCategoryCode`	CIS `TARIFF_CAT` -> translate via a small lookup	Your codes (`LT1A`, `HT2C` etc.) -> IES codes
`customerProfile.energyResources[].id`	`did:web:<your-domain>:assets:meter:<MET_SLNO>`	Compose at issuance time

This table **is** the mapping. The code that follows is mechanical.

Task 2 — Write the field transformations

Most fields are pass-through. A few need:

- **Unit packaging** — your `5` becomes `{ "value": 5, "unit": "kW" }` per QuantitativeValue.
- **Tariff code translation** — your internal `LT1A` becomes a stable IES code; keep the lookup in one place.
- **Date formatting** — ISO 8601 with timezone offset (`2026-01-10T00:00:00+05:30`).
- **DID composition** — `did:web:<your-domain>:assets:meter:<existing-serial>` (your serial is preserved verbatim).

Task 3 — Validate against the schema

Use the canonical JSON Schema for the use case you're shipping:

```
python3 scripts/validate_schema.py \
  schemas/ElectricityCredential/v1.2/schema.json \
  my-test-output.json
```

The repo ships this validator and a `make validate` target. Examples to validate against live in `schemas/<family>/<version>/examples/`.

This proves repository-local structural conformance against the schema shown above — see Conformance Checklist — What this checklist proves for exactly what it does and doesn't establish before you call a use case done.

Task 4 — Wire into your engine (ONIX and/or OpenCred)

Beckn-network use cases — wire into ONIX. ONIX delivers a typed Beckn message to a handler endpoint you register. The handler does Tasks 1–3 in real time:

1. Receive the inbound message.
2. Extract scope (which meter, which date range, which credential).
3. Query your internal systems.
4. Build the IES-shaped response.
5. Validate against the schema.
6. Return.

The Beckn wiring (subscriber id, callback URL, route registration) is in ONIX config; the handler endpoint is your code.

Credential use cases — wire into OpenCred. Your mapping assembles the credential subject from CIS / MDM data (Tasks 1–3) and POSTs it to OpenCred's `/v1/credentials/issue`; OpenCred signs with your key and returns the credential for delivery to the holder. The issue / verify / revoke walkthrough is in **Issuing Credentials**.

Task 5 — Test end-to-end

Beckn-network use cases. Use the Data Exchange devkit to send your handler real Beckn messages from a sandbox counterparty. Iterate until all required schema fields are populated correctly, signatures verify on the receiving side, and the end-to-end round trip succeeds for at least one realistic record.

Credential use cases. There is no Beckn counterparty — test the issuance path directly: have your mapping assemble the subject from a real record and POST it to OpenCred's `/v1/credentials/issue`, then run the §3.9 smoke test (issue -> resolve your `did:web` -> verify -> revoke -> re-check). Iterate until it issues a schema-valid, signature-verifiable credential for at least one realistic record.

Then graduate from sandbox -> testnet -> prod with the same adapter.

Reuse across use cases

The biggest win comes from doing this once. After your first use case is live, adding the next typically means:

- A new mapping table (Task 1) — usually one page of YAML
- A few new transformations (Task 2) — units, codes, identifier composition
- A new handler endpoint (Task 4) — ~50 lines of code
- A new schema validator config (Task 3) — one line

No new identity. No new ONIX deployment. No new DeDi namespace. No new network onboarding.

Checklist

For the first use case:

- [ ] Use case picked (from the table above)
- [ ] Schema field -> internal-system mapping table written
- [ ] Field transformations implemented (units, codes, dates, DIDs)
- [ ] Output validates against the schema for at least three real records
- [ ] Mapping wired into its engine — BPP handler registered with ONIX (Beckn-network use case) or issuance feed calling OpenCred (credential-only use case)
- [ ] One realistic record exchanged or issued end-to-end with a counterparty / verifier

When all six are ticked, the adapter for your first use case is done. Move on to the **Conformance Checklist**, then publish.

For the second and subsequent use cases, only Tasks 1–4 repeat. The sandbox test (Task 5) reruns quickly, and conformance and all the identity / network setup are reused.

Security Model

How IES treats security: as a first-class design concern, layered on top of the power sector's existing operational-security standards, with software-layer guarantees — signed services, verifiable endpoints, cryptographic agility — that the specifications require of every participant.

This page states the security posture of the IES specifications in three parts. Detailed, testable requirements live with the building blocks they secure — Register, Discover, Exchange and Verifiable Credentials — and in the schema definitions themselves.

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1. Security is a first-class citizen

Security in IES is not a hardening step applied after the fact; it is built into the architecture's core primitives. Every participant's identity is a cryptographic identity (a W3C DID) before it is anything else. Every interaction between participants — a discovery call, a data exchange, a credential issued to a consumer — is signed by construction, so every message is attributable to a registered, verifiable actor. There is no anonymous or unauthenticated path through the stack: the same Register -> Discover -> Exchange sequence that makes data flow possible is also what makes each flow authenticated, authorised and auditable.

2. IES builds on the sector's existing security — it does not replace it

The power sector already operates under established operational- and device-level security regimes: **IEC 62351** (security for power-system communication protocols), **IEC 62443** (security for industrial automation and control systems), the **DLMS/COSEM security suites** protecting meter communications, and the **CEA cyber-security guidelines** that regulate Indian power-sector IT and OT systems. IES does not reimplement, weaken or bypass any of these.

IES operates one layer above: at the point where structured data, already produced inside a utility's secured perimeter, is shared with another organisation or with a consumer. A DISCOM's HES/MDM chain, SCADA systems and meter fleet keep their existing protections; the IES adapter sits at the utility's edge and adds interoperable, verifiable exchange on top of them. Sector security standards govern how data is produced and held; IES governs how it is shared.

3. What IES adds at the software layer

On top of the sector's existing controls, the IES specifications bring the security techniques of modern digital public infrastructure to every exchange:

- **Every service is signed.** There are no unsigned APIs in IES. Beckn-protocol messages carry participant signatures, and credentials carry W3C proofs — a receiver can always verify who sent a payload and that it was not altered in transit.
- **Every endpoint is verifiable.** A participant's `did:web` identity and its DeDi directory entry let any counterparty resolve, and cryptographically check, who it is talking to before any data moves — trust is established by lookup and verification, not by bilateral arrangement.
- **Verifiability travels with the data.** A verifiable credential can be checked by a bank, a marketplace or a regulator that has never met the issuer, because the proof is inside the artifact itself rather than dependent on a trusted channel.
- **Cryptographic agility.** Signature suites and key material are referenced indirectly (through DID documents and proof metadata), not hard-coded, so algorithms can be rotated and upgraded — including migration to post-quantum schemes as standards mature — without redesigning the stack.

Conformance Checklist

> **The final check before you publish.** Verify your interface is correct end-to-end and (for Beckn participants) submit for IES network membership in production. About 1 day.

Before you start

- Setting up Register complete for your role.
- The engine(s) your use cases need are running: Issuing Credentials (B2C) and/or Setting up Discover & Exchange (B2B).
- At least one use case wired through Build your Internal-facing Adapter.

Only the checks for the rails you actually use apply. The checklist below is tagged: **[all]** applies to everyone, **[credential]** only to credential issuers (B2C), **[beckn]** only to Beckn-network participants (B2B). A credential-only issuer skips every **[beckn]** item and the production-network submission at the end; a pure data-exchange participant skips every **[credential]** item.

What this checklist proves — and what it doesn't

Every item below is either something you can run and check yourself, or a step in a documented network-membership process. Neither is the same thing as full conformance. Three boundaries matter beyond schema validation (see What schema validation proves further down for that boundary specifically):

- **JSON-LD conformance** — that your `@context` resolves and the payload expands as valid JSON-LD, not merely valid JSON — is a separate check from JSON Schema validation and is not asserted by anything on this page.
- **Beckn/network protocol interoperability** — a real `discover`/`confirm` round-trip against a live counterparty on a reachable network — is only as good as the sandbox peer you actually reach it against, and is distinct from schema validation.
- **Certification and production participation** are a dated determination made by the IES Secretariat (see "Submitting for production" below), not something any local script or checklist item confers by itself. Whether that process, the sandbox, or any IES network is currently reachable is not something this repository can verify — see where things stand on the Getting Started page.

Clearing every box below means your interface is well-formed and internally consistent by the checks this repository can run. It is not, on its own, proof of external-schema conformance, JSON-LD conformance, protocol interoperability, certification, or production readiness.

What conformance means

Identity is correctly published [all]

- [ ] `did:web` resolves from outside your network (`curl https://<your-domain>/.well-known/did.json` returns valid JSON)
- [ ] The DID document declares your current signing key
- [ ] Key rotation procedure documented (who, how, on what trigger)
- [ ] DeDi namespace verified (DNS TXT proof) and listed on `explore.dedi.global`

Network membership is in place [beckn]

- [ ] Beckn subscriber record published in your DeDi namespace (Ed25519 key, callback URL, BAP/BPP role)
- [ ] Reference entry written by the IES Secretariat into the IES network registry
- [ ] Sandbox loop succeeds (`discover` or `confirm` round-trip with a peer)

At least one use case is wired and validates [all]

- [ ] One use-case implementation guide (e.g. Consumer Energy Passport or Smart Meter Data Exchange) implemented end-to-end
- [ ] Output validates against the canonical JSON Schema (`make validate` or `python3 scripts/validate_schema.py …`)
- [ ] **[beckn]** At least one real record exchanged with a counterparty on the testnet, **or [credential]** at least one credential issued, verified, and revoked (the §3.9 smoke test)

- [ ] Schema validation enforced on every outgoing payload

Signatures and trust chain check out

- [ ] **[credential]** Every credential you issue carries a valid signature (W3C VC Data Model 2.0 proof block)
- [ ] **[credential]** Revocation registry (`vc-revocation-registry`) exists in your DeDi namespace (OpenCred auto-creates it; if not using OpenCred, create it manually)
- [ ] **[credential]** Verification rehearsed by at least one relying party (peer DISCOM, bank, marketplace) resolving your `did:web`
- [ ] **[beckn]** Every Beckn message you send carries a valid Ed25519 signature
- [ ] **[beckn]** Signature verification rehearsed by at least one counterparty (peer DISCOM, AMISP)

Operations

- [ ] Logging captures every inbound and outbound Beckn message (subscriber, action, message id, timestamp)
- [ ] Monitoring / alerting on adapter health
- [ ] Key rotation runbook
- [ ] Incident escalation path defined (within IES Secretariat process)

Security and privacy

- [ ] Signing keys held in KMS / HSM where available; never in source control
- [ ] TLS on all callback endpoints
- [ ] For consumer-facing credentials: identity-proofing procedure documented and privacy-reviewed
- [ ] For consumer-facing credentials: explicit consent capture on every issuance / disclosure

---

What schema validation proves — and what it doesn't

`scripts/validate_schema.py` (and `make validate`) check **repository-local structural and semantic validation** — your payload against the IES-authored `schema.json` for that family, under Draft 2020-12. For schema families that are fully self-contained — every `ElectricityCredential` version, `MeterDataRequest`, `ArrFiling`, `MeterData/v0.5` — that check is complete against everything this repository defines.

Other families reference externally-hosted Beckn/DEG structures that are not mirrored in this repository, and the local validator resolves every one of them through a permissive stub (`{"type": "object"}`) rather than the real upstream schema — so a passing result does not check the stubbed structure against its actual definition. The unchecked surface differs by family, and is not uniformly small:

- **`MeterData/v0.6`** references two external substructures, `Address` and `GeoJSONGeometry`, both stubbed.
- **`OutageNotification/v0.1`** references one external substructure, `GeoJSONGeometry`, stubbed.
- **`MeterDataCredential/v0.6`** and **`MeterDataRequestCredential/v0.1`** each reference the external `EnergyCredential/v2.0` credential envelope — the entire Beckn VC wrapper (`issuer`, `proof`, `credentialSubject` shape, and everything else it constrains) — and that whole envelope is stubbed permissively. A passing result on these two families only checks the local `meterData` / `meterDataRequest` payload nested inside the credential; it does not check the envelope fields against their actual upstream definitions at all. Both families also nest a local `$ref` to `MeterData/v0.6` or `MeterDataRequest/v0.6` respectively, so `MeterDataCredential/v0.6` additionally inherits `MeterData/v0.6`'s own unchecked `Address`/`GeoJSONGeometry` substructures.

Passing local validation on any of these families means "well-formed except for the unchecked externally-hosted structure(s) listed above for that family," not full external-schema conformance.

Treat a schema-validation pass as scoped to what this repository defines — see What this checklist proves above for the JSON-LD, protocol, and certification boundaries that sit outside schema validation entirely.

How to confirm conformance

Beckn participants (B2B)

Run the conformance suite that ships with the devkit you cloned in Setting up Discover & Exchange:

```
cd DEG/devkits/data-exchange/conformance
./run.sh --subscriber did:web:ies.discom.example --usecase consumer-energy-passport
```

The script exercises every path on the checklist above against a sandbox peer and emits a signed report. For use cases not yet in the suite, do the equivalent manually: generate one record with your adapter, validate it against the canonical JSON Schema (`scripts/validate_schema.py`), send it to a sandbox counterparty, and have them verify the signature and contents.

Credential-only issuers (B2C)

You have no Beckn devkit and no counterparty, so these checks are self-contained — run the §3.9 smoke test end-to-end:

1. Issue a credential with your adapter (or OpenCred directly).
2. Validate the credential document (`out.json`) against the canonical credential-root JSON Schema (`python3 scripts/validate_schema.py schemas/<Credential>/<version>/schema.json out.json`) — repository-local structural validation, per the scope note above.
3. Resolve your own `did:web` over HTTPS and confirm the published key verifies the credential's `proof`.
4. Check revocation status (not revoked), revoke, re-check (revoked).

If all four pass, your issuance has cleared the B2C checks on this page: it is schema-valid within the scope described above, and independently verifiable by anyone who resolves your `did:web`. There is nothing to submit to a network — verification happens bilaterally, whenever a relying party checks your `did:web`, not through a central certification step. This is not, by itself, a claim of JSON-LD conformance or of conformance against externally-hosted schema substructures your payload references.

Submitting for production (Beckn participants only)

This is the documented network-membership process. Whether the IES Secretariat, the sandbox, and the production/test network registries referenced below are currently staffed and reachable is not something this repository can verify on its own — see where things stand on the Getting Started page before relying on this as a live, working pipeline.

When all the above pass on the sandbox / testnet:

1. Email the IES Secretariat (IES.Secretariat@fsrglobal.org, alternate ies@recindia.com) with: your `did:web`, your DeDi namespace, your testnet conformance report, and the use cases you have implemented.
2. The Secretariat re-runs spot checks on your sandbox endpoints.
3. The Secretariat writes a production reference entry pointing at your subscriber record.

Once that reference entry is written, your subscriber record is queryable as in-network by any counterparty that looks it up, without a further bilateral arrangement — that is what "referenced into the network" means (see Setting up Register §1.7). That determination is made by the Secretariat; it is not something this checklist, `scripts/validate_schema.py`, or any other local script confers by itself, and it is a statement about network registration — not, on its own, a statement about JSON-LD conformance or protocol interoperability with every possible counterparty. (Credential-only issuers skip this step — there is no central network registry in the credential-only path; the B2C checks above are what make your issuances independently verifiable.)

After conformance

- **Add new use cases** by repeating only Build your Internal-facing Adapter for the new schema. Identity, engines, network membership and trust foundation are reused.

- **Stay current with schema versions.** New versions are announced via the ies-accelerator repository; migration guides ship alongside.
- **Contribute** — propose a new schema or a change to an existing one by raising a discussion on the repository (see Propose a Schema). The IES Cell governance process handles ratification.

Appendix — Schemas Reference

The field reference for each IES schema family, at its current version: structure, attribute tables, and worked examples. This appendix mirrors the same content published on GitBook under **Schemas**; plain-language walkthroughs of each family appear earlier, under **What IES Provides -> Schemas Overview**.

ElectricityCredential

A W3C Verifiable Credential, issued per meter or connection by a distribution licensee, that carries the static facts of a consumer's connection and the energy resources behind the meter.

> **Mirror.** This directory is a verbatim mirror of the Beckn DEG `ElectricityCredential` specification. Upstream is authoritative; open an issue if you find a discrepancy.

Status: Stable — In Pilot (current: **v1.2**) · **Issued by** DISCOMs · **Consumed by** consumers (holder-bound), grid operators and aggregators (bearer), banks / subsidy portals / DER marketplaces (as verifiers)

What it records

For one connection: the non-PII **customer profile** (utility CA number, tariff and load terms per meter via `consumptionProfiles[]`), the **energy resources behind the meter** (`energyResources[]` — a discriminated union of seven kinds: meter, generator, storage, EV charger, inverter, controllable load, network equipment, each with make, ratings, commissioning and inspection facts), and optionally the PII **customer details** (name, installation address) when issued to the consumer. It records static, structural facts only — live readings belong to MeterData.

Two issuance variants of the same record: **bearer** (no `credentialSubject.id`; grid-side, non-PII issuance) and **holder-bound** (`credentialSubject.id` = the consumer's wallet DID; the Consumer Energy Passport pattern).

How things are identified

The issuer is a `did:web` on the DISCOM's domain, optionally paired with a regulator licensing reference (`issuer.idRef`). Assets carry `did:web` paths under the issuer's domain (e.g. `did:web:ies.discom.example:assets:meter:<slno>`) — existing serial numbers are preserved, never replaced. Topology is expressed through `parentResources[]` / `subResources[]` links, which lets one credential cover anything from a single domestic meter to a multi-meter building with tenant sub-metering or parallel import/export metering.

The schema is a **composition**: its own `attributes.yaml` defines only the envelope and profile containers, pulling in `EnergyCredential/v2.0` (base), `EnergyResource/v2.1`, `MeterServiceProfile/v1.1`, `CustomerDetails/v1.0` and `IdRef/v1.0` by reference.

Standards basis

- **IS 16444** (smart meter spec; DLMS/COSEM per IS 15959) for meter protocol fields; **IS 16221** / IEC 61727 for PV capacity.
- **CEA Connectivity Regulations 2013 (amended 2018)** with **IEEE 1547-2018 Cl. 11** for the per-asset inspection record.
- **IEC 61968 / 61970 (CIM)** class-level alignment for every resource kind; **IEEE 2030.5**, SunSpec DER models, OCPP / ISO 15118 for DER, inverter and V2G attributes.

- Where no Indian standard exists (aggregator enrolment, ride-through categories, DR control protocols), international standards fill the gap — flagged per field in the reference.

How it fits together

One schema behind two use cases: as the **Consumer Energy Passport** it is issued holder-bound into the consumer's wallet — one credential per consumer connection, never combining consumers. For **DER Visibility**, its `EnergyResource` and `ConsumptionProfile` building blocks are published PII-free as per-feeder arrays, giving operators a checkable view of what is connected. It inherits its envelope from `EnergyCredential`:

`beckn:Credential` └── `EnergyCredential` └── `ElectricityCredential`

Open points

- Deprecated type aliases `SOLAR` -> `SOLAR_PV` and `BATTERY` -> `BESS` remain valid during transition; `ratedPower` is kept for backward compatibility, with `maxExport` / `maxImport` preferred.
- The composition pins five sibling schemas; confirm exact sibling versions before treating "v1.2" as a complete wire-format spec.

Versions

Version	Status	Notes
v1.2	**Current**	Composable EnergyResource kinds (7 typed discriminants); directional power fields; EV charger kind
v1.1	Previous	Unified `energyResources[]`, multi-meter topology support
v1.0	Deprecated	Separate consumption/ generation/storage profile arrays

v1.2

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

Upstream authority and mirror status. `beckn/DEG/specification/schema/ElectricityCredential/v1.2/` is canonical. Run `python -X utf8 -B scripts/verify_ec_mirror.py` from the repository root to compare this directory with pinned commit `84a4de0d749ce17b58ec37d18c4a13deace9c023`, selected as the immutable 2026-06-26 v1.2 upstream tree current when this launch gate was introduced. As of 2026-07-26, that verifier reports drift in seven of eight files (the bundled `schema.json` matches); this directory must not be represented as a synchronized verbatim mirror until an independently reviewed upstream refresh makes the verifier pass.

W3C Verifiable Credential (VC Data Model 2.0) issued per meter by electricity distribution utilities.

v1.2 introduces a **composable EnergyResource hierarchy**: each entry in `energyResources[]` is discriminated by `type` into one of seven typed kinds, each with a typed `attributes` bag. All power and capacity fields use **QuantitativeValue `{value, unit}`** with short unit aliases (`kW`, `kWh`, `kVA`, `kVAR`, `kV`, `MW`, `MWh`, `MVA`, `MVAR`, `V`, `W`) mapped to QUDT IRIs via JSON-LD context.

Structure

```
credentialSubject
├── id                          (optional — customer DID)
├── customerProfile             (required — non-PII)
│   ├── customerNumber          (required — CA number)
│   ├── idRef                   (optional — external identity reference)
```

```
|   ├── energyResources[]      (required — all physical assets, min 1)
|   |   ├── id                 (meter serial for METER; stable id for DERs)
|   |   ├── type               (discriminator — see kinds below)
|   |   ├── attributes         (kind-specific bag inheriting EnergyResourceCommonAttributes)
|   |   ├── subResources[]     (child resource ids or inline objects)
|   |   └── parentResources[]  (parent resource ids — e.g. the meter a DER sits behind)
|   └── consumptionProfiles[]  (optional — tariff/load per meter, linked via meterId)
└── customerDetails            (optional — PII)
    ├── fullName               (PII — only here)
    ├── installationAddress
    └── serviceConnectionDate
```

EnergyResource kinds

Kind	`type` values	CIM class (IEC 61970/61968)
`EnergyResourceMeter`	`METER`	`cim:Meter` / `cim:EndDevice` (IEC 61968-9)
`EnergyResourceGenerator`	`SOLAR_PV`, `WIND`, `HYDRO`, `BIOGAS`, `CHP`, `FUEL_CELL`	`cim:GeneratingUnit` subtypes (IEC 61970-302)
`EnergyResourceStorage`	`BESS`	`cim:BatteryUnit` (IEC 61970-302)
`EnergyResourceEVCharger`	`EV_CHARGER`, `EV_V2G`	`cim:ElectricVehicleChargingStation` (CIM17+)
`EnergyResourceInverter`	`INVERTER`	`cim:PowerElectronicsConnection` (IEC 61970-302)
`EnergyResourceLoad`	`SMART_HVAC`, `SMART_WATER_HEATER`, `CONTROLLABLE_LOAD`	`cim:EnergyConsumer` / `cim:ConformLoad` (IEC 61970-301)
`EnergyResourceNetwork`	`DT`, `BUS`, `FEEDER`, `MICROGRID`	`cim:PowerTransformer`, `cim:BusbarSection`, `cim:Feeder`, `cim:Substation` (IEC 61970-301)

Deprecated type aliases (still valid for backward compatibility): `SOLAR` -> use `SOLAR_PV`; `BATTERY` -> use `BESS`.

v1.1 -> v1.2 migration

Change	v1.1 (scalar)	v1.2 (QuantitativeValue)
Power fields	`ratedPowerKw`, `maxExportKw`, `maxImportKw` (number)	`ratedPower`, `maxExport`, `maxImport` (`{value, unit: W|kW|MW}`)
Generator nameplate	`nominalPowerKw` (number)	`nominalPower` (`{value, unit: W|kW|MW}`)
Storage capacity	`storageCapacityKwh` (number)	`storageCapacity` (`{value, unit: kWh|MWh}`)
Apparent power	`ratedApparentPowerKva` (number)	`ratedApparentPower` (`{value, unit: kVA|MVA}`)
Reactive power	`maxReactivePowerKvar`, `minReactivePowerKvar` (number)	`maxReactivePower`, `minReactivePower` (`{value, unit: kVAR|MVAR}`)
Network voltage	`nominalVoltageKv` (number)	`nominalVoltage` (`{value, unit: V|kV}`)
Tariff load	`sanctionedLoadKw`, `contractMaxDemandKw` (number)	`sanctionedLoad`, `contractMaxDemand` (`{value, unit: W|kW|MW}`)

Files

File	Description
`attributes.yaml`	OpenAPI 3.1.1 schema with composable EnergyResource hierarchy
`schema.json`	Bundled JSON Schema (draft 2020-12) — self-contained
`context.jsonld`	JSON-LD context with unit aliases (kW->qudt:KiloW, kWh->qudt:KiloW-HR, etc.)
`vocab.jsonld`	RDF vocabulary with CIM class alignments
`examples/example.json`	Single meter + SOLAR_PV + WIND + 2× BESS
`examples/example-submetering.json`	Building main meter + 2 tenant sub-meters + rooftop solar
`examples/example-parallel-metering.json`	Import meter + export meter (solar FIT)

For full documentation see the upstream README.

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing ***** is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

ElectricityCredential v1.2

Field	Type	Description
`id` *	uri	—
`type` *	list of text	—
`issuer` *	object	—
`validFrom` *	date-time	—
`validUntil`	date-time	—
`credentialStatus`	object	—
`credentialSubject` *	object	—
`proof`	object	—

CustomerDetails

Field	Type	Description
`fullName` *	text	**Based on** CIM (IEC 61968-1 Customer.name). Full name of the customer as per ID proof.
`careOf`	object	Care-of (c/o) reference person used to uniquely identify / disambiguate a customer who may share the same fullName with others in a locality (e.g. a village). Pairs a name with an optional, gender-neutral relationship.
`installationAddress` *	Location	**Based on** GeoJSON RFC 7946; schema.org PostalAddress; CIM (IEC 61968-1 ServiceLocation). Physical location of the metered installation. geo (GeoJSON Point, coordinates [longitude, latitude]) is required; address (schema.org PostalAddress fields) is optional.
`serviceConnectionDate` *	date-time	**Based on** CIM (IEC 61968-1 ServiceLocation activation date). Date and time the service connection was activated, with timezone offset (ISO 8601).

Location

Field	Type	Description
`geo` *	GeoJSONGeometry	—
`address`	Address	—

GeoJSONGeometry

Field	Type	Description
`bbox`	list of number	Optional bounding box `[west, south, east, north]` in degrees.
`coordinates`	list of —	Coordinates per RFC 7946 for all types **except** GeometryCollection. Order is **[lon, lat, (alt)]**. For Polygons, this is an array of linear rings; each ring is an array of positions.

Field	Type	Description
`geometries`	list of GeoJSONGeometry	Member geometries when `type` is **GeometryCollection**.
`type` *	`Point` / `LineString` / `Polygon` / `MultiPoint` / `MultiLineString` / `MultiPolygon` / `GeometryCollection`	—

Address

Field	Type	Description
`addressCountry`	text	Country name or ISO-3166-1 alpha-2 code.
`addressLocality`	text	City/locality.
`addressRegion`	text	State/region/province.
`extendedAddress`	text	Address extension (apt/suite/floor, C/O).
`postalCode`	text	Postal/ZIP code.
`streetAddress`	text	Street address (building name/number and street).

CustomerProfile

Field	Type	Description
`customerNumber` *	text	Utility customer account (CA) number.
`idRef`	IdRef	—
`energyResources` *	list of EnergyResource	All physical energy assets for this account. Each entry is discriminated by 'type' into one of seven composable kinds.
`consumptionProfiles`	list of ConsumptionProfile	Tariff and load characteristics per meter connection. Each entry links to a METER via meterId.

IdRef

Field	Type	Description
`issuedBy` *	uri	DID or URI of the issuing authority.
`subjectId` *	text	Subject identifier in authority-domain:id-value format.

EnergyResourceMeter

Field	Type	Description
`id` *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
`type` *	text	Asset-class discriminator. Must be "METER". CIM cim:Meter (IEC 61968-9).
`subResources`	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
`parentResources`	list of text	Upward topology — ids of parent resources.
`attributes`	EnergyResourceMeterAttributes	—

EnergyResourceCommon

Field	Type	Description
`id`	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
`type`	text	Asset class discriminator. Constrained to a kind-specific enum or const in each typed kind schema.
`subResources`	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
`parentResources`	list of text	Upward topology — ids of parent resources.
`attributes`	EnergyResourceCommonAttributes	Attribute bag. Inherits EnergyResourceCommonAttributes via allOf plus kind-specific fields.

EnergyResourceCommonAttributes

Field	Type	Description
`make`	text	Manufacturer (free text).
`model`	text	Model (free text).
`ratedPower`	QVPower (`W` / `kW` / `MW`)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.

Field	Type	Description
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.

QVPower

Field	Type	Description
value *	number	—
unit *	W / kW / MW	—

EnergyResourceMeterAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
meterCapability	Electromechanical / CMRI / AMR / AMI	**Based on** CIM (IEC 61968-9 AmiBillingReadyKind). Communication/automation generation. Electromechanical: induction-disc. CMRI: manual optical-port (India legacy). AMR: one-way automated read. AMI: two-way smart meter.
energyDirection	Forward / Reverse / Bidirectional / Net	**Based on** CIM (FlowDirectionKind); ESPI NAESB REQ.21. Energy flow direction metered at this point.
functions	list of ToU / NetMetering / MaxDemand / LoadControl / TamperDetection / PowerQuality / EventLogging	**Based on** CIM (IEC 61968-9 EndDeviceFunction). Active meter capabilities.
feeder	text	Feeder identifier this meter is supplied from.
bus	text	Busbar identifier at the meter's connection point.
communicationTechnology	PLC / RF_Mesh / GPRS / NB-IoT / LoRa / ZigBee / Other	Last-mile physical-layer communication technology.
applicationProtocol	DLMS_COSEM / ANSI_C12_18 / IEC_61850 / Modbus / Other	Application-layer protocol for meter data. DLMS_COSEM: IEC 62056, mandatory for India AMI per BIS IS 16444. ANSI_C12_18: North American. Orthogonal to communicationTechnology (physical layer).

EnergyResourceGenerator

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	SOLAR_PV / SOLAR / WIND / HYDRO / BIOGAS / CHP / FUEL_CELL	Asset-class discriminator. SOLAR is deprecated — use SOLAR_PV.
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceGeneratorAttributes	—

EnergyResourceGeneratorAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
nominalPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.nominalP). Nominal (nameplate) power output. Use when distinct from maxExport (peak).
efficiency	number	Conversion efficiency as a percentage (0–100). Most relevant for FUEL_CELL and CHP resources.
dcArrayCapacity	QVPower (W / kW / MW)	**Based on** IS 16221 (PV module qualification); IEC 61727 (PV grid interface). DC-side nameplate capacity of a photovoltaic array at Standard Test Conditions (industry term: "kWp"). For PV systems this is typically larger than the AC-side maxExport because of inverter clipping and DC-to-AC ratios. Relevant for SOLAR_PV resources. The unit is the standard QUDT power alias kW — the STC/peak semantic is documented here, not encoded in the unit string.

EnergyResourceStorage

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	BESS / BATTERY	Asset-class discriminator. BATTERY is deprecated — use BESS. CIM: BatteryUnit (IEC 61970-302).
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceStorageAttributes	—

EnergyResourceStorageAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.

Field	Type	Description
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
storageCapacity	QVEnergy (kWh / MWh)	**Based on** CIM (IEC 61970-302 BatteryUnit.ratedE). Rated stored-energy capacity. Replaces storageCapacityKwh from v1.0.
storageType	LithiumIon / LeadAcid / FlowBattery / NaS / NiCd / Flywheel / Other	Battery storage technology type.
stateOfHealthPct	number	Battery state-of-health as a percentage (0–100).
roundTripEfficiencyPct	number	**Based on** IEC 62933-2-1 (performance test method). AC-to-AC round-trip efficiency as a percentage (0–100): the fraction of energy returned to the grid relative to energy drawn during a full charge/discharge cycle. Distinct from stateOfHealthPct (cumulative life indicator) and from inverter conversion efficiency.

QVEnergy

Field	Type	Description
value *	number	—
unit *	kWh / MWh	—

EnergyResourceEVCharger

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	EV_CHARGER / EV_V2G	Asset-class discriminator. EV_CHARGER: EVSE hardware. EV_V2G: Vehicle-to-Grid capable EVSE with ISO 15118-20 / OCPP 2.1 BPT.
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceEVChargerAttributes	—

EnergyResourceEVChargerAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.

Field	Type	Description
connectorType	Type1 / Type2 / CCS1 / CCS2 / CHAdeMO / GB_T / NACS / Other	Physical connector standard. Type1: IEC 62196-2 Type 1 (J1772). Type2: IEC 62196-2 Type 2 (Mennekes). CCS1/CCS2: Combined Charging System DC fast charge. CHAdeMO: CHAdeMO DC fast charge. GB_T: GB/T 20234. NACS: SAE J3400.
controlProtocol	OCPP_1.6 / OCPP_2.0.1 / OCPP_2.1 / ISO_15118_2 / ISO_15118_20 / Other	EVSE control and smart-charging protocol.
v2xProtocol	CHAdeMO_V2G / CCS_BPT / ISO_15118_20_AC_BPT / ISO_15118_20_DC_BPT / Other	Vehicle-to-Grid / V2X protocol. Present only for EV_V2G resources.

EnergyResourceInverter

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	text	Asset-class discriminator. Must be “INVERTER”. CIM: PowerElectronicsConnection (IEC 61970-302).
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceInverterAttributes	—

EnergyResourceInverterAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
ratedApparentPower	QVApparentPower (kVA / MVA)	**Based on** SunSpec DER Model 702 (maxVA); CIM (IEC 61970-302 PowerElectronicsConnection.ratedS). Rated apparent power. Replaces ratedApparentPowerKva from v1.0.
maxReactivePower	QVReactivePower (kVAR / MVAR)	**Based on** SunSpec DER Model 702 (maxVar); CIM (IEC 61970-302 PowerElectronicsConnection.maxQ). Maximum reactive power injection (leading / over-excited). Replaces maxReactivePowerKvar from v1.0.
minReactivePower	QVReactivePower (kVAR / MVAR)	**Based on** SunSpec DER Model 702 (maxVarNeg); CIM (IEC 61970-302 PowerElectronicsConnection.minQ). Maximum reactive power absorption (lagging / under-excited). Value is typically negative. Replaces minReactivePowerKvar from v1.0.
rideThroughCategory	CategoryI / CategoryII / CategoryIII	**Based on** IEEE 1547-2018 (ride-through category). Abnormal operating performance category. CategoryI: basic. CategoryII: enhanced for distribution-connected DER. CategoryIII: advanced for large/ transmission-connected DER.
operatingMode	GridFollowing / GridForming / Standby	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.inverterMode). Inverter grid-interaction mode. GridFollowing: PLL-based sync. GridForming: own V/f reference (microgrid islanding, black-start). Standby: energised but not injecting.
voltVarEnabled	yes / no	**Based on** IEEE 2030.5 (opModVoltVar); SunSpec DER Model 705. Volt-VAr curve active.
freqDroopEnabled	yes / no	**Based on** IEEE 1547-2018; SunSpec DER Model 711. Frequency-Watt droop active.
enterServiceRampTimeSec	number	**Based on** SunSpec DER Model 703 (ESRmpTms). Seconds to ramp from 0 to rated power after reconnection.

QVApparentPower

Field	Type	Description
value *	number	—
unit *	kVA / MVA	—

QVReactivePower

Field	Type	Description
value *	number	—
unit *	kVAR / MVAR	—

EnergyResourceLoad

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	SMART_HVAC / SMART_WATER_HEATER / CONTROLLABLE_LOAD	Asset-class discriminator. CIM: EnergyConsumer / ConformLoad (IEC 61970-301).
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceLoadAttributes	—

EnergyResourceLoadAttributes

Field	Type	Description
make	text	Manufacturer (free text).
model	text	Model (free text).
ratedPower	QVPower (W / kW / MW)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
maxExport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
maxImport	QVPower (W / kW / MW)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
telemetryProvider	text	Vendor API / data source identifier for telemetry.
commissioningDate	date-time	ISO 8601 date-time the asset was commissioned.
location	Location	Physical location of this asset.
serialNumber	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
inspection	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
aggregator	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
controlProtocol	OpenADR_2.0b / OCPP_2.0.1 / SunSpec_Modbus / EEBus / Modbus / Other	Demand-response / control protocol supported by this load device.
loadCategory	Heating / Cooling / WaterHeating / Lighting / EV / Industrial / Other	**Based on** CIM (IEC 61970-301 ConformLoad classification). Functional category of this load.

EnergyResourceNetwork

Field	Type	Description
id *	text	Stable identifier for this resource. For METER resources the meter serial number is conventional.
type *	DT / BUS / FEEDER / MICROGRID	Asset-class discriminator. DT -> PowerTransformer, BUS -> BusbarSection, FEEDER -> Feeder (EquipmentContainer), MICROGRID -> Substation / microgrid container (IEC 61970-301).
subResources	list of text or EnergyResource	Topology — child resources. Each item is EITHER a bare id string OR an inline EnergyResource object.
parentResources	list of text	Upward topology — ids of parent resources.
attributes	EnergyResourceNetworkAttributes	—

EnergyResourceNetworkAttributes

Field	Type	Description
`make`	text	Manufacturer (free text).
`model`	text	Model (free text).
`ratedPower`	QVPower (`W` / `kW` / `MW`)	**Based on** CIM (IEC 61968-9 EndDeviceInfo.ratedPower; IEC 61970 GeneratingUnit.maxOperatingP). Manufacturer-rated peak power (nameplate value in principal direction). Kept for backward compatibility — prefer maxExport.
`maxExport`	QVPower (`W` / `kW` / `MW`)	**Based on** CIM (IEC 61970 GeneratingUnit.maxOperatingP; IEC 61970-302 PowerElectronicsConnection.maxP, injection). Maximum power this resource injects to the grid (generates/discharges). Always >=0. For bidirectional resources (BESS, V2G) this is the max discharge rate. Supersedes ratedPower.
`maxImport`	QVPower (`W` / `kW` / `MW`)	**Based on** CIM (IEC 61970-302 PowerElectronicsConnection.maxP, absorption). Maximum power this resource draws from the grid (absorbs/charges). Always >=0. For bidirectional resources (BESS, V2G) this is the max charge rate.
`telemetryProvider`	text	Vendor API / data source identifier for telemetry.
`commissioningDate`	date-time	ISO 8601 date-time the asset was commissioned.
`location`	Location	Physical location of this asset.
`serialNumber`	text	**Based on** CIM (IEC 61968-9 EndDeviceInfo.serialNumber). Manufacturer-assigned device serial number from the equipment nameplate. Distinct from id (which is the network-issued DID).
`inspection`	object	**Based on** IEEE 1547-2018 Cl. 11 (commissioning); CEA Connectivity Regs 2013 (amended 2018). Commissioning / safety inspection record for the asset. Captured by the distribution licensee at energisation and on re-certification events.
`aggregator`	object	**Based on** IEEE 2030.5; IEC 61850-7-420 (DER control roles). Third-party flexibility / demand-response enrolment for this asset. Present when an aggregator is authorised to dispatch or observe the resource. Controllability flag is asset-level; the asset may still be observable even when controllable is false.
`nominalVoltage`	QVVoltage (`V` / `kV`)	**Based on** CIM (IEC 61970-301 BaseVoltage.nominalVoltage). Nominal operating voltage. Replaces nominalVoltageKv from v1.0.
`zone`	text	Operating zone or region identifier used by the utility.
`substationId`	text	Parent substation identifier per utility records.
`feederCode`	text	Feeder code per utility records. Relevant for FEEDER and DT resources.

QVVoltage

Field	Type	Description
`value` *	number	—
`unit` *	`V` / `kV`	—

ConsumptionProfile

Field	Type	Description
`meterId` *	text	Matches the id of a METER entry in customerProfile.energyResources[].
`sanctionedLoad` *	QVPower (`W` / `kW` / `MW`)	**Based on** CIM (IEC 61968-9 UsagePoint). Sanctioned/approved import load.
`sanctionedExportLoad`	QVPower (`W` / `kW` / `MW`)	Sanctioned/approved grid export limit.
`billingCycleDay`	integer	Day of month on which the billing cycle resets.
`contractMaxDemand`	QVPower (`W` / `kW` / `MW`)	Maximum demand contracted with the utility for this connection.
`tariffCategoryCode` *	text	**Based on** CIM (IEC 61968-9 UsagePoint.serviceCategory). Billing/tariff category code assigned by the utility.
`premisesType`	`Residential` / `Commercial` / `Industrial` / `Agricultural`	Type of premises at the metering point.
`connectionType`	`Single-phase` / `Three-phase`	**Based on** CIM (IEC 61968-9 UsagePoint.phaseCode). Electrical connection type.
`paymentMode`	`POSTPAID` / `PREPAID`	**Based on** CIM (IEC 61968-9 AmiBillingReadyKind, ESPI). Billing/payment modality. POSTPAID: consume now, pay later. PREPAID: pay-before-use.
`serviceStatus`	`active` / `suspended` / `closed`	**Based on** CIM (IEC 61968-9 UsagePoint.status). Lifecycle state of the service connection (the UsagePoint), not of the meter device itself. 'active' = currently energised and billable; 'suspended' = temporarily disconnected (non-payment, inspection, fault) with the contract still on record; 'closed' = permanently terminated. Distinct from the meter device's operational state.

MeterData

A lightweight, high-throughput telemetry payload for exchanging smart-meter readings, events and alarms — cheap to parse and store at Head-End-System scale, without cryptographic wrapping.

Status: Stable — In Pilot (current: **v0.6**) · **Issued by** AMISPs, HES/MDM systems, DISCOMs · **Consumed by** DISCOMs, regulators, TSPs and consented third parties

What it records

Nine record shapes, discriminated by `profileType`:

Profile	Description
`CUSTOMER`	Slow-changing customer metadata, service points, meter installations (PII sub-object omissible for anonymised exchange)
`INTERVAL`	Block load survey at high-resolution intervals (15 or 30 min)
`DAILY`	Daily accumulated load survey
`MONTHLY`	Monthly billing resets — ToU buckets, MD, cumulative registers
`BILL_DETAILS`	Computed billing details — amount, due date, payment status
`INSTANTANEOUS`	Real-time snapshot of voltages, currents, powers
`EVENT`	IS 15959 diagnostic codes and tamper events
`ALARM`	Active alerts — tamper, low prepayment credit, voltage sag, overload
`DESCRIPTOR`	Shared dictionary (the Data Descriptor Engine) — register metadata and compact column layouts declared once, not per row

Readings distinguish `READING` (register value) from `USAGE` (delta over a period) and carry a `validationStatus` and `source`. The compact matrix form keeps a `DailyProfile` interval row as small as `{"id": 0, "payloads": [24512.4, 25134.8]}`. The schema carries no signature — when provenance attestation is needed, the payload is wrapped in a MeterDataCredential.

How things are identified

A generic `Identifier{scheme, value, namespace}` object; schemes include `METER_SERIAL`, `CONSUMER_NUMBER`, `OBIS`, `MRID` (from CIM — the one field with no Indian equivalent), and `DID`. Existing identifiers are carried verbatim. Individual registers are named by OBIS code or short code; the crosswalk (75 registers, plus IS 15959 event and alarm taxonomies) lives in the schema's own `IES codes.json` registry, cited per register down to the IS 15959 part/table/clause.

Standards basis

- **IS 15959 (Parts 1–3)** — OBIS codes, event IDs and meter categories, cited at individual-register granularity in `IES codes.json`.
- **IS 16444** — the AC smart meter specification the profiles are built to carry.

How it fits together

The primary telemetry payload of **Smart Meter Data Exchange**, carried B2B over the IES Beckn network; the payload itself is wire-agnostic and consumer-facing delivery need not involve Beckn. The companion MeterDataRequest family (capabilities -> authorisation -> request) sits in front of the exchange; the MeterDataCredential wrapper adds provenance where needed (the **Consumer Meter Digest** pattern). A PII-redacted `CustomerProfile` doubles as an anonymised grid-topology index for TSPs.

Open points

- A documented overlap with ElectricityCredential (consumer and DER-capacity concepts) is pending reconciliation in a future major version.
- `attributes.yaml`'s internal `info.version` still reads `0.5.0` under the `v0.6` directory — a labelling lag in the source.

Versions

Version	Status	Notes
v0.6	**Current**	Adds ALARM profile, anonymised topology, reading mode, split billing
v0.5	Previous	Six profiles: CUSTOMER, INTERVAL, DAILY, BILLING, INSTANTANEOUS, EVENT

v0.6

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

This directory contains the **Meter Data Compact Profiles** (v0.6) schema definitions and semantic models for telemetry data exchange. It is tailored for high-efficiency, data-only transmissions of smart meter readings and events, omitting heavy cryptographic wrapping when only raw telemetry payload is being exchanged.

The `MeterData` schema standardizes telemetry exchanges into **eight compact profile shapes** representing different read cadences and data structures from smart meters:

1. **`CUSTOMER`** — Slow-changing customer metadata, service points, and meter installations.
2. **`INTERVAL`** — Block load survey profile at high-resolution intervals (e.g., 15-minute or 30-minute blocks).
3. **`DAILY`** — Daily accumulated load survey profiles.
4. **`MONTHLY`** — Monthly billing resets (billing history cumulative total energy registers, ToU buckets, and Maximum Demand).
5. **`BILL_DETAILS`** — Utility billing computed details (billing amount, bill number, due dates, currency, prepaid balance, and payment status).
6. **`INSTANTANEOUS`** — Real-time snapshots of electrical quantities (voltages, currents, active/reactive powers) at a specific moment.
7. **`EVENT`** — Event logs matching standard IS 15959 diagnostic codes and tamper events.
8. **`ALARM`** — Real-time active alerts representing immediate state conditions (tamper, low prepayment credit, voltage sag, overload) from the meter.

Directory Structure and Files

File	Description
`attributes.yaml`	Canonical OpenAPI schema source of truth, describing all attributes, models, and types.
`schema.json`	Compiled Draft 2020-12 JSON Schema for standard validation of data payloads.
`context.jsonld`	Compiled JSON-LD context mapping telemetry attributes to semantic terms under `ies:` or `schema:` namespaces.
`vocab.jsonld`	Compiled JSON-LD vocabulary (ontology graph) defining all classes and properties.
`IES codes.json`	Canonical registry of OBIS and Short Codes mapping physical quantities to units, categories, and categories of meters.
`examples/`	Standard, JSON-LD augmented telemetry payload example files for all profile types.

Documentation Guides

- User Guide: Detailed guidelines on HES, MDM, and Billing system interactions, advertising system capabilities via request mechanisms, and verification scenarios.
- Reference Guide: Top-down reference documentation detailing different profile roles, centralised codes, and telemetry vs. non-telemetry handling.

Data Descriptor Engine & The Array Envelope

In v0.6, the telemetry architecture uses an optimized, strict inheritance structure powered by the **Data Descriptor Engine**.

Instead of transmitting bulky metadata inline with every physical reading, telemetry can be wrapped in a **JSON Array**.

Centralized `PayloadDescriptorProfile` A JSON array payload can contain one or more `PayloadDescriptorProfile` objects. This serves as the dictionary for all compact profiles inside the payload array: * **`payloadDescriptors`**: Defines the metadata for specific registers (e.g., `readingType`, `unit`, `flowDirection`, `category`, and `multiplier`). * **`compactSequences`**: Defines the physical column layout of the dense numerical payload matrices.

Dynamic `SequenceItem` Columns (`attribute`) A sequence defines what each column in the `payloads` array represents using the **`attribute`** property: * `value`: The actual meter reading number. * `occurredAt`: The timestamp string representing exactly when a specific demand or reading occurred. * `openingValue` / `closingValue`: To provide mathematical proofs for `USAGE` deltas. * `validationStatus`: Standard validation flags (e.g. `ESTIMATED`, `VALID`).

Because `payloads` allows mixed arrays of `number` and `string`, metadata is mapped densely alongside values, while sparse overrides are used for quality indications (such as validation status or fail codes in specific intervals).

Strict Derived Profiles Inside the array, individual profiles inherit strictly from `BaseProfile`. `BaseProfile` only contains identity (`meterRefs`, `customerRefs`, `serviceDeliveryPointRefs`). Derived profiles (like `IntervalProfile`) strictly allow **only** their relevant properties (`intervals`, `payloadDescriptorSetRef`, `intervalPeriod`).

> [!NOTE] ### Rationale for MonthlyProfile not using the compact matrix The compact form is prevented natively by the schema for `MonthlyProfile`. It strictly requires `readings` and `touBuckets`, and does not permit `intervals`. The compact matrix is designed for dense, highly symmetrical time-series data. A `MonthlyProfile` typically captures a single sparse snapshot of total registers, ToU buckets, and Maximum Demand peaks at the end of the month. Encoding sparse, highly variable metadata (e.g., MD timestamps) into a flat numerical array offers negligible compression and becomes extremely brittle when dealing with ad-hoc billing resets or meter swaps mid-cycle.
>
> *See MonthlyProfile_MultipleResets.json and Billing_MeterChange.json for examples of how the elaborated representation cleanly handles these edge cases.*

Identifier Flexibility (OBIS vs. Short Codes)

The schema relies on `IES codes.json` as the canonical registry for interpreting physical quantities. The exact mapping of OBIS codes to physical units, phases, and categories is defined in this file.

When transmitting telemetry, you specify a simple **`readingType`** string. This can be: * **Using OBIS Codes**: `"1.0.1.8.0.255"` * **Using Short Names**: `"kWh imp"` * **Using Canonical Links**: Payload descriptors can optionally include an `obis` string parameter to provide a direct physical mapping when a short name is used as the `readingType`.

See the MultiMeterBulkDataset.json (OBIS Codes) and MultiMeterBulkDatasetShortCodes.json (Short Codes) for bulk examples of both representations.

TelemetryMode (READING vs USAGE)

The schema natively distinguishes between raw register readings and calculated usage: - **`READING`**: Represents the physical register value at a point in time. For cumulative registers, this strictly increases dynamically. - **`USAGE`**: Represents the delta or consumed amount over a period. Demand registers inherently use `USAGE` mode. For energy, this represents the exact block consumption.

Modes are indicated inside the payload descriptor only via the `reportedMode` property.

Schema Generation and Compilation

The `schema.json`, `context.jsonld`, and `vocab.jsonld` files are compiled automatically from the core `attributes.yaml`.

To Compile Schemas To recompile schemas following modifications in `attributes.yaml`, run the following python scripts from the repository root:

```
python scripts/generate_schema_permissive.py schemas/MeterData/v0.6
```

Examples

These examples cover a wide variety of scenarios, organized by the system they typically originate from:

MDM (Meter Data Management) Examples MDM datasets are meter-centric and intentionally omit `customerRefs` to decouple the technical metering domain from the commercial billing domain. * `IntervalProfile.json`: Block load survey. * `DailyProfile.json`: Daily midnight snapshots. * `MDM_MonthlyProfile.json`: End of billing cycle snapshots across multiple meters. * `InstantaneousProfile.json`: Snapshot of voltage, current, power factor, etc. * `EventProfile.json`: Meter tampers, diagnostics, power outages. * `AlarmProfile.json`: Critical thresholds being crossed. * `MultiMeterBulkDataset.json`: Large scale transmission of intervals across many meters.

CIS / Billing Examples Billing and CIS examples enrich technical meter data with `customerRefs`, Tou buckets, and monetary calculations. * `Billing_MonthlyProfile.json`: Usage-centric monthly consumption linked to consumers. * `CustomerProfile.json`: Linking meters, service delivery points, and customers. * `BillDetails.json`: Computed monetary bill with due dates and calculated consumption. * `CustomerBillingSummary.json`: Historic billing trends for a consumer dashboard.

Support data used by the example-generation utility is deliberately kept out of the schema-validated payload directory:

- `support/CustomerMapping.json`: illustrative mapping of meter serial numbers to commercial customer accounts used by `scripts/generate_billing_from_mdm.py`; it is not a `MeterData` payload.

Highlighted Examples

- **Meter Swap Mid-Cycle**: `Billing_MeterChange.json` demonstrates how to handle a meter replacement in the middle of a billing period by chaining multiple monthly profiles under different meter serials.
- **Multiple Monthly Resets**: `MonthlyProfile_MultipleResets.json` demonstrates how to represent multiple billing resets triggered in the same calendar month.
- **Identifier Representation Comparison**: Compare `MultiMeterBulkDataset.json` (using raw OBIS codes) with `MultiMeterBulkDatasetShortCodes.json` (using Short Names) to see how the payload descriptor engine abstracts identifier models.

Telemetry Verification & Validation

The validator is not limited to example files; it can be used to validate any `MeterData` payload. In this repository, all example JSON files are validated for structural compliance against `schema.json` and semantic compliance against `IES codes.json` using the v0.6 validator.

To Validate Payloads Run the following command from the `validation/` directory:

```
python validator.py <path_to_json_file_or_directory>
```

The validator dynamically parses JSON files. If they are arrays containing a `PayloadDescriptorProfile`, it matches `payloadDescriptorSetRef` against the array's descriptors, and asserts both matrix arity and strict column types based on the `attribute` enum.

Overlaps and Differences with ElectricityCredential

- **`meterType` vs `meterCategory`**:
  - `meterType` describes the physical communication/technology capability of the meter (e.g. `AMI`, `AMR`, `Prepaid`, `NetMeter`).
  - `meterCategory` describes the regulatory standards category under IS 15959 (e.g. `A`, `B`, `C`, `D1`–`D4`).
  - These represent distinct aspects of the physical/standard classification, and both are kept.
- **`premisesType` vs `consumerCategory`**:
  - `premisesType` (Residential, Commercial, etc. defined in `ElectricityCredential/v1.2`) classifies the physical type of the premises.
  - `consumerCategory` (LT2A, NDS, HT, etc. defined in `MeterData/v0.6`) represents the utility-specific tariff category.
  - To prevent duplication and maintain backwards compatibility, the pre-existing `consumerCategory` property was kept, and `premisesType` was excluded. This overlap is marked here for future reconciliation.
- **`energyResources` vs `meters`/`associations` (Export & Storage modeling)**:
  - `ElectricityCredential v1.2` models all physical assets (meters, solar PV arrays, battery storage) using a unified `energyResources` list of typed `EnergyResource` objects. This allows rich modeling of child DER/storage assets (`SOLAR_PV`, `BESS`) directly detailing attributes like `ratedPower` and `storageCapacity` — each a QuantitativeValue `{value, unit}` pair — linked via topology references.
  - `MeterData v0.6` retains a lightweight representation focusing strictly on billing/metering associations, lacking top-level asset representations. To bridge this gap for grid planners, three lightweight properties were introduced in `v0.6`:
    - **`sanctionedExportLoadKw`** (on the `Customer` schema) to explicitly specify approved net-metering solar export limits.
    - **`generationCapacityKw`** (on the `Association` schema) to denote the rated solar/generation inverter capacity connected.
    - **`storageCapacityKw`** (on the `Association` schema) to denote the connected battery storage energy capacity in kWh.
  - This structural divergence is documented here for future alignment in subsequent major version releases.

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing ***** is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

CustomerProfile

Field	Type	Description
`profileType` *	text	—
`customerRefs`	IdentifierList	—
`timeZone`	text	IANA time-zone, e.g. Asia/Kolkata.
`customerDetails`	CustomerDetails	—
`customer` *	Customer	—
`serviceDeliveryPoints` *	list of ServiceDeliveryPoint	—
`meters` *	list of Meter	—
`associations` *	list of Association	—

BaseProfile

Field	Type	Description
`profileType` *	text	—
`customerRefs`	IdentifierList	—
`meterRefs` *	IdentifierList	—
`serviceDeliveryPointRefs`	IdentifierList	—
`payloadDescriptorSetRef`	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.

IntervalProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
compactSequenceRef	text	Name of the compact sequence to use from the payloadDescriptorSets.
intervalPeriod *	IntervalPeriod	—
intervals	list of Interval	—
readings	list of Reading	—

DailyProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
compactSequenceRef	text	Name of the compact sequence to use from the payloadDescriptorSets.
intervalPeriod *	IntervalPeriod	—
intervals	list of Interval	—
readings	list of Reading	—

MonthlyProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
timePeriod *	TimePeriod	—
readings *	list of Reading	—
touBuckets	list of TouBucket	—

BillDetails

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
timePeriod *	TimePeriod	—
readings	list of Reading	—
touBuckets	list of TouBucket	—
billNumber	text	—
billDate	date	—
dueDate	date	—
currency	text	ISO 4217 (INR, USD, …).
amountDue *	number	—
energyCharges	number	Charges for active/reactive energy consumption.
fixedCharges	number	Fixed charges/demand charges.
otherCharges	number	Other taxes, duties, surcharges, or adjustments.
prepaidBalance	number	Prepaid remaining balance/credit amount on the account/meter, if applicable.
paymentStatus	text	Status of the payment, e.g. PAID, UNPAID, PARTIAL.

InstantaneousProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—

Field	Type	Description
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
timestamp *	date-time	—
readings *	list of Reading	—

AlarmProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
timestamp *	date-time	—
alarms *	list of MeterAlarm	—

EventProfile

Field	Type	Description
profileType *	text	—
customerRefs	IdentifierList	—
meterRefs *	IdentifierList	—
serviceDeliveryPoint Refs	IdentifierList	—
payloadDescriptorSet Ref	text	Reference ID matching a previously exchanged PayloadDescriptorProfile's payloadDescriptorSet.
timePeriod *	TimePeriod	—
events *	list of MeterEvent	—

Identifier

Field	Type	Description
scheme *	METER_SERIAL / METER_BADGE / MRID / OBIS / SHORT_CODE / CONSUMER_NUMBER / SERVICE_DELIVERY_POINT / DID / ORG / OTHER	—
value *	text	—
namespace	text	—

TimePeriod

Field	Type	Description
start *	date-time	—
duration *	duration	—

ReadingDefinition

Field	Type	Description
readingType *	text	—
supportedModes *	list of READING / USAGE	—
defaultMode *	READING / USAGE	—
name	text	—
shortLabel	text	—
unit	kWh / kVAh / kvarh / kW / kvar / kVA / V / A / Hz / PF / NONE / INR / USD	—
phase	NONE / R / Y / B / ABC	—
flowDirection	IMPORT / EXPORT / NONE	—
accumulation Behaviour	CUMULATIVE / DELTA / INSTANTANEOUS / SUMMATION / INDICATING	—
touZone	integer	—

PayloadDescriptorSet

Field	Type	Description
name *	text	—
payloadDescriptors *	list of PayloadDescriptor	—
compactSequences	list of CompactSequence	—

CompactSequence

Field	Type	Description
name *	text	—
sequenceItems *	list of SequenceItem	—

SequenceItem

Field	Type	Description
readingType *	text	—
attribute	value / occurredAt / openingValue / closingValue / validationStatus	—

PayloadDescriptor

Field	Type	Description
readingType *	text	—
obis	text	Optional canonical OBIS code when readingType uses a short code.
name	text	—
unit	kWh / kVAh / kvarh / kW / kvar / kVA / V / A / Hz / PF / NONE / INR / USD	—
flowDirection	IMPORT / EXPORT / NONE	—
category	text	—
reportedMode	READING / USAGE	—
touZone	integer	—
multiplier	number	Decimal scaling factor (e.g. 0.001 for milli, 1000 for kilo). Default value is 1.
accuracy	number	Accuracy class or precision value, applied after the multiplier.

IntervalPeriod

Field	Type	Description
start *	date-time	—
duration *	duration	—

Interval

Field	Type	Description
id *	integer	—
intervalPeriod	IntervalPeriod	—
payloads	list of number / string / boolean	—
readings	list of Reading	—
overrides	list of Override	—

Override

Field	Type	Description
descriptorIndex *	integer	—
occurredAt	date-time	—
zone	integer	—
validationStatus	VALID / ESTIMATED / MANUAL / SUSPECT / REJECTED	—
source	METER / HES / ESTIMATED / MANUAL / IMPORT / MDM_COMPUTED / CIS_COMPUTED	—
changeMethod	text	—
failCode	text	—

Reading

Field	Type	Description
readingType *	text	—
value *	number	—
openingValue	number	MUST ONLY be provided when the associated payload descriptor specifies reportedMode=USAGE.
closingValue	number	MUST ONLY be provided when the associated payload descriptor specifies reportedMode=USAGE.
integrationPeriod	duration	Demand integration period, e.g. PT30M or PT15M.
timePeriod	TimePeriod	—
occurredAt	date-time	—
validationStatus	VALID / ESTIMATED / MANUAL / SUSPECT / REJECTED	—
source	METER / HES / ESTIMATED / MANUAL / IMPORT / MDM_COMPUTED / CIS_COMPUTED	—
changeMethod	text	—
failCode	text	—

TouBucket

Field	Type	Description
zone *	integer	—
readings *	list of Reading	—

Customer

Field	Type	Description
id *	Identifier	—
name	text	—
consumerCategory	text	Utility tariff category (e.g., LT2A, NDS, HT).
sanctionedLoadKw	number	—
billingCycleDay	integer	—
paymentMode	PREPAID / POSTPAID	Indicates if the connection is prepaid or postpaid.
connectionType	Single-phase / Three-phase	Type of connection (Single-phase or Three-phase).
contractMaxDemandKw	number	Maximum demand contracted with the utility for this connection, in kW.
tariffCategoryCode	text	Billing/tariff category code assigned by the utility.
sanctionedExportLoadKw	number	Sanctioned/approved grid export limit in kW.

ServiceDeliveryPoint

Field	Type	Description
id *	Identifier	—
address	Address	—
geo	GeoJSONGeometry	—

Meter

Field	Type	Description
id *	Identifier	—
make	text	Make of the meter.
model	text	Model of the meter.
meterType	AMR / AMI / Electromechanical / Forward / Reverse / Bidirectional / Prepaid / NetMeter / Other	—
meterCategory	A / B / C / D1 / D2 / D3 / D4	—
serviceKind	ELECTRICITY / GAS / WATER / HEAT	—

Association

Field	Type	Description
serviceDeliveryPointRefs *	IdentifierList	—
meterRefs *	IdentifierList	—
parentResources	list of Identifier	List of parent resources (such as feeders or DTs) for this association, replacing feederId and dtId.
telemetryProvider	text	Telemetry provider for this meter-service association.

Field	Type	Description
commissioningDate	date-time	Commissioning date of the meter association.
generationCapacityKw	number	Rated power generation capacity (e.g. solar PV inverter capacity) in kW.
storageCapacityKw	number	Rated energy storage capacity in kWh.

MeterEvent

Field	Type	Description
timestamp *	date-time	—
eventId *	integer	—
eventName	text	—
phase	NONE / R / Y / B / ABC	—
sequence	integer	—
magnitude	number	—
duration	duration	—

MeterAlarm

Field	Type	Description
timestamp *	date-time	—
alarmId *	integer	—
alarmName	text	—
status *	ACTIVE / CLEARED	—
severity	CRITICAL / WARNING / INFO	—

PayloadDescriptorProfile

Field	Type	Description
profileType *	text	—
id *	text	Unique identifier for this specific configuration state.
payloadDescriptorSets *	list of PayloadDescriptorSet	—

CustomerDetails

Field	Type	Description
name	text	Full name of the customer as per ID proof.

MeterDataCredential

A signed, tamper-evident wrapper that attests who delivered a set of smart-meter readings, and that the readings have not been altered since.

Status: Stable — In Pilot (current: **v0.6**, aligned with MeterData v0.6) · **Issued by** data providers (AMISPs, MDM systems, or a DISCOM in that role) · **Consumed by** DISCOMs, consumers' wallets, and downstream verifiers

What it records

Structurally thin by design: the schema-specific surface is one object, `credentialSubject`, with `id` (DID of the consumer or asset the data is about) and `meterData` (the attested MeterData payload — a single profile or an array opening with a descriptor). Everything else — `issuer` (DID, name, licence number), `validFrom`/`validUntil`, `credentialStatus`, `proof` — is inherited from **EnergyCredential v2.0**. A new instance is issued per delivery, superseded rather than amended.

`beckn:Credential └─ EnergyCredential └─ MeterDataCredential`

How things are identified

The issuer is a DID (a `did:web` in the worked examples), with the signing key referenced from `proof.verificationMethod`. Revocation is a DeDi registry (`credentialStatus.type: dediregistry`) namespaced under the issuer's own DID — each provider runs its own revocation list. Meters, service points and readings inside the payload use MeterData's own `Identifier` scheme.

Standards basis

- **W3C VC Data Model 2.0** and **W3C DID Core** for the envelope (no Indian standard governs the VC envelope itself).
- The wrapped payload transitively carries MeterData's **IS 15959** / OBIS (IEC 62056) basis.

How it fits together

The credential stands on its own: issued, held and verified against the provider's published key over any channel — a wallet (DigiLocker), a portal, a file. A verifier checks type, validity window, revocation, the `proof` signature, and that the embedded profiles fall within the originally contracted scope. In the **Consumer Meter Digest** use case it is issued holder-bound to the consumer's wallet DID, so the consumer can re-present verified meter history without returning to the DISCOM. Its request-side counterpart is MeterDataRequestCredential (proves the right to ask; this credential attests what was delivered).

Open points

- `credentialSubject.id` is optional at the schema level; whether an unscoped-subject credential (e.g. feeder-level aggregate data) is an intended case is not stated.
- Expected revocation timelines after a meter fault or privacy request, and holder notification, are not detailed in the schema files.

Versions

Version	Status	Notes
v0.6	**Current**	Wraps MeterData v0.6; DeDi-registry revocation (`credentialStatus.type: dediregistry`)

v0.6

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

A **W3C Verifiable Credential (VC Data Model 2.0)** that wraps a `MeterData v0.6` payload to attest the authenticity and provenance of delivered smart meter telemetry.

Purpose

In a Beckn data-exchange flow, a **provider** (e.g., AMISP, MDM system) responds to a confirmed `MeterDataRequest` with an `on_status` message. The provider wraps the telemetry payload in a `MeterDataCredential` so that:

1. **Identity is bound** — the data is cryptographically tied to the issuing provider's DID.
2. **Provenance is verifiable** — any downstream system can verify who delivered the data and when.
3. **Tamper-evidence** — the payload cannot be altered without invalidating the signature.
4. **Revocation is supported** — the DeDi registry allows the provider to invalidate a credential if data is recalled (e.g., due to meter fault or privacy request).

The receiver checks: credential type is `MeterDataCredential`, `validUntil` has not passed, status is not revoked, and the embedded `meterData` profiles are within the contracted scope of the originating `MeterDataRequest`.

Inheritance

`MeterDataCredential` is a **subclass of `EnergyCredential v2.0`** (defined in the DEG specification):

```
beckn:Credential
  └── EnergyCredential (https://schema.nfh.global/EnergyCredential/v2.0)
        └── MeterDataCredential
```

The common VC envelope is **inherited, not duplicated**:

Inherited from EnergyCredential	Defined here
`issuer` (id, name, `licenseNumber`)	`credentialSubject.meterData`
`validFrom` / `validUntil`	`MeterDataCredential` type
`credentialStatus` (DeDi registry)	
`proof` (Ed25519Signature2020)	

In `attributes.yaml` this is expressed as `allOf: - $ref: https://schema.nfh.global/EnergyCredential/v2.0`, matching the same pattern used by `MeterDataRequestCredential`, `ConsumptionProfileCredential`, and other DEG energy credentials.

The `@context` of a credential instance must include: 1. `https://www.w3.org/ns/credentials/v2` 2. `https://schema.nfh.global/EnergyCredential/v2.0/context.jsonld` 3. `https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataCredential/v0.6/context.jsonld`

Schema Files

File	Purpose
`attributes.yaml`	OpenAPI 3.1.1 source of truth with `x-jsonld` annotations
`schema.json`	JSON Schema Draft 2020-12 for validation
`context.jsonld`	JSON-LD context (semantic term mappings)
`vocab.jsonld`	RDF vocabulary / ontology definitions
`examples/example-customer-profile.json`	VC wrapping a single CustomerProfile
`examples/example-interval-profile.json`	VC wrapping a PayloadDescriptorProfile + IntervalProfile array
`examples/example-monthly-profile.json`	VC wrapping a MonthlyProfile with ToU buckets

Credential Structure

```
MeterDataCredential (W3C VC 2.0)
├── @context        – W3C credentials/v2 + EnergyCredential + MeterDataCredential context
├── id              – urn:uuid:...
├── type            – ["VerifiableCredential", "MeterDataCredential"]
├── issuer
│   ├── id          – DID of the data provider (AMISP, MDM, DISCOM)
│   ├── name        – Human-readable name
│   └── licenseNumber – Regulatory licence (e.g., SERC-AMISP-2025-007)
├── validFrom       – Issuance datetime
├── validUntil      – Expiry (typically 1 year)
├── credentialStatus – DeDi revocation registry reference
├── credentialSubject
│   ├── id          – DID of the consumer or asset entity
│   └── meterData   – MeterData v0.6 payload
│       ├── (single profile)  – one of CUSTOMER / INTERVAL / DAILY / MONTHLY /
│       │                         BILL_DETAILS / INSTANTANEOUS / EVENT / ALARM / DESCRIPTOR
│       └── (array of profiles) – mixed profile types, typically starting with DESCRIPTOR
└── proof           – Ed25519Signature2020 (or equivalent)
```

meterData payload shapes

The `meterData` value follows the `MeterData v0.6` schema — either a **single** `EnergyData` profile object or an **array** of profile objects. All standard profile types are supported:

`profileType`	Description
`CUSTOMER`	Slow-changing customer + service-point + meter-list summary
`INTERVAL`	Block Load Survey — PT15M / PT30M cadence interval blocks
`DAILY`	Daily Load Profile — P1D interval blocks
`MONTHLY`	Monthly billing readings (with optional ToU buckets)
`BILL_DETAILS`	Computed billing amount, bill number, due dates
`INSTANTANEOUS`	Snapshot of electrical quantities at one moment
`EVENT`	IS 15959 event log over a time period
`ALARM`	Real-time active alarm indicators
`DESCRIPTOR`	PayloadDescriptor sets (exchanged before or alongside compact interval data)

Arrays typically start with a `DESCRIPTOR` profile followed by one or more data profiles that reference it via `payloadDescriptorSetRef`.

Relationship to MeterDataRequestCredential

`MeterDataCredential` is the **response** counterpart to `MeterDataRequestCredential`:

	`MeterDataRequestCredential`	`MeterDataCredential`
Issued by	Requester (DISCOM)	Provider (AMISP / MDM)
Wraps	`MeterDataRequest`	`MeterData` payload
Used in	`confirm` — proves authorisation to request	`on_status` — attests delivered data
Subject	Requesting entity DID	Consumer / asset DID

Usage in Beckn Data Exchange (UC1)

Provider side — `on_status` The provider delivers the telemetry as a `MeterDataCredential` in `dataPayload`. The embedded `meterData` must cover the profile types and time window originally requested in the `MeterDataRequest`.

Receiver side — verification

1. Resolve the issuer DID and retrieve the public key at `verificationMethod`.
2. Verify the `proof` signature over the canonical serialisation of the credential.
3. Check `validUntil` has not passed and `credentialStatus` is not revoked via the DeDi registry.
4. Validate `meterData` against the MeterData v0.6 schema.

Examples

File	Description
`example-customer-profile.json`	DISCOM AMISP attests CustomerProfile for consumer `RR-1234`
`example-interval-profile.json`	DISCOM AMISP attests PT15M interval data (with PayloadDescriptor)
`example-monthly-profile.json`	DISCOM AMISP attests January 2026 monthly readings with ToU buckets

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Changelog

Version	Date	Notes
v0.6	2026-06-06	Initial draft — aligns with MeterData v0.6

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing ***** is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

MeterDataCredential

Field	Type	Description
`credentialSubject`	MeterDataCredentialSubject	The subject of the credential: the consumer or asset entity whose meter data is attested, plus the MeterData payload.

MeterDataCredentialSubject

Field	Type	Description
`id`	uri	DID of the consumer or asset entity whose meter data is being delivered.
`meterData` *	schema.json	The attested meter data payload. May be a single EnergyData profile or an array of profiles per MeterData v0.6.

MeterDataRequest

The query, capability and authorisation shapes a party uses to ask for smart-meter telemetry — not the telemetry itself.

Status: Stable — In Pilot (current: **v0.6**) · **Published by** providers (DISCOMs, AMISPs) · **Used by** requesters (DISCOMs, TSPs, consented third parties)

What it records

Three composable models under one `oneOf` payload root, replacing v0.5's single flat request object:

Model	Purpose
`MeterDataCapabilities`	What a provider can serve — profile types (optionally down to individual registers), supported query scopes, maximum history window
`MeterDataAuthorisation`	Who authorised whom — grantor, grantee, purpose, validity window, and the granted capability slice (all six fields required)

Model	Purpose
`MeterDataRequest`	The actual query — target consumers/resources, scope (`ResourceOnly` / `ResourceAndChildren` / `ChildrenOnly`), time window, consent references, backing authorisation, requested capability slice, record cap

How things are identified

Consumers and resources are URIs/DIDs; the schema mints no identifier scheme of its own. Registers are named by OBIS code or short code (`kWh imp`), resolved against the MeterData family's `IES codes.json` registry. A shipped semantic validator cross-checks what JSON Schema alone cannot: that a requested register exists, is permitted under the stated profile type, supports the requested telemetry mode, and that authorisation windows are coherent.

Standards basis

The request/authorisation envelope is IES-native. Its one standards touchpoint is the register space in `ValueCapability.value`, anchored to **IS 15959** (via the per-register part/table/clause citations in `IES codes.json`), with IEC 62056/OBIS as the second-order reference.

How it fits together

The request leg of **Smart Meter Data Exchange**: a provider advertises capabilities, a requester obtains an authorisation naming its slice, then issues per-query requests — answered by a separate MeterData payload. Where authorisation must be proven cryptographically, the request is wrapped in a MeterDataRequestCredential.

Open points

- `authorisation` accepts an inline object or a URI reference; a documented convention for when each form is expected (and how a URI-referenced grant is verified) is pending.
- Every shipped example uses short codes; the raw-OBIS form of `ValueCapability.value` is currently unexercised.

Versions

Version	Status	Notes
v0.6	**Current**	Split into three composable models: MeterDataCapabilities, MeterDataAuthorisation, MeterDataRequest
v0.5	Previous	Single unified request schema

v0.6

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

This directory contains the schemas, context mapping, and vocabularies for querying smart meter telemetry and profiles. In version 0.6, the request layer has been split into three distinct, composable models: 1. **MeterDataCapabilities** 2. **MeterDataAuthorisation** 3. **MeterDataRequest**

---

1. MeterDataCapabilities

Represents the data sharing capabilities supported by a meter data provider (e.g. DISCOM). It acts as the blueprint of what data points are available, specifying what profile types, register keys, telemetry modes, and query boundaries are supported.

- **Main Schema**: `schema.json#/$defs/MeterDataCapabilities`

- **JSON-LD Type**: `ies:MeterDataCapabilities`
- **Link to Example**: `Capabilities_Example.json`

---

2. MeterDataAuthorisation

Represents the cryptographic or digital grant/token allowing an entity (such as a Technical Service Provider) to access a specific subset of data capabilities. It defines: - **`grantor`**: The authorizing entity (e.g. Utility/DISCOM). - **`grantee`**: The authorized entity (e.g. TSP). - **`purpose`**: The explicit reason for accessing the data (e.g. ESG reporting). - **`validFrom` / `validUntil`**: The validity period. - **`capabilities`**: The allowed `MeterDataCapabilities` object.

- **Main Schema**: `schema.json#/$defs/MeterDataAuthorisation`
- **JSON-LD Type**: `ies:MeterDataAuthorisation`
- **Link to Example**: `Authorisation_Example.json`

---

3. MeterDataRequest

Represents the actual data query payload sent by the authorized entity to pull telemetry. It contains query criteria and proves authorization and user consent: - **`consumers`** (optional): List of target consumer DIDs/URIs. - **`resources`** (optional): List of target meter serials or service point URIs. - **`scope`**: Hierarchy of query (`ResourceOnly`, `ResourceAndChildren`, `ChildrenOnly`). - **`from` / `duration`**: The start and duration of the query time window. - **`consumerConsent`** (optional): Array of consumer consent DIDs or receipt hashes. - **`authorisation`**: Inline `MeterDataAuthorisation` object or URL pointing to the grant. - **`capabilitiesRequested`**: The requested `MeterDataCapabilities` subset.

- **Main Schema**: `schema.json` (root schema)
- **JSON-LD Type**: `ies:MeterDataRequest`
- **Link to Example**: `Request_Example.json`

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing * is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

MeterDataRequest

Field	Type	Description
`consumers`	list of uri	List of consumer URIs/DIDs for whom the data is requested (optional).
`resources`	list of uri	List of resource URIs/DIDs (e.g. meter DIDs, service point IDs) to query (optional).
`scope`	`ResourceOnly` / `ResourceAndChildren` / `ChildrenOnly`	—
`from` *	date-time	ISO 8601 UTC date-time indicating the start time of the requested data window.
`duration` *	duration	ISO 8601 duration string representing the length of the requested data window (e.g., PT15M, P1D, P30D).
`consumerConsent`	list of text	Consent references/receipts proving authorization (optional).
`authorisation`	MeterDataAuthorisation or uri	Inline authorization object or URI reference to it.
`capabilitiesRequested` *	MeterDataCapabilities	—
`maxRecordsShared`	integer	Maximum number of records that should be shared or returned in a single batch/page.

MeterDataAuthorisation

Field	Type	Description
`grantor` *	uri	URI/DID of the entity authorizing access.
`grantee` *	uri	URI/DID of the authorized entity.
`purpose` *	text	Reason/purpose for data access.
`validFrom` *	date-time	ISO 8601 UTC date-time indicating when the authorization becomes valid.
`validUntil` *	date-time	ISO 8601 UTC date-time indicating when the authorization expires.
`capabilities` *	MeterDataCapabilities	—

MeterDataCapabilities

Field	Type	Description
`profiles` *	list of ProfileCapability	List of profiles and their specific values/modes.
`supportedScopes`	list of `ResourceOnly` / `ResourceAndChildren` / `ChildrenOnly`	Hierarchical scopes supported by the query processor.
`maxHistoryDuration`	duration	Maximum length of historical data window supported by the provider.

ProfileCapability

Field	Type	Description
`profileType` *	`CustomerProfile` / `IntervalProfile` / `DailyProfile` / `MonthlyProfile` / `BillDetails` / `InstantaneousProfile` / `EventProfile` / `AlarmProfile`	—
`readings`	list of ValueCapability	Granular capabilities for specific registers and metrics. If omitted, all readings under this profile are supported.

ValueCapability

Field	Type	Description
`value` *	text	The OBIS code (e.g. 1.0.1.8.0.255) or short code (e.g. kWh imp) representing the register.
`mode`	`READING` / `USAGE`	The telemetry mode (READING, USAGE) supported or requested for this register.
`multiplier`	number	Optional decimal multiplier scaling factor (e.g., 0.001 or 1000).
`accuracy`	number	Optional accuracy class or precision value.

MeterDataRequestCredential

A W3C Verifiable Credential a data requester attaches to a Beckn request to prove it is authorised to ask for smart-meter telemetry.

Status: Stable — In Pilot (current: **v0.1**) · **Issued by** the requester (a DISCOM, or a third-party aggregator/TSP) · **Consumed by** the data provider (AMISP or MDM system)

What it records

The requester's identity (DID at `issuer.id` and `credentialSubject.id`, plus a regulatory licence number inherited from **EnergyCredential v2.0**) bound to a scoped, time-bounded query: the embedded `credentialSubject.meterDataRequest` is a MeterDataRequest object (targets, scope, window, consent references, backing authorisation, requested capability slice). It carries no telemetry — it only authorises a request for it. The credential's validity window and the historical data window it requests are two independent time ranges; both must be checked.

`beckn:Credential` └─ `EnergyCredential` └─ `MeterDataRequestCredential`

Standards basis

- **W3C VC Data Model 2.0** and **W3C DID Core**, as a subclass of DEG **EnergyCredential v2.0** — the same inheritance pattern as its sibling energy credentials.
- The engineering substance (registers, profile types, modes) lives in the wrapped MeterDataRequest, anchored to the OBIS register space (**IEC 62056**, via IS 15959 citations in the MeterData code registry).

How it fits together

The optional authorisation attachment on the request leg of **Smart Meter Data Exchange**. A provider's offer policy (`offerAttributes.policy.requiredCredentials`) can require it at Beckn `confirm`; the seeker attaches the signed credential, and the provider verifies the type, validity window, revocation status (DeDi registry in `credentialStatus`), and that the embedded request is a subset of the provider's advertised capability profile. Its delivery-side counterpart is MeterDataCredential — the two bracket the same exchange from opposite ends.

Open points

- When a provider requires this credential versus accepting a plain request is a per-provider policy decision.
- `consumerConsent` entries are plain strings; the expected format (DID, receipt hash, other) is undefined at the schema level.

Versions

Version	Status	Notes
v0.1	**Current**	Initial release; wraps MeterDataRequest v0.6

v0.1

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

A **W3C Verifiable Credential (VC Data Model 2.0)** that wraps a `MeterDataRequest` to prove a data requester's authorisation for accessing smart meter telemetry.

Purpose

In a Beckn data-exchange flow, a **seeker** (e.g., DISCOM) discovers an AMI dataset offered by a **provider** (e.g., AMISP). The provider's offer policy requires the seeker to attach a `MeterDataRequestCredential` at `confirm` time. This credential:

1. **Identifies** the requesting entity (DID + regulatory licence number).
2. **Scopes** the request — which meters/feeders, what hierarchy, what time window, which profile types.
3. **Is cryptographically signed**, so the provider can verify it without a round-trip to the issuer.
4. **Can be revoked** via the DeDi registry if authorisation is withdrawn.

The provider checks: credential type is `MeterDataRequestCredential`, `validUntil` has not passed, status is not revoked, and the embedded `MeterDataRequest.resources` are within the contracted scope.

Inheritance

`MeterDataRequestCredential` is a **subclass of `EnergyCredential v2.0`** (defined in the DEG specification):

```
beckn:Credential
  └── EnergyCredential (https://schema.nfh.global/EnergyCredential/v2.0)
        └── MeterDataRequestCredential
```

The common VC envelope is **inherited, not duplicated**:

Inherited from EnergyCredential	Defined here
`issuer` (id, name, `licenseNumber`)	`credentialSubject.meterDataRequest`
`validFrom` / `validUntil`	`MeterDataRequestCredential` type
`credentialStatus` (DeDi registry)	
`proof` (Ed25519Signature2020)	

In `attributes.yaml` this is expressed as `allOf: - $ref: https://schema.nfh.global/EnergyCredential/v2.0`, matching the same pattern used by `ConsumptionProfileCredential`, `GenerationProfileCredential`, and other DEG energy credentials.

The `@context` of a credential instance must include: 1. `https://www.w3.org/ns/credentials/v2` 2. `https://schema.nfh.global/EnergyCredential/v2.0/context.jsonld` 3. `https://india-energy-stack.github.io/ies-accelerator/schemas/MeterDataRequestCredential/v0.1/context.jsonld`

Schema Files

File	Purpose
`attributes.yaml`	OpenAPI 3.1.1 source of truth with `x-jsonld` annotations
`schema.json`	JSON Schema Draft 2020-12 for validation
`context.jsonld`	JSON-LD context (semantic term mappings)
`vocab.jsonld`	RDF vocabulary / ontology definitions
`examples/example.json`	Complete W3C VC 2.0 example

Credential Structure

```
MeterDataRequestCredential (W3C VC 2.0)
├── @context        – W3C credentials/v2 + IES MeterDataRequestCredential context
├── id              – urn:uuid:...
├── type            – ["VerifiableCredential", "MeterDataRequestCredential"]
├── issuer
│   ├── id          – DID of the requesting entity (e.g., DISCOM)
│   ├── name        – Human-readable name
│   └── licenseNumber – Regulatory licence (e.g., SERC-DISCOM-2025-001)
├── validFrom       – Issuance datetime
├── validUntil      – Expiry (short window, e.g., 30 days)
├── credentialStatus – DeDi revocation registry reference
├── credentialSubject
│   ├── id          – DID of the requesting entity
│   └── meterDataRequest (MeterDataRequest v0.6)
│       ├── consumers      – Optional list of consumer DIDs
│       ├── resources      – Optional list of meter/feeder DIDs to query
│       ├── scope          – ResourceOnly | ResourceAndChildren | ChildrenOnly
│       ├── from           – Start of data window (ISO 8601 UTC)
│       ├── duration       – Length of data window (ISO 8601 duration)
│       ├── consumerConsent – Optional list of consumer consent DIDs
│       ├── authorisation  – Reference or inline authorisation grant
│       ├── capabilitiesRequested – The requested capabilities (MeterDataCapabilities)
│       └── maxRecordsShared – Optional cap on returned records
└── proof           – Ed25519Signature2020 (or equivalent)
```

Usage in Beckn Data Exchange (UC1)

Provider side — `catalog/publish` The provider advertises: 1. **`ies:dataSchema`** in `resourceAttributes` — URL to the MeterData v0.6 schema used for response payloads. 2. **`ies:capabilityProfile`** in `resourceAttributes` — a `MeterDataRequest` object describing the maximum scope the provider can serve (think of it as the provider's data-access envelope). 3. **`offerAttributes.policy.requiredCredentials`** — declares that a valid `MeterDataRequestCredential` must be attached at `confirm`.

Seeker side — `confirm` The seeker includes the signed credential in `commitmentAttributes.ies:meterDataRequest`. The embedded `MeterDataRequest` must be a subset of the provider's `capabilityProfile`.

Provider side — `on_status` The response `dataPayload` contains **IES MeterData v0.6** profile objects (e.g., `IntervalProfile`, `CustomerProfile`) — the schema advertised in `ies:dataSchema`.

Example

See `examples/example.json` for a complete W3C VC 2.0 instance where the DISCOM requests Q1 2026 interval + daily + monthly data for all meters under feeder `IN-KA-BLR-ZONE-A`.

Changelog

Version	Date	Notes
v0.1	2026-05-26	Initial draft

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing ***** is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

MeterDataRequestCredential

Field	Type	Description
`credentialSubject`	MeterDataRequestCredentialSubject	The subject of the credential: the requesting entity and the specific MeterDataRequest they are authorised to make.

MeterDataRequestCredentialSubject

Field	Type	Description
`id`	uri	DID of the requesting entity (e.g., the DISCOM).
`meterDataRequest` *	schema.json	The scoped, time-bounded data request. Specifies the meter resources, hierarchical scope, time window, and profile types being requested. The provider validates this against its capability profile before fulfilling delivery.

ArrFiling (WIP)

A structured, machine-readable version of the Aggregate Revenue Requirement (ARR) filing a distribution licensee submits to its state electricity regulator.

Status: Work in progress (current: **v0.5**) · **Filed by** DISCOMs · **Received by** SERCs / Joint Electricity Regulatory Commissions

What it records

One regulatory filing built from three nested models:

Model	Purpose
`ArrFiling`	Root — filing ID, licensee, regulator, filing type (`MYT` / `ANNUAL` / `TRUE_UP` / `REVISED`), control period, currency (INR) + document-wide `unitScale`, status
`ArrFiscalYear`	One fiscal year — `yearType` (base / control-period / historical) crossed with `amountBasis` (audited / approved / proposed / trued-up / not-filed)
`ArrLineItem`	One cost, revenue, subtotal or adjustment row — category and cross-DISCOM `subCategory`, form heading, amount (nullable; negative = credit), roll-up via `componentOf` and optional human-readable `formula`

The shape absorbs real cross-state variation: MYT wide-format and long-run historical filings, differing line-item granularity (one O&M line vs. three), and cost heads that appear or consolidate across years — without forcing filers into one fixed table.

How things are identified

Plain regulatory references, not DIDs: the `filingId` the regulator already assigns, and stable kebab-case `lineItemId`s (`power-purchase-cost`) reused across fiscal years so lines can be compared year over year. Roll-ups are expressed through `componentOf` references rather than nesting.

Standards basis

Each state's own **SERC tariff regulations** define the filing forms; ArrFiling's `category`/`subCategory` enums are an IES superset across them, still converging. No Indian or international standard predates a machine-readable ARR format — the gap this schema fills is structural.

How it fits together

The payload of the **DISCOM Regulatory Filing** use case: the DISCOM (as provider) delivers the filing as a signed, schema-validated payload over the IES Beckn network to the SERC (as consumer), which ingests it directly instead of a PDF. Supporting workbooks ride as separate signed datasets linked by `filingId`.

Open points

- The `category`/`subCategory` superset is expected to gain additions as more states are mapped; per-DISCOM mapping from finance-system categories needs sign-off before cross-DISCOM comparison.
- `formula` is optional and inconsistently populated — tooling that recomputes roll-ups must handle its absence.

Versions

Version	Status	Notes
v0.5	**Current**	Three relational models: ArrFiling, ArrFiscalYear, ArrLineItem

v0.5

Files — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/

This directory contains the **Aggregate Revenue Requirement (ARR) Filing** (v0.5) schema definitions and semantic models for utility regulatory data exchange. It is designed to standardize the structure of ARR filings made by distribution licensees (DISCOMs) to State Electricity Regulatory Commissions (SERCs) in India.

Overview

The `ArrFiling` schema unifies diverse state-level regulatory filing formats by structuring them into three relational classes:

1. **`ArrFiling` (Root)** — Describes global metadata about a filing, including filing ID, licensee/DISCOM details, receiving regulatory commission, control period, currency, and unit scale.
2. **`ArrFiscalYear`** — Connects a specific fiscal year to its baseline/control-period classification and amount basis (e.g., Audited actuals, SERC-Approved values, or proposed projections).
3. **`ArrLineItem`** — Represents a single cost, revenue, subtotal, or true-up adjustment line item matching the regulatory sub-forms (e.g. Power Purchase Cost, O&M Expenses, Net ARR).

Directory Structure and Files

File	Description
`attributes.yaml`	Canonical OpenAPI 3.1.0 schema source of truth, describing all attributes, models, and types.
`schema.json`	Compiled Draft 2020-12 JSON Schema for standard validation of data payloads.
`context.jsonld`	Compiled JSON-LD context mapping ARR attributes to semantic terms under the `ies:` prefix.
`vocab.jsonld`	Compiled JSON-LD vocabulary ontology graph defining classes and properties.
`examples/`	JSON-LD absolute-linked regulatory example files.

Schema Generation and Compilation

The JSON Schema and JSON-LD context/vocabulary are compiled automatically from the core `attributes.yaml` using our generic Python build script.

To Compile Schemas To recompile schemas following modifications in `attributes.yaml`, run the following command from the repository root:

```
python scripts/generate_schema.py schemas/ArrFiling/v0.5
```

Payload Verification & Validation

Example JSON files are validated for structural compliance against `schema.json` using our dedicated validation script.

To Validate Example Payloads Run the following command from the repository root:

```
python scripts/validate_schema.py schemas/ArrFiling/v0.5/schema.json schemas/ArrFiling/v0.5/examples
```

JSON-LD Integration

All example payloads in the `examples/` directory have been augmented to support standard Linked Data semantic resolution: 1. **`@context`** — Points to the canonical absolute URL `https://india-energy-stack.github.io/ies-accelerator/schemas/ArrFiling/v0.5/context.jsonld` to resolve terms. 2. **`@type`** — Indicates the profile class shape (e.g., `ArrFiling`), aligning the regulatory datasets to the broader India Energy Stack ecosystem.

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing * is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

ArrFiling

Field	Type	Description
`objectType` *	`ARR_FILING`	—
`id`	text	Unique identifier for this filing.
`filingId` *	text	Regulatory filing reference number.
`filingDate`	date	—
`filingType`	`MYT` / `ANNUAL` / `TRUE_UP` / `REVISED`	MYT - Multi-Year Tariff control period filing (multiple years, mix of actual/proposed) ANNUAL - single year or historical year-by-year approved data TRUE_UP - reconciliation of actuals vs previously approved amounts REVISED - amended filing with corrections
`licensee` *	text	Full name of the distribution licensee.

Field	Type	Description
licenseeCode	text	Short code for the licensee.
stateProvince	text	—
regulatoryCommission *	text	SERC or Joint ERC that receives the filing.
controlPeriodStart	text	Start fiscal year of MYT control period (only for MYT filings).
controlPeriodEnd	text	End fiscal year of MYT control period.
currency *	INR	—
unitScale *	CRORE / LAKH / ABSOLUTE	Scale of all amounts in the filing.
status	DRAFT / SUBMITTED / UNDER_REVIEW / APPROVED / REJECTED	—
formReference	text	Regulatory form identifier.
notes	list of text	Footnotes, regulatory order references, and explanatory notes.
fiscalYears *	list of ArrFiscalYear	—

ArrFiscalYear

Field	Type	Description
fiscalYear *	text	Fiscal year label.
yearType	BASE_YEAR / CONTROL_PERIOD / HISTORICAL	BASE_YEAR - reference year in an MYT filing (typically the year before the control period) CONTROL_PERIOD - a year within the MYT control period being filed for HISTORICAL - a past year with finalized data (used in annual/historical filings)
amountBasis *	AUDITED / APPROVED / PROPOSED / TRUED_UP / NOT_FILED	AUDITED - actual costs verified by auditors APPROVED - amounts approved by the SERC in a tariff order PROPOSED - amounts requested by the DISCOM, pending SERC approval TRUED_UP - reconciled amounts after comparing actuals to approved NOT_FILED - placeholder year in a control period, no data yet
lineItems *	list of ArrLineItem	—

ArrLineItem

Field	Type	Description
lineItemId *	text	Stable identifier for this line item across years. Use kebab-case, e.g., “power-purchase-cost”, “interest-working-cap”.
serialNumber	integer	Display order as shown in the regulatory form.
category *	VARIABLE / FIXED / INCOME / SUB_TOTAL / ARR / ADJUSTMENT	VARIABLE - costs varying with energy volume (power purchase) FIXED - costs independent of volume (O&M, depreciation, interest, return) INCOME - revenue credits that reduce the ARR (negative amounts) SUB_TOTAL - computed aggregation of other line items ARR - the final net/aggregate revenue requirement ADJUSTMENT - true-up corrections, pass-throughs, FPPCA adjustments
subCategory	POWER_PURCHASE / NETWORK_COST / O_AND_M / DEPRECIATION / INTEREST / RETURN_ON_EQUITY / PROVISIONAL / OTHER / NON_TARIFF_INCOME / REVENUE_CREDIT / TOTAL / NET_ARR	Functional sub-classification for analysis and comparison across DISCOMs.
head *	text	Short heading as used in the regulatory form. Varies by DISCOM — use the original text from the filing.
particulars	text	Detailed description or the “Particulars” column value. Useful when head alone is ambiguous (e.g., “Others” head with “Incentive/Disincentive on achievement of norms” as particulars).
amount *	number / null	Amount in the filing’s currency and unitScale. Null means the line item exists in the form but was not filed / not applicable. Negative values represent credits, deductions, or adjustments that reduce ARR.
formReference	text	Reference to the supporting sub-form or schedule.
componentOf	text	lineItemId of the parent subtotal this item contributes to.
formula	text	Human-readable computation formula expressed as references to other lineItemIds. Only present on SUB_TOTAL and ARR items.

OutageNotification (WIP)

> [warn] **WORK IN PROGRESS — subject to change.** Early draft family (v0.1) for discussion; field names, enums and interpretation are not final. Do not depend on it for production integrations yet.

A machine-readable payload a distribution licensee publishes to describe one planned or unplanned electricity outage — usable both as a public outage-map feed and as a push alert to affected consumers.

Status: Work in progress (current: **v0.1**) · **Issued by** DISCOMs (or their OMS / MDMS / SCADA / RTDAS) · **Consumed by** outage-map feeds, subscribed consumers, mass-alerting networks, reliability reporting

What it records

One notice per outage, updated in place over its life (ALERT -> UPDATE -> RESTORED/CANCELLED, each version independently signed): classification (`outageClass` — planned / breakdown / rostering — separate from lifecycle `status`), a three-layer **cause** (IEEE 1782 category + the vendor's own fault code carried verbatim + free text), **affected assets** (feeders, DTs, substations, with voltage level and optional GeoJSON geometry) and **area**, **impact** (customers affected; de-energised/energised connection points reserved for an authenticated tier), **timing** (window, estimated and actual restoration — the inputs to IEEE 1366 indices — and a 240-minute SLA target), field **response**, localized **public-facing text** (CAP-shaped), and **provenance** back to the smart-meter signal and MeterData `AlarmProfile` that auto-detected it.

Model	Purpose
`OutageNotification`	Root — class, status, cause, assets, area, timing, impact, response, public info, provenance, extensions
`OutageAsset`	Affected feeder/DT/substation with smart-meter ref and optional GeoJSON geometry
`OutageProvenance`	Detecting system, AMISP code, raw feeder-status signal, alarm refs into MeterData

How things are identified

A single `Identifier{scheme, value, namespace}` object throughout; the notice's `id` (the DISCOM's own breakdown/ shutdown ID) stays constant across updates. Assets ideally carry a stable `GIS_FEATURE_ID` or `MRID` so consumers holding the DISCOM's GIS layer can join on id and pull geometry from GIS; inline `geo` is the fallback.

Standards basis

No Indian standard yet covers outage publication; the schema composes one layer per standard: **IEC 61968-3 (CIM)** for outage/UsagePoint semantics and the planned/unplanned split, **OASIS CAP v1.2** for the push-alert envelope, **IEEE 1782-2022** (§4.4/§4.5) for the cause taxonomy, **IEEE 1366** for reliability-index inputs, plus the **ODIN** two-tier publication pattern, **MultiSpeak** detection flow and **GeoJSON (RFC 7946)**. A shipped crosswalk maps a real DISCOM's 31-code fault-reason pick-list onto the IEEE taxonomy.

How it fits together

Downstream of a two-layer pipeline: a thin **detection/ingest** layer (an AMISP-operated substation meter's digital-input signal, batched into the `FeederStatusIngest` API) and a **publication** layer — the same notice serves a `GET /outages.geojson` map feed, a CAP feed for alerting (with SACHET/NDMA Cell Broadcast as the intended wide-area channel), and per-connection lookup. A coarse public tier carries no PII; an authenticated tier adds affected connection points. No IES use-case guide exists yet.

Design note

- v0.1/OutageNotification_Design.md — full rationale: standards survey, OMS/CAP/CIM mapping tables, GIS readiness, enum provenance
- v0.1/FeederStatusIngest.openapi.yaml — feeder-status ingest API (detection layer)

Open points

- "RS" in the source OMS shorthand is interpreted as Roster Shutdown (restoration modelled as a status transition, not a class) — pending confirmation with the source utility.
- Several enums are deliberately local (`outageClass`, `status`, `assetLevel`, `consumerCategory`, `source`); whether to propose any for standardisation is open, and vendor feeder-status codes are not yet mapped to the IS 15959 event/OBIS list.
- The public-tier vs. authenticated-tier granularity of `impact.deEnergized[]` is an open privacy question.

Versions

Version	Status	Notes
v0.1	**Draft (WIP)**	Initial model: OutageNotification + sub-models; provenance links to MeterData AlarmProfile

v0.1

> **Files** — attributes.yaml · schema.json · context.jsonld · vocab.jsonld · examples/
>
> [warn] **WORK IN PROGRESS — subject to change.** This is an early draft (v0.1) published for discussion. Models, field names, enums, and the SD/BD/RS interpretation are not final and may change without notice before a stable release. Do not depend on it for production integrations yet.

Machine-readable schema for a distribution licensee (DISCOM) to publish **planned and unplanned electricity outage** details. One canonical object that is publishable as a **website / outage-map feed** (pull) and **pushable to affected or subscribed consumers** (push). Designed to be **GIS-ready** and **future-proof**.

Overview

`OutageNotification` describes a single outage (planned shutdown, breakdown, roster/load-shedding) with its cause, affected assets, area, timing, impact, field response, and public-facing text. Where an outage is auto-detected, `provenance` links back to the originating smart-meter signal and the IES `MeterData AlarmProfile`.

It composes proven standards by layer:

- **IEC 61968-3 (CIM)** — outage/UsagePoint data semantics, planned vs unplanned.
- **ODIN** (US DOE/ORNL) — publication pattern; coded + free-text cause.
- **OASIS CAP v1.2** (ITU-T X.1303) — push-alert envelope (`msgType`, `references`, `severity`, area + polygon, headline/instruction).
- **MultiSpeak** — AMI/MDMS -> OMS detection flow and provenance.
- **GeoJSON (RFC 7946)** — GIS geometry, WGS84, for outage maps.
- **IEEE 1366 / RDSS** — restoration time + customer counts for SAIDI/SAIFI.

Design rationale, the full standards survey, mapping tables (OMS/DISCOM/CAP/CIM), the GIS/outage-map readiness section, and the feeder-status ingest (detection) layer are in the design note alongside this schema: OutageNotification_Design.md (and the ingest stub FeederStatusIngest.openapi.yaml).

Enum provenance (borrowed vs local)

Each enum declares its origin in its `description` (and a machine-readable `$comment`); enums borrowed wholesale from a standard are also semantically anchored in `context.jsonld` via their term IRI.

Enum	Origin
`msgType`	**Borrowed** — OASIS CAP v1.2 `alert/msgType` (anchored `urn:oasis:names:tc:emergency:cap:1.2#msgType`)
`severity`	**Borrowed** — OASIS CAP v1.2 `info/severity` (anchored `urn:oasis:names:tc:emergency:cap:1.2#severity`)
`cause.category` / `cause.subcategory`	**Aligned** — IEEE 1782-2022 §4.4 (ten cause categories) and §4.5 (subcategories), used with IEEE 1366
`Identifier.scheme`	**Mixed** — `MRID` from CIM (IEC 61968/61970), `DID` from W3C DID Core; remaining values local
`feederStatus`	**Local** normalization; energized/de-energized align with CIM UsagePoint semantics
`outageClass`	**Local** — DISCOM OMS “Down Info” value set

Enum	Origin
`status`, `category`, `assetLevel`, `consumerCategory`, `source`	**Local** — not from a published standard

Note: `category` (outage category) is **not** CAP's `info/category` — it is a local value set with a deliberately different membership.

Models

Model	Purpose
`OutageNotification`	Root — class, status, cause, assets, area, timing, impact, response, public info, provenance
`OutageCause`	IEEE 1782 cause category + subcategory, vendor code (verbatim), free-text reason
`OutageAsset`	Affected asset (feeder/DT/substation/line) with meter ref + optional geometry
`OutageNetworkContext`	DISCOM org hierarchy (Discom > Zone > Circle > Division > Subdivision > Substation)
`OutageAffectedArea`	CAP area — text + admin areas + GeoJSON geometry
`OutageImpact`	Customers affected + de-/energized UsagePoints (CIM)
`OutageTiming`	Window (TimePeriod), ETR, actual restoration, SLA target
`OutageResponse`	Back-feed, resource status, complaint count
`OutagePublicInfo`	CAP info block for web/push
`OutageProvenance`	Source, AMISP code, raw signal, alarm refs into MeterData

---

Directory Structure and Files

File	Description
`attributes.yaml`	Canonical OpenAPI 3.1.0 source of truth describing all models, attributes, and types.
`schema.json`	Compiled Draft 2020-12 JSON Schema for payload validation.
`context.jsonld`	Compiled JSON-LD context mapping attributes to `ies:` terms.
`vocab.jsonld`	Compiled JSON-LD vocabulary (classes and properties).
`examples/`	JSON-LD example payloads (planned multi-feeder shutdown; unplanned breakdown with provenance; restoration UPDATE).
`OutageNotification_Design.md`	Design note — standards survey, mapping tables, GIS readiness, enum provenance.
`FeederStatusIngest.openapi.yaml`	Real-time feeder-status ingest API (detection layer).
`fault_reason_crosswalk.json`	DISCOM FAULT_REASON master (FORM_ID 8312) -> `cause.category`/ `cause.subcategory` (IEEE 1782 §4.4/§4.5) crosswalk.
`OutageNotification_ Implementation.md`	Step-by-step guide to build a production pull/push outage-notification system.
`tools/`	DISCOM planned-shutdown CSV + transformer script (CSV -> OutageNotification JSON).

Schema Generation and Validation

`schema.json`, `context.jsonld`, and `vocab.jsonld` are **compiled** from `attributes.yaml` — do not edit them by hand.

```
# from this directory
make            # generate + validate
# or, from the repo root:
python3 scripts/generate_schema_permissive.py schemas/OutageNotification/v0.1
python3 scripts/validate_schema.py schemas/OutageNotification/v0.1/schema.json
 ↪  schemas/OutageNotification/v0.1/examples
```

Field reference

*Auto-generated from `schema.json`. A field name in **bold** with a trailing ***** is required; all others are optional. **Type** shows units for QuantitativeValue models. Where a field is derived from a standard, its description begins with **Based on** and the standard reference (from the field's `x-standard` annotation). One table per object.*

OutageNotification Payload

Field	Type	Description
`objectType` *	`OUTAGE_NOTIFICATION`	—
`id` *	Identifier	Stable notice identity; reused across UPDATE/CANCEL messages.
`outageClass` *	`PLANNED` / `BREAKDOWN` / `SCHEDULED_ROSTERING` / `EMERGENCY_ROSTERING`	Outage class, from the DISCOM OMS "Down Info" set (local, additive enum). Rostering = rotational load-shedding (scheduled vs emergency). Restoration is a status transition (status=RESTORED), not a class.
`status` *	`SCHEDULED` / `ACTIVE` / `PARTIALLY_RESTORED` / `RESTORED` / `CANCELLED`	Outage lifecycle state. LOCAL enum (not from a published standard); loosely aligned with the CIM Outage lifecycle.
`msgType`	`ALERT` / `UPDATE` / `CANCEL`	**Based on** OASIS CAP v1.2 (alert/msgType). Message type; UPDATE/CANCEL refer to a prior notice via `references`.
`references`	list of Identifier	**Based on** OASIS CAP v1.2 (alert/references). Prior notice ids this message updates or cancels (CAP `references`).
`severity`	`EXTREME` / `SEVERE` / `MODERATE` / `MINOR` / `UNKNOWN`	**Based on** OASIS CAP v1.2 (info/severity). Severity of the outage.
`category`	`MAINTENANCE` / `UPGRADE` / `FAULT` / `LOAD_SHEDDING` / `WEATHER` / `SAFETY` / `OTHER`	Coarse descriptor for display/filtering (local enum; not CAP `info/category`). For analytics, prefer the standardized `cause.category`.
`cause`	OutageCause	—
`forceMajeure`	yes / no	OMS "Force Majeure" flag.
`issuedBy`	Party	—
`issuedAt`	date-time	—
`detectedAt`	date-time	When MDMS/SCADA first detected the outage (unplanned).
`lastUpdatedAt`	date-time	—
`network`	OutageNetworkContext	—
`affectedAssets` *	list of OutageAsset	—
`affectedArea`	OutageAffectedArea	—
`impact`	OutageImpact	—
`timing` *	OutageTiming	—
`response`	OutageResponse	—
`publicInfo`	OutagePublicInfo	—
`provenance`	OutageProvenance	—
`extensions`	object	Namespaced DISCOM-specific fields, e.g. { "discom": { "breakdownId": "6357257", "downType": "FEEDER" } }.

OutageCause

Field	Type	Description
`category`	`EQUIPMENT` / `LIGHTNING` / `PLANNED` / `POWER_SUPPLY` / `PUBLIC` / `VEGETATION` / `WEATHER` / `WILDLIFE` / `UNKNOWN` / `OTHER`	**Based on** IEEE 1782-2022 §4.4 (with IEEE 1366). Standardized interruption cause category, for reliability benchmarking. Load-shedding/rostering has no standard category — map scheduled rostering to PLANNED, emergency to OTHER, and carry the nature in `outageClass`.
`subcategory`	`DEGRADATION` / `EQUIPMENT_ERROR` / `ENVIRONMENTAL` / `DIRECT_STRIKE` / `INDIRECT_STRIKE` / `NEW_CONSTRUCTION` / `MAINTENANCE` / `CUSTOMER_REQUEST` / `OTHER_UTILITY_REQUEST` / `GENERATION` / `TRANSMISSION` / `SUBTRANSMISSION` / `DISTRIBUTION` / `DISTRIBUTED_GENERATION_STORAGE` / `OTHER_UTILITY_SUPPLY` / `DIG_IN` / `FOREIGN_CONTACT` / `FIRE_POLICE` / `VEHICLE` / `WITHIN_CLEARANCE_ZONE` / `OUTSIDE_CLEARANCE_ZONE` / `PRECIPITATION` / `ICE` / `WIND` / `EXTREME_TEMPERATURE` / `MAMMAL` / `BIRD` / `REPTILE_AMPHIBIAN` / `NO_SPECIFIC_CAUSE_FOUND` / `UTILITY_ERROR` / `OTHER_UTILITY_INITIATED` / `OTHER`	**Based on** IEEE 1782-2022 §4.5. Standardized cause subcategory; must be valid for the chosen `category` (see $comment for the mapping). Leave unset if the field finding is inconclusive.
`faultType`	text	Vendor asset/voltage level of the fault, e.g. 33KV, 11KV, DT, LT.
`code`	text	Vendor/OMS fault-reason code, carried verbatim (e.g. a DISCOM FAULT_REASON) — not standardized; map to `category/subcategory` for interoperability (see fault_reason_crosswalk.json). Set `codeNamespace` where the code's authority matters.

Field	Type	Description
codeNamespace	text	Authority/vendor that defines code (e.g. the OMS vendor or DISCOM).
text	text	Free-text reason; may be localized (e.g. Hindi).

OutageAsset

Field	Type	Description
id *	Identifier	Asset id/name; ideally a stable GIS feature id or MRID for map join.
assetLevel *	SUBSTATION / FEEDER / DT / LINE_SEGMENT / SERVICE_POINT	Network level of the affected asset (local enum; aligns with CIM Equipment/ UsagePoint).
voltageLevel	text	e.g. 33kV, 11kV, LT, DT.
consumerCategory	URBAN / RURAL / AGRICULTURE / INDUSTRIAL / MIXED / OTHER	Consumer mix on the asset (local enum; DISCOM "Feeder Type").
meterRef	Identifier	Feeder/substation smart-meter number — join key to MDMS/MeterData.
parentRef	Identifier	Parent asset (e.g. a feeder's substation, a DT's feeder).
geo	GeoJSONGeometry	**Based on** GeoJSON (RFC 7946), WGS84. Optional inline geometry (WGS84): Point (substation/DT), LineString (feeder route), Polygon (service area).

OutageNetworkContext

Field	Type	Description
discom	Identifier	—
zone	text	—
circle	text	—
division	text	—
subdivision	text	—
substation	Identifier	—
district	text	—

OutageAffectedArea

Field	Type	Description
text	text	Free-text localities (CAP areaDesc), e.g. "SEC-14, 9, 11 Rajnagar".
adminAreas	list of text	Named localities / wards / villages.
geo	GeoJSONGeometry	**Based on** GeoJSON (RFC 7946); CAP area. Polygon/MultiPolygon service area, or Point/circle (CAP). WGS84.

OutageImpact

Field	Type	Description
customersAffected	integer	—
deEnergized	list of Identifier	**Based on** CIM (IEC 61968-3 UsagePoint). Optional SDP/UsagePoint refs (authenticated tier).
energized	list of Identifier	**Based on** CIM (IEC 61968-3 UsagePoint). Optional points confirmed still energized.

OutageTiming

Field	Type	Description
period *	TimePeriod	Outage window (Down From + duration).
estimatedRestoration	date-time	ETR (OMS "Estimated Time").
actualRestoration	date-time	Set when status=RESTORED; feeds IEEE 1366 indices.
slaTargetMinutes	integer	SLA target, e.g. 240 (4 hrs).

OutageResponse

Field	Type	Description
backFeedGiven	yes / no	Partial restoration via alternate feed (OMS "Back-Feed given").
resourceStatus	text	e.g. PENDING, RESOURCE_ALLOCATED, IN_PROGRESS.
complaintCount	integer	Linked consumer complaints (OMS "No. of Complaints").

OutagePublicInfo

Field	Type	Description
language	text	BCP-47, e.g. en, hi.
headline	text	—
description	text	—
instruction	text	What the consumer should do.

OutageProvenance

Field	Type	Description
source	MDMS / OMS / SCADA / RTDAS / AMI / MANUAL	System that raised this notice (local enum; RTDAS = Real-Time Data Acquisition System).
amispCode	text	AMI Service Provider that supplied the detection signal (feeder-status ingest API).
detectionRef	Identifier	Reference to the real-time detection record that raised the outage (e.g. a DISCOM RTDAS_DATA_ID). Absence or a "0" sentinel indicates a manually entered outage (source=MANUAL).
signal	OutageSignal	—
alarmRefs	list of OutageAlarmRef	References into MeterData AlarmProfile records.

OutageSignal

Field	Type	Description
eventId	text	Idempotency key of the originating feeder-status event.
feederStatus	ENERGIZED / DE_ENERGIZED / UNKNOWN	Normalized feeder status (local enum); raw vendor code in rawCode. Energized/de-energized align with CIM UsagePoint semantics.
diPort	text	Digital-input port on the substation meter that carried the signal.
rawCode	text	Vendor-native status code as received (e.g. FEEDER_STATUS=102), for traceability.

OutageAlarmRef

Field	Type	Description
meterRef *	Identifier	—
alarmId	integer	—
timestamp	date-time	—

Party

Field	Type	Description
id	Identifier	—
name	text	—
contact	text	Phone/email/URL for queries (e.g. 1912).

Identifier

Field	Type	Description
scheme *	METER_SERIAL / MRID / GIS_FEATURE_ID / SERVICE_DELIVERY_POINT / FEEDER / SUBSTATION / DT / CONSUMER_NUMBER / ORG / DID / OTHER	**Based on** MRID: CIM (IEC 61968/61970); DID: W3C DID Core. Identifier scheme (mixed provenance). MRID and DID are borrowed from their standards; the rest are local vendor/DISCOM master keys.
value *	text	—
namespace	text	—

TimePeriod

Field	Type	Description
start *	date-time	—
duration *	duration	ISO-8601, e.g. PT2H.

External Schemas

Some schemas used across IES are **defined and published outside this repository** — served canonically at schema.nfh.global — and documented here so the book carries a complete field reference. Each schema links to its canonical definition; each domain links to its use-case guide and deployable devkit.

These tables are **auto-generated** from the schema sources by `scripts/generate_external_field_tables.py` (version `v2.0`). Don't edit by hand — re-run the generator to refresh. Where a field derives from a recognised standard, its description begins with **Based on** and the standard reference (from the source's `x-standard` annotation).

Energy Trading (P2P)

Use case · Devkit

A peer-to-peer trade is modelled as a `P2PTrade` contract (a beckn `Contract` subclass) whose energy-specific attributes are carried by the offer, customer, order-item, settlement and resource schemas below. `EnergyResource`, `DEGContract`, `RevenueFlow` and `BecknTimeSeries` are shared primitives — Demand Flexibility reuses them.

P2PTrade

Defined at `P2PTrade/v2.0` — P2PTrade.

*Inherits all fields from **EnergyContract**.*

EnergyTradeOffer

Defined at `EnergyTradeOffer/v2.0` — EnergyTradeOffer — Offer Attributes (v2.0).

Field	Type	Description
`validityWindow`	TimePeriod	Time window during which this offer can be selected/accepted. Typically set to expire before the earliest delivery starts. Present in catalog publish; may be omitted in downstream messages.
`contractAttributes`	object	JSON-LD container for contract terms attached at catalog publish time. Only roles with a non-null participantId are included (unknown parties are omitted at publish time). MUST carry @context and @type (typically DEGContract). Present in catalog publish; omitted in downstream messages where Contract.contractAttributes carries the full party list.
`commitmentAttributes`	object	Seller's BecknTimeSeries published at catalog time. Declares the full schema of the contract table via payloadDescriptors, each annotated with `insertedBy` (the role responsible for populating that column). Carries the seller's initial interval data (PRICE_PER_KWH + AVAILABLE_QTY per slot). Immutable after publish — repeated verbatim in all downstream messages as the authoritative offer definition. MUST carry @context (BecknTimeSeries/v1.0/context.jsonld) and @type: TimeSeries. Standard payloadDescriptors and their insertedBy: PRICE_PER_KWH (EVENT) — insertedBy: seller AVAILABLE_QTY (EVENT) — insertedBy: seller REQUESTED_QTY (EVENT) — insertedBy: buyer BUYER_DISCOM_ALLOC (REPORT) — insertedBy: buyerDiscom BUYER_DISCOM_STATUS (REPORT) — insertedBy: buyerDiscom SELLER_DISCOM_ALLOC (REPORT) — insertedBy: sellerDiscom SELLER_DISCOM_STATUS (REPORT) — insertedBy: sellerDiscom FINAL_ALLOC (REPORT) — insertedBy: sellerDiscom

Object: `EnergyTradeOffer.contractAttributes`

Field	Type	Description
`@type` *	text	—

Object: `EnergyTradeOffer.commitmentAttributes`

Field	Type	Description
`@type` *	`TimeSeries`	—

EnergyCustomer

Defined at `EnergyCustomer/v2.0` — EnergyCustomer — Customer Attributes (v2.0).

Field	Type	Description
meterId *	text	Meter identifier in DER address format (der://meter/{id}). Used for customer identification and energy delivery tracking in P2P trading flows.
sanctionedLoad	number	Sanctioned load capacity in kilowatts (kW). Optional field representing the approved electrical load capacity for the customer's connection. Used for load management and regulatory compliance.
utilityCustomerId	text	Customer's account identifier with the utility company (e.g., PG&E customer number). Used for billing, service identification, and utility system integration.
utilityId	text	Utility/DISCOM identifier for the customer's service territory (e.g., "TPDDL-DL", "BESCOM-KA"). Used in inter-utility / inter-DISCOM P2P trading to identify which utility serves this customer.
platformUrl	uri	Base URL (scheme + host[:port]) of the Beckn trading platform that represents this customer on the network. The platform exposes the standard Beckn paths under this base: `/bap/caller`, `/bap/receiver`, `/bpp/caller`, `/bpp/receiver`. Used by downstream nodes to address this customer's platform for cascade sub-transactions — e.g. a seller-side adapter cascading a /status query into its own discom ledger and needing the discom's response routed back to its own /bap/receiver. The path appended depends on the action's BAP<->BPP direction (/bap/receiver for on_*, /bpp/receiver for request actions).

EnergyOrderItem

Defined at `EnergyOrderItem/v2.0` — EnergyOrderItem.

Field	Type	Description
providerAttributes *	object	Provider/customer information for this order item, including meter ID and utility account.
fulfillment Attributes	object	Optional fulfillment tracking for this order item. Only populated in on_status and on_update responses (not in init/confirm flows). Contains delivery status, meter readings, and energy allocation data.

Object: `EnergyOrderItem.providerAttributes`

Field	Type	Description
@type	EnergyCustomer	Type identifier for provider attributes

Object: `EnergyOrderItem.fulfillmentAttributes`

Field	Type	Description
@type	EnergyTradeDelivery	Type identifier for fulfillment attributes

RevenueFlow

Defined at `RevenueFlow/v2.0` — RevenueFlow — Consideration Attributes (v2.0).

Field	Type	Description
revenueFlows *	list of object	Per-role signed revenue-flow entries computed by the contract policy after settlement. Sum MUST be zero across the array.

Object: `RevenueFlow.revenueFlows[]`

Field	Type	Description
role *	text	—
value *	number	Signed amount. Positive = role receives; negative = role pays.
currency *	text	**Based on** ISO 4217. ISO-4217 currency code.
description	text	Human-readable rationale for the flow.

DEGContract

Defined at `DEGContract/v2.0` — DEG — Contract (v2.0).

Field	Type	Description
roles *	list of object	Contract roles. The participantId key is required on every role entry, but its value MAY be null in the catalog/offer (buyer and buyerDiscom unknown until select/init) and is bound to a concrete participant id at select/init. Identity attributes (meter, utility, utilityCustomerId) for each bound role live at Contract.participants[*].participantAttributes.
policy *	object	OPA/Rego policy governing this contract.
revenueFlows	list of object	Optional. Settlement-time per-role flows written by the `contractpolicyenforcer` plugin when it is configured with outputMode: raw and outputPath ending at this property (the legacy demand-flex shape). Wave2 P2P trading instead uses Contract.consideration[*].considerationAttributes (RevenueFlow JSON-LD) and leaves this array unset.

Object: `DEGContract.roles[]`

Field	Type	Description
role *	text	—
participantId *	text (nullable)	Concrete participant id once bound (buyerPlatform/sellerPlatform use the utilityId-meter compound id, e.g. "TPDDL-DL-seller-001"; discom roles use the discom-ledger participant id). Null at catalog publish for sides not yet bound.

Object: `DEGContract.policy`

Field	Type	Description
url *	uri	—
queryPath *	text	—

Object: `DEGContract.revenueFlows[]`

Field	Type	Description
role *	text	—
value *	number	—
currency *	text	**Based on** ISO 4217.
description	text	—

EnergyResource

Defined at `EnergyResource/v2.0` — EnergyResource — Resource Attributes (v2.0).

Discriminated union of all typed EnergyResource kinds. Dispatched by the 'type' field. Each kind lives in its own schema at schema.nfh.global//v1.0 and is referenced above. EnergyResourceCommon (id, type, subResources, parentResources, attributes) and EnergyResourceCommonAttributes (make, model, maxExportKw, maxImportKw, ratedPowerKw, telemetryProvider, commissioningDate, location) are defined in EnergyResourceCommon/v1.0 and inherited by all kinds via allOf external $ref. For P2P-trading: minimal usage is {id, type}. For demand-flex and credentials: add attributes fields as needed. For targeted validation: $ref the specific kind directly.

Discriminated union (`oneOf`) of: `EnergyResourceMeter`, `EnergyResourceGenerator`, `EnergyResourceStorage`, `EnergyResourceEVCharger`, `EnergyResourceInverter`, `EnergyResourceLoad`, `EnergyResourceNetwork`.

BecknTimeSeries

Defined at `BecknTimeSeries/v1.0` — BecknTimeSeries — Time-Series Payload (v1.0).

*Based on **OpenADR 3.1.0**.*

A time-series payload. `intervalPeriod` sets the default temporal bounds; each `interval` carries one or more typed `payloads` (valuesMap rows). `payloadDescriptors` declare units/currency/ reading-type for each `payloadType` referenced in the rows; every type used in `intervals` SHOULD be declared in `payloadDescriptors`. `payloadType` follows OpenADR's open-string convention — consumer profiles (e.g. `DemandFlexPerformance`) MAY restrict it to a domain-specific set and add cross-field membership checks.

Field	Type	Description
`@type`	`TimeSeries`	Optional JSON-LD type discriminator. Present when the embedder wants to surface the type in the payload; the parent schema's `$ref` to BecknTimeSeries is what drives validation.
intervalPeriod *	intervalPeriod	Default temporal bounds of the series — ISO 8601 `start` and `duration`. An individual interval may override these.
payloadDescriptors *	list of eventPayloadDescriptor or reportPayloadDescriptor	Sidecar describing each `payloadType` carried by the rows (units, currency, reading type, accuracy/confidence). Every `intervals[*].payloads[*].type` SHOULD appear here; consumer profiles enforce that as a cross-field constraint. Each descriptor must be either an OpenADR3 eventPayloadDescriptor (objectType: EVENT_PAYLOAD_DESCRIPTOR — for offer/bid signals) or a reportPayloadDescriptor (objectType: REPORT_PAYLOAD_DESCRIPTOR — for telemetry/meter readings). The objectType field is the discriminator.

Field	Type	Description
`intervals` *	list of interval	Series rows. Each row carries an integer `id` and a list of typed `payloads` (valuesMap). Per-row `intervalPeriod` is optional and overrides the default.
`resourceName`	text	Identifies the single resource these readings are about — the data source. Mirrors OpenADR 3.1.0's `Report.resources[].resourceName`. Use when the series captures telemetry from a specific physical device or asset; in energy domains this is typically the meter id. Omit (or set to `"0"`, following OpenADR convention) for aggregate / unattributed series. Optional, so existing payloads continue to validate without change.
`clientName`	text	Identifies the reporting client — the party authoring this time-series and pushing it onto the wire. Mirrors OpenADR 3.1.0's `Report.clientName` (VEN name). In energy domains this is typically the utility / DISCOM / aggregator subscriber id. Optional.

Demand Flexibility

Guide · Devkit

A flexibility programme advertises a `DemandFlexNeed`, contracts via a `DemandFlexBuyOffer`, and reports measurement & verification with `DemandFlexPerformance` (per-meter telemetry as a `BecknTimeSeries`). It reuses the shared `EnergyResource`, `DEGContract`, `RevenueFlow` and `BecknTimeSeries` schemas listed under Energy Trading.

DemandFlexNeed

Defined at `DemandFlexNeed/v2.0` — Demand Flex — Need Attributes (v2.0).

*Based on **OpenADR 3.1.0 (event time series)**.*

Buyer-authored demand-flex procurement schedule as an OpenADR-aligned time series — one interval per tranche. This is the single, unified Need shape for every demand-flex market, from full price discovery (uc2 pay-as-clear auction: many aggregators bid, market clears) down to its monopsony degenerate (uc1: a single buyer, the DISCOM, fixes one clearing price for any quantity — a flat, self-cleared curve). The column set is intentionally NOT fixed by this schema. Each market profile carries different columns, and the governing CONTRACT POLICY REGO imposes the exact required set as a hard const: uc1 (deg.contracts.demand_flex) — CAPACITY_REQUESTED, PRICE, SHORTFALL_PENALTY uc2 (deg.contracts.demand_flex_pac) — CAPACITY_REQUESTED

Field	Type	Description
`@type`	`DemandFlexNeed`	JSON-LD type discriminator.
`location`	Location	**Based on** GeoJSON (RFC 7946). Geographic area where flex is needed. Beckn Location — GeoJSON geometry (geo) plus optional address.
`intervalPeriod` *	intervalPeriod	Default temporal bounds of the schedule — ISO 8601 `start` and `duration` (e.g. PT30M). Interval `id` 0..n-1 index sequential tranches off this grid. This exact grid MUST be shared by the seller's CAPACITY_OFFERED series and by every meter's telemetry.
`payloadDescriptors` *	list of eventPayloadDescriptor	Buyer-authored column descriptors (OpenADR eventPayloadDescriptor). The set is profile-specific and NOT fixed here — the governing contract policy rego imposes the exact required columns as a hard const (uc1: CAPACITY_REQUESTED / PRICE / SHORTFALL_PENALTY; uc2: CAPACITY_REQUESTED). Downstream series extend the grid with CAPACITY_OFFERED (seller, on commitmentAttributes) and BASELINE / USAGE (meter telemetry).
`intervals` *	list of interval	One row per tranche. Each row carries integer `id` and typed `payloads` — CAPACITY_REQUESTED, PRICE, SHORTFALL_PENALTY.

DemandFlexBuyOffer

Defined at `DemandFlexBuyOffer/v2.0` — Demand Flex — Buy Offer Attributes (v2.0).

DemandFlexBuyOffer

Field	Type	Description
`contractAttributes`	object	Portable DEGContract template. Present only in catalog/discover. Promoted to Contract.contractAttributes at select/init.
`inputs` *	list of DemandFlexRoleInput	One entry per role. participantId is null until the role is bound (e.g. seller is null at catalog, filled at init). inputs object is optional when participantId is null, required when participantId is set.

DemandFlexRoleInput

Field	Type	Description
`role` *	`buyer` / `seller`	—
`participantId` *	text (nullable)	Participant bound to this role. Null until bound.

Field	Type	Description
`inputs`	object	Role-specific scalar inputs. Per-slot commercial terms are NOT here — they ride the DemandFlexNeed time series (buyer columns CAPACITY_REQUESTED / PRICE / SHORTFALL_PENALTY) and the commitment series (seller column CAPACITY_OFFERED); these inputs carry only what does not vary by interval. For buyer: currency and baselineMethodology (a negative PRICE column pays for increased consumption, so there is no scalar incentive/penalty field). For seller: participatingMeters (or participatingMetersRef when the cohort is bulk), energyResources (or energyResourcesRef) — EnergyResource objects each linked to a participatingMeters[*] entry via meterId — and reportDescriptors (see below). The seller's per-slot CAPACITY_OFFERED is added as a column on Commitment.commitmentAttributes at confirm, not here. Bulk delivery: the offer is bound at confirm and cannot span Beckn messages, so when the seller's cohort would stretch a single confirm payload beyond a reasonable wire budget (e.g. > 10k meters) the *Ref siblings replace the inline arrays. participatingMetersRef / energyResourcesRef each carry a BecknResourceRef pointing at a content-addressed BPP-hosted bundle. participatingMetersDigest is an on-protocol tamper-evidence anchor that survives even if the ref URL later goes 404 — Beckn signing of the confirm payload commits the contract to that exact cohort.

Object: `DemandFlexRoleInput.inputs`

Field	Type	Description
`reportDescriptors`	BecknReportDescriptors (nullable)	Seller-side declaration of what telemetry will be delivered for this contract. Set to `null` when the contract requires no telemetry. When non-null, the seller commits to delivering a BecknTimeSeries (per device, in performance.meters[]. telemetry on the on_status reply) whose payloadDescriptors cover every entry in this array. See BecknReportDescriptors for the canonical payloadType / units / cardinality table (USAGE, POWER, SOC_END, GPS_LAT, GPS_LON, …).
`participatingMetersRef`	BecknResourceRef	Optional. Off-protocol delivery of the participatingMeters cohort when it's too large to inline. Mutually exclusive with `participatingMeters`. Consumers MUST treat the fetched body as the authoritative meter list once the receiver has verified `contentHash` and `count` against the body's canonicalized form.
`energyResources`	list of EnergyResource	Optional. EnergyResources enrolled in this contract. Each entry is an EnergyResource object whose `meterId` field references one of the meter URIs in `participatingMeters[*]` (many resources MAY share a meter). Identity and rated dimensioning live here; per-event state lives in BecknTimeSeries telemetry on the performance record's meters[]. EnergyResource is the canonical, technology-neutral asset class — EV chargers, batteries, solar PV, smart HVAC, etc. — shared with P2P-trading and any future DEG domain.
`energyResourcesRef`	BecknResourceRef	Optional. Off-protocol delivery of the energyResources cohort when it's too large to inline. Mutually exclusive with `energyResources`. Same verification rules as `participatingMetersRef`.
`participatingMetersDigest`	object	Optional on-protocol tamper-evidence anchor for the meter cohort, regardless of whether it was delivered inline (`participatingMeters`) or by reference (`participatingMetersRef`). Computed as SHA-256 of the ASCII-sorted, newline-joined meter IDs. Beckn-signing of the confirm payload then commits the seller to this exact cohort permanently — useful for downstream audit long after any off-protocol bundle URL has expired.

Object: `DemandFlexRoleInput.inputs.participatingMetersDigest`

Field	Type	Description
`count` *	integer	Total meter count in the cohort.
`sha256OfSortedIds` *	text	**Based on** SHA-256 (FIPS 180-4). `sha256:<64-hex>` of the cohort's meter IDs after they have been ASCII-sorted and joined with a single `\n` separator (no trailing newline).

DemandFlexPerformance

Defined at `DemandFlexPerformance/v2.0` *— Demand Flex — Performance Attributes (v2.0).*

Performance attributes for demand-flex M&V (Measurement & Verification). Attached to Performance.performanceAttributes in on_status callbacks.

Field	Type	Description
`eventId`	text	Identifier of the flex event being measured.
`methodology`	text	Baseline methodology used across all meters (e.g., "5of10" means average of 5 highest-consumption days out of last 10).

Field	Type	Description
`meters`	list of object	Per-meter M&V data. Each entry binds a meter ID to its time-series telemetry for the event window. When the cohort fits in a single message the BPP SHOULD inline the full array and omit `pageInfo` entirely. For larger cohorts the BPP either ships paged slices (sibling `pageInfo` present) or substitutes `metersRef` and omits `meters[]` from this message. Settlement rego runs only on the assembled view — receivers MUST NOT fire settlement-dependent actions until `pageInfo.isLast == true` or the `metersRef` body has been fetched and verified.
`pageInfo`	BecknPageInfo	Optional. Present only when the `meters[]` collection has been split across multiple on_status messages (paged inline delivery). Receivers assemble all pages of the same (transactionId, performance.id) by `pageInfo.sequence` and defer settlement until `pageInfo.isLast == true`. Omit entirely when the whole cohort fits in one message — absence of `pageInfo` is the signal that this message is self-contained.
`metersRef`	BecknResourceRef	Optional. Off-protocol delivery of the per-meter telemetry bundle when even paged-inline (`pageInfo`) is too heavy. Mutually exclusive with `meters[]` and `pageInfo` on the same message — the BPP substitutes this reference for the entire collection. Receivers fetch `uri`, verify against `contentHash` and `count`, then run settlement against the fetched body.

Object: `DemandFlexPerformance.meters[]`

Field	Type	Description
`meterId` *	text	Meter identifier.
`telemetry` *	BecknTimeSeries	BecknTimeSeries carrying BASELINE (always) and — once the event has completed — USAGE payloads, aligned to the same interval grid. Validated inline via `$ref`, so the embedded payload only needs `@type` (optional) — `@context` is declared once at the envelope level via `context.schemaContext[]`.